1 Appendix A: Barcode Metric, Threshold-Gap Repair, and Repaired-Zero Transport

1.1 Scope and relation to Chapter 2

Appendix A is the detailed persistence-theoretic Foundation Appendix for Chapter 2 in the AK-HPDST v20.0.0 *Finite Proof Compression, Higher Obstruction Bridges, and Problem Demonstration Release*. The legacy transposed acronym AK-HDPST may occur in historical artifacts, but it has no normative use in this appendix.

This appendix preserves the complete repaired v19 metric and crop calculus at exactly its proved strength and supplies the additional mathematical and audit-facing components required by the v20 finite-certificate architecture. It does not enlarge the Core beyond the Chapter 2 contract and does not create an external-domain theorem.

The inherited metric repair is the following.

> For a positive threshold $\tau > 0$, hard barcode threshold deletion T_τ^{bar} is neither a global exact Serre localization nor a globally non-expansive operation under the standard bottleneck metric. Metric transport through hard deletion is licensed only by the sharp two-sided threshold-gap-safe theorem or by a separately proved exact comparison at its declared strength.

The v20 conservative extension is the following.

> A bottleneck bound between repaired profiles does not preserve arbitrary pointwise detector values. It preserves the exact zero/nonzero predicate only inside the τ-reduced image and under the sharp post-threshold separation condition $2\varepsilon \le \tau$. The operational v20 contract uses the strict reserve $2\varepsilon < \tau$.

The inherited crop repair is also retained.

> Window restriction and hard threshold deletion commute exactly when every globally retained bar, including every infinite bar, is either disjoint from the window or remains retained after clipping. A criterion that checks only finite globally retained bars is false on bounded windows.

Accordingly, this appendix establishes and records:

(i) constructible one-parameter persistence and barcode normal forms over a field;

(ii) the standard finite/infinite bottleneck convention and interleaving isometry;

(iii) failure of the positive-threshold Serre property;

(iv) failure of global non-expansiveness for hard threshold deletion;

(v) sharp two-sided threshold-gap-safe stability with closed band $[\tau - \varepsilon, \tau + \varepsilon]$;

(vi) the identity $\mathrm{gap}_\tau = \mathrm{clear}_\tau$, with no factor $1/2$;

(vii) typed scalarization of a quantale-valued metric ledger, together with evidence that the scalarized value bounds the actual bottleneck discrepancy consumed by the proof;

(viii) the τ-reduced image of hard deletion and the exact distance-to-zero formula;

(ix) sharp zero separation for finite τ-reduced profiles at $2\varepsilon \leq \tau$;

(x) metric-safe repaired-zero transport and its strict scalarized operational form;

(xi) detector-certified zero transport, restricted to the already certified zero predicate;

(xii) the explicit failure of bottleneck stability for raw pointwise detector ranks;

(xiii) the fixed order window $\to$ pre-threshold profile $\to$ hard deletion;

(xiv) the distinction between exact comparison, metric threshold-gap-certified comparison, and post-threshold zero-separation;

(xv) the infinite-bar obstruction to the former finite-only commutation condition;

(xvi) an exact necessary-and-sufficient crop–collapse commutation criterion;

(xvii) conditional functoriality and exactness only in declared τ-split or exact-admissible regimes;

(xviii) actual-morphism, terminal-tameness, and monitored-scope requirements for Type IV diagnostics;

(xix) the boundary that object-distance control does not automatically control kernel or cokernel defect profiles;

(xx) the distinction between manifest completeness and mathematical certification.

Remark 1.1 (Conservative extension relative to v18 and v19)**.** Earlier v18 formulations correctly rejected the overstrong Serre-localization interpretation of positive-threshold deletion, but still retained language suggesting global barcode non-expansiveness. The v19 repair removed that overclaim and replaced it by Theorem 1.33. The v20 additions below are consequences on the image of hard deletion. They do not make hard deletion globally non-expansive, exact, functorial, or detector-rank stable.

Division of responsibility. Appendix A proves metric transport and post-threshold zero separation. It does not prove detector soundness, completeness, birth or germ exhaustiveness, stencil coverage, family membership, MinCert bounds, or finite-to-infinite reduction. Those are separate Chapter 3 and downstream Foundation obligations. A zero-looking finite observation therefore remains undefined until the governing detector certificate is complete.

Remark 1.2 (What this appendix does not do)**.** This appendix is a Foundation Appendix for the persistence-theoretic Core. It does not prove a higher Ext-edge, an arithmetic, PDE, geometric, categorical, or cryptographic theorem, or an external Return Theorem. Any external-domain conclusion requires a separately proved and registered Problem-Bridge chain.

1.2 Constructible one-parameter persistence

Fix a field k. Let

$$\mathsf{Pers}_k := [(\mathbb{R}, \leq), \mathsf{Vect}_k]$$

be the category of one-parameter persistence modules over k, indexed by the ordered real line.

Definition 1.3 (Constructible persistence module)**.** A persistence module $M \in \mathsf{Pers}_k$ is called *constructible* if:

(i) $M(t)$ is finite-dimensional over k for every $t \in \mathbb{R}$;

(ii) on every bounded interval $J \subset \mathbb{R}$, there is a finite critical set outside which all structure maps are isomorphisms;

(iii) there is no accumulation of critical parameters inside a bounded interval.

The full subcategory is denoted by

$$\mathsf{Pers}_k^{\mathrm{cons}} \subset \mathsf{Pers}_k.$$

Remark 1.4 (Working regime)**.** All Core statements in this appendix are made in the pointwise finite-dimensional constructible one-parameter regime over k, or for finite barcode profiles obtained from declared windowed Core evaluations. Wild, nonconstructible, infinite-rank, or uncontrolled multiparameter data are outside scope unless an Interface theorem reduces them to declared constructible one-parameter profiles.

Definition 1.5 (Finite-type filtered chain complex)**.** Let $\mathsf{Ch}^b(k)$ be the category of bounded chain complexes of finite-dimensional k-vector spaces. A *finite-type filtered chain complex* is a pair

$$F = (C_\bullet, \{F^t C_\bullet\}_{t\in\mathbb{R}})$$

such that:

(i) $F^t C_\bullet \subseteq F^{t'} C_\bullet$ whenever $t \le t'$;

(ii) the filtration is exhaustive and left-bounded;

(iii) for every degree i, the persistence module

$$\mathbf{P}_i(F) : \mathbb{R} \longrightarrow \mathsf{Vect}_k, \qquad t \longmapsto H_i(F^t C_\bullet)$$

belongs to $\mathsf{Pers}_k^{\mathrm{cons}}$.

The category of such objects and filtration-preserving chain maps is denoted by

$$\mathsf{FiltCh}(k)\,.$$

Remark 1.6 (Persistence layer first)**.** The Core reads F through monitored persistence profiles

$$\mathbf{P}_i(F).$$

Any filtered-level or derived-level use requires an additional representative policy. Barcode-level deletion does not by itself supply a global strict filtered lift.

1.3 Barcode normal forms and bar lengths

Definition 1.7 (Interval module)**.** For an interval $I \subseteq \mathbb{R}$, the interval module k_I is defined by

$$k_I(t) = \begin{cases} k, & t \in I, \\ 0, & t \notin I, \end{cases}$$

with identity structure maps when both parameters lie in I, and zero otherwise.

Definition 1.8 (Bar and barcode)**.** An interval summand k_I is a *bar*. A barcode of $M \in \mathsf{Pers}_k^{\mathrm{cons}}$ is an interval decomposition

$$M \cong \bigoplus_{\lambda\in\Lambda} k_{I_\lambda}$$

with locally finite support on bounded intervals. The multiset of bars is

$$\bar{}(M) = \{I_\lambda\}_{\lambda\in\Lambda}.$$

Theorem 1.9 (Interval decomposition)**.** *Every $M \in \mathsf{Pers}_k^{\mathrm{cons}}$ admits a locally finite interval decomposition*

$$M \cong \bigoplus_{\lambda\in\Lambda} k_{I_\lambda}.$$

The barcode is unique up to permutation of interval summands.

Proof sketch. This is the standard interval-decomposition theorem for pointwise finite-dimensional one-parameter persistence modules over a field. Constructibility gives local finite type, and uniqueness follows from the Krull–Schmidt property in the interval-decomposable regime. □

Definition 1.10 (Length of a bar)**.** If I is finite with endpoint values $a < b$, define

$$\ell(I) := b - a.$$

If $I = [a, \infty)$, up to harmless endpoint decoration at a, define

$$\ell(I) := +\infty$$

for thresholding purposes.

Remark 1.11 (Endpoint independence)**.** The value $\ell(I)$ depends only on endpoint values, not on finite-endpoint open/closed decoration. Every threshold rule below is therefore independent of that bookkeeping.

1.4 Standard bottleneck metric and infinite-bar convention

Definition 1.12 (Standard bottleneck distance with infinite bars)**.** Let B, C be finite barcode profiles, or locally finite constructible profiles restricted to a declared bounded audit range. The bottleneck distance $d_B(B, C)$ is computed by partial matchings of bars and diagonal copies. For finite bars with endpoint values

$$I = [a, b), \qquad J = [a', b'),$$

up to the declared finite-endpoint decoration convention, the finite-to-finite matching cost is

$$\mathrm{cost}(I, J) := \max\{|a - a'|,\ |b - b'|\}.$$

The factor $1/2$ appears only in the finite-bar-to-diagonal cost below; it does not divide the endpoint L^∞-cost between two finite bars.

The infinite-bar convention is:

(i) a finite bar I may be matched to the diagonal with cost $\ell(I)/2$;

(ii) an infinite bar is never matched to the diagonal at finite cost;

(iii) a finite bar and an infinite bar never match at finite cost;

(iv) $[a, \infty)$ and $[a', \infty)$ may match with cost

$$|a - a'|.$$

Lemma 1.13 (Infinite-bar threshold invariance)**.** *For every finite $\tau > 0$,*

$$T_\tau^{\mathrm{bar}}([a, \infty)) = [a, \infty).$$

Moreover, an infinite bar never lies in a finite threshold band

$$[\tau - \varepsilon, \tau + \varepsilon].$$

Proof. An infinite bar has threshold length $+\infty$, whereas hard deletion removes only finite bars of length at most τ. □

Remark 1.14 (Bottleneck–interleaving isometry)**.** For q-tame or constructible one-parameter persistence modules over k, the isometry theorem gives

$$d_{\mathrm{int}}(M, N) = d_B(\bar{(}M), \bar{(}N)).$$

The equality is interpreted under Definition 1.12. Consequently, a d_B-statement may be transferred to d_{int} only when the persistence objects and barcode profiles compared are exactly those appearing in the certified record.

1.5 Positive-threshold short bars are not Serre in general

Proposition 1.15 (Positive-threshold short-bar objects are not Serre in general)**.** *Let $\tau > 0$. The class of constructible persistence modules all of whose barcode summands have finite length at most τ is not extension-closed in general. Hence it is not a Serre subcategory of $\mathsf{Pers}_k^{\mathrm{cons}}$ in general.*

Proof. For $\tau = 1$, consider

$$0 \longrightarrow k_{[1,2)} \longrightarrow k_{[0,2)} \longrightarrow k_{[0,1)} \longrightarrow 0.$$

The subobject and quotient have length 1, while the middle interval has length 2. □

Corollary 1.16 (No global exact reflector claim)**.** *Hard positive-threshold deletion is not asserted to arise globally from a Serre localization, an abelian exact reflector, or a global exact endofunctor on all of $\mathsf{Pers}_k^{\mathrm{cons}}$.*

Remark 1.17 (Why this guard-rail matters)**.** Without this restriction, exactness, functoriality, filtered lifts, and tower comparison maps would inherit unsupported categorical strength. Stronger calculus must be proved in a declared τ-split or exact-admissible regime.

1.6 Hard barcode threshold deletion

Definition 1.18 (Hard barcode threshold deletion)**.** For $\tau > 0$, define

$$T_\tau^{\mathrm{bar}}(B) := \{\, I \in B \mid \ell(I) > \tau \text{ or } \ell(I) = +\infty \,\}.$$

Thus finite bars of length $\leq \tau$ are deleted, finite bars of length $> \tau$ are retained, and infinite bars are retained. For a module M, the notation $T_\tau^{\mathrm{bar}}(M)$ means that the operation is applied to $\bar{(}M)$ at barcode-profile level.

Remark 1.19 (Status of T_τ^{bar})**.** In the general Core, T_τ^{bar} is an operation on barcode normal forms, hence on isomorphism classes. It is not asserted to be a strict exact endofunctor on all morphisms.

Proposition 1.20 (Basic properties of hard threshold deletion)**.** *For every $\tau > 0$, T_τ^{bar} is:*

(i) endpoint-independent;

(ii) idempotent;

(iii) infinite-bar preserving;

(iv) threshold-monotone.

Explicitly,

$$T_\tau^{\mathrm{bar}} T_\tau^{\mathrm{bar}}(B) = T_\tau^{\mathrm{bar}}(B).$$

Proof. Each assertion follows directly from deletion by the predicate $\ell(I) \le \tau$ on finite bars. □

Proposition 1.21 (Threshold composition)**.** *For $\tau, \sigma > 0$,*

$$T_\sigma^{\mathrm{bar}} T_\tau^{\mathrm{bar}}(B) = T_{\max\{\tau,\sigma\}}^{\mathrm{bar}}(B).$$

Proof. The composite deletes exactly the finite bars whose length is at most one of the two thresholds, equivalently at most their maximum. □

1.7 Counterexample to global non-expansiveness

Example 1.22 (Failure of global non-expansiveness)**.** Fix $\tau > 0$ and $0 < \eta < \tau/4$. Let

$$B = \{[0, \tau + \eta)\}, \qquad C = \{[0, \tau - \eta)\}.$$

The direct finite-to-finite matching has cost

$$\max\big\{|0 - 0|, |(\tau + \eta) - (\tau - \eta)|\big\} = 2\eta.$$

Matching both bars to the diagonal has bottleneck cost

$$\max\left\{\frac{\tau + \eta}{2}, \frac{\tau - \eta}{2}\right\} = \frac{\tau + \eta}{2} > 2\eta,$$

where the final inequality follows from $\eta < \tau/4$. Hence

$$d_B(B, C) = 2\eta.$$

After deletion,

$$T_\tau^{\mathrm{bar}}(B) = \{[0, \tau + \eta)\}, \qquad T_\tau^{\mathrm{bar}}(C) = \varnothing,$$

and therefore

$$d_B\big(T_\tau^{\mathrm{bar}}(B), T_\tau^{\mathrm{bar}}(C)\big) = \frac{\tau + \eta}{2} > 2\eta.$$

Thus global non-expansiveness fails.

Remark 1.23 (Meaning of the counterexample)**.** A small endpoint perturbation can move a finite bar across the hard classification boundary $\ell = \tau$. The correct stability theorem must therefore exclude both input barcodes from a neighborhood of the threshold.

Corollary 1.24 (Rejected global non-expansiveness claim)**.** *Global non-expansiveness of hard positive-threshold deletion under the standard bottleneck metric was rejected by the v19 repair and remains unavailable for v20 theorem use.*

1.8 Threshold clearance and gap

Definition 1.25 (Threshold clearance)**.** For a barcode profile B, define

$$\mathrm{clear}_\tau(B) := \inf_{I \in B_{\mathrm{fin}}} |\ell(I) - \tau|.$$

If B has no finite bars, set $\mathrm{clear}_\tau(B) = +\infty$. If B contains a finite bar of length exactly τ, then $\mathrm{clear}_\tau(B) = 0$.

Definition 1.26 (Threshold gap as ledger alias)**.** For one profile,

$$\mathrm{gap}_\tau(B) := \mathrm{clear}_\tau(B).$$

For a finite family,

$$\mathrm{gap}_\tau(B_1, \dots, B_m) := \min_j \mathrm{clear}_\tau(B_j).$$

Thus gap_τ is a ledger-facing alias, not a second invariant, and there is no factor $1/2$.

Definition 1.27 (Typed bottleneck scalarization)**.** Let

$$(\mathsf{V}, \oplus, 0_{\mathsf{V}}, \preceq)$$

be the declared metric-ledger quantale. A bottleneck scalarization is a monotone, subadditive map

$$\mathrm{val}_{\mathrm{B}} : \mathsf{V} \longrightarrow [0, +\infty]$$

with

$$\mathrm{val}_{\mathrm{B}}(0_{\mathsf{V}}) = 0, \qquad \mathrm{val}_{\mathrm{B}}(q \oplus q') \le \mathrm{val}_{\mathrm{B}}(q) + \mathrm{val}_{\mathrm{B}}(q').$$

For $\mathfrak{d}_{\mathrm{met}} \in \mathsf{V}$, define

$$\varepsilon_{\mathrm{B}}(\mathfrak{d}_{\mathrm{met}}) := \mathrm{val}_{\mathrm{B}}(\mathfrak{d}_{\mathrm{met}}).$$

A certified use additionally contains a theorem, verified computation, or artifact establishing that the relevant actual discrepancy satisfies

$$d_B(B, C) \le \varepsilon_{\mathrm{B}}(\mathfrak{d}_{\mathrm{met}}).$$

The shorthand

$$\Sigma\delta < \mathrm{gap}_\tau$$

means the typed real inequality

$$\varepsilon_{\mathrm{B}}(\mathfrak{d}_{\mathrm{met}}) < \mathrm{gap}_\tau(\mathfrak{B})$$

for the declared protected family $\mathfrak{B}$.

Only metric defects belong to $\mathfrak{d}_{\mathrm{met}}$. Representative existence, actual morphisms, eligibility, detector applicability, endpoint convention, completeness provenance, terminal tameness, threshold reconciliation, higher coherence, claim status, manifest consistency, and artifact identity are non-scalar obligations.

Remark 1.28 (Sharp threshold-safety margin versus scalarization tightness)**.** Under an ε-bottleneck matching, a finite bar length may change by at most 2ε. Nevertheless, for a two-sided certificate the sharp forbidden band is

$$[\tau - \varepsilon, \tau + \varepsilon],$$

not

$$[\tau - 2\varepsilon, \tau + 2\varepsilon].$$

The word *sharp* refers to the optimal half-width for a true bottleneck bound; it does not assert that a chosen scalarization is numerically tight.

Definition 1.29 (Windowed pre-threshold profile)**.** For $F \in \mathsf{FiltCh}(k)$, monitored degree i, and window W, define

$$B_{i,W}(F) := \mathsf{W}_W \mathbf{P}_i(F).$$

Threshold clearance and threshold gap are measured on this profile. The repaired evaluation is

$$\mathrm{Eval}_{i,W,\tau}(F) = T_\tau^{\mathrm{bar}}(B_{i,W}(F)).$$

Remark 1.30 (Clearance is measured before deletion, after windowing)**.** Clearance is measured on

$$B_{i,W}(F) = \mathsf{W}_W \mathbf{P}_i(F),$$

not on the already deleted profile. Once a bar has been deleted, its threshold proximity can no longer be audited.

Definition 1.31 (Threshold-gap record)**.** A threshold-gap record for a metric comparison of two repaired evaluations contains:

(i) immutable identifiers for the two windowed pre-threshold profiles B and C;

(ii) a finiteness witness for the compared barcode profiles, or a declared bounded audit-range restriction producing the finite profiles to which the matching theorem is applied;

(iii) the monitored degree, common window, and threshold;

(iv) the finite-bar length data or certified summaries for both profiles;

(v) $\mathrm{clear}_\tau(B)$, $\mathrm{clear}_\tau(C)$, and $\mathrm{gap}_\tau(B,C)$;

(vi) the metric-ledger element $\mathfrak{d}_{\mathrm{met}}$;

(vii) the scalarization $\mathrm{val}_{\mathrm{B}}$ and

$$\varepsilon_{\mathrm{B}} := \mathrm{val}_{\mathrm{B}}(\mathfrak{d}_{\mathrm{met}});$$

(viii) evidence that

$$d_B(B,C) \le \varepsilon_{\mathrm{B}};$$

(ix) confirmation that neither finite-bar family meets

$$[\tau - \varepsilon_{\mathrm{B}}, \tau + \varepsilon_{\mathrm{B}}],$$

equivalently

$$\varepsilon_{\mathrm{B}} < \mathrm{gap}_\tau(B,C);$$

(x) every crop, window-edge, endpoint, or threshold-reconciliation component used to derive the windowed bound;

(xi) artifact references sufficient to reconstruct the comparison;

(xii) the specialized status

`gap_certified`, `gap_failed`, `gap_pending`.

The record is two-sided by definition. The specialized statuses map respectively to the atomic statuses `pass`, `reject`, and `undefined`. Here `gap_failed` means that a well-typed required gap condition is proved or artifact-evidenced to fail. Missing, incomplete, stale, or pending evidence is `gap_pending`, never `gap_failed`.

For a single profile, a clearance record may be stored as a readiness record, but it does not by itself certify a comparison.

Remark 1.32 (Length exactly τ is not gap-safe)**.** A finite bar of length exactly τ is deleted by the rule $\ell(I) \leq \tau$, but it gives $\mathrm{clear}_\tau = 0$. The evaluation is defined, whereas a positive robustness certificate at that threshold fails.

1.9 Threshold-gap-safe stability

Theorem 1.33 (Metric-repair theorem: sharp threshold-gap-safe stability)**.** *Let B, C be finite barcodes, possibly with infinite bars, under Definition 1.12. Let $\tau > 0$, $\varepsilon \geq 0$, and suppose*

$$d_B(B, C) \leq \varepsilon.$$

Assume no finite bar of either barcode has length in

$$[\tau - \varepsilon, \tau + \varepsilon].$$

Then

$$d_B\big(T_\tau^{\mathrm{bar}}(B), T_\tau^{\mathrm{bar}}(C)\big) \leq \varepsilon.$$

Proof. If $\varepsilon = 0$, the barcodes agree and deterministic deletion gives the result.

Assume $\varepsilon > 0$ and choose a bottleneck matching of cost at most ε, which exists for finite barcodes.

Infinite bars match only infinite bars and are retained on both sides.

Let I be a retained finite bar of B. The closed-band hypothesis gives

$$\ell(I) > \tau + \varepsilon.$$

If I is matched to the diagonal, retain that diagonal match. If it is matched to a finite bar $J \in C$, then

$$|\ell(I) - \ell(J)| \leq 2\varepsilon.$$

If J were deleted, then the closed-band hypothesis on C would force

$$\ell(J) < \tau - \varepsilon,$$

hence

$$\ell(I) - \ell(J) > 2\varepsilon,$$

a contradiction. Thus J is retained.

The symmetric argument applies to retained bars of C. After removing pairs deleted on both sides, the original matching therefore restricts, together with retained-to-diagonal matches, to a matching of the retained barcodes of cost at most ε. □

Example 1.34 (Sharpness of the ε-band)**.** Let $0 < c < \varepsilon$, choose $0 < \eta < \varepsilon - c$, and choose τ so large that the diagonal costs below exceed ε. Set

$$L_+ = \tau + c + \eta, \qquad L_- = \tau - c - \eta$$

and take centered bars I, J of lengths L_+, L_-. Both lengths avoid the smaller band $[\tau - c, \tau + c]$, and

$$d_B(\{I\}, \{J\}) = c + \eta < \varepsilon.$$

But I is retained, J is deleted, and the post-deletion distance is $L_+/2 > \varepsilon$. Thus no smaller two-sided half-width $c < \varepsilon$ suffices in general.

Remark 1.35 (Two-sided certificate)**.** It is insufficient for only one barcode to avoid the threshold band. A one-sided 2ε-type margin may be used as a separately declared conservative policy, but it is not the sharp inherited two-sided theorem.

Corollary 1.36 (Scalarized-ledger-safe threshold classification)**.** *Let*

$$\varepsilon_{\mathrm{B}} := \mathrm{val}_{\mathrm{B}}(\mathfrak{d}_{\mathrm{met}}).$$

Suppose the record proves

$$d_B(B, C) \le \varepsilon_{\mathrm{B}}$$

and

$$\varepsilon_{\mathrm{B}} < \mathrm{gap}_\tau(B, C).$$

Then

$$d_B\big(T_\tau^{\mathrm{bar}}(B), T_\tau^{\mathrm{bar}}(C)\big) \le \varepsilon_{\mathrm{B}}.$$

Proof. The strict inequality implies two-sided avoidance of the closed band with half-width ε_{B}. Apply Theorem 1.33. □

1.10 Sharp zero separation inside the repaired image

Definition 1.37 (τ-reduced barcode profile)**.** A barcode profile P is *τ-reduced* if

$$T_\tau^{\mathrm{bar}}(P) = P.$$

Equivalently, every finite bar of P has length strictly greater than τ, while every infinite bar is retained. Every repaired profile $T_\tau^{\mathrm{bar}}(B)$ is τ-reduced.

Definition 1.38 (Zero radius of a finite barcode profile)**.** For a finite barcode profile P, define

$$\mathrm{zrad}(P) := d_B(P, 0),$$

where 0 denotes the empty barcode. Under the standard convention,

$$\mathrm{zrad}(P) = \begin{cases} 0, & P = 0, \\ \dfrac{1}{2} \max\limits_{I \in P} \ell(I), & P \neq 0 \text{ and every bar of } P \text{ is finite}, \\ +\infty, & P \text{ contains an infinite bar.} \end{cases}$$

Lemma 1.39 (Distance-to-zero formula)**.** *Definition 1.38 gives the exact bottleneck distance from P to the zero profile.*

Proof. If every bar is finite, every bar must be matched to the diagonal, and the cost of the unique type of admissible matching is the maximum of the diagonal costs $\ell(I)/2$. If an infinite bar is present, it cannot be matched to the diagonal at finite cost. The empty profile has distance zero from itself. □

Theorem 1.40 (Sharp zero separation for τ-reduced profiles)**.** *Let P, Q be finite τ-reduced barcode profiles, possibly including infinite bars. Let $\varepsilon \geq 0$, and suppose*

$$d_B(P, Q) \leq \varepsilon, \qquad 2\varepsilon \leq \tau.$$

Then

$$P = 0 \iff Q = 0.$$

The constant is sharp: if $2\varepsilon > \tau$, the conclusion can fail.

Proof. Assume $P = 0$. If Q contained an infinite bar, then $d_B(P, Q) = +\infty$, contradicting the hypothesis. If $Q \neq 0$ and all bars of Q were finite, τ-reducedness would give

$$\ell(I) > \tau \geq 2\varepsilon \qquad (I \in Q).$$

Lemma 1.39 would then imply

$$d_B(0, Q) = \frac{1}{2} \max_{I \in Q} \ell(I) > \varepsilon,$$

again a contradiction. Hence $Q = 0$. The converse is symmetric.

For sharpness, suppose $2\varepsilon > \tau$. Choose L with

$$\tau < L \leq 2\varepsilon,$$

let $P = 0$, and let Q consist of one finite bar of length L. Then Q is τ-reduced and

$$d_B(P, Q) = L/2 \leq \varepsilon,$$

but $P = 0$ and $Q \neq 0$. □

Remark 1.41 (Strict operational reserve)**.** Theorem 1.40 is valid at the closed boundary $2\varepsilon = \tau$, because every retained finite bar has length strictly greater than τ. The v20 audit contract uses

$$2\varepsilon < \tau$$

to retain a positive reserve and to avoid equality-sensitive scalarization, rounding, or finite-precision policies. The strict policy is conservative, not mathematically sharp.

Proposition 1.42 (Exact zero transport)**.** *If $P \cong Q$ as persistence profiles, then*

$$P = 0 \iff Q = 0.$$

No metric ledger or threshold-gap certificate is required for this exact claim. The actual isomorphism and its immutable proof artifact are required.

Proof. The zero object is invariant under isomorphism. □

Theorem 1.43 (Metric-safe repaired-zero transport)**.** *Let B, C be finite barcode profiles, possibly including infinite bars. Let $\tau > 0$, $\varepsilon \geq 0$, and suppose*

$$d_B(B, C) \leq \varepsilon.$$

Assume that no finite bar of either profile has length in

$$[\tau - \varepsilon, \tau + \varepsilon],$$

and assume

$$2\varepsilon \leq \tau.$$

Then

$$T_\tau^{\mathrm{bar}}(B) = 0 \iff T_\tau^{\mathrm{bar}}(C) = 0.$$

Proof. Theorem 1.33 gives

$$d_B\big(T_\tau^{\mathrm{bar}}(B), T_\tau^{\mathrm{bar}}(C)\big) \le \varepsilon.$$

Both repaired profiles are τ-reduced by Definition 1.37. Apply Theorem 1.40. □

Corollary 1.44 (Scalarized metric-safe repaired-zero transport)**.** *Let* $\mathfrak{d}_{\mathrm{met}} \in \mathsf{V}$*, set*

$$\varepsilon_{\mathrm{B}} := \mathrm{val}_{\mathrm{B}}(\mathfrak{d}_{\mathrm{met}}),$$

and suppose the record proves

$$d_B(B, C) \le \varepsilon_{\mathrm{B}}, \qquad \varepsilon_{\mathrm{B}} < \mathrm{gap}_\tau(B, C), \qquad 2\varepsilon_{\mathrm{B}} < \tau.$$

Then

$$T_\tau^{\mathrm{bar}}(B) = 0 \quad \Longleftrightarrow \quad T_\tau^{\mathrm{bar}}(C) = 0.$$

This is the strict operational form consumed by the v20 metric-safe detector chain.

Proof. The strict gap inequality supplies the two-sided forbidden-band hypothesis. Apply Theorem 1.43. □

Definition 1.45 (Post-threshold zero-separation record)**.** A post-threshold zero-separation record contains:

(i) immutable identifiers for $P = T_\tau^{\mathrm{bar}}(B)$ and $Q = T_\tau^{\mathrm{bar}}(C)$;

(ii) the common threshold τ, the proved bound $d_B(P, Q) \le \varepsilon$, and its derivation;

(iii) evidence that P and Q are τ-reduced;

(iv) the mathematical inequality $2\varepsilon \le \tau$, or the stricter operational inequality $2\varepsilon < \tau$;

(v) the theorem reference Theorem 1.40;

(vi) exact source, scalarization, computation, proof, and artifact identities.

Missing required evidence yields `undefined`. An evidenced false required inequality yields `reject`. This record creates no independent status algebra.

1.11 Detector-zero transport and the rank-instability boundary

Definition 1.46 (Metric-safe detector-zero transport record)**.** A metric-safe detector-zero transport record from B to C contains:

(i) a complete Chapter 3 detector certificate establishing $T_\tau^{\mathrm{bar}}(B) = 0$, including its exact profile stage;

(ii) the two-sided threshold-gap comparison record for B and C;

(iii) the repaired-profile distance bound produced by Theorem 1.33;

(iv) the post-threshold zero-separation record of Definition 1.45;

(v) the conclusion $T_\tau^{\mathrm{bar}}(C) = 0$;

(vi) immutable detector, comparison, theorem, and replay artifacts.

The record transports only the exact zero predicate. It does not transport detector sample values, ranks, birth sets, germ sets, sample counts, coverage witnesses, or certificate sizes.

Corollary 1.47 (Detector-certified metric zero transport)**.** *Assume a complete detector certificate proves*

$$T_\tau^{\mathrm{bar}}(B) = 0.$$

If the hypotheses of Corollary 1.44 hold for (B, C), then

$$T_\tau^{\mathrm{bar}}(C) = 0.$$

The converse holds after exchanging B and C. Detector completeness is required only for a profile whose zero statement is obtained from detector data; the metric theorem transports an already certified zero predicate.

Proof. The detector certificate supplies the source zero statement. Apply Corollary 1.44. □

Definition 1.48 (Pointwise span-rank observation)**.** For a persistence module M, a point t, and a span $s > 0$, define

$$r_{t,s}(M) := \dim_k \operatorname{im}\big(M(t) \longrightarrow M(t+s)\big),$$

when the displayed structure map is defined. This is a finite observation, not by itself a complete detector certificate.

Example 1.49 (Arbitrarily small perturbation can change a pointwise detector rank)**.** Fix $s > 0$ and $L > 0$. For any

$$0 < \delta < \min\{L, (s+L)/2\},$$

let

$$M_\delta := k_{[0,s+L)}, \qquad N_\delta := k_{[\delta,s+L+\delta)}.$$

Their one-bar barcode profiles B_δ, C_δ satisfy

$$d_B(B_\delta, C_\delta) = \delta.$$

Both bars have the same length $s+L$, and therefore have the same clearance from every threshold that depends only on bar length. Nevertheless,

$$r_{0,s}(M_\delta) = 1, \qquad r_{0,s}(N_\delta) = 0.$$

Thus pointwise span-rank observations can change under arbitrarily small bottleneck perturbations, even when no bar crosses a length threshold.

Proposition 1.50 (No bottleneck stability theorem for raw pointwise detector ranks)**.** *There is no general implication*

$$d_B(B, C) \le \varepsilon \quad \Longrightarrow \quad r_{t,s}(B) = r_{t,s}(C)$$

for a fixed sample point t and span s, even for one-bar finite barcodes and arbitrarily small positive ε. Here the rank notation is read on the interval modules represented by the barcodes.

Proof. Apply Example 1.49 with $0 < \delta \le \varepsilon$. □

Remark 1.51 (Zero predicate, not raw detector output)**.** The v20 metric chain preserves the repaired zero/nonzero predicate under the sharp separation hypotheses. It does not preserve the raw output of a fixed-point detector. Birth, germ, endpoint, cofinality, coverage, mesh, family-membership, excess-bound, and profile-stage conditions remain separate non-scalar obligations and cannot be replaced by metric slack.

1.12 Window restriction and window-edge budget

Definition 1.52 (Right-open window)**.** A domain window is a nonempty right-open interval

$$W = [u, u'), \qquad u \in \mathbb{R}, \qquad u' \in \mathbb{R} \cup \{+\infty\}, \qquad u < u'.$$

A finite u' gives a bounded audit window; $u' = +\infty$ gives a terminal audit window.

Definition 1.53 (Window restriction)**.** Define $\mathsf{W}_W(M)$ by restriction to W, with extension by zero outside W whenever a globally indexed module is required. At barcode level, bars are clipped to $I \cap W$, and empty intersections are discarded.

Remark 1.54 (Infinite bars under window restriction)**.** On a bounded window, an infinite bar $[a, \infty)$ with nonempty intersection is clipped to

$$[\max\{a, u\}, u')$$

and therefore becomes finite; it is then classified by its clipped finite length. On a terminal window it remains infinite and is retained by every finite threshold.

Proposition 1.55 (No unconditional crop–collapse commutation)**.** *In general,*

$$\mathsf{W}_W T_\tau^{\mathrm{bar}}(B) \neq T_\tau^{\mathrm{bar}} \mathsf{W}_W(B).$$

Proof. Take $B = \{[0, 10)\}$, $W = [0, 1)$, and $\tau = 2$. Global deletion retains the length-10 bar, which clips to $[0, 1)$. Windowing first produces a length-1 bar, which is then deleted. □

Definition 1.56 (Window-edge clearance)**.** A window-edge clearance record supplies a lower bound on the distance of relevant endpoints or critical parameters from the active window boundaries. Its role is to justify that cropping does not silently gain or lose relevant bars under a source perturbation.

Definition 1.57 (Crop budget)**.** A crop budget

$$\delta_{\mathrm{crop}}(i, W)$$

is a metric-ledger component bounding discrepancy introduced by restriction, clipping, endpoint uncertainty, or moving window boundaries. A source-level perturbation may be transported to a windowed pre-threshold bound only through a theorem proving zero crop error or through a ledger that includes the crop contribution.

Remark 1.58 (Gap-safe stability consumes windowed profile distance)**.** Theorem 1.33 consumes

$$d_B(B, C) \leq \varepsilon$$

for the actual profiles to which hard deletion is applied. In AK applications these are the windowed pre-threshold profiles. It does not itself prove source-to-window metric stability.

Example 1.59 (Infinite-bar obstruction to the former finite-only condition)**.** Let

$$B = \{[0, +\infty)\}, \qquad W = [0, 1), \qquad \tau = 2.$$

Global threshold deletion retains the infinite bar, after which cropping gives $[0, 1)$. Windowing first turns the bar into a finite length-1 bar, which is deleted. Thus infinite globally retained bars must be checked.

Definition 1.60 (τ-window-adapted condition)**.** The pair (B, W) is *τ-window-adapted* if every bar retained globally by T_τ^{bar}, including every infinite bar, is either:

(i) disjoint from W; or

(ii) clipped to a bar retained by threshold deletion on W.

Proposition 1.61 (Exact crop–collapse commutation criterion)**.** *For a barcode B, right-open window W, and $\tau > 0$, the following are equivalent:*

(i) (B, W) *is τ-window-adapted;*

(ii)

$$\mathsf{W}_W T_\tau^{\mathrm{bar}}(B) = T_\tau^{\mathrm{bar}} \mathsf{W}_W(B)$$

as barcode multisets, including multiplicity.

Proof. A globally deleted finite bar has length at most τ, and clipping cannot increase its length, so it is deleted in both orders. A globally retained bar contributes equally in both orders exactly when it is disjoint from the window or its nonempty crop remains retained. Multiplicity makes this criterion necessary as well as sufficient. □

Remark 1.62 (Default policy and audit use)**.** The Core does not assume τ-window-adaptedness. The default order is window first, threshold second. Commutation may be used only when Definition 1.60 is verified for every globally retained bar and recorded as an artifact-backed non-scalar obligation.

1.13 Core windowed evaluation

Definition 1.63 (Windowed repaired evaluation)**.** For $F \in \mathsf{FiltCh}(k)$, monitored degree i, window W, and $\tau > 0$, define

$$\mathrm{Eval}_{i,W,\tau}(F) := T_\tau^{\mathrm{bar}}\big(\mathsf{W}_W \mathbf{P}_i(F)\big).$$

Remark 1.64 (Fixed order)**.** The notation means the ordered pipeline

$$F \longmapsto \mathbf{P}_i(F) \longmapsto B_{i,W}(F) \longmapsto T_\tau^{\mathrm{bar}}(B_{i,W}(F)).$$

Definition 1.65 (Repaired evaluation record)**.** A repaired evaluation record contains:

(i) immutable source-object and profile identifiers;

(ii) degree i, window W, and threshold τ;

(iii) the pre-threshold profile $B_{i,W}(F)$;

(iv) the repaired profile $\mathrm{Eval}_{i,W,\tau}(F)$;

(v) the comparison mode

`exact`, `metric_gap_certified`, `not_compared`;

(vi) reconstructible artifact references.

Definition 1.66 (Threshold-gap-certified repaired evaluation record)**.** A repaired evaluation record is threshold-gap-certified only when every metric comparison transporting through hard deletion is supplemented by the two-sided record of Definition 1.31.

An exact equality or proved profile isomorphism may instead be stored with comparison mode `exact`. Such a record does not automatically certify robustness under a positive perturbation.

Remark 1.67 (Evaluation is definable without certification)**.** The repaired profile is definitionally available once the pre-threshold profile is available. Threshold-gap certification is required whenever a metric discrepancy is transported through hard deletion. If the transported conclusion is an exact repaired-zero predicate, the post-threshold record of Definition 1.45 is additionally required; threshold-gap stability alone bounds distance and does not identify zero and nonzero profiles. No metric certificate is fabricated for a purely exact identity.

1.14 Overlap Gate: post-threshold comparison and pre-threshold gap

Definition 1.68 (Overlap pre-threshold profiles)**.** For local inputs F_α, F_β and

$$W_{\alpha\beta} = W_\alpha \cap W_\beta,$$

define

$$B^{\alpha}_{i,W_{\alpha\beta}} := \mathsf{W}_{W_{\alpha\beta}}\mathbf{P}_i(F_\alpha),$$
$$B^{\beta}_{i,W_{\alpha\beta}} := \mathsf{W}_{W_{\alpha\beta}}\mathbf{P}_i(F_\beta).$$

The repaired overlap profiles are obtained by applying T^{bar}_{τ} to these two inputs.

Remark 1.69 (Two-level structure of the Overlap Gate)**.** The threshold-gap theorem requires a discrepancy bound on the *pre-threshold* profiles. The theorem then transports that bound to the *post-threshold* repaired profiles. A post-threshold comparison alone is not a substitute for the pre-threshold hypothesis.

Proposition 1.70 (Overlap transfer under a scalarized threshold-gap certificate)**.** *Let*

$$\varepsilon_{\alpha\beta,i} := \mathrm{val}_{\mathrm{B}}(\mathfrak{d}_{\alpha\beta,i}).$$

Suppose the record proves

$$d_B\big(B^{\alpha}_{i,W_{\alpha\beta}}, B^{\beta}_{i,W_{\alpha\beta}}\big) \le \varepsilon_{\alpha\beta,i}$$

and

$$\varepsilon_{\alpha\beta,i} < \mathrm{gap}_{\tau}\big(B^{\alpha}_{i,W_{\alpha\beta}}, B^{\beta}_{i,W_{\alpha\beta}}\big).$$

Then

$$d_B\big(\mathrm{Eval}_{i,W_{\alpha\beta},\tau}(F_\alpha), \mathrm{Eval}_{i,W_{\alpha\beta},\tau}(F_\beta)\big) \le \varepsilon_{\alpha\beta,i}.$$

In the declared constructible regime, the same bound may be read using d_{int}.

Proof. Apply Corollary 1.36 to the two pre-threshold overlap profiles. □

Corollary 1.71 (Overlap repaired-zero equivalence)**.** *Under the hypotheses of Proposition 1.70, assume in addition*

$$2\varepsilon_{\alpha\beta,i} \le \tau.$$

Then

$$\mathrm{Eval}_{i,W_{\alpha\beta},\tau}(F_\alpha) = 0 \quad\Longleftrightarrow\quad \mathrm{Eval}_{i,W_{\alpha\beta},\tau}(F_\beta) = 0.$$

For operational audit consumption, the strict reserve $2\varepsilon_{\alpha\beta,i} < \tau$ is used.

Proof. Proposition 1.70 supplies the repaired-profile bottleneck bound. Both repaired profiles are τ-reduced. Apply Theorem 1.40. □

1.15 Exact-admissible regimes

Definition 1.72 (τ-split object)**.** An object $M \in \mathsf{Pers}_k^{\mathrm{cons}}$ is τ-split if it is equipped with declared data

$$M \cong M_{\le\tau} \oplus M_{>\tau}$$

such that finite bars in $M_{\le\tau}$ have length $\le \tau$, finite bars in $M_{>\tau}$ have length $> \tau$, and all infinite bars lie in $M_{>\tau}$.

Definition 1.73 (τ-split morphism)**.** A morphism $f : M \to N$ between τ-split objects is τ-split if

$$f(M_{\le\tau}) \subseteq N_{\le\tau}, \qquad f(M_{>\tau}) \subseteq N_{>\tau}.$$

Definition 1.74 (τ-split regime)**.** A τ-split regime is a declared subcategory or workflow domain in which every object and morphism under consideration is τ-split with the chosen decompositions as part of the data.

Proposition 1.75 (Functorial deletion in a τ-split regime)**.** *Inside a declared τ-split regime, the assignment*

$$M \longmapsto M_{>\tau}$$

is functorial and agrees with $T_\tau^{\mathrm{bar}}(M)$ on barcode profiles.

Proof. Every declared morphism restricts to the retained summands. □

Definition 1.76 (Exact-admissible collapse regime)**.** An exact-admissible collapse regime is a τ-split regime in which the retained-summand assignment is additionally proved to preserve every exact sequence used by the relevant argument.

Remark 1.77 (Exactness is conditional)**.** Exactness is not a global Core theorem. A proof object invoking it records:

(i) the τ-split decomposition policy;

(ii) the morphism-compatibility policy;

(iii) the exact sequences being tested;

(iv) the proof or immutable artifact establishing exact preservation.

1.16 Filtered representatives of windowed collapse

Definition 1.78 (Windowed collapsed persistence profile)**.** For $F \in \mathsf{FiltCh}(k)$, define

$$\mathcal{P}_{i,W,\tau}(F) := \mathrm{Eval}_{i,W,\tau}(F).$$

Definition 1.79 (Admissible filtered representative)**.** A filtered object $C_{\tau,W}F$ is an admissible representative if, for every monitored degree i,

$$\mathbf{P}_i(C_{\tau,W}F) \cong \mathcal{P}_{i,W,\tau}(F)$$

at the declared profile-equivalence strength. Existence is an independent non-scalar obligation.

Remark 1.80 (No global strict lift by default)**.** The Core does not assert a global strict endofunctor

$$C_\tau : \mathsf{FiltCh}(k) \longrightarrow \mathsf{FiltCh}(k)\,.$$

The default notation $C_{\tau,W}F$ denotes a declared window-relative representative.

Definition 1.81 (Representative equivalence)**.** Two representatives are equivalent for a declared gate if their monitored persistence profiles agree in every monitored degree under the declared barcode or persistence-profile equivalence policy. This does not automatically imply chain isomorphism, filtered quasi-isomorphism, or derived equivalence.

Remark 1.82 (Filtered quasi-isomorphism)**.** Filtered quasi-isomorphism may be required when the argument uses filtered-chain or derived constructions. It is not required merely to define a barcode-level repaired profile.

1.17 Consequences for B-Gate$^+$

Definition 1.83 (Windowed B-side input)**.** The B-side input is

$$\mathrm{Eval}_{i,W,\tau}(F),$$

or a verified representative $C_{\tau,W}F$ realizing that profile.

Definition 1.84 (Windowed topological clause)**.** The topological clause is

$$\mathrm{Eval}_{1,W,\tau}(F) = 0.$$

If representative compatibility is verified, it may equivalently be recorded as

$$\mathrm{PH}_1(C_{\tau,W}F) = 0.$$

Definition 1.85 (Windowed categorical clause)**.** The categorical clause may be invoked only when an admissible, realization-compatible representative exists and the eligible regime, including

$$R(C_{\tau,W}F) \in D^{[-1,0]}(k\text{-mod}),$$

is verified. Only then is

$$\mathrm{Ext}^1(R(C_{\tau,W}F), k) = 0$$

a well-typed Core clause.

Definition 1.86 (Comparison-typed B-Gate$^+$ input)**.** A B-Gate$^+$ input records:

(i) the post-threshold repaired profile $\mathrm{Eval}_{1,W,\tau}(F)$;

(ii) the pre-threshold profile $B_{1,W}(F)$;

(iii) the comparison mode for every comparison involving that profile;

(iv) in mode `metric_gap_certified`, the two compared pre-threshold profiles, their two-sided threshold-gap comparison record, and every ledger component needed to justify the actual bottleneck bound, including crop and window-edge contributions;

(v) in mode `exact`, the actual equality or profile-isomorphism proof and its immutable artifact;

(vi) every required non-scalar obligation.

A single-profile clearance value is readiness information only and is not substituted for a two-sided metric comparison record. No purely exact comparison is assigned a fabricated positive-radius reserve.

Remark 1.87 (PH before Ext)**.** The repaired persistence profile is primary. The Ext clause is read only through an eligible representative and the one-way bridge. No reverse implication is supplied here.

1.18 Tower comparison and typed Type IV input

Definition 1.88 (Windowed collapsed tower objects)**.** For a tower

$$F_0 \to F_1 \to F_2 \to \cdots,$$

write

$$M_{n;i,W,\tau} := \mathrm{Eval}_{i,W,\tau}(F_n).$$

The sequence of barcode profiles is not by itself a directed system: actual profile morphisms and composition laws must be declared, unless a registered τ-split or exact-admissible theorem supplies them.

Definition 1.89 (Tower comparison artifact)**.** A tower comparison artifact contains:

(i) an actual repaired profile system and its transition morphisms;

(ii) a source profile $\mathrm{ColimProfile}_{i,W,\tau}(F_\bullet)$;

(iii) an apex profile $\mathrm{ApexProfile}_{i,W,\tau}(F_\infty)$;

(iv) an actual morphism

$$\phi_{i,W,\tau} : \mathrm{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \mathrm{ApexProfile}_{i,W,\tau}(F_\infty);$$

(v) kernel/cokernel typing, terminal-tameness evidence, and immutable artifact references.

Barcode multisets or bottleneck matchings without actual morphisms are not a tower comparison artifact.

Remark 1.90 (No automatic tower comparison)**.** The comparison morphism is not supplied automatically by hard threshold deletion, a barcode matching, or the mere coexistence of a source tower and an apex.

Definition 1.91 (Typed Type IV scope status)**.** Every run records exactly one Type IV status:

`not_invoked`, `undefined`, `certified_zero`, `obstructed`.

They mean respectively:

(i) tower semantics are outside the claim;

(ii) tower semantics are in scope but required data are absent or ill-typed;

(iii) a valid terminal-generic diagnostic is defined and zero;

(iv) a valid terminal-generic diagnostic is defined and nonzero.

A missing artifact is `undefined`, never a numerical zero and never `not_invoked`, unless the declared scope policy explicitly excludes tower semantics.

Definition 1.92 (Collapse diagnostics notation contract)**.** For a valid, terminal-tame comparison artifact, define

$$\mu_{i,W,\tau} := \mathrm{tgdim}_W(\ker \phi_{i,W,\tau}), \qquad \nu_{i,W,\tau} := \mathrm{tgdim}_W(\mathrm{coker}\, \phi_{i,W,\tau}).$$

On a right-unbounded window, tgdim_W is the eventual generic dimension and counts defect bars cofinal to $+\infty$. On a bounded window $W = [u, u')$, for a terminal-tame defect profile D,

$$\mathrm{tgdim}_W(D) := \lim_{t \uparrow u'} \dim_k D(t),$$

so it counts defect bars cofinal to the right endpoint and is not replaced by the dimension after zero extension to $+\infty$.

For one fixed declared pair (W, τ) and a declared finite monitored degree set I_{mon}, define

$$\mu_{\mathrm{Collapse}}(W, \tau) := \sum_{i \in I_{\mathrm{mon}}} \mu_{i,W,\tau}, \qquad \nu_{\mathrm{Collapse}}(W, \tau) := \sum_{i \in I_{\mathrm{mon}}} \nu_{i,W,\tau}.$$

The pair (W, τ) and I_{mon} are part of the diagnostic scope. Diagnostics from different windows or thresholds are not silently summed. An infinite monitored degree set requires a separately proved aggregation or summability theorem.

The legacy symbols $u_{i,W,\tau}$ and u_{Collapse} are deprecated.

Remark 1.93 (Type IV remains Core, but zero is typed)**.** For a fixed declared scope, the shorthand equality

$$(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) = (0, 0)$$

means

$$\big(\mu_{\mathrm{Collapse}}(W, \tau), \nu_{\mathrm{Collapse}}(W, \tau)\big) = (0, 0)$$

and may be used only under status `certified_zero`. It means absence of monitored terminal-generic defect relative to the declared artifact. It does not imply full isomorphism on finite interior bars.

Remark 1.94 (Object-distance control does not automatically control defect profiles)**.** A bottleneck bound between the source and target objects of two comparison morphisms does not, by itself, produce bottleneck bounds between their kernels or cokernels. Metric transport of a detector on `kernel_defect` or `cokernel_defect` therefore requires actual comparison morphisms between the defect profiles, or a separately proved kernel/cokernel stability theorem with the required commutative data. Object proximity cannot manufacture Type IV or transient-defect evidence.

1.19 Manifest consequences

Definition 1.95 (Collapse policy manifest entry)**.** A collapse-policy entry contains:

(i) immutable run, object, profile, comparison, theorem, and artifact identifiers;

(ii) the threshold τ, monitored degrees, window, and window type;

(iii) the evaluation order

$$\mathbf{P}_i \longrightarrow \mathsf{W}_W \mathbf{P}_i \longrightarrow T_\tau^{\mathrm{bar}}(\mathsf{W}_W \mathbf{P}_i);$$

(iv) the collapse status

`barcode_level`, `tau_split`, `exact_admissible`;

(v) exactly one comparison mode for every consumed comparison;

(vi) the repaired evaluation records;

(vii) for every finite-barcode theorem invocation, a finiteness witness or bounded audit-range record identifying the exact finite profiles consumed by the proof;

(viii) for every metric discrepancy transported through hard deletion, the complete two-sided threshold-gap comparison record;

(ix) for every exact comparison, the actual equality or profile-isomorphism proof and immutable artifact;

(x) for every metric transport of a repaired-zero predicate, the post-threshold zero-separation record, including $2\varepsilon \leq \tau$ or the stricter operational reserve;

(xi) for every detector-derived zero statement, the complete detector certificate and, when transported, the metric-safe detector-zero transport record;

(xii) the metric ledger, bottleneck scalarization, scalarized budget, and actual discrepancy upper-bound evidence whenever metric mode is invoked;

(xiii) crop and window-edge records required to reach the actual windowed pre-threshold comparison;

(xiv) any τ-window-adapted commutation certificate, including explicit verification of globally retained infinite bars on bounded windows;

(xv) representative and eligibility records when invoked;

(xvi) Type IV status, monitored scope, terminal-generic input, and actual comparison artifact when invoked;

(xvii) all non-scalar obligations, including detector applicability, completeness provenance, profile stage, and artifact identity;

(xviii) repair ancestry and Claim Register references.

Definition 1.96 (Metric, zero-transport, and crop repair record)**.** An Appendix A repair record contains:

(i) the deprecated global non-expansiveness claim being avoided;

(ii) the replacement theorem Theorem 1.33;

(iii) the counterexample Example 1.22;

(iv) the protected pre-threshold profiles and their two-sided threshold-gap comparison record;

(v) the metric ledger, scalarization, and actual discrepancy upper-bound evidence;

(vi) every crop or window-edge component required to obtain the windowed profile comparison;

(vii) if crop–collapse commutation is used, the exact criterion of Proposition 1.61;

(viii) explicit confirmation that globally retained infinite bars were included in that criterion;

(ix) every post-threshold zero-separation record consumed by the proof;

(x) every detector-zero transport record consumed by the proof;

(xi) explicit confirmation that no raw pointwise detector rank, birth or germ set, coverage witness, sample count, or certificate-size stability was inferred from bottleneck proximity;

(xii) Type IV scope and actual artifact references when tower semantics are invoked;

(xiii) confirmation that canonical ν-notation is used and legacy u-notation is confined to deprecation metadata.

Remark 1.97 (Audit consequence)**.** A proof object that merely says "apply T_τ" is incomplete. It must identify whether T_τ means barcode-level hard threshold deletion, a τ-split functorial model, or an exact-admissible regime.

A metric discrepancy transported through hard deletion must cite the two-sided scalarized threshold-gap record rather than global non-expansiveness. A transported repaired-zero conclusion must additionally cite a post-threshold zero-separation record. Threshold-gap stability alone bounds distance; it does not identify zero and nonzero profiles.

A separately proved exact comparison is stored in exact mode and is not misrepresented as a positive-radius robustness certificate. Crop–collapse commutation must verify all globally retained bars, including infinite bars clipped by bounded windows. Type IV use requires an actual comparison morphism. Detector use requires a complete detector certificate at the exact declared profile stage.

Manifest completeness is necessary for certification but is not sufficient for mathematical truth. A complete manifest may faithfully record a failed comparison, a pending detector, or an obstructed tower.

1.20 Operating summary

For each monitored (F, i, W, τ), the Foundation pipeline is

$$F \longmapsto \mathbf{P}_i(F) \longmapsto B_{i,W}(F) = \mathsf{W}_W \mathbf{P}_i(F) \longmapsto T_\tau^{\mathrm{bar}}(B_{i,W}(F)) = \mathrm{Eval}_{i,W,\tau}(F).$$

Threshold clearance is measured after windowing and before deletion. For a metric comparison, the repaired-zero chain is

$$\mathfrak{d}_{\mathrm{met}} \xrightarrow{\mathrm{val}_{\mathrm{B}}} \varepsilon_{\mathrm{B}}, \qquad d_B(B, C) \le \varepsilon_{\mathrm{B}}, \qquad \varepsilon_{\mathrm{B}} < \mathrm{gap}_\tau(B, C),$$

hence

$$d_B\big(T_\tau^{\mathrm{bar}}(B), T_\tau^{\mathrm{bar}}(C)\big) \le \varepsilon_{\mathrm{B}},$$

and, under

$$2\varepsilon_{\mathrm{B}} \le \tau,$$

$$T_\tau^{\mathrm{bar}}(B) = 0 \quad \Longleftrightarrow \quad T_\tau^{\mathrm{bar}}(C) = 0.$$

The mathematically sharp zero-separation boundary is $2\varepsilon_{\mathrm{B}} \le \tau$. The v20 operational contract uses the strict reserve $2\varepsilon_{\mathrm{B}} < \tau$.

A complete detector may establish the source zero predicate. Appendix A transports that already certified predicate through the displayed metric chain. It does not transport raw detector ranks, sample values, birth or germ sets, coverage, certificate size, or completeness.

The default order is window first and threshold second. Source-level perturbations require crop and window-edge control before they become bounds on the actual pre-threshold profiles. Filtered or derived arguments require declared representatives. Tower and transient-defect arguments require actual comparison morphisms. Object-level bottleneck proximity does not automatically control kernel or cokernel defect profiles. Exactness requires a declared exact-admissible regime.

1.21 Foundation package

Theorem 1.98 (Appendix A v20 metric-repair and repaired-zero foundation package)**.** *Within the constructible one-parameter regime and the standard infinite-bar convention:*

(i) constructible modules admit interval decompositions;

(ii) *hard deletion removes exactly finite bars of length $\leq \tau$, preserves infinite bars, is idempotent, threshold-compositional, and threshold-monotone;*

(iii) *positive-threshold short-bar objects are not extension-closed in general;*

(iv) *hard deletion is not globally non-expansive;*

(v) *the sharp two-sided threshold-gap theorem holds;*

(vi) $\mathrm{gap}_\tau = \mathrm{clear}_\tau$*;*

(vii) *scalarized budgets are usable only with actual discrepancy upper-bound evidence;*

(viii) *every repaired profile is τ-reduced;*

(ix) *the bottleneck distance to zero is given exactly by Lemma 1.39;*

(x) *finite τ-reduced profiles have the sharp zero-separation boundary $2\varepsilon \leq \tau$;*

(xi) *metric-safe repaired-zero transport follows from the inherited threshold-gap theorem plus zero separation;*

(xii) *a complete detector certificate may be composed with the metric zero-transport theorem, but only its certified zero predicate is transported;*

(xiii) *raw pointwise detector ranks are not bottleneck-stable in general;*

(xiv) *clearance is measured after windowing and before deletion;*

(xv) *crop and hard deletion do not commute in general;*

(xvi) *they commute exactly under the all-retained-bars criterion;*

(xvii) *overlap repaired-zero equivalence requires both the inherited threshold-gap bound and post-threshold zero separation;*

(xviii) *functoriality and exactness are conditional on declared split or exact-admissible regimes;*

(xix) *representatives are window-relative declared representatives, not a global collapse functor;*

(xx) *Type IV requires actual morphisms, terminal tameness, finite or otherwise justified aggregation scope, and typed status;*

(xxi) *object-distance control does not automatically control kernel or cokernel defect profiles;*

(xxii) *manifest completeness does not replace mathematical validity.*

Proof. Items (i)–(ii) follow from Theorem 1.9, Lemma 1.13, and Propositions 1.20–1.21. Item (iii) is Proposition 1.15. Item (iv) is Example 1.22. Item (v) is Theorem 1.33 and Example 1.34. Items (vi)–(vii) are Definitions 1.26 and 1.27. Items (viii)–(ix) are Definition 1.37 and Lemma 1.39. Item (x) is Theorem 1.40. Item (xi) is Theorem 1.43 and Corollary 1.44. Item (xii) is Corollary 1.47. Item (xiii) is Example 1.49 and Proposition 1.50. Item (xiv) follows from Definitions 1.29, 1.56, and 1.57. Items (xv)–(xvi) are Proposition 1.55, Example 1.59, and Proposition 1.61. Item (xvii) is Corollary 1.71. Item (xviii) is Proposition 1.75 and Definition 1.76. Item (xix) is Definition 1.79. Item (xx) is Definitions 1.89, 1.91, and 1.92. Item (xxi) is Remark 1.94. Item (xxii) is Remark 1.97. □

1.22 Explicit non-claims

The following are not asserted.

(i) global non-expansiveness of hard positive-threshold deletion under the standard bottleneck metric;

(ii) a positive-threshold Serre subcategory in general;

(iii) a global exact endofunctor, abelian reflector, or Serre localization;

(iv) unconditional crop–collapse commutation;

(v) sufficiency of a finite-only commutation criterion;

(vi) source-to-window metric transport without a crop theorem, clearance argument, or crop budget;

(vii) direct comparison of an unscalarized quantale value with real clearance;

(viii) scalarization without evidence that it bounds the actual consumed bottleneck discrepancy;

(ix) numerical repair of a missing non-scalar obligation;

(x) one-sided threshold-gap certification;

(xi) replacement of a two-sided comparison record by a single-profile readiness value;

(xii) gap safety for a finite bar of length exactly τ;

(xiii) a sharp 2ε-band for the inherited two-sided theorem;

(xiv) $\mathrm{gap}_\tau = \frac{1}{2}\,\mathrm{clear}_\tau$;

(xv) a global strict filtered lift;

(xvi) an automatic tower comparison map;

(xvii) a kernel or cokernel defined merely by a barcode matching;

(xviii) identifying undefined Type IV input with zero;

(xix) fabricating numerical zero for a no-tower claim;

(xx) full comparison isomorphism from terminal-generic zero alone;

(xxi) an aggregate Type IV value over an undeclared infinite scope;

(xxii) identifying a bounded-window cofinal bar with a global infinite bar;

(xxiii) a well-typed Ext clause without an eligible representative;

(xxiv) identifying manifest completeness with proof;

(xxv) replacing a Core obligation by a Search artifact, advisory score, or AI ranking;

(xxvi) repaired-zero equivalence from threshold-gap stability alone without post-threshold zero separation;

(xxvii) metric zero transport when $2\varepsilon > \tau$;

(xxviii) bottleneck stability of raw pointwise detector ranks, birth sets, germ sets, sample counts, coverage witnesses, or certificate sizes;

(xxix) promotion of detector soundness to completeness by metric proximity or repeated zero observations;

(xxx) silent change of detector profile stage under metric transport;

(xxxi) automatic metric control of kernel or cokernel defect profiles from object-level bottleneck bounds;

(xxxii) proof of a Chapter 3 finite detector certificate merely because its already certified zero predicate can be transported metrically;

(xxxiii) any higher Ext, arithmetic, PDE, geometric, cryptographic, or external Return theorem.

1.23 Provenance and status

Interval decomposition, barcode uniqueness, and the bottleneck/interleaving isometry are standard one-parameter persistence results. The inherited v19 Core contribution is the metric and crop repair: rejection of global non-expansiveness, replacement by the sharp two-sided threshold-gap theorem, and the all-retained-bars crop criterion including infinite bars clipped by bounded windows. The v20 conservative extension adds the exact distance-to-zero formula, sharp zero separation in the τ-reduced image, metric-safe repaired-zero transport, detector-zero composition, and the explicit pointwise-rank instability boundary.

Status labels for Appendix A. *Standard persistence theorem:* Theorem 1.9.

Inherited metric-repair theorems: Theorem 1.33, Corollary 1.36.

v20 repaired-zero transport theorems: Theorem 1.40, Proposition 1.42, Theorem 1.43, Corollary 1.44, Corollary 1.47, Corollary 1.71.

Foundation closure theorem: Theorem 1.98.

Core correction and boundary results: Proposition 1.15, Corollary 1.16, Example 1.22, Corollary 1.24, Proposition 1.55, Example 1.59, Proposition 1.61, Example 1.49, Proposition 1.50.

Core definitions: Definitions 1.3, 1.5, 1.7, 1.8, 1.10, 1.12, 1.18, 1.25, 1.26, 1.27, 1.29, 1.37, 1.38, 1.48, 1.52, 1.53, 1.60, 1.63, 1.78, 1.79, 1.81, 1.83, 1.84, 1.85, 1.88, 1.89, 1.91, 1.92.

Operational policy and records: Definitions 1.31, 1.45, 1.46, 1.56, 1.57, 1.65, 1.66, 1.68, 1.86, 1.95, 1.96; Remarks 1.64, 1.62, 1.51, 1.90, 1.93, 1.94, 1.97.

Exact-admissible optional regime: Definitions 1.72, 1.73, 1.74, 1.76; Proposition 1.75.

Explicit exclusions: Subsection 1.22.

2 Appendix B: Finite Detector Calculus and Filtered Representatives of Windowed Barcode Collapse

2.1 Scope and dependence on the repaired Core

Appendix B is the detailed Foundation Appendix for the v20 finite-certificate layer between the inherited metric repair of Chapter 2 and Appendix A and the higher eligible realization calculus of Chapter 4. It preserves the complete v19 filtered-representative discipline and adds the v20 finite zero-reflecting detector and proof-compression calculus at exactly the strength proved in Chapter 3.

The canonical project acronym is AK-HPDST. The legacy transposed acronym AK-HDPST may occur in historical artifacts, but it has no normative use in this appendix.

The Core does not treat positive-threshold deletion as a global exact Serre reflector, a globally non-expansive endofunctor, or a strict functor on all persistence morphisms. Its primary data are the declared windowed pre-threshold profile

$$B_{i,W}(F) := \mathsf{W}_W \mathbf{P}_i(F)$$

and the repaired barcode-level evaluation

$$\mathrm{Eval}_{i,W,\tau}(F) := T_\tau^{\mathrm{bar}}(B_{i,W}(F)).$$

Appendix B has two coupled but non-interchangeable responsibilities.

(i) It proves when finitely many structure maps of a declared pre-threshold profile give a sound or complete certificate for the exact zero/nonzero predicate of the repaired profile.

(ii) It specifies when a family of repaired profiles may be represented by a filtered chain complex for filtered, derived, higher-Ext, or tower-level use.

The central rules are:

> A finite observation is not a finite proof certificate until its source profile stage, endpoint convention, soundness, completeness or coverage, family membership, excess-bound source, and immutable artifacts are certified.
>
> A filtered representative does not define collapse and does not retroactively supply a detector certificate. It realizes a declared repaired persistence profile, at a declared strength, for a declared monitored scope.

Accordingly, Appendix B separates:

(i) windowed pre-threshold and repaired post-threshold profiles;

(ii) detector stages `windowed_prethreshold`, `repaired_postthreshold`, `kernel_defect`, and `cokernel_defect`;

(iii) finite-stencil soundness from detector-zero completeness;

(iv) left-closed birth detection from decoration-invariant right-germ detection;

(v) externally certified coverage from family-mesh completeness;

(vi) mathematical completeness from meaningful proof compression;

(vii) family-relative MinCert, sharp mesh bounds, and the right-unbounded no-go regime;

(viii) metric transport of a certified zero predicate from stability of raw detector ranks;

(ix) object-level realization of a finite barcode family;

(x) monitored profile equivalence, levelwise filtered quasi-isomorphism, and realization-compatible equivalence;

(xi) actual representative morphism and tower-comparison artifacts;

(xii) detector preservation at an exact source stage from mere representative admissibility.

Remark 2.1 (Conservative-extension boundary)**.** This appendix does not restore a global exact collapse functor, a Serre localization, global non-expansiveness, automatic tower comparison, or a reverse higher-Ext-to-persistence bridge. It does not assert pointwise detector-rank stability under bottleneck perturbation. It does not identify a finite detector with a natural higher Ext-edge.

The object-level realization theorem concerns finite barcode profile families only. It does not provide functoriality on arbitrary morphisms, preserve additional geometric structure, establish higher eligibility, or prove an external Return Theorem.

Remark 2.2 (Dependency direction)**.** Appendix B consumes the barcode, window, threshold, repaired-zero transport, and metric conventions of Appendix A. Chapter 3 supplies the canonical Main statements whose detailed Foundation proofs are recorded here. Appendix C consumes the realization-compatible representative contract of Appendix B for the degree-n edge calculus. Appendix D may consume representative tower data and defect-stage detector records, but an actual tower comparison morphism and the kernel/cokernel profiles are never generated by Appendix B alone.

Division of responsibility. Appendix A proves metric-safe transport of an already certified repaired-zero predicate. Appendix B proves the finite detector characterizations and completeness conditions that can certify the source zero predicate. Appendix C proves the higher eligible realization and Ext-edge statements. Appendix D proves terminal/transient defect and finite-to-infinite tower results. No one layer silently discharges another.

2.2 Filtered chain complexes and monitored persistence

Definition 2.3 (Finite-type filtered chain complex)**.** A *finite-type filtered chain complex* over a field k is a bounded chain complex $C_\bullet$ of finite-dimensional k-vector spaces, together with an increasing filtration by subcomplexes

$$F^t C_\bullet \subseteq F^{t'} C_\bullet \qquad (t \le t'),$$

such that:

(i) the filtration is exhaustive and left-bounded;

(ii) only finitely many filtration-critical values occur;

(iii) for every homological degree i, the persistence module

$$\mathbf{P}_i(F) : t \longmapsto H_i(F^t C_\bullet)$$

belongs to $\mathsf{Pers}_k^{\mathrm{cons}}$.

The category of such objects and filtration-preserving chain maps is denoted by

$$\mathsf{FiltCh}(k)\,.$$

Definition 2.4 (Persistence functor)**.** For every homological degree i, define

$$\mathbf{P}_i : \mathsf{FiltCh}(k) \longrightarrow \mathsf{Pers}_k^{\mathrm{cons}}$$

by

$$\mathbf{P}_i(F)(t) := H_i(F^t C_\bullet).$$

Definition 2.5 (Monitored scope)**.** A *monitored scope* is a finite set of homological degrees

$$I_{\mathrm{mon}} \subset \mathbb{Z}.$$

All representative assertions in this appendix are relative to a declared monitored scope unless a full-degree statement is explicitly proved.

Remark 2.6 (Monitored-degree convention)**.** Agreement in I_{mon} does not imply agreement in unmonitored degrees. A proof object must not silently upgrade a monitored representative to a full filtered-chain equivalence.

2.3 Three levels of representative equivalence

Definition 2.7 (Monitored persistence equivalence)**.** Two filtered objects $F, G \in \mathsf{FiltCh}(k)$ are *monitored-persistence equivalent*, written

$$F \simeq_{\mathrm{PH}, I_{\mathrm{mon}}} G,$$

if for every $i \in I_{\mathrm{mon}}$ there is an isomorphism

$$\mathbf{P}_i(F) \cong \mathbf{P}_i(G)$$

in $\mathsf{Pers}_k^{\mathrm{cons}}$.

Definition 2.8 (Levelwise filtered quasi-isomorphism)**.** A filtration-preserving chain map

$$f : F \longrightarrow G$$

is a *levelwise filtered quasi-isomorphism* if, for every parameter t, the map of chain complexes

$$F^t C_\bullet \longrightarrow G^t D_\bullet$$

is a quasi-isomorphism.

Such a map induces

$$\mathbf{P}_i(F) \cong \mathbf{P}_i(G)$$

for every degree i, and hence monitored-persistence equivalence for every monitored scope.

Definition 2.9 (Realization-compatible equivalence)**.** Fix a realization functor or realization procedure

$$\mathcal{R} : \mathsf{FiltCh}(k) \dashrightarrow D^{\mathrm{b}}(k\text{-mod})$$

on a declared domain.

Two representatives F, G are *$\mathcal{R}$-equivalent* if:

(i) both $\mathcal{R}(F)$ and $\mathcal{R}(G)$ are defined;

(ii) an actual isomorphism

$$\mathcal{R}(F) \cong \mathcal{R}(G)$$

is supplied in $D^{\mathrm{b}}(k\text{-mod})$;

(iii) the isomorphism is compatible with every structure consumed by the relevant Ext or bridge argument.

Proposition 2.10 (Implications between equivalence levels)**.** *A levelwise filtered quasi-isomorphism implies monitored-persistence equivalence. Monitored-persistence equivalence alone does not imply a levelwise filtered quasi-isomorphism, chain-homotopy equivalence, or $\mathcal{R}$-equivalence.*

Proof. The first statement follows by taking homology at every filtration level and using naturality of the structure maps.

For the second statement, monitored persistence records only the declared homology persistence modules. It forgets chain-level extension data, unmonitored degrees, and any extra structure consumed by $\mathcal{R}$. Hence no stronger equivalence follows without an additional theorem or artifact. □

Remark 2.11 (Why the equivalence hierarchy matters)**.** Levelwise filtered quasi-isomorphism is sufficient for every monitored persistence clause, but it is stronger than mere profile agreement. The weaker profile relation is therefore used only where the proof consumes no additional chain-level or derived structure.

Remark 2.12 (No automatic localization)**.** The v20 Core does not require a global localization of $\mathsf{FiltCh}(k)$ at monitored-persistence equivalences. If a later construction uses a localized category, it must declare the class of weak equivalences, prove that the localization is suitable for the intended argument, and record the actual morphisms used there.

Definition 2.13 (Localized representative category)**.** When a declared workflow has fixed a class $\mathcal{W}$ of filtered quasi-isomorphisms for which the required localization exists, write

$$\mathrm{Ho}_{\mathcal{W}}\big(\mathsf{FiltCh}(k)\big)$$

for that declared localization.

This notation is optional and has no default global meaning in Appendix B.

2.4 Windowed repaired persistence profiles

Recall the fixed evaluation order from Chapter 1 and Appendix A:

$$F \longmapsto \mathbf{P}_i(F) \longmapsto \mathsf{W}_W\mathbf{P}_i(F) \longmapsto T_\tau^{\mathrm{bar}}\big(\mathsf{W}_W\mathbf{P}_i(F)\big).$$

Definition 2.14 (Windowed repaired persistence profile)**.** For $F \in \mathsf{FiltCh}(k)$, $i \in I_{\mathrm{mon}}$, a right-open window W, and $\tau > 0$, define

$$\mathcal{P}_{i,W,\tau}(F) := \mathrm{Eval}_{i,W,\tau}(F) = T_\tau^{\mathrm{bar}}\big(\mathsf{W}_W\mathbf{P}_i(F)\big).$$

Definition 2.15 (Profile family over the monitored scope)**.** For a finite monitored scope I_{mon}, define

$$\mathcal{P}_{W,\tau}^{I_{\mathrm{mon}}}(F) := \big\{\mathcal{P}_{i,W,\tau}(F)\big\}_{i\in I_{\mathrm{mon}}}.$$

Remark 2.16 (The profile is primary)**.** The family $\mathcal{P}_{W,\tau}^{I_{\mathrm{mon}}}(F)$ is the primary Core object. A filtered representative is secondary evidence used only when a chain-level, derived, Ext, or tower-level construction requires it.

Definition 2.17 (Comparison mode for profile claims)**.** Every comparison between representative profiles is declared as exactly one of the following:

$$\texttt{exact}, \qquad \texttt{metric\_gap\_certified}, \qquad \texttt{not\_compared}.$$

An exact comparison is supported by an actual isomorphism or equality. A metric comparison is supported by a two-sided threshold-gap record and an artifact-backed bottleneck bound on the relevant windowed pre-threshold profiles. The mode `not_compared` is permitted only when the governing claim consumes no comparison inference; it cannot discharge detector transport, representative comparison, overlap, Restart, or tower obligations.

2.5 Detector profile stages and effective domains

Definition 2.18 (Detector profile stage)**.** Every detector record declares exactly one source stage:

$$\begin{gathered}\texttt{windowed\_prethreshold},\\ \texttt{repaired\_postthreshold},\\ \texttt{kernel\_defect}, \qquad \texttt{cokernel\_defect}.\end{gathered}$$

The standard finite stencil, birth, germ, and family-mesh theorems below are proved at the stage

$$\texttt{windowed\_prethreshold}.$$

A theorem proved at one stage is not reused at another stage without an explicit stage-transport theorem or exact source-stage isomorphism. The defect stages are available only after an actual comparison morphism and well-typed kernel or cokernel profile have been supplied by Chapter 5 and Appendix D.

Definition 2.19 (Effective detector domain)**.** Let

$$W = [a, b), \qquad b \in \mathbb{R} \cup \{+\infty\}, \qquad \tau > 0.$$

Define

$$W_\tau := \{t \in W \mid t + \tau \in W\}.$$

Thus

$$W_\tau = \begin{cases} [a, b - \tau), & b < +\infty \text{ and } b - a > \tau, \\ \varnothing, & b < +\infty \text{ and } b - a \le \tau, \\ [a, +\infty), & b = +\infty. \end{cases}$$

Only points of W_τ may be sampled by the exact τ-stencil detector.

Remark 2.20 (Window-relative endpoint and cofinal policy)**.** Every bar in this detector calculus is a bar of the already windowed profile. A bar reaching the right endpoint of a bounded window is window-cofinal but finite for threshold-length purposes. Its length is the clipped window-relative length and it is included in every coverage, excess, and boundary calculation. A bar is genuinely infinite only on a right-unbounded window.

2.6 Exact finite stencil and active-start regions

Definition 2.21 (Exact finite stencil detector)**.** Let B be a constructible persistence module on W, let $T \subseteq W_\tau$ be finite, and fix the left-closed right-open interval registration convention. Define

$$\mathcal{D}^{\mathrm{stc}}_{T,\tau}(B) := \bigoplus_{t \in T} \mathrm{im}\big(B(t) \longrightarrow B(t+\tau)\big).$$

Its value is zero exactly when every displayed image is zero. The direct sum is finite and retains vector-space multiplicity.

Definition 2.22 (Retained bars and active-start region)**.** Let $\bar{\ }(B)$ be the barcode of a declared windowed profile and define, with multiplicity,

$$\mathrm{Ret}_{\tau,W}(B) := \{I \in \bar{\ }(B) \mid \ell_W(I) > \tau\}.$$

For an interval summand I, define its τ-active-start region by

$$\mathrm{Act}_\tau(I) := \{t \in W_\tau \mid t, t+\tau \in I\}.$$

If $I = [p, q)$ in the left-closed right-open convention, then

$$\mathrm{Act}_\tau(I) = \begin{cases} [p, q-\tau), & q < +\infty \text{ and } q - p > \tau, \\ [p, +\infty), & q = +\infty, \\ \varnothing, & q - p \le \tau. \end{cases}$$

Thus $\mathrm{Act}_\tau(I) \neq \varnothing$ exactly when I survives T^{bar}_τ.

Proposition 2.23 (Interval decomposition of a stencil image)**.** *Under Definition 2.21,*

$$\dim_k \mathrm{im}\big(B(t) \to B(t+\tau)\big)$$

is the number, counted with multiplicity, of interval summands I for which $t \in \mathrm{Act}_\tau(I)$. Consequently,

$$\mathcal{D}^{\mathrm{stc}}_{T,\tau}(B) \neq 0 \iff \exists I \in \mathrm{Ret}_{\tau,W}(B) \text{ with } T \cap \mathrm{Act}_\tau(I) \neq \varnothing.$$

Proof. Decompose B into interval modules. On an interval summand k_I, the structure map from t to $t+\tau$ is the identity exactly when both parameters lie in I, and is zero otherwise. Images and finite direct sums decompose componentwise in the pointwise finite-dimensional constructible regime. □

Theorem 2.24 (Finite-stencil soundness)**.** *For every finite $T \subseteq W_\tau$,*

$$\mathcal{D}^{\mathrm{stc}}_{T,\tau}(B) \neq 0 \implies T^{\mathrm{bar}}_\tau(B) \neq 0.$$

Equivalently,

$$T^{\mathrm{bar}}_\tau(B) = 0 \implies \mathcal{D}^{\mathrm{stc}}_{T,\tau}(B) = 0.$$

No coverage or mesh hypothesis is required for this direction.

Proof. A nonzero detector component supplies a retained interval whose active-start region contains a sample point, by Proposition 2.23. □

Remark 2.25 (Soundness is not completeness)**.** A zero stencil may miss a retained bar whose active-start region contains no sample point. The converse of Theorem 2.24 therefore requires a birth, germ, externally certified coverage, or family-mesh theorem. Missing completeness evidence makes a detector-based zero verdict `undefined`; it does not manufacture profile vanishing.

2.7 Birth and right-germ characterizations

Definition 2.26 (Exhaustive birth set)**.** For a finite left-closed right-open windowed barcode

$$\bar{(}B) = \{[p_\lambda, q_\lambda)\}_{\lambda\in\Lambda},$$

define

$$T_{\mathrm{birth}}(B) := \{p_\lambda \mid \lambda \in \Lambda\}, \qquad T_{\mathrm{birth},\tau}(B) := T_{\mathrm{birth}}(B) \cap W_\tau.$$

A birth-set certificate is exhaustive when it proves that every interval summand has its left endpoint in the recorded finite set. Only the effective subset is sampled. Multiplicity remains part of the barcode artifact even when repeated birth values are sampled once.

Theorem 2.27 (Birth detector characterization)**.** *Let B be a finite windowed barcode in the left-closed right-open convention and let its exhaustive birth set be certified. Then*

$$\mathcal{D}^{\mathrm{stc}}_{T_{\mathrm{birth},\tau}(B),\tau}(B) \neq 0 \quad\Longleftrightarrow\quad T^{\mathrm{bar}}_\tau(B) \neq 0.$$

Equivalently,

$$\mathcal{D}^{\mathrm{stc}}_{T_{\mathrm{birth},\tau}(B),\tau}(B) = 0 \quad\Longleftrightarrow\quad T^{\mathrm{bar}}_\tau(B) = 0.$$

Proof. Soundness is Theorem 2.24. Conversely, a retained interval $I = [p, q)$ satisfies $p + \tau < q$, so its structure map at its certified birth point is nonzero. □

Remark 2.28 (Endpoint dependence of the point-birth theorem)**.** Theorem 2.27 evaluates $B(p)$ and therefore consumes the left-closed registration convention. It is not decoration-invariant. Decorated intervals are handled by the right-germ theorem below or by another separately proved endpoint-aware theorem.

Definition 2.29 (Stable right germ and shifted germ map)**.** Let B be constructible and let $p \in \mathbb{R}$. Choose $\delta_p > 0$ such that every structure map

$$B(p+\epsilon) \longrightarrow B(p+\epsilon') \qquad (0 < \epsilon \le \epsilon' < \delta_p)$$

is an isomorphism. The *stable right germ* $B(p^+)$ is the resulting canonical isomorphism class of these stable values. Equivalently, after fixing any $0 < \epsilon_p < \delta_p$, one may take

$$B(p^+) := B(p + \epsilon_p),$$

with independence of the choice supplied by the displayed transition isomorphisms.

The traditional notation

$$B(p^+) = \varinjlim_{\epsilon\downarrow 0} B(p+\epsilon)$$

is used only as shorthand for this constructible stable-right-value construction. It is not an unrestricted colimit over all $\epsilon > 0$.

For $\tau > 0$, choose $\delta > 0$ small enough that the stable-right conditions hold simultaneously at p and $p+\tau$. For any $0 < \epsilon < \delta$, the structure map

$$B(p+\epsilon) \longrightarrow B(p+\tau+\epsilon)$$

induces a choice-independent map

$$\gamma_{p,\tau} : B(p^+) \longrightarrow B((p+\tau)^+).$$

Naturality and the stable transition isomorphisms make $\gamma_{p,\tau}$ independent of the auxiliary ϵ.

Definition 2.30 (Germ detector)**.** Let $T_{\mathrm{gbirth}}(B)$ be the finite set of geometric lower endpoint values of the decorated interval summands, and put

$$T_{\mathrm{gbirth},\tau}(B) := T_{\mathrm{gbirth}}(B) \cap W_\tau.$$

Define

$$\mathcal{D}_\tau^{\mathrm{germ}}(B) := \bigoplus_{p \in T_{\mathrm{gbirth},\tau}(B)} \mathrm{im}(\gamma_{p,\tau}).$$

An exhaustive geometric-birth certificate is part of the detector record.

Theorem 2.31 (Decoration-invariant germ characterization)**.** *Let B be a finite decorated windowed barcode with an exhaustive geometric-birth set. Then, independently of finite-endpoint open or closed decoration,*

$$\mathcal{D}_\tau^{\mathrm{germ}}(B) \neq 0 \iff T_\tau^{\mathrm{bar}}(B) \neq 0.$$

Equivalently,

$$\mathcal{D}_\tau^{\mathrm{germ}}(B) = 0 \iff T_\tau^{\mathrm{bar}}(B) = 0.$$

Proof. For a decorated interval with geometric endpoints $p < q$, the shifted right germ is nonzero exactly when points immediately to the right of p and $p + \tau$ remain in the interval. This is equivalent to $p + \tau < q$, hence to length strictly greater than τ. At equality $q - p = \tau$, the target right germ lies strictly to the right of q and is zero, including when q is closed. Summing over the exhaustive geometric-birth set proves the result. □

2.8 Externally certified stencil completeness

Definition 2.32 (Stencil coverage certificate)**.** A stencil coverage certificate for (B, W, τ, T) is an artifact-backed proof of

$$\forall I \in \mathrm{Ret}_{\tau,W}(B), \qquad T \cap \mathrm{Act}_\tau(I) \neq \varnothing.$$

The record declares the source of the coverage evidence and includes bounded-window cofinal bars. Coverage is a non-scalar obligation.

Theorem 2.33 (Certified stencil completeness)**.** *Let $T \subseteq W_\tau$ be finite and suppose a valid stencil coverage certificate is supplied. Then*

$$\mathcal{D}_{T,\tau}^{\mathrm{stc}}(B) = 0 \iff T_\tau^{\mathrm{bar}}(B) = 0.$$

Proof. The implication from right to left is Theorem 2.24. If the repaired profile is nonzero, choose a retained interval. Coverage supplies a sample point in its active-start region, and Proposition 2.23 makes the stencil nonzero. □

Remark 2.34 (Logical validity versus proof compression)**.** A genuine coverage proof establishes mathematical completeness even if it was obtained by reconstructing the complete target barcode. A claim of proof compression additionally requires a non-circular source, such as a family-generation theorem, finite critical-data theorem, endpoint lattice, separation theorem, filtration constraint, or external theorem. A posteriori full-profile enumeration does not by itself establish a compression gain.

2.9 Family threshold excess, augmented mesh, and sharpness

Definition 2.35 (Window-relative threshold excess)**.** For a windowed barcode B, define

$$\mathrm{exc}_{\tau,W}(B) := \inf_{I \in \mathrm{Ret}_{\tau,W}(B)} \big(\ell_W(I) - \tau\big),$$

with $\inf \varnothing := +\infty$. A genuinely infinite retained interval contributes $+\infty$; a bounded-window cofinal interval contributes its finite clipped excess.

For a nonempty barcode family $\mathcal{F}$, a uniform excess lower-bound certificate is a number

$$e_{\tau,W}(\mathcal{F}) > 0$$

with artifact-backed proof that

$$e_{\tau,W}(\mathcal{F}) \le \mathrm{exc}_{\tau,W}(B) \qquad (B \in \mathcal{F}).$$

Remark 2.36 (Independent source of family excess)**.** When family-mesh completeness is advertised as proof compression, the lower bound must be obtained independently of complete retrospective reading of the target barcode. Permitted sources include a model-class theorem, endpoint lattice, Lipschitz or separation theorem, filtration constraint, external-domain theorem, or certified family generator. An absent or unverifiable source makes family-mesh completeness `undefined`.

Definition 2.37 (Augmented mesh)**.** Assume $W = [a, b)$ is bounded and $b - a > \tau$. Put

$$c := b - \tau, \qquad W_\tau = [a, c), \qquad L_{\mathrm{eff}} := c - a = b - a - \tau.$$

For

$$T = \{t_1 < \cdots < t_m\} \subseteq [a, c),$$

define

$$\mathrm{mesh}_{W_\tau}(T) := \max\{t_1 - a, t_2 - t_1, \ldots, t_m - t_{m-1}, c - t_m\}.$$

For $T = \varnothing$, set

$$\mathrm{mesh}_{W_\tau}(\varnothing) := L_{\mathrm{eff}}.$$

The endpoints a, c are virtual boundary points for measuring gaps; they are not automatically detector samples.

Lemma 2.38 (Augmented-mesh hitting lemma)**.** *Let $J = [p, r) \subseteq [a, c)$. If*

$$r - p > \mathrm{mesh}_{W_\tau}(T),$$

then $J \cap T \neq \varnothing$.

Proof. If $J \cap T = \varnothing$, then J lies in one component between two successive augmented boundary or sample points. Its length is at most that augmented gap, hence at most the mesh. □

Theorem 2.39 (Family-mesh completeness)**.** *Let $W = [a, b)$ be bounded, let $\mathcal{F}$ be a nonempty family of left-closed right-open windowed barcodes, and suppose a valid uniform excess lower bound $e_{\tau,W}(\mathcal{F}) > 0$ is supplied. If a finite stencil $T \subseteq W_\tau$ satisfies*

$$\mathrm{mesh}_{W_\tau}(T) < e_{\tau,W}(\mathcal{F}),$$

then T is complete for every $B \in \mathcal{F}$:

$$\forall B \in \mathcal{F}, \qquad \mathcal{D}^{\mathrm{stc}}_{T,\tau}(B) = 0 \iff T^{\mathrm{bar}}_{\tau}(B) = 0.$$

If $b - a \le \tau$, then $W_\tau = \varnothing$ and every finite windowed bar has length at most τ, so the finite-bar zero conclusion is trivial.

Proof. Let $I = [p, q)$ be retained. Its active-start region has length

$$q - p - \tau \geq e_{\tau,W}(\mathcal{F}) > \mathrm{mesh}_{W_\tau}(T).$$

The hitting lemma gives a sample in every retained active-start region, so Theorem 2.33 applies. □

Theorem 2.40 (Strict mesh sharpness and equality boundary)**.** *Fix a bounded effective domain $W_\tau = [a, c)$ and a finite stencil T.*

(i) If $\mathrm{mesh}_{W_\tau}(T) > e > 0$, there exists a one-bar barcode B satisfying

$$\mathrm{exc}_{\tau,W}(B) > e, \qquad T_\tau^{\mathrm{bar}}(B) \neq 0, \qquad \mathcal{D}_{T,\tau}^{\mathrm{stc}}(B) = 0.$$

(ii) The strict hypothesis in Theorem 2.39 cannot uniformly be replaced by $\mathrm{mesh}_{W_\tau}(T) \leq e$. For example, when $t_1 - a = e$, the retained interval

$$I = [a, a + \tau + e)$$

has active region $[a, t_1)$, containing no detector sample.

(iii) At equality $\mathrm{mesh}_{W_\tau}(T) = e$, completeness is sensitive to boundary convention and the location of the maximal gap. No universal pass or fail conclusion follows without additional placement data.

Proof. For (i), choose an augmented gap of length $\mathrm{mesh}_{W_\tau}(T)$, choose $e < e' < \mathrm{mesh}_{W_\tau}(T)$, and place a half-open active-start interval $[p, p + e')$ inside the sample-free gap. The bar $[p, p + \tau + e')$ lies in W, survives thresholding, and is missed by T. Item (ii) is the displayed left-boundary construction. Item (iii) follows because internal and boundary equality gaps have different half-open endpoint behavior. □

2.10 Detector certificates and two-axis status

Definition 2.41 (Finite detector certificate record)**.** A finite detector certificate record contains at least:

(i) detector, theorem, run, and source-profile identifiers;

(ii) the exact source object and profile stage;

(iii) monitored degree, window, threshold, endpoint convention, and window-cofinal policy;

(iv) detector mode

`birth`, `germ`, `certified_stencil`, `family_mesh`;

(v) the finite stencil, birth set, or geometric-birth set;

(vi) soundness evidence;

(vii) completeness, characterization, or coverage evidence;

(viii) family membership and independently sourced excess evidence when family mesh is invoked;

(ix) exact or metric transport records when the consumed predicate belongs to a compared profile;

(x) certificate size, family-relative MinCert data, and immutable replay artifacts;

(xi) separate certificate and mathematical-value statuses.

Definition 2.42 (Two-axis detector status)**.** Detector certification and mathematical value are independent fields. The certificate axis is

`detector_certified`, `detector_failed`, `detector_pending`,

mapping respectively to the Chapter 7 atomic statuses

`pass`, `reject`, `undefined`.

Here `detector_failed` means that a well-typed required detector condition is evidenced false. Missing, incomplete, stale, or pending evidence is `detector_pending`, not failure.

The value axis is

`zero`, `nonzero`, `undefined`.

A zero value discharges a zero clause only with certified completeness. A certified nonzero value rejects a required zero clause. A raw zero-looking computation with missing completeness remains undefined. When detector semantics are outside the declared scope, the atomic status is `not_invoked`; no certificate or value token is fabricated.

Proposition 2.43 (Detector invariance under exact source-stage isomorphism)**.** *Let*

$$\eta : B \xrightarrow{\cong} C$$

be an isomorphism of persistence modules on the declared window. For every finite $T \subseteq W_\tau$,

$$\mathcal{D}^{\mathrm{stc}}_{T,\tau}(B) \cong \mathcal{D}^{\mathrm{stc}}_{T,\tau}(C).$$

The corresponding right-germ detectors are isomorphic when the same geometric-birth convention is used. Hence detector zero/nonzero is invariant under an actual isomorphism of the exact source-stage profiles.

Proof. Naturality gives a commutative square for every structure map from t to $t + \tau$. Vertical isomorphisms induce isomorphisms of image spaces, and finite direct sums preserve them. Passing through the canonical stable right-neighborhood identifications gives the germ assertion. □

2.11 Metric-safe detector composition and robust support

Theorem 2.44 (Metric zero-verdict composition)**.** *Let B, C be windowed pre-threshold profiles. Suppose:*

(i) complete detector certificates establish

$$\mathcal{D}^{\mathrm{stc}}_{T_B,\tau}(B) = 0 \Longleftrightarrow T^{\mathrm{bar}}_\tau(B) = 0$$

and

$$\mathcal{D}^{\mathrm{stc}}_{T_C,\tau}(C) = 0 \Longleftrightarrow T^{\mathrm{bar}}_\tau(C) = 0;$$

(ii) $d_B(B, C) \le \varepsilon$;

(iii) the two-sided threshold-gap hypotheses of Appendix A hold at the same ε;

(iv) $2\varepsilon \le \tau$.

Then

$$\mathcal{D}^{\mathrm{stc}}_{T_B,\tau}(B) = 0 \quad \Longleftrightarrow \quad \mathcal{D}^{\mathrm{stc}}_{T_C,\tau}(C) = 0.$$

The operational v20 certificate uses the strict reserve $2\varepsilon < \tau$.

Proof. Theorem 1.43 gives

$$T^{\mathrm{bar}}_\tau(B) = 0 \Longleftrightarrow T^{\mathrm{bar}}_\tau(C) = 0.$$

Compose this with the two complete detector characterizations. □

Corollary 2.45 (One-way transport of a detector-certified zero)**.** *If a complete detector certificate proves* $T^{\mathrm{bar}}_\tau(B) = 0$ *and the metric hypotheses of Theorem 2.44 hold, then*

$$T^{\mathrm{bar}}_\tau(C) = 0.$$

A second detector certificate is required only when the target conclusion is to be restated as a target detector value.

Remark 2.46 (Zero predicate, not raw detector-rank stability)**.** The metric theorem transports a certified repaired-zero predicate. It does not compare raw ranks at common sample points, preserve birth or germ sets, transport coverage witnesses, or preserve certificate size. Appendix A supplies explicit counterexamples to pointwise rank stability.

Definition 2.47 (Boundary-safe robust stencil)**.** Let B, C be equipped with actual ε-interleaving morphisms. Define

$$W^{\mathrm{rob}}_{\tau,\varepsilon} := \{t \mid t - \varepsilon,\ t + \tau + \varepsilon \in W\}.$$

For finite $T \subseteq W^{\mathrm{rob}}_{\tau,\varepsilon}$, define

$$\mathcal{D}^{\mathrm{rob}}_{T,\tau,\varepsilon}(B) := \bigoplus_{t\in T} \mathrm{im}\big(B(t-\varepsilon) \to B(t+\tau+\varepsilon)\big).$$

For $W = [a, b)$, the boundary-safe domain is $[a + \varepsilon, b - \tau - \varepsilon)$ when nonempty.

Proposition 2.48 (Robust-to-exact sandwich)**.** *Let* B, C *carry specified* ε*-interleaving morphisms on the required boundary-safe domain. Then*

$$\mathcal{D}^{\mathrm{rob}}_{T,\tau,\varepsilon}(B) \neq 0 \quad \Longrightarrow \quad \mathcal{D}^{\mathrm{stc}}_{T,\tau}(C) \neq 0,$$

and symmetrically with B, C *exchanged.*

Proof. For each $t \in T$, the structure map

$$B(t-\varepsilon) \longrightarrow B(t+\tau+\varepsilon)$$

factors, by naturality and the interleaving identities, through

$$B(t-\varepsilon) \longrightarrow C(t) \longrightarrow C(t+\tau) \longrightarrow B(t+\tau+\varepsilon).$$

A nonzero composite forces the middle exact structure map of C to be nonzero. □

Remark 2.49 (Supporting status of the robust sandwich)**.** The sandwich is one-way and requires actual interleaving morphisms, not merely a scalar distance. It is a supporting detector theorem, not a substitute for completeness or for the repaired-zero metric chain.

2.12 Family-relative minimal certificates

Definition 2.50 (Detector information class)**.** A detector information class is a tuple

$$\Theta = (\mathcal{F}, W, \tau, \mathrm{stage}, \mathrm{endpt}, \mathrm{critical}, \mathrm{coverage}),$$

where $\mathcal{F}$ is a nonempty declared barcode family and the remaining fields fix the detector stage, endpoint, critical-data, and coverage policies. A finite set T is complete for Θ when the governing theorem proves, for every $B \in \mathcal{F}$,

$$\mathcal{D}^{\mathrm{stc}}_{T,\tau}(B) = 0 \quad \Longleftrightarrow \quad T^{\mathrm{bar}}_{\tau}(B) = 0.$$

Definition 2.51 (Family-relative minimal certificate)**.** Define

$$\mathrm{MinCert}(\Theta) := \min\{|T| \mid T \text{ is a complete finite stencil for } \Theta\},$$

when such a finite stencil exists, and set

$$\mathrm{MinCert}(\Theta) := +\infty$$

otherwise. Weighted proof size and compression ratios are Technical quantities unless separately promoted. The empty stencil for a single already-known zero object does not define a nontrivial family-relative certificate bound.

Remark 2.52 (Minimum versus infimum notation)**.** The release-level contract may write the same quantity as an infimum. Since every finite stencil size lies in $\mathbb{N}$, whenever a complete finite stencil exists the nonempty set of attainable sizes has a least element. Thus the minimum in Definition 2.51 equals that infimum. No substantive claim depends on the notational choice.

Proposition 2.53 (Birth-set upper bound)**.** *Suppose*

$$T_{\mathrm{birth},\tau}(\mathcal{F}) := \bigcup_{B \in \mathcal{F}} T_{\mathrm{birth},\tau}(B)$$

is finite and exhaustiveness is certified uniformly over the family. Then

$$\mathrm{MinCert}(\Theta) \le |T_{\mathrm{birth},\tau}(\mathcal{F})|.$$

For an object-relative certificate, the corresponding upper bound is $|T_{\mathrm{birth},\tau}(B)|$.

Proof. The union contains the effective birth set of every family member, so Theorem 2.27 detects every retained bar in the family. ◻

Definition 2.54 (Minimal augmented-mesh stencil number)**.** For $L_{\mathrm{eff}} \ge 0$ and $\gamma > 0$, define

$$m_{\mathrm{mesh}}(L_{\mathrm{eff}}, \gamma) := \max\left\{0, \left\lceil \frac{L_{\mathrm{eff}}}{\gamma} \right\rceil - 1\right\}.$$

Theorem 2.55 (Exact stencil count for an augmented-mesh target)**.** *Fix $\gamma > 0$. Let $W_\tau = [a, c)$ have length $L_{\mathrm{eff}} = c - a$. Then:*

(i) there exists $T \subseteq [a, c)$ with

$$|T| = m_{\mathrm{mesh}}(L_{\mathrm{eff}}, \gamma), \qquad \mathrm{mesh}_{W_\tau}(T) \le \gamma;$$

(ii) every stencil satisfying $\mathrm{mesh}_{W_\tau}(T) \le \gamma$ *has*

$$|T| \ge m_{\mathrm{mesh}}(L_{\mathrm{eff}}, \gamma).$$

Thus the displayed number is exact for the augmented-mesh design problem.

Proof. Put $N = \lceil L_{\mathrm{eff}}/\gamma \rceil$. If $N = 0$, the effective domain is empty. If $N \ge 1$, divide it into N equal gaps and place $N - 1$ internal samples. Conversely, m samples create $m + 1$ augmented gaps whose lengths sum to L_{eff}; if all are at most γ, then $L_{\mathrm{eff}} \le (m + 1)\gamma$. □

Corollary 2.56 (Family-mesh upper bound for MinCert)**.** *Under Theorem 2.39, choose*

$$0 < \gamma < e_{\tau,W}(\mathcal{F}).$$

Then

$$\mathrm{MinCert}(\Theta) \le m_{\mathrm{mesh}}(L_{\mathrm{eff}}, \gamma).$$

This need not equal the true MinCert *of a family with additional structure.*

2.13 Right-unbounded-window no-go theorem

Theorem 2.57 (Uniform finite-detector no-go on an unbounded birth family)**.** *Let*

$$W = [a, +\infty), \qquad \tau > 0, \qquad e > 0,$$

and consider

$$\mathcal{U}_{a,\tau,e} := \{B_p = \{[p, p + \tau + e)\} \mid p \ge a\}.$$

For every finite stencil $T \subseteq W_\tau$*, there exists* $B_p \in \mathcal{U}_{a,\tau,e}$ *such that*

$$\mathcal{D}^{\mathrm{stc}}_{T,\tau}(B_p) = 0, \qquad T^{\mathrm{bar}}_\tau(B_p) \ne 0.$$

For the corresponding information class $\Theta_{\mathcal{U}}$*,*

$$\mathrm{MinCert}(\Theta_{\mathcal{U}}) = +\infty.$$

The same conclusion holds for every family admitting retained active-start regions arbitrarily far to the right of every finite sample set.

Proof. A finite T is bounded above. Choose $p > \max T$, with the empty-stencil case arbitrary. The retained bar has active-start region $[p, p + e)$, disjoint from T, while its length $\tau + e$ exceeds the threshold. □

Remark 2.58 (Admissible escape conditions)**.** The theorem does not exclude finite detectors when additional structure is proved, such as a finite known birth set, global birth bound, eventual zero, eventual stabilization, finite-to-infinite reduction theorem, or separately certified countable cofinal detector. Finite observed stability alone establishes none of these conditions.

2.14 Legacy whole-profile degree-one detector

Definition 2.59 (Residual degree-one bar mass)**.** Let $\mathcal{B}^{\mathrm{res}}_{1,W,\tau}(F)$ denote the residual barcode of $\mathrm{Eval}_{1,W,\tau}(F)$, counted with multiplicity. Define

$$E_1(F;W,\tau) := \sum_{I\in(\mathcal{B}^{\mathrm{res}}_{1,W,\tau}(F))_{\mathrm{fin}}} \ell(I).$$

If an infinite residual bar occurs, set $E_1 = +\infty$, unless a separately declared truncated functional is used.

Proposition 2.60 (Whole-profile E_1 characterization)**.** *For a finite residual barcode under the stated convention,*

$$E_1(F;W,\tau) = 0 \iff \mathrm{Eval}_{1,W,\tau}(F) = 0.$$

Proof. Every nonempty finite interval has positive length and every infinite interval forces $+\infty$. The zero barcode has empty sum. □

Remark 2.61 (Why E_1 is not finite proof compression)**.** The functional reads the already reconstructed residual barcode. It is a whole-profile vanishing functional, not a finite-map stencil certificate, not a bottleneck-Lipschitz quantity, and not a threshold-gap budget. It cannot be counted as a proof-compression result.

2.15 Explicit realization of finite barcode profile families

The following construction turns the object-level existence of a bare filtered representative into a theorem in the finite barcode regime. It does not assert functoriality or preservation of additional structures.

Definition 2.62 (Elementary interval complex)**.** Fix a degree i.

For a finite right-open interval $I = [b,d)$ with $b < d$, define the filtered chain complex $E_i(I)$ by generators

$$x_I \in C_i, \qquad y_I \in C_{i+1},$$

with

$$\partial y_I = x_I, \qquad \partial x_I = 0,$$

and filtration values

$$\mathrm{fil}(x_I) = b, \qquad \mathrm{fil}(y_I) = d.$$

For an infinite interval $I = [b,\infty)$, define $E_i(I)$ to have one generator $x_I \in C_i$, with $\partial x_I = 0$ and $\mathrm{fil}(x_I) = b$.

Lemma 2.63 (Persistence of an elementary interval complex)**.** *For every interval I in Definition 2.62,*

$$\mathbf{P}_i(E_i(I)) \cong k_I,$$

and

$$\mathbf{P}_j(E_i(I)) = 0 \qquad (j \neq i).$$

Proof. For $I = [b,d)$, no generator is present before b. For $b \le t < d$, only x_I is present, so it represents one nonzero class in degree i. At $t \ge d$, the generator y_I appears and $\partial y_I = x_I$, so the pair is acyclic. This gives the interval module $k_{[b,d)}$ in degree i and no other homology.

For $I = [b,\infty)$, the class x_I appears at b and is never killed, producing $k_{[b,\infty)}$. □

Definition 2.64 (Bare barcode realization)**.** Let I_{mon} be finite, and let

$$\mathcal{B} = \{B_i\}_{i \in I_{\mathrm{mon}}}$$

be a family of finite barcodes written in right-open interval normal form. Define

$$\mathrm{Rep}_{\mathrm{bar}}(\mathcal{B}) := \bigoplus_{i \in I_{\mathrm{mon}}} \bigoplus_{I \in B_i} E_i(I).$$

Theorem 2.65 (Finite barcode family realization)**.** *Let I_{mon} be finite and let each B_i be a finite barcode. Then $\mathrm{Rep}_{\mathrm{bar}}(\{B_i\})$ is a finite-type filtered chain complex and, for every $i \in I_{\mathrm{mon}}$,*

$$\mathbf{P}_i\big(\mathrm{Rep}_{\mathrm{bar}}(\{B_j\})\big) \cong \bigoplus_{I \in B_i} k_I$$

as persistence modules. Equivalently,

$$\bar{}\Big(\mathbf{P}_i\big(\mathrm{Rep}_{\mathrm{bar}}(\{B_j\})\big)\Big) = B_i$$

as barcode multisets, including multiplicity.

Consequently, every finite windowed repaired profile family

$$\mathcal{P}^{I_{\mathrm{mon}}}_{W,\tau}(F)$$

admits at least one bare filtered representative.

Proof. The direct sum is finite because both the monitored scope and every barcode are finite. It is bounded and finite-dimensional. The persistence homology of a finite direct sum is the direct sum of the persistence homologies of its summands. The result follows from Lemma 2.63. □

Remark 2.66 (Strength and limitation of the realization theorem)**.** Theorem 2.65 proves object-level existence of a filtered complex realizing the declared finite profiles. It does not supply:

- a canonical choice independent of barcode decomposition data;
- a functor on arbitrary persistence morphisms;
- compatibility with the original filtered object F;
- geometric, arithmetic, sheaf-theoretic, or categorical structure;
- $\mathcal{R}$-equivalence between different representatives;
- the amplitude or edge-realization hypotheses needed for Appendix C.

Those are independent obligations.

2.16 Admissible filtered representatives and invocation status

Definition 2.67 (Admissible filtered representative)**.** Let $F \in \mathsf{FiltCh}(k)$, let W be a right-open window, let $\tau > 0$, and let I_{mon} be finite.

A filtered object

$$C_{\tau,W}F \in \mathsf{FiltCh}(k)$$

is an *admissible filtered representative* if it is equipped with actual profile isomorphisms

$$\rho_i : \mathbf{P}_i(C_{\tau,W}F) \xrightarrow{\cong} \mathcal{P}_{i,W,\tau}(F)$$

for every $i \in I_{\mathrm{mon}}$.

The tuple

$$\left(C_{\tau,W}F, I_{\mathrm{mon}}, \{\rho_i\}_{i \in I_{\mathrm{mon}}}\right)$$

is part of the representative artifact.

Definition 2.68 (Representative compatibility)**.** An admissible representative is *profile-compatible* if its declared isomorphisms ρ_i identify its monitored persistence exactly with the windowed repaired profile family.

Proposition 2.69 (Compatibility is exactly the representative condition)**.** *A filtered object is an admissible representative in the sense of Definition 2.67 if and only if it is supplied with profile-compatibility isomorphisms for every monitored degree.*

Proof. This is the defining condition, with the requirement that the isomorphisms and monitored scope are explicit artifact data. □

Definition 2.70 (Representative lifecycle and canonical record status)**.** The optional lifecycle axis is

$$\texttt{not\_required}, \quad \texttt{constructed}, \quad \texttt{verified}, \quad \texttt{failed}, \quad \texttt{missing}.$$

It records workflow progress only.

The canonical representative-record status consumed by the v20 verifier is exactly one of

$$\texttt{not\_invoked}, \qquad \texttt{undefined}, \qquad \texttt{certified}.$$

A well-typed supplied representative that is evidenced to violate a required compatibility condition produces the Chapter 7 atomic status `reject`; it is not relabeled as missing evidence.

When a representative is required:

$$\texttt{verified} \mapsto \texttt{certified} \mapsto \texttt{pass},$$

$$\texttt{constructed} \text{ or } \texttt{missing} \mapsto \texttt{undefined},$$

while `failed` maps to `reject`. The lifecycle value `not_required` maps to `not_invoked` only under an explicit scope-exclusion policy.

Remark 2.71 (Lifecycle status versus canonical audit status)**.** Lifecycle labels do not replace the canonical atomic status calculus. Missing, incomplete, stale, or pending evidence remains `undefined`; an evidenced false compatibility condition is `reject`; and a clause is `not_invoked` only when it is explicitly outside scope.

Remark 2.72 (Representative, not functor)**.** The notation $C_{\tau,W}F$ denotes a chosen representative artifact. It does not assert that $C_{\tau,W}$ is globally defined, canonical, or functorial.

2.17 Implementable range, non-uniqueness, and invariance

Definition 2.73 (Implementable representative range)**.** The *implementable representative range* is the declared class of tuples

$$(F, W, \tau, I_{\mathrm{mon}})$$

for which the proof object supplies:

(i) the finite repaired profile family $\mathcal{P}^{I_{\mathrm{mon}}}_{W,\tau}(F)$;

(ii) a constructed representative;

(iii) verified profile-compatibility isomorphisms;

(iv) immutable artifact references;

(v) every stronger compatibility required by the invoked downstream clause.

Remark 2.74 (Why an implementable range is needed)**.** The bare realization theorem proves that finite barcode profiles can be realized by filtered complexes. It does not prove that a representative preserves the extra structure required by a particular derived, geometric, or tower-level argument. The implementable range records exactly where those stronger obligations have also been discharged.

Axiom 2.1 (Structure-compatible representative availability)**.** *A run may claim membership in the implementable representative range only after an actual representative artifact has been constructed and verified.*

The finite barcode realization theorem discharges bare object-level existence. Any stronger requirement—for example realization compatibility, geometric compatibility, functoriality, or tower coherence—must be discharged separately.

Remark 2.75 (Status of the availability axiom)**.** Axiom 2.1 is an audit admissibility rule, not an assertion that every repaired profile has a unique canonical or structure-compatible representative.

Proposition 2.76 (Non-uniqueness of filtered representatives)**.** *An admissible filtered representative is generally not unique as a strict filtered chain complex.*

Proof. A monitored persistence profile does not determine chain-level contractible summands, unmonitored degrees, bases, or additional structure. Different filtered complexes can therefore realize the same monitored profile family. □

Definition 2.77 (Admissible representative choice)**.** An admissible representative choice consists of:

(i) the representative $C_{\tau,W}F$;

(ii) the finite monitored scope;

(iii) the profile isomorphisms ρ_i;

(iv) the comparison mode;

(v) the declared equivalence strength;

(vi) any realization-compatibility data;

(vii) artifact identifiers and hashes.

Proposition 2.78 (Representative invariance of persistence-level clauses)**.** *If $C_{\tau,W}F$ and $C'_{\tau,W}F$ are admissible representatives of the same profile family, then every Core clause depending only on the monitored persistence modules has the same value for both representatives.*

Proof. For each monitored degree,

$$\mathbf{P}_i(C_{\tau,W}F) \cong \mathcal{P}_{i,W,\tau}(F) \cong \mathbf{P}_i(C'_{\tau,W}F).$$

Hence every predicate invariant under persistence-module isomorphism has the same value. □

Remark 2.79 (No derived invariance from profile invariance)**.** Proposition 2.78 applies only to persistence-level clauses. It does not imply invariance of $\mathcal{R}$, Ext, products, higher operations, or tower comparison maps. Those require stronger compatibility artifacts.

Definition 2.80 (Stage-compatible detector transport)**.** A stage-compatible detector transport from a profile B to a profile C consists of:

(i) the same declared detector stage on source and target;

(ii) an actual isomorphism between the exact source-stage persistence modules;

(iii) identical window, threshold, endpoint, and sample conventions, or a separately proved transport of those conventions;

(iv) artifact-backed compatibility with every birth, germ, coverage, family-membership, and excess record consumed by the target claim.

Proposition 2.81 (Detector preservation by an exact stage-compatible isomorphism)**.** *A stage-compatible detector transport preserves the exact stencil and germ detector zero/nonzero predicates. It preserves a complete detector certificate only when the certificate's completeness and artifact fields are also transported or independently reverified.*

Proof. The zero/nonzero assertion follows from Proposition 2.43. Completeness is an additional theorem-and-artifact obligation and therefore does not follow from value invariance alone. □

Remark 2.82 (An admissible repaired representative does not preserve the standard pre-threshold detector)**.** The standard finite stencil reads $B_{i,W}(F) = \mathsf{W}_W\mathbf{P}_i(F)$ at stage `windowed_prethreshold`. An admissible representative supplies only

$$\mathbf{P}_i(C_{\tau,W}F) \cong \mathrm{Eval}_{i,W,\tau}(F),$$

which lies at stage `repaired_postthreshold`. Therefore representative admissibility alone does not transport the standard pre-threshold stencil, birth set, active-start regions, coverage certificate, or MinCert data. A separate stage-transport theorem is mandatory.

2.18 Global representative notation and its restricted use

Definition 2.83 (Global representative construction)**.** A *global representative construction* at threshold τ is a declared assignment

$$F \longmapsto C_\tau F$$

with actual profile isomorphisms

$$\mathbf{P}_i(C_\tau F) \cong T_\tau^{\mathrm{bar}}\big(\mathbf{P}_i(F)\big)$$

for every declared monitored degree.

Remark 2.84 (Global representative construction is not default)**.** Definition 2.83 is optional. It does not assert functoriality on morphisms or compatibility with window restriction.

Definition 2.85 (Window compatibility of a global representative)**.** A global representative construction is *window-compatible on W* if for every monitored degree there is an actual isomorphism

$$\mathsf{W}_W \mathbf{P}_i(C_\tau F) \cong T_\tau^{\mathrm{bar}}\big(\mathsf{W}_W \mathbf{P}_i(F)\big).$$

Proposition 2.86 (Restriction under window compatibility)**.** *If a global representative construction is window-compatible on W, then the restricted object $(C_\tau F)|_W$ is an admissible representative of the windowed repaired profile.*

Proof. By the definition of restriction to the declared window,

$$\mathbf{P}_i\big((C_\tau F)|_W\big) \cong \mathsf{W}_W \mathbf{P}_i(C_\tau F).$$

The defining window-compatibility isomorphism therefore gives

$$\mathbf{P}_i\big((C_\tau F)|_W\big) \cong T_\tau^{\mathrm{bar}}\big(\mathsf{W}_W \mathbf{P}_i(F)\big),$$

which is precisely the monitored profile isomorphism required by Definition 2.67. □

Remark 2.87 (Why restriction is not the default)**.** Appendix A, Proposition 1.55, proves that

$$\mathsf{W}_W T_\tau^{\mathrm{bar}} \not\cong T_\tau^{\mathrm{bar}} \mathsf{W}_W$$

in general. The infinite-bar obstruction is exhibited in Example 1.59, and the exact window-adapted criterion is Definition 1.60 together with Proposition 1.61. Therefore

$$C_{\tau,W} F = (C_\tau F)|_W$$

may be used only under a verified window-compatibility or τ-window-adapted criterion.

2.19 Profile algebra at the representative level

Definition 2.88 (Window-normalized representative)**.** An admissible representative $C_{\tau,W}F$ is *window-normalized* if its monitored persistence profiles are supported on W with the same extension-by-zero convention used in Appendix A.

Proposition 2.89 (Windowed representative idempotence)**.** *Let $C_{\tau,W}F$ be a verified window-normalized representative. Suppose a second representative is constructed using the same window, monitored scope, and profile-equivalence policy. Then*

$$C_{\tau,W}(C_{\tau,W}F) \simeq_{\mathrm{PH},I_{\mathrm{mon}}} C_{\tau,W}F.$$

Proof. Window normalization ensures that reapplying W_W does not clip the monitored profile further. Every finite bar in the first repaired profile has length strictly greater than τ, and every infinite bar is retained. Barcode-level idempotence therefore gives identical monitored repaired profiles. □

Remark 2.90 (Idempotence is profile-level)**.** The proposition does not assert strict equality of filtered complexes or the existence of a globally idempotent endofunctor.

Proposition 2.91 (Windowed threshold composition)**.** *Let $\tau, \sigma > 0$. Suppose all relevant representatives exist, are window-normalized, and use the same monitored scope and window. Then*

$$C_{\sigma,W}(C_{\tau,W}F) \simeq_{\mathrm{PH},I_{\mathrm{mon}}} C_{\max\{\tau,\sigma\},W}F.$$

Proof. On every monitored profile,

$$T^{\mathrm{bar}}_{\sigma}T^{\mathrm{bar}}_{\tau} = T^{\mathrm{bar}}_{\max\{\tau,\sigma\}}.$$

Window normalization prevents a second crop from creating new short bars. Thus both representatives realize the same monitored profile family. □

Remark 2.92 (Threshold sweeps)**.** Threshold composition is a statement about repaired profiles and verified representatives. It does not create a strict nested family of filtered complexes or actual transition morphisms between threshold levels.

2.20 Representative morphisms and conditional functoriality

Definition 2.93 (Repaired profile morphism artifact)**.** Let $f : F \to G$ be filtration-preserving. A *repaired profile morphism artifact* for (i, W, τ) is an actual persistence-module morphism

$$\overline{f}_{i,W,\tau} : \mathcal{P}_{i,W,\tau}(F) \longrightarrow \mathcal{P}_{i,W,\tau}(G)$$

with declared provenance and compatibility with the intended source map.

A bottleneck matching or a small bottleneck distance is not such a morphism.

Definition 2.94 (Representative morphism datum)**.** Let $C_{\tau,W}F$ and $C_{\tau,W}G$ be admissible representatives. A *representative morphism datum* consists of an actual filtration-preserving chain map, or a declared morphism in a justified localized category,

$$C_{\tau,W}f : C_{\tau,W}F \longrightarrow C_{\tau,W}G,$$

such that the induced monitored persistence morphisms commute with the profile isomorphisms and the repaired profile morphisms $\overline{f}_{i,W,\tau}$.

Definition 2.95 (Weak representative functoriality)**.** A representative construction is *weakly functorial* on a declared subcategory if:

(i) every object has a verified representative;

(ii) every morphism has a repaired profile morphism artifact and a representative morphism datum;

(iii) identity morphisms are represented by identity data up to the declared equivalence;

(iv) composition is compatible up to a specified coherent representative equivalence.

Proposition 2.96 (Weak functoriality is conditional)**.** *Weak representative functoriality is available only relative to the declared subcategory, actual morphism artifacts, and composition-coherence data. It is not automatic in the general Core.*

Proof. Barcode-level hard deletion is not a globally defined exact functor on all persistence morphisms, and arbitrary representative choices do not determine chain maps. The listed artifacts are therefore additional data, not consequences of object-level profile realization. □

Remark 2.97 (No strict global functor)**.** Appendix B does not assert a strict global endofunctor

$$C_{\tau} : \mathsf{FiltCh}(k) \longrightarrow \mathsf{FiltCh}(k)$$

or a strict bifunctorial construction $(F, W) \mapsto C_{\tau,W}F$.

2.21 Compatibility with realization and Ext-level use

Definition 2.98 (Realization-compatible representative)**.** An admissible representative $C_{\tau,W}F$ is *realization-compatible* for a declared realization $\mathcal{R}$ if:

(i) $\mathcal{R}(C_{\tau,W}F)$ is defined in $D^{\mathrm{b}}(k\text{-mod})$;

(ii) the representative artifact records the exact realization used;

(iii) every alternative representative used interchangeably is supplied with an actual $\mathcal{R}$-equivalence;

(iv) the degree-n amplitude, edge-extractor, edge-realization, naturality, and eligibility predicates required by Appendix C can be evaluated;

(v) all additional structure consumed by the Ext argument is preserved.

Proposition 2.99 (Realization invariance under stronger equivalence)**.** *If two admissible representatives are $\mathcal{R}$-equivalent, then their realized objects are isomorphic in $D^{\mathrm{b}}(k\text{-mod})$, and every isomorphism-invariant derived predicate has the same value on both.*

Proof. This is immediate from the supplied isomorphism in the derived category. □

Remark 2.100 (Profile equivalence is insufficient for Ext)**.** Monitored-persistence equivalence alone does not justify replacing one representative by another inside Ext. The degree-n persistence-to-Ext discharge consumes a realization-compatible representative and the corresponding n-eligible regime of Appendix C.

2.22 Relation to B-Gate^{+}

Definition 2.101 (Detector and representative input to B-Gate^{+})**.** A degreewise input contains:

(i) F, the monitored degree n, W, τ, and the finite monitored scope;

(ii) the exact pre-threshold profile $B_{n,W}(F)$;

(iii) the repaired profile $\mathcal{P}_{n,W,\tau}(F)$;

(iv) the detector stage, mode, certificate status, and mathematical value;

(v) every soundness, completeness, endpoint, coverage, family, excess-bound, and artifact obligation;

(vi) the comparison mode and every exact or metric-safe transport record;

(vii) a verified admissible representative when a filtered, derived, higher-Ext, or tower clause is invoked;

(viii) realization-compatible and n-eligible data when the degree-n Ext clause is invoked;

(ix) canonical atomic statuses and immutable artifact references.

Definition 2.102 (Degreewise topological zero clause)**.** The primary degree-n topological clause is

$$\mathrm{Eval}_{n,W,\tau}(F) = 0.$$

It may be discharged directly from the repaired profile or through a complete finite detector certificate at the exact permitted stage. If a verified admissible representative is supplied, the same repaired profile may be read as

$$\mathbf{P}_n(C_{\tau,W}F) = 0$$

through the declared profile isomorphism. The degree-one clause is the specialization $n = 1$.

Definition 2.103 (Degreewise categorical clause via representative)**.** The categorical clause

$$\operatorname{Ext}^n\big(\mathcal{R}_n(C_{\tau,W}F), k\big) = 0$$

may be invoked only when the representative is certified, realization-compatible, and n-eligible under Appendix C. Appendix B does not prove this implication; it supplies the required representative and detector-side input records.

Remark 2.104 (No representative, no derived clause)**.** Barcode-level evaluation and detector certification can proceed without a filtered representative. A missing or unverified in-scope representative makes a required filtered, derived, or higher-Ext clause `undefined`; it is not a pass and cannot be repaired by metric slack or a favorable detector value.

2.23 Representative towers and Type IV input

Definition 2.105 (Representative tower)**.** Let

$$F_0 \longrightarrow F_1 \longrightarrow F_2 \longrightarrow \cdots$$

be a directed tower.

A *representative tower* for (i, W, τ) consists of:

(i) verified representatives $C_{\tau,W}F_n$;

(ii) actual representative morphism data between consecutive levels;

(iii) identity and composition coherence;

(iv) compatibility with the repaired profile system.

Definition 2.106 (Representative tower comparison artifact)**.** A representative tower comparison artifact is declared data specifying:

(i) a repaired directed profile system;

(ii) a source profile $\mathrm{ColimProfile}_{i,W,\tau}(F_\bullet)$;

(iii) an apex profile $\mathrm{ApexProfile}_{i,W,\tau}(F_\infty)$;

(iv) an actual comparison morphism

$$\phi_{i,W,\tau} : \mathrm{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \mathrm{ApexProfile}_{i,W,\tau}(F_\infty);$$

(v) kernel, cokernel, terminal-tameness, monitored-scope, and provenance data required by Chapter 5 and Appendix D.

Remark 2.107 (Type IV input)**.** Representative existence, representative tower existence, and comparison morphism existence are three distinct obligations. The first two do not supply the third.

A barcode matching, small bottleneck distance, expected colimit map, or sequence of barcode multisets is not an actual Type IV comparison artifact.

2.24 Certificate, obligation, and manifest requirements

Definition 2.108 (Appendix B certificate and manifest extension)**.** A proof object invoking Appendix B records, as applicable:

(i) source object, monitored degrees, windows, thresholds, and exact pre-/post-threshold profile identities;

(ii) detector stage, endpoint and cofinal policies, detector mode, finite sample or critical set, and exact computation artifacts;

(iii) soundness theorem identity and separate completeness, characterization, or coverage theorem identity;

(iv) exhaustive birth/germ records, coverage witnesses, family membership, augmented mesh, and every independently sourced excess lower bound;

(v) detector certificate and mathematical-value statuses on separate axes;

(vi) certificate size, information class, MinCert bounds, sharpness records, no-go records, and any proved escape condition;

(vii) comparison mode and exact or metric-safe detector-zero transport records, including the Appendix A post-threshold separation record;

(viii) actual interleaving morphisms and boundary-safe domain when the robust sandwich is used;

(ix) representative identifier, canonical status, lifecycle metadata, monitored profile isomorphisms, and equivalence strength;

(x) exact source-stage isomorphisms and preservation evidence whenever a detector record is transported through a representative;

(xi) window-normalization, realization compatibility, and degree-n eligibility when invoked;

(xii) actual representative morphisms and coherence when functoriality or a tower is invoked;

(xiii) actual tower comparison artifacts when Type IV is invoked;

(xiv) claim status, dependency closure, repair ancestry, schema versions, artifact hashes, and canonical atomic status for every obligation.

Definition 2.109 (Appendix B non-scalar obligation bundle)**.** The non-scalar bundle includes:

(i) detector applicability and exact source-stage typing;

(ii) endpoint and window-cofinal conventions;

(iii) soundness and completeness theorem applicability;

(iv) birth/germ exhaustiveness, coverage, family membership, and excess-bound provenance;

(v) representative existence and monitored profile compatibility;

(vi) equivalence strength and detector-preservation scope;

(vii) realization compatibility and degree-n eligibility when used;

(viii) actual morphisms and composition coherence when used;

(ix) tower comparison existence when Type IV is used;

(x) claim status, manifest consistency, and artifact identity.

None of these obligations is a numerical metric-ledger element.

Remark 2.110 (Appendices F and G handoff)**.** Appendix F elaborates status, dependency closure, failure routing, and non-compensation. Appendix G serializes the fields and artifact references. A complete record does not prove its mathematical assertions; theorem applicability and artifact validity remain separate checks.

2.25 Finite-detector and filtered-representative foundation package

Theorem 2.111 (Appendix B v20 finite-detector and representative package)**.** *Within the declared constructible one-parameter and finite-certificate regime:*

(i) the primary objects are the pre-threshold profiles $B_{i,W}(F)$ and repaired profiles $\mathcal{P}_{i,W,\tau}(F)$;

(ii) finite-stencil nonzero is sound for repaired-profile nonzero;

(iii) exhaustive birth sets give an exact zero/nonzero characterization in the left-closed right-open convention;

(iv) exhaustive right germs give a decoration-invariant characterization;

(v) externally certified active-start coverage gives stencil completeness;

(vi) a strict augmented-mesh bound below an independently certified family excess gives uniform family completeness;

(vii) the strict mesh boundary is sharp and equality is placement-sensitive;

(viii) detector certificate status and mathematical value are separate;

(ix) exact source-stage profile isomorphisms preserve detector values, but certificate completeness requires its own preservation evidence;

(x) metric transport composes complete detector zero-reflection with the Appendix A repaired-zero theorem and does not imply raw rank stability;

(xi) family-relative MinCert *admits birth and mesh upper bounds and is infinite on the unrestricted right-unbounded translation family;*

(xii) every finite repaired barcode profile family admits a bare finite-type filtered-chain realization;

(xiii) an admissible representative is a filtered object plus explicit monitored profile isomorphisms and is generally non-unique;

(xiv) monitored profile equivalence preserves only profile-level clauses and does not imply derived, detector-stage, or realization equivalence;

(xv) representative idempotence and threshold composition hold only at profile level under stated window-normalization hypotheses;

(xvi) weak functoriality requires actual profile morphisms, representative morphisms, and coherence;

(xvii) higher-Ext use requires a realization-compatible representative and the degree-n eligible regime of Appendix C;

(xviii) a representative tower does not automatically supply a Type IV comparison morphism;

(xix) every detector, representative, morphism, completeness, and artifact requirement is a non-scalar obligation.

Proof. Items (i)–(ii) follow from Definitions 2.14, 2.21, and Theorem 2.24. Items (iii)–(iv) are Theorems 2.27 and 2.31. Item (v) is Theorem 2.33. Items (vi)–(vii) are Theorems 2.39 and 2.40. Items (viii)–(ix) are Definition 2.42 and Propositions 2.43 and 2.81. Item (x) is Theorem 2.44 and Remark 2.46. Item (xi) follows from Definition 2.51, Proposition 2.53, Corollary 2.56, and Theorem 2.57. Item (xii) is Theorem 2.65. Items (xiii)–(xiv) are Definition 2.67, Propositions 2.76, 2.78, and Remark 2.82. Item (xv) is Propositions 2.89 and 2.91. Item (xvi) is Proposition 2.96. Item (xvii) is Definition 2.98. Item (xviii) is Definitions 2.105 and 2.106. Item (xix) is Definition 2.109. □

2.26 Explicit exclusions

The following are not asserted in Appendix B.

(i) No finite observation is treated as a complete zero certificate without an applicable completeness theorem.

(ii) No detector theorem is silently moved between profile stages.

(iii) No point-birth theorem is treated as decoration-invariant.

(iv) No zero stencil implies repaired-profile zero without birth, germ, coverage, family-mesh, or another proved completeness route.

(v) No family-mesh theorem is invoked without a strictly smaller mesh than a positive, independently sourced family excess.

(vi) No equality-boundary mesh case is assigned a universal pass.

(vii) No bounded-window cofinal bar is omitted from coverage or excess.

(viii) No retrospective full-barcode enumeration is counted automatically as proof compression.

(ix) No object-relative empty stencil defines a nontrivial family-relative MinCert.

(x) No uniform finite complete detector exists for the unrestricted right-unbounded translation family.

(xi) No finite observed stabilization defeats the right-unbounded no-go theorem without a proved escape condition.

(xii) No bottleneck bound implies pointwise detector-rank, birth-set, germ-set, coverage, family-membership, or certificate-size stability.

(xiii) No scalar bottleneck distance replaces actual interleaving morphisms in the robust sandwich.

(xiv) No whole-profile E_1 functional is counted as finite proof compression.

(xv) No strict global endofunctor $C_\tau : \mathsf{FiltCh}(k) \to \mathsf{FiltCh}(k)$ is asserted.

(xvi) No strict global functor $(F, W) \mapsto C_{\tau,W}F$ is asserted.

(xvii) No uniqueness or hidden canonicality of a representative is asserted.

(xviii) No monitored-profile equivalence is automatically promoted to a filtered quasi-isomorphism, derived equivalence, detector preservation, or realization equivalence.

(xix) No admissible repaired representative automatically transports the standard pre-threshold detector.

(xx) No bare barcode realization is automatically compatible with source geometry, arithmetic, sheaves, categories, or higher realization.

(xxi) No unconditional identity $C_{\tau,W}F = (C_\tau F)|_W$ is asserted.

(xxii) No global exactness, Serre localization, reflector, or global non-expansiveness claim is restored.

(xxiii) No profile idempotence becomes strict filtered-complex idempotence.

(xxiv) No threshold-composition identity creates transition morphisms.

(xxv) No bottleneck matching is treated as a persistence morphism.

(xxvi) No representative morphism follows from object-level representative existence.

(xxvii) No higher-Ext clause is invoked without realization compatibility and degree-n eligibility.

(xxviii) No detector output is identified with a natural higher Ext-edge.

(xxix) No representative tower automatically supplies a colimit, apex, or comparison morphism.

(xxx) No missing detector, representative, morphism, completeness theorem, or artifact is repaired by numerical slack.

(xxxi) No lifecycle vocabulary replaces the canonical audit statuses.

(xxxii) No complete manifest replaces mathematical verification.

(xxxiii) No external-domain theorem or Return Theorem is created here.

2.27 Provenance and status

The barcode-level operation T_τ^{bar}, fixed window-first order, two-sided threshold-gap repair, sharp repaired-zero separation, and metric-safe zero transport are supplied by Chapter 2 and Appendix A. The current canonical Appendix A references include Theorem 1.40, Theorem 1.43, and Corollary 1.47. Crop–collapse noncommutation is Proposition 1.55; the infinite-bar obstruction is Example 1.59; and the all-retained-bars criterion is Definition 1.60 with Proposition 1.61.

The elementary interval-complex construction is the standard finite barcode realization. The inherited v19 contribution is the typed separation of profile realization, filtered and derived equivalence, morphisms, tower artifacts, and audit status. The v20 conservative extension adds finite-stencil soundness, birth and germ characterizations, externally certified coverage, family-mesh completeness, strict sharpness, family-relative MinCert, the right-unbounded no-go theorem, metric-safe detector composition, and the exact stage boundary for representative-based detector transport.

Status labels for Appendix B. *Finite-detector Core theorems:* Theorems 2.24, 2.27, 2.31, 2.33, 2.39, 2.40, 2.44, 2.55, and 2.57.

Foundation realization theorem: Theorem 2.65.

Foundation closure theorem: Theorem 2.111.

Detector definitions and records: Definitions 2.18, 2.19, 2.21, 2.22, 2.26, 2.29, 2.30, 2.32, 2.35, 2.37, 2.41, 2.42, 2.47, 2.50, 2.51, and 2.54.

Detector propositions, lemmas, and corollaries: Proposition 2.23, Lemma 2.38, Propositions 2.43, 2.48, 2.53, 2.60, and Corollaries 2.45 and 2.56.

Representative Core definitions: Definitions 2.3, 2.4, 2.5, 2.7, 2.8, 2.9, 2.13, 2.14, 2.15, 2.17, 2.62, 2.64, 2.67, 2.68, 2.70, 2.73, 2.77, 2.80, 2.83, 2.85, 2.88, 2.93, 2.94, 2.95, 2.98, 2.101, 2.102, 2.103, 2.105, 2.106, 2.108, and 2.109.

Representative propositions and lemmas: Lemma 2.63; Propositions 2.10, 2.69, 2.76, 2.78, 2.81, 2.86, 2.89, 2.91, 2.96, and 2.99.

Operational rule: Axiom 2.1.

Explicit exclusions: Subsection 2.26.

3 Appendix C: Higher Eligible Realization, Canonical Ext-Edges, and Zero-Reflection Boundaries

3.1 Scope and conservative-extension boundary

Appendix C supplies the detailed derived-category, higher-eligibility, zero-reflection, and audit foundation for Chapter 4 of AK-HPDST v20.0.0. It consumes:

(i) the inherited metric and repaired-zero transport calculus of Appendix A;

(ii) the finite detector, profile-stage, proof-compression, and filtered representative discipline of Appendix B;

(iii) the Chapter 4 higher-obstruction contract.

The canonical project acronym is AK-HPDST. The legacy transposed acronym AK-HDPST may occur in historical artifacts, but it has no normative use in this appendix.

The complete v19 degree-one bridge is inherited unchanged at its original strength. The v20 conservative extension applies to every invoked degree $n \geq 1$. Its unconditional algebraic input is the field-coefficient identification

$$\operatorname{Ext}^n(A, k) := \operatorname{Hom}_{D^{\mathrm{b}}(k\text{-mod})}(A, k[n]) \xrightarrow{\sim} \operatorname{Hom}_k\bigl(H^{-n}(A), k\bigr),$$

canonical and natural for every bounded derived object A.

The persistence-to-derived interface is separate:

$$H^{-n}\bigl(A_{n;F,W,\tau}\bigr) \xrightarrow{\sim} \operatorname{Edge}_n\bigl(\operatorname{Eval}_{n,W,\tau}(F)\bigr).$$

It is an artifact-backed realization condition, not a consequence of a barcode, a detector value, an amplitude declaration, or a desired verdict.

Under zero compatibility, the default Core direction is

$$\operatorname{Eval}_{n,W,\tau}(F) = 0 \quad \Longrightarrow \quad \operatorname{Ext}^n\bigl(A_{n;F,W,\tau}, k\bigr) = 0.$$

A converse is available only after a separately proved zero-reflection package on a declared profile class. The resulting statement is an equivalence of vanishing predicates, not an object-level identification of persistence and Ext objects.

Remark 3.1 (Conservative extension of the degree-one package)**.** For $n = 1$, the standard v20 record specializes to the inherited v19 eligible regime

$$A_{1;F,W,\tau} \in D^{[-1,0]}(k\text{-mod}), \qquad H^{-1}(A_{1;F,W,\tau}) \cong \mathrm{Edge}_1\big(\mathrm{Eval}_{1,W,\tau}(F)\big).$$

No v19 theorem is weakened or retroactively reinterpreted.

Remark 3.2 (Division of responsibility)**.** Appendix C proves the field-coefficient higher Ext edge and the conditional higher discharge. It does not prove detector completeness, metric transport, transient-defect detection, finite-to-infinite reduction, an external realization theorem, or an external Return Theorem. Those remain independent obligations in Appendices A–B, D, and the Problem Interface layer.

Remark 3.3 (No external-domain upgrade)**.** Nothing in this appendix proves an arithmetic, PDE, geometric, motivic, Fukaya-categorical, cryptographic, or other external-domain theorem. A Core Ext clause has external force only through a separately proved Interface–Core–Reduction–Return chain.

3.2 Derived category setting and canonical higher Ext algebra

Fix a field k. Let

$$D^{\mathrm{b}}(k\text{-mod})$$

denote the bounded derived category of finite-dimensional k-vector spaces, equipped with the standard t-structure.

Definition 3.4 (Bounded derived object)**.** An object

$$A \in D^{\mathrm{b}}(k\text{-mod})$$

is represented by a bounded cochain complex of finite-dimensional k-vector spaces. Its cohomology objects are denoted by $H^j(A)$.

Definition 3.5 (Amplitude)**.** An object A has amplitude contained in $[a, b]$ if

$$H^j(A) = 0 \qquad (j \notin [a, b]).$$

We write

$$A \in D^{[a,b]}(k\text{-mod}).$$

Definition 3.6 (Inherited two-term eligible amplitude)**.** An object has the inherited degree-one eligible amplitude when

$$A \in D^{[-1,0]}(k\text{-mod}).$$

This is the specialization of the standard degree-n policy to $n = 1$.

Definition 3.7 (Ext^1 in the derived category)**.** For $A \in D^{\mathrm{b}}(k\text{-mod})$, define

$$\mathrm{Ext}^1(A, k) := \mathrm{Hom}_{D^{\mathrm{b}}(k\text{-mod})}(A, k[1]).$$

Definition 3.8 (Higher derived Ext)**.** For $n \geq 0$ and $A \in D^{\mathrm{b}}(k\text{-mod})$, define

$$\mathrm{Ext}^n(A, k) := \mathrm{Hom}_{D^{\mathrm{b}}(k\text{-mod})}(A, k[n]).$$

Lemma 3.9 (Semisimplicity over a field)**.** *The abelian category of finite-dimensional k-vector spaces is semisimple. Consequently,*

$$\mathrm{Ext}^p_k(V, k) = 0 \qquad (p > 0)$$

for every finite-dimensional vector space V.

Proof. Every vector space over a field is projective and injective, so every short exact sequence splits and all positive derived Ext groups vanish. □

Lemma 3.10 (Derived splitting over a field)**.** *Every $A \in D^{\mathrm{b}}(k\text{-mod})$ admits an isomorphism*

$$A \cong \bigoplus_{j \in \mathbb{Z}} H^j(A)[-j].$$

Such a direct-sum isomorphism generally depends on choices and is not used as the source of the canonical naturality below.

Proof. Choose splittings of cycles, boundaries, and cohomology in a bounded complex representing A. The resulting cohomology complex with zero differential is quasi-isomorphic to A. The chosen splittings need not be functorial. □

Theorem 3.11 (Canonical higher field-coefficient Ext edge)**.** *Let $A \in D^{\mathrm{b}}(k\text{-mod})$ and $n \geq 0$. There is a canonical isomorphism, natural in A,*

$$\mathrm{Ext}^n(A, k) \xrightarrow{\sim} \mathrm{Hom}_k\big(H^{-n}(A), k\big).$$

No amplitude restriction is required for this algebraic theorem.

Proof. The standard t-structure supplies the natural hyper-Ext spectral sequence

$$E_2^{p,q} = \mathrm{Ext}^p_k\big(H^{-q}(A), k\big) \quad \Longrightarrow \quad \mathrm{Ext}^{p+q}(A, k).$$

By Lemma 3.9, all terms with $p > 0$ vanish. For total degree n, the unique possibly nonzero term is

$$E_2^{0,n} = \mathrm{Hom}_k\big(H^{-n}(A), k\big).$$

There are no nontrivial incoming or outgoing differentials and the induced filtration on $\mathrm{Ext}^n(A, k)$ has one nonzero graded piece. The resulting edge map is therefore a canonical isomorphism. Naturality follows from the naturality of the spectral sequence. □

Corollary 3.12 (Higher edge naturality)**.** *For every morphism $f : A \to A'$, the canonical square*

$$\begin{tikzcd}
\mathrm{Ext}^n(A', k) \arrow[r] \arrow[d, "\sim"] & \mathrm{Ext}^n(A, k) \arrow[d, "\sim"] \\
\mathrm{Hom}_k(H^{-n}(A'), k) \arrow[r] & \mathrm{Hom}_k(H^{-n}(A), k)
\end{tikzcd}$$

commutes. Both horizontal maps are contravariantly induced by f.

Proof. This is the naturality assertion of Theorem 3.11. □

Corollary 3.13 (Higher edge vanishing criterion)**.** *For $A \in D^{\mathrm{b}}(k\text{-mod})$,*

$$H^{-n}(A) = 0 \quad \Longrightarrow \quad \mathrm{Ext}^n(A, k) = 0.$$

Conversely, because $H^{-n}(A)$ is finite-dimensional,

$$\mathrm{Ext}^n(A, k) = 0 \quad \Longrightarrow \quad H^{-n}(A) = 0.$$

Proof. The first direction follows directly from Theorem 3.11. For the converse, a nonzero finite-dimensional vector space has a nonzero linear functional, so $\mathrm{Hom}_k(H^{-n}(A), k) = 0$ implies $H^{-n}(A) = 0$. □

Corollary 3.14 (Two-term splitting)**.** *If $A \in D^{[-1,0]}(k\text{-mod})$, then there exists an isomorphism*

$$A \cong H^{-1}(A)[1] \oplus H^0(A).$$

The displayed splitting is explanatory and need not be natural.

Proof. Apply Lemma 3.10 and the amplitude restriction. □

Theorem 3.15 (Canonical two-term Ext^1-edge identification)**.** *Let $A \in D^{[-1,0]}(k\text{-mod})$. Then there is a canonical isomorphism, natural in A,*

$$\mathrm{Ext}^1(A, k) \xrightarrow{\sim} \mathrm{Hom}_k\big(H^{-1}(A), k\big).$$

Proof. The standard t-structure gives the canonical truncation triangle

$$H^{-1}(A)[1] \longrightarrow A \longrightarrow H^0(A) \xrightarrow{+1} H^{-1}(A)[2].$$

Apply the contravariant cohomological functor

$$\mathrm{Hom}_{D^{\mathrm{b}}(k\text{-mod})}(-, k[1]).$$

The resulting long exact sequence contains

$$\begin{aligned} \mathrm{Hom}_D(H^0(A), k[1]) &\longrightarrow \mathrm{Hom}_D(A, k[1]) \\ &\longrightarrow \mathrm{Hom}_D(H^{-1}(A)[1], k[1]) \\ &\longrightarrow \mathrm{Hom}_D(H^0(A), k[2]), \end{aligned}$$

where $D = D^{\mathrm{b}}(k\text{-mod})$. Semisimplicity gives

$$\mathrm{Hom}_D(H^0(A), k[1]) = 0, \qquad \mathrm{Hom}_D(H^0(A), k[2]) = 0.$$

Hence the middle arrow is an isomorphism, and

$$\mathrm{Hom}_D(H^{-1}(A)[1], k[1]) \cong \mathrm{Hom}_k(H^{-1}(A), k).$$

Naturality follows from the canonical truncation triangle. □

Corollary 3.16 (Naturality square)**.** *For every morphism $f : A \to A'$ in $D^{[-1,0]}(k\text{-mod})$, the square*

$$\begin{array}{ccc} \mathrm{Ext}^1(A', k) & \longrightarrow & \mathrm{Ext}^1(A, k) \\ \downarrow\wr & & \downarrow\wr \\ \mathrm{Hom}_k(H^{-1}(A'), k) & \longrightarrow & \mathrm{Hom}_k(H^{-1}(A), k) \end{array}$$

commutes.

Proof. This is the naturality assertion of Theorem 3.15. □

Corollary 3.17 (Edge vanishing criterion)**.** *If $A \in D^{[-1,0]}(k\text{-mod})$ and $H^{-1}(A) = 0$, then*

$$\mathrm{Ext}^1(A, k) = 0.$$

Proof. Apply Theorem 3.15. ◻

Remark 3.18 (Amplitude is an eligibility policy, not an algebraic necessity)**.** The standard v20 n-eligible regime uses

$$A_{n;F,W,\tau} \in D^{[-n,0]}(k\text{-mod})$$

as a conservative typing and audit policy. Theorem 3.11 itself is valid for every bounded derived object and does not require this amplitude condition.

Remark 3.19 (No reverse information from the algebraic edge)**.** The canonical Ext edge reflects the vector space $H^{-n}(A)$, not the full persistence profile. Reverse recovery of a repaired profile requires a separate zero-reflection theorem for the declared persistence edge.

Remark 3.20 (Why the field hypothesis matters)**.** Over a general coefficient ring, the terms

$$\operatorname{Ext}^p\big(H^{-q}(A), k\big) \qquad (p > 0)$$

need not vanish. The spectral sequence need not collapse to one row, so the displayed higher-edge isomorphism requires separate homological hypotheses. Non-field extensions belong to downstream Technical Extension material.

Remark 3.21 (Splitting versus canonical naturality)**.** Lemma 3.10 explains the semisimple shape of the answer, but a chosen splitting is generally noncanonical. The canonical identification used by the Core is the natural edge map of Theorem 3.11.

3.3 Repaired higher-degree input and admissible representatives

Definition 3.22 (Repaired windowed profile)**.** For $F \in \mathsf{FiltCh}(k)$, degree i, right-open window W, and $\tau > 0$, define

$$\operatorname{Eval}_{i,W,\tau}(F) := T_\tau^{\mathrm{bar}}\big(\mathsf{W}_W \mathbf{P}_i(F)\big).$$

This is the repaired post-threshold profile. Threshold clearance, whenever metric transport is invoked, is measured on the windowed pre-threshold profile

$$\mathsf{W}_W \mathbf{P}_i(F).$$

Definition 3.23 (Admissible representative)**.** An object

$$C_{\tau,W}F \in \mathsf{FiltCh}(k)$$

is an admissible representative on a declared finite monitored degree set when it carries the explicit profile isomorphisms required by Appendix B, Definition 2.67.

Definition 3.24 (Degree-one compatible representative)**.** An admissible representative is degree-one compatible if it carries an actual isomorphism

$$\rho_{1;F,W,\tau} : \mathbf{P}_1(C_{\tau,W}F) \xrightarrow{\sim} \operatorname{Eval}_{1,W,\tau}(F).$$

When this condition is part of the record, the notation

$$\mathrm{PH}_1(C_{\tau,W}F) = 0$$

is used only as shorthand for the vanishing of the repaired degree-one profile.

Definition 3.25 (Degree-n compatible representative)**.** Fix $n \geq 1$. An admissible representative is *degree-n compatible* if it carries an actual isomorphism

$$\rho_{n;F,W,\tau} : \mathbf{P}_n(C_{\tau,W}F) \xrightarrow{\sim} \mathrm{Eval}_{n,W,\tau}(F).$$

The notation

$$\mathrm{PH}_n(C_{\tau,W}F) = 0$$

is licensed only as shorthand for the vanishing of this declared repaired profile.

Remark 3.26 (Representative, not collapse functor)**.** The notation $C_{\tau,W}F$ denotes a chosen representative artifact. It does not define a global strict collapse functor and does not imply

$$C_{\tau,W}F = (C_\tau F)|_W.$$

Remark 3.27 (Detector stage and representative stage)**.** An admissible representative normally realizes the `repaired_postthreshold` profile. It does not automatically preserve a `windowed_prethreshold`, `kernel_defect`, or `cokernel_defect` detector. Any detector-to-representative transport requires the exact stage-compatible data of Appendix B.

3.4 Higher realization data and edge extractors

3.4.1 Inherited degree-one realization interface

Definition 3.28 (Realization functor)**.** A realization functor or realization procedure is a declared functor

$$\mathcal{R} : \mathcal{C}_{\mathrm{rep}} \longrightarrow D^{\mathrm{b}}(k\text{-mod}),$$

where $\mathcal{C}_{\mathrm{rep}}$ is a declared representative subcategory. Its domain, version, morphism policy, and representative-equivalence policy are part of the record.

Definition 3.29 (Realization-compatible representative)**.** A degree-one compatible representative $C_{\tau,W}F$ is realization-compatible when:

(i) it belongs to the domain of $\mathcal{R}$;

(ii) $\mathcal{R}(C_{\tau,W}F)$ is well-defined;

(iii) every alternative representative used interchangeably is related by an actual realization equivalence;

(iv) the inherited amplitude and edge-realization conditions can be audited.

Definition 3.30 (Eligible realized object)**.** For a realization-compatible degree-one representative define

$$A_{F,W,\tau} := \mathcal{R}(C_{\tau,W}F).$$

Remark 3.31 (Why realization compatibility is required)**.** Persistence profiles and realized derived objects live in different categories. Profile equivalence alone does not identify realizations or their Ext groups. A fixed representative commitment or an actual realization equivalence is therefore mandatory.

Definition 3.32 (Residual degree-one persistence edge)**.** A residual degree-one edge extractor is a declared vector-space-valued assignment

$$\mathrm{Edge}_1 : \mathfrak{P}_1 \longrightarrow \mathsf{Vect}_k$$

on a declared class containing the zero profile, with

$$\mathrm{Edge}_1(0) = 0.$$

No faithfulness, conservativity, or zero-reflection is inferred.

Definition 3.33 (Edge-realization condition)**.** An inherited degree-one edge-realization certificate is an actual isomorphism

$$\epsilon_{F,W,\tau} : H^{-1}(A_{F,W,\tau}) \xrightarrow{\sim} \mathrm{Edge}_1\big(\mathrm{Eval}_{1,W,\tau}(F)\big).$$

The record declares whether this is objectwise or belongs to a natural family.

Remark 3.34 (Persistence-to-vector-space type correction)**.** The repaired profile is a persistence module, while $H^{-1}(A_{F,W,\tau})$ is a finite-dimensional vector space. They are not directly identifiable. The declared edge extractor is the required type-changing interface.

Definition 3.35 (Eligible regime)**.** The inherited degree-one eligible regime consists of:

(i) the repaired degree-one evaluation record;

(ii) a degree-one compatible admissible representative;

(iii) a realization-compatible $\mathcal{R}$;

(iv) the object $A_{F,W,\tau}$;

(v) an amplitude certificate

$$A_{F,W,\tau} \in D^{[-1,0]}(k\text{-mod});$$

(vi) a fixed residual edge extractor with $\mathrm{Edge}_1(0) = 0$;

(vii) the edge-realization certificate;

(viii) the field coefficient declaration;

(ix) representative- and realization-change policies;

(x) complete claim, manifest, proof, and artifact records.

Eligibility is independent of whether the repaired profile is zero or nonzero. Missing required data make an invoked categorical clause `undefined`.

Remark 3.36 (Meaning of the edge-realization condition)**.** Amplitude alone does not connect persistence to derived cohomology. The edge-realization isomorphism is an independent, non-scalar obligation that records exactly which vector-space edge is realized.

Remark 3.37 (No raw persistence-to-derived shortcut)**.** The edge extractor consumes the repaired post-threshold profile, not the raw unwindowed persistence module. No raw barcode, detector number, or metric budget supplies the edge-realization certificate.

3.4.2 Higher realization interface

Remark 3.38 (Eligibility is a gate condition)**.** Eligibility licenses use of the inherited categorical edge. Missing eligibility is neither categorical vanishing nor rejection; an invoked incomplete clause is `undefined`.

Definition 3.39 (Declared repaired-profile class)**.** For each invoked degree $n \geq 1$, a repaired-profile class

$$\mathfrak{P}_n$$

is a declared category or class of constructible repaired degree-n profiles, together with:

(i) its object and morphism scopes, including the zero profile;

(ii) its profile stage;

(iii) endpoint, window-cofinal, and threshold conventions;

(iv) a membership test or proof obligation;

(v) immutable scope and policy identifiers.

The class is fixed independently of the desired categorical verdict.

Definition 3.40 (Higher realization functor)**.** A degree-n realization is a declared functor or procedure

$$\mathcal{R}_n : C_{\mathrm{rep}}^{(n)} \longrightarrow D^{\mathrm{b}}(k\text{-mod})$$

on a declared representative subcategory. Its domain, morphism policy, version, representative-change policy, and artifact identity are part of the realization record.

Definition 3.41 (Realization-compatible representative)**.** A degree-n compatible representative is *realization-compatible* when:

(i) it lies in the domain of $\mathcal{R}_n$;

(ii) the realized object is well-defined;

(iii) every alternative representative used interchangeably is related by an actual realization equivalence in the sense of Appendix B, Definition 2.9;

(iv) the amplitude, edge-realization, and naturality records required below can be audited;

(v) all additional structures consumed by the categorical clause are preserved at the declared strength.

Definition 3.42 (Eligible realized object)**.** For a realization-compatible representative define

$$A_{n;F,W,\tau} := \mathcal{R}_n(C_{\tau,W}F).$$

This object must be independently meaningful for the declared realization purpose. It may not be defined solely by encoding the desired detector output unless the record explicitly classifies the construction as synthetic.

Definition 3.43 (Higher edge extractor)**.** A degree-n edge extractor is a fixed vector-space-valued functor or objectwise assignment

$$\mathrm{Edge}_n : \mathfrak{P}_n \longrightarrow \mathsf{Vect}_k$$

with declared object and morphism semantics. For the one-way bridge it must satisfy zero compatibility:

$$P = 0 \quad \Longrightarrow \quad \mathrm{Edge}_n(P) = 0.$$

Functoriality, faithfulness, conservativity, and zero-reflection are not inferred from zero compatibility.

Definition 3.44 (Edge-realization certificate)**.** An edge-realization certificate is an actual isomorphism

$$\epsilon_{n;F,W,\tau} : H^{-n}\big(A_{n;F,W,\tau}\big) \xrightarrow{\sim} \mathrm{Edge}_n\big(\mathrm{Eval}_{n,W,\tau}(F)\big).$$

The record states whether this is:

$$\texttt{objectwise} \quad \text{or} \quad \texttt{natural\_family}.$$

A natural-family assertion additionally contains the compatible morphism class and the commutative naturality squares.

Definition 3.45 (Non-adaptive policy)**.** The realization, edge extractor, profile class, detector policy, and edge-realization theorem are *non-adaptive* when their identifiers and scope are fixed before the desired pass/reject value is inspected. An adaptively chosen zero extractor, realization, sample point, or profile class cannot support theorem evidence.

Definition 3.46 (Standard n-eligible regime)**.** Fix $n \geq 1$. A standard n-eligible record for $(F; W, \tau)$ contains:

(i) the repaired degree-n profile and its exact profile identity;

(ii) a degree-n compatible admissible representative;

(iii) a realization-compatible $\mathcal{R}_n$;

(iv) the independently meaningful realized object $A_{n;F,W,\tau}$;

(v) the conservative amplitude certificate

$$A_{n;F,W,\tau} \in D^{[-n,0]}(k\text{-mod}),$$

or a separately proved and registered admissible replacement;

(vi) the declared profile class $\mathfrak{P}_n$ and membership evidence;

(vii) a fixed edge extractor Edge_n;

(viii) zero compatibility;

(ix) the edge-realization certificate;

(x) the non-adaptive policy record;

(xi) the field coefficient declaration;

(xii) representative-change and realization-change policies;

(xiii) claim-status, manifest, proof, and immutable artifact data.

If an invoked item is missing or ill-typed, the degree-n categorical clause is `undefined`.

Remark 3.47 (Eligibility is verdict-independent)**.** A standard n-eligible record types the categorical object and its edge. It does not require the repaired profile to vanish. Exact or metric evidence establishing a zero statement belongs to the discharge record, not to eligibility. This separation is necessary for nonzero direct computations, reject certificates, and the reverse-failure examples below.

Remark 3.48 (Standard amplitude is conservative)**.** The amplitude

$$D^{[-n,0]}$$

is the standard release typing policy. It makes the intended obstruction edge and other cohomological degrees explicit. It is not a necessary hypothesis for Theorem 3.11.

Remark 3.49 (Non-scalar eligibility obligations)**.** Representative existence, profile-class membership, realization compatibility, amplitude, edge naturality, edge realization, detector completeness, non-adaptivity, claim status, and artifact identity are non-scalar obligations. No metric reserve or ledger slack can manufacture them.

3.5 The higher one-way repaired bridge

Proposition 3.50 (Repaired degree-n vanishing implies edge vanishing)**.** *Let $(F; W, \tau)$ have a complete standard n-eligible record. If*

$$\mathrm{Eval}_{n,W,\tau}(F) = 0,$$

then

$$H^{-n}\big(A_{n;F,W,\tau}\big) = 0.$$

Proof. The edge-realization certificate gives

$$H^{-n}\big(A_{n;F,W,\tau}\big) \cong \mathrm{Edge}_n\big(\mathrm{Eval}_{n,W,\tau}(F)\big).$$

Zero compatibility sends the zero profile to the zero vector space. □

Theorem 3.51 (Conditional higher repaired bridge)**.** *Let $n \geq 1$, and let $(F; W, \tau)$ have a complete standard n-eligible record. Then*

$$\mathrm{Eval}_{n,W,\tau}(F) = 0 \quad \Longrightarrow \quad \mathrm{Ext}^n\big(A_{n;F,W,\tau}, k\big) = 0.$$

Proof. By Proposition 3.50,

$$H^{-n}\big(A_{n;F,W,\tau}\big) = 0.$$

Apply Corollary 3.13. □

Proposition 3.52 (Repaired degree-one vanishing implies edge vanishing)**.** *Let $(F; W, \tau)$ have a complete inherited eligible record. If*

$$\mathrm{Eval}_{1,W,\tau}(F) = 0,$$

then

$$H^{-1}(A_{F,W,\tau}) = 0.$$

Proof. The inherited edge-realization certificate gives

$$H^{-1}(A_{F,W,\tau}) \cong \mathrm{Edge}_1\big(\mathrm{Eval}_{1,W,\tau}(F)\big).$$

Zero compatibility gives $\mathrm{Edge}_1(0) = 0$. □

Theorem 3.53 (One-way repaired bridge)**.** *Let $(F; W, \tau)$ have a complete inherited eligible record. If*

$$\mathrm{Eval}_{1,W,\tau}(F) = 0,$$

then

$$\mathrm{Ext}^1\big(\mathcal{R}(C_{\tau,W}F), k\big) = 0.$$

Equivalently, when the degree-one representative record licenses the notation,

$$\mathrm{PH}_1(C_{\tau,W}F) = 0 \quad \Longrightarrow \quad \mathrm{Ext}^1\big(\mathcal{R}(C_{\tau,W}F), k\big) = 0.$$

Proof. Set $A_{F,W,\tau} = \mathcal{R}(C_{\tau,W}F)$. Eligibility gives $A_{F,W,\tau} \in D^{[-1,0]}$. Proposition 3.52 gives $H^{-1}(A_{F,W,\tau}) = 0$. Apply Corollary 3.17. □

Remark 3.54 (Where the topological input enters)**.** The sole topological input is exact vanishing of the repaired degree-one profile. The edge extractor and its realization certificate translate that statement into $H^{-1} = 0$.

Remark 3.55 (Metric transport is independent)**.** Appendix C does not reprove threshold-gap or repaired-zero transport. If the topological zero statement is transported from another profile, the record must cite the Appendix A two-sided gap theorem, the actual discrepancy bound, post-threshold zero separation, and the applicable strict operational reserve.

Remark 3.56 (Discharge theorem)**.** The inherited one-way theorem discharges one categorical gate clause. It does not reconstruct persistence from Ext vanishing and does not supply any missing metric, representative, or realization artifact.

Remark 3.57 (Independence from the metric repair proof)**.** Metric repair controls transport of an exact repaired-zero statement between profiles. Eligibility and the Ext edge control categorical discharge after that exact zero statement has been established. Neither proof replaces the other.

3.6 Natural zero-reflection and vanishing equivalence

Definition 3.58 (Zero-reflecting edge extractor)**.** An edge extractor Edge_n is *zero-reflecting* on $\mathfrak{P}_n$ when a theorem proves

$$\mathrm{Edge}_n(P) = 0 \quad \Longrightarrow \quad P = 0 \qquad (P \in \mathfrak{P}_n).$$

Together with zero compatibility, this gives

$$\mathrm{Edge}_n(P) = 0 \quad \Longleftrightarrow \quad P = 0$$

on the declared class. Zero-reflection is weaker than full faithfulness and does not identify objects with their edge values.

Definition 3.59 (Natural zero-reflection package)**.** A natural zero-reflection package contains:

(i) a profile class $\mathfrak{P}_n$ fixed independently of the verdict;

(ii) a fixed functorial edge extractor Edge_n;

(iii) a natural-family edge-realization isomorphism;

(iv) a zero-reflection theorem on $\mathfrak{P}_n$;

(v) representative-change compatibility;

(vi) realization-change compatibility;

(vii) profile-class membership evidence;

(viii) non-adaptivity, proof, claim, and immutable artifact records.

An objectwise edge certificate alone is not a natural zero-reflection package.

Theorem 3.60 (Natural zero-reflecting vanishing equivalence)**.** *Let $n \geq 1$. Suppose $(F; W, \tau)$ has a complete standard n-eligible record and a complete natural zero-reflection package. Then*

$$\mathrm{Eval}_{n,W,\tau}(F) = 0 \quad \Longleftrightarrow \quad \mathrm{Ext}^n\big(A_{n;F,W,\tau}, k\big) = 0.$$

This is an equivalence of zero predicates only.

Proof. The forward direction is Theorem 3.51.

Conversely, suppose the Ext group vanishes. By Corollary 3.13,

$$H^{-n}\big(A_{n;F,W,\tau}\big) = 0.$$

The edge-realization isomorphism gives

$$\mathrm{Edge}_n\big(\mathrm{Eval}_{n,W,\tau}(F)\big) = 0.$$

Zero-reflection on $\mathfrak{P}_n$ yields

$$\mathrm{Eval}_{n,W,\tau}(F) = 0.$$

□

Remark 3.61 (No object-level PH_n-to-Ext^n isomorphism)**.** Theorem 3.60 does not assert

$$\mathrm{PH}_n \cong \mathrm{Ext}^n$$

or an equivalence of persistence and derived categories. The two zero predicates are connected through a typed extractor and realization interface.

3.7 Detector-induced zero-reflection at the exact profile stage

Definition 3.62 (Fixed-threshold stencil edge functor)**.** Fix a right-open window W, a threshold $\tau > 0$, and a finite stencil

$$T \subseteq W_\tau$$

in the valid Appendix B detector domain. For a repaired post-threshold profile P supported on W, define

$$\mathrm{Edge}_n^{T,\tau}(P) := \bigoplus_{t \in T} \mathrm{im}\big(P(t) \longrightarrow P(t+\tau)\big).$$

Proposition 3.63 (Functoriality of the fixed-threshold stencil edge)**.** *On every declared category of repaired profiles and actual persistence morphisms on the fixed domain* (W, τ, T)*, the construction* $\mathrm{Edge}_n^{T,\tau}$ *is a functor to finite-dimensional vector spaces.*

Proof. For each $t \in T$, naturality of a persistence morphism gives a commutative square between the structure maps at t and $t + \tau$. The target component sends the source image into the target image. These induced maps preserve identities and composition, and finite direct sum preserves functoriality. □

Definition 3.64 (Family-mesh repaired profile class)**.** Let $W = [a, b)$ be bounded. Let $\mathcal{F}$ be a nonempty family of left-closed right-open windowed barcodes with a valid Appendix B uniform excess certificate

$$e_{\tau,W}(\mathcal{F}) > 0.$$

Let $T \subseteq W_\tau$ be finite and satisfy

$$\mathrm{Mesh}_{W_\tau}(T) < e_{\tau,W}(\mathcal{F}).$$

Define

$$\mathfrak{P}_n^{\mathcal{F},T,\tau} := \left\{ T_\tau^{\mathrm{bar}}(B) \,\middle|\, B \in \mathcal{F} \right\} \cup \{0\},$$

with actual persistence morphisms on the fixed window. When nontriviality is advertised, the record also proves that some $B \in \mathcal{F}$ has $T_\tau^{\mathrm{bar}}(B) \neq 0$.

Theorem 3.65 (Non-synthetic family-mesh zero-reflection)**.** *On the class* $\mathfrak{P}_n^{\mathcal{F},T,\tau}$*, the fixed-threshold stencil edge is natural and zero-reflecting:*

$$\mathrm{Edge}_n^{T,\tau}(P) = 0 \quad \iff \quad P = 0.$$

If the nontriviality clause in Definition 3.64 is discharged, this is a non-synthetic natural zero-reflection theorem on a declared class containing a nonzero profile.

Proof. Naturality is Proposition 3.63. The implication $P = 0 \Rightarrow \mathrm{Edge}_n^{T,\tau}(P) = 0$ is immediate.

Conversely, let

$$P = T_\tau^{\mathrm{bar}}(B), \qquad B \in \mathcal{F},$$

and suppose $P \neq 0$. Choose a retained interval I of B. Theorem 2.39 gives a point

$$t \in T \cap \mathrm{Act}_\tau(I).$$

The interval I is a summand of P, and its structure map

$$P(t) \longrightarrow P(t + \tau)$$

is nonzero. Hence $\mathrm{Edge}_n^{T,\tau}(P) \neq 0$, proving zero-reflection. □

Corollary 3.66 (Family-mesh higher Ext equivalence)**.** *Assume:*

(i) standard n-eligibility;

(ii) membership of the repaired profile in $\mathfrak{P}_n^{\mathcal{F},T,\tau}$*;*

(iii) a natural-family, independently meaningful edge-realization isomorphism

$$H^{-n}(A_{n;F,W,\tau}) \xrightarrow{\sim} \mathrm{Edge}_n^{T,\tau}\big(\mathrm{Eval}_{n,W,\tau}(F)\big);$$

(iv) representative-change and realization-change compatibility.

Then

$$\mathrm{Eval}_{n,W,\tau}(F) = 0 \iff \mathrm{Ext}^n(A_{n;F,W,\tau}, k) = 0.$$

Proof. Theorem 3.65 supplies a proved natural zero-reflection package. Apply Theorem 3.60. ◻

Proposition 3.67 (Complete detector higher Ext discharge)**.** *Let a complete Appendix B detector certificate, on the exact source profile stage and certified profile class, establish*

$$\mathcal{D}\big(\mathrm{Eval}_{n,W,\tau}(F)\big) = 0 \implies \mathrm{Eval}_{n,W,\tau}(F) = 0.$$

If its detector value is certified zero and the object is standard n-eligible, then

$$\mathrm{Ext}^n\big(A_{n;F,W,\tau}, k\big) = 0.$$

Proof. Detector completeness gives repaired-profile vanishing. Apply Theorem 3.51. ◻

Remark 3.68 (Birth and germ certificates)**.** The Appendix B birth and germ characterizations may supply objectwise zero-reflection on their certified endpoint and profile classes. They become part of a *natural* zero-reflection package only after a functorial family construction and natural edge realization have been proved. Object-dependent birth sets do not become a natural transformation merely because each objectwise certificate is complete.

Remark 3.69 (Stage discipline)**.** A detector proved complete on `windowed_prethreshold` data does not automatically become a zero-reflecting edge on the repaired profile. Likewise, kernel- and cokernel-defect detectors have different source objects. Every detector edge must cite its exact Appendix B stage-transport theorem.

Remark 3.70 (Detector output is not a metric ledger)**.** A stencil, birth, germ, bar-mass, or other finite-dimensional detector is not a bottleneck bound, a threshold-gap certificate, or a metric-ledger component merely because its value is numerical or finite-dimensional.

3.8 Inherited interaction with the E_1 detector

Definition 3.71 (Imported residual degree-one bar-mass detector)**.** The symbol

$$E_1(F; W, \tau)$$

denotes the canonical non-truncated residual degree-one bar-mass detector of the inherited Core: it is the sum of the lengths of all finite residual bars in $\mathrm{Eval}_{1,W,\tau}(F)$, counted with multiplicity, and has value $+\infty$ when a residual infinite bar is present. It is an operational exact-zero detector, not a complete persistence invariant or metric certificate.

Proposition 3.72 (Inherited E_1-first discharge)**.** *If*

$$E_1(F; W, \tau) = 0$$

and the degree-one eligible record is complete, then

$$\mathrm{Ext}^1\big(\mathcal{R}(C_{\tau,W}F), k\big) = 0.$$

Proof. The canonical non-truncated definition excludes both finite and infinite residual bars when its value is zero, hence

$$\mathrm{Eval}_{1,W,\tau}(F) = 0.$$

Apply Theorem 3.53. □

Remark 3.73 (No metric, stability, or converse role for E_1)**.** The preceding proposition does not make E_1 bottleneck-stable, a threshold-gap record, a metric-ledger component, a complete profile invariant, or a reverse Ext-to-persistence theorem. A truncated variant requires its own theorem and artifact policy.

3.9 Natural nonfaithful edge and reverse-failure boundary

Proposition 3.74 (Higher reverse inference requires zero-reflection)**.** *Under standard n-eligibility alone,*

$$\mathrm{Ext}^n\big(A_{n;F,W,\tau}, k\big) = 0$$

implies only

$$\mathrm{Edge}_n\big(\mathrm{Eval}_{n,W,\tau}(F)\big) = 0.$$

It does not imply repaired-profile vanishing without zero-reflection.

Proof. Ext vanishing implies $H^{-n}(A_{n;F,W,\tau}) = 0$ by Corollary 3.13. The edge-realization certificate then gives edge-value zero. No conclusion about the full profile follows without Definition 3.58. □

Proposition 3.75 (Higher failure of the reverse bridge for a natural nonfaithful edge)**.** *For every $n \geq 1$, there is a fixed non-adaptive functorial realization, a fixed functorial edge extractor, and a natural edge-realization family on a declared category for which standard n-eligibility holds, but*

$$\mathrm{Ext}^n(A_{n;F,W,\tau}, k) = 0 \qquad \text{and} \qquad \mathrm{Eval}_{n,W,\tau}(F) \neq 0.$$

Proof. Choose $0 < \tau < L$ and $\delta > 0$, and set

$$W = [0, L + \delta), \qquad t_* = L + \frac{\delta}{2}.$$

Let

$$C_I := E_n([0, L))$$

be the elementary interval representative from Appendix B, and set

$$F := C_I, \qquad C_{\tau,W}F := C_I.$$

Because W contains $[0, L)$ and $L > \tau$, this is an admissible degree-n representative with repaired profile

$$P_I := \mathrm{Eval}_{n,W,\tau}(F) = k_{[0,L)}.$$

Let C_* be the declared representative category generated by the zero representative and C_I, with their actual filtration-preserving chain maps. Let $\mathfrak{P}_*$ be the corresponding repaired-profile category. For a filtered complex C, write $C_{\leq t_*}$ for its filtration subcomplex at t_*. Fix, independently of the verdict,

$$\mathrm{Edge}_n(P) := P(t_*)$$

and

$$\mathcal{R}_n(C) := H_n(C_{\leq t_*})[n].$$

Both are functorial. The canonical identity

$$H^{-n}\big(\mathcal{R}_n(C)\big) = H_n(C_{\le t_*}) = \mathbf{P}_n(C)(t_*),$$

combined with the degree-n representative isomorphism, forms a natural edge-realization family.

For $C = C_I$,

$$\mathcal{R}_n(C_I) \in D^{[-n,-n]}(k\text{-mod}) \subseteq D^{[-n,0]}(k\text{-mod}),$$

but $t_* > L$, so

$$\mathrm{Edge}_n(P_I) = P_I(t_*) = 0.$$

Therefore

$$\mathrm{Ext}^n\big(\mathcal{R}_n(C_I), k\big) = 0,$$

while

$$P_I = \mathrm{Eval}_{n,W,\tau}(F) \neq 0.$$

Thus naturality, non-adaptivity, and standard eligibility do not imply zero-reflection. □

Proposition 3.76 (Reverse implication fails in the inherited degree-one regime)**.** *Under the inherited degree-one eligibility axioms,*

$$\mathrm{Ext}^1\big(\mathcal{R}(C_{\tau,W}F), k\big) = 0$$

does not imply

$$\mathrm{Eval}_{1,W,\tau}(F) = 0.$$

Proof. Apply Proposition 3.75 with $n = 1$, using $\mathcal{R}_1 = \mathcal{R}$ and $\mathrm{Edge}_1 = \mathrm{Edge}_1$. □

Remark 3.77 (Why the counterexample is stronger than a constant-zero edge)**.** The realization and edge above are fixed functorially at scope level and have ordinary mathematical meaning as point evaluation. The failure is therefore not caused by selecting a constant-zero policy after seeing the verdict. It is caused by genuine information loss: a point evaluation can miss a transient retained bar.

Remark 3.78 (What a reverse theorem requires)**.** A reverse theorem requires a separately proved zero-reflection or faithfulness condition on the declared profile class, together with exact compatibility between that edge and the realized H^{-n}. Neither standard eligibility nor Ext vanishing supplies this condition.

3.10 Synthetic shifted detector objects and non-promotion

Definition 3.79 (Synthetic shifted detector object)**.** Let $\mathcal{D}(P)$ be a finite-dimensional detector value. Define the shifted detector encoding

$$\mathrm{Syn}_{n,\mathcal{D}}(P) := \mathcal{D}(P)[n].$$

Then

$$H^{-n}\big(\mathrm{Syn}_{n,\mathcal{D}}(P)\big) = \mathcal{D}(P).$$

Unless an independent realization theorem identifies this object with an externally or internally meaningful $A_{n;F,W,\tau}$, its classification is

`synthetic` or `toy`.

Proposition 3.80 (Synthetic detector duality)**.** *For every finite-dimensional detector value,*

$$\mathrm{Ext}^n\big(\mathrm{Syn}_{n,\mathcal{D}}(P), k\big) \cong \mathrm{Hom}_k\big(\mathcal{D}(P), k\big)$$

canonically.

Proof. Apply Theorem 3.11 to $\mathrm{Syn}_{n,\mathcal{D}}(P) = \mathcal{D}(P)[n]$. □

Theorem 3.81 (No synthetic promotion)**.** *Proposition 3.80 does not by itself establish:*

(i) a natural persistence-to-derived realization theorem;

(ii) representative-independent categorical meaning;

(iii) an object-level PH_n-to-Ext^n bridge;

(iv) a problem-specific external obstruction theorem.

Promotion to the principal natural higher bridge requires an independently defined realization, natural edge certificate, representative compatibility, and all applicable Interface and Return Theorems.

Proof. The shifted object is defined from the detector output itself. The algebraic identity therefore expresses the tautological duality of that encoding. It supplies none of the independent semantic, naturality, representative, or external-transport obligations listed above. □

3.11 Degreewise categorical status and gate integration

Definition 3.82 (Ext-edge discharge record)**.** An inherited degree-one discharge record contains:

(i) the exact repaired evaluation artifact establishing $\mathrm{Eval}_{1,W,\tau}(F) = 0$;

(ii) the comparison mode and any exact or metric transport record used;

(iii) the complete inherited eligible record;

(iv) the theorem identifier for Theorem 3.53;

(v) the canonical two-term Ext edge and its naturality data;

(vi) the resulting Ext-vanishing conclusion;

(vii) immutable claim, proof, manifest, and artifact references.

Definition 3.83 (Typed Ext-edge clause status)**.** The inherited degree-one categorical clause has one of

`not_invoked`, `undefined`, `pass`, `reject`.

A passing clause requires a well-typed Ext group evidenced to vanish by the bridge theorem or a separately verified direct computation. A rejecting clause requires evidenced nonvanishing. Missing data are `undefined`, never zero.

Definition 3.84 (Typed degree-n Ext-clause status)**.** For each invoked degree n, the categorical clause has exactly one canonical atomic meaning:

(i) `not_invoked`: an explicit scope policy excludes the degree-n clause;

(ii) `undefined`: the clause is invoked but eligibility, detector completeness, naturality, comparison, claim, or artifact evidence is missing or ill-typed;

(iii) `pass`: the well-typed Ext group is evidenced to vanish;

(iv) `reject`: the well-typed Ext group is evidenced to be nonzero.

Missing data are never encoded as a zero group.

Definition 3.85 (Finite monitored degree set and aggregate categorical status)**.** Let

$$N_{\mathrm{mon}} \subseteq \mathbb{Z}_{\geq 1}$$

be finite. The aggregate higher categorical clause:

(i) is `not_invoked` when every degree is explicitly out of scope;

(ii) is `reject` when at least one invoked, well-typed degree is evidenced nonzero;

(iii) is `undefined` when no reject is forced but at least one invoked required degree is undefined;

(iv) is `pass` when every invoked degree passes and all out-of-scope degrees are explicitly recorded.

No numerical sum of Ext dimensions replaces this conjunctive status.

Definition 3.86 (Higher Ext discharge record)**.** A degree-n discharge record contains:

(i) the exact repaired profile and zero-evidence artifact;

(ii) the standard n-eligible record;

(iii) the theorem identifier Theorem 3.51;

(iv) the higher Ext edge and naturality data;

(v) the resulting categorical conclusion;

(vi) the degreewise and aggregate statuses;

(vii) exact or metric comparison evidence used to obtain the topological zero statement;

(viii) detector identities and completeness evidence when detector-derived;

(ix) immutable Claim, CM, TB, proof, hash, and provenance references.

Proposition 3.87 (Degree-one B-Gate discharge)**.** *Under the inherited degree-one eligible regime,*

$$\mathrm{Eval}_{1,W,\tau}(F) = 0$$

discharges the degree-one categorical clause:

$$\mathrm{Ext}^1\big(\mathcal{R}(C_{\tau,W}F), k\big) = 0.$$

Proof. Apply Theorem 3.53. □

Remark 3.88 (Non-compensation)**.** A passing categorical clause does not cancel a failed or undefined detector, metric, representative, Type IV, transient-defect, finite-to-infinite, manifest, claim-status, or external Return-Theorem obligation.

3.12 Representative-change and realization-change compatibility

Definition 3.89 (Realization-equivalent eligible representatives)**.** Two inherited degree-one eligible representatives are realization-equivalent when an actual isomorphism

$$\alpha : \mathcal{R}(C_{\tau,W}F) \xrightarrow{\sim} \mathcal{R}(C'_{\tau,W}F)$$

is supplied and the edge-realization certificates commute with $H^{-1}(\alpha)$.

Proposition 3.90 (Eligibility is invariant under admissible representative equivalence)**.** *The inherited amplitude, edge-realization, and categorical typing transport across a declared degree-one realization equivalence.*

Proof. Derived isomorphisms preserve cohomology and amplitude, and compatibility of the edge certificates transports the inherited edge condition. ◻

Proposition 3.91 (One-way bridge is representative-independent)**.** *For realization-equivalent inherited eligible representatives, the Ext groups are identified contravariantly and the truth value of degree-one categorical vanishing is unchanged.*

Proof. Apply $\mathrm{Hom}_{D^{\mathrm{b}}(k\text{-mod})}(-, k[1])$ to the recorded realization isomorphism and use Corollary 3.16. ◻

Definition 3.92 (Higher realization-equivalent representatives)**.** Two standard n-eligible representatives are *realization-equivalent* when an actual isomorphism

$$\alpha : A_{n;F,W,\tau} \xrightarrow{\sim} A'_{n;F,W,\tau}$$

is supplied in $D^{\mathrm{b}}(k\text{-mod})$, and their edge-realization certificates are compatible with $H^{-n}(\alpha)$. For a natural zero-reflection package, the declared profile class and edge functor must also be preserved.

Proposition 3.93 (Higher eligibility invariance)**.** *Under a declared realization equivalence, amplitude, cohomology, edge-realization, degreewise categorical typing, and the applicable zero-reflection package transport to the second representative.*

Proof. Derived isomorphisms preserve cohomology and amplitude. Compatibility with $H^{-n}(\alpha)$ transports the edge isomorphism. The explicitly required profile-class and edge compatibility transport the zero-reflection data. ◻

Proposition 3.94 (Representative independence of the higher discharge)**.** *The realization isomorphism induces an identification*

$$\mathrm{Ext}^n(A'_{n;F,W,\tau}, k) \xrightarrow{\sim} \mathrm{Ext}^n(A_{n;F,W,\tau}, k),$$

and the truth value of Ext vanishing is independent of the representative choice relative to the recorded equivalence.

Proof. Apply the contravariant functor

$$\mathrm{Hom}_{D^{\mathrm{b}}(k\text{-mod})}(-, k[n])$$

to α. Corollary 3.12 identifies this with dualizing $H^{-n}(\alpha)$. ◻

Remark 3.95 (Relative independence, not hidden canonicality)**.** The proposition gives independence only relative to the recorded realization equivalence. It does not make the filtered representative, realization functor, or edge extractor unique.

Remark 3.96 (Profile equivalence remains weaker)**.** Monitored persistence-profile equivalence from Appendix B does not by itself imply realization equivalence or detector-edge preservation.

3.13 Higher eligibility manifest and obligation handoff

Definition 3.97 (Eligibility manifest extension)**.** An inherited degree-one eligibility entry records at least:

(i) repaired and pre-threshold profile identities;

(ii) the representative and actual degree-one profile isomorphism;

(iii) the realization functor and realized object;

(iv) the two-term amplitude certificate;

(v) the fixed degree-one edge extractor;

(vi) the edge-realization certificate and its objectwise/natural scope;

(vii) coefficient field, equivalence policy, status, theorem, claim, manifest, hash, and provenance records;

(viii) exact or metric evidence only when used to establish the separate topological zero statement.

Definition 3.98 (Eligible-realization obligation bundle)**.** The inherited non-scalar obligation bundle contains representative availability, degree-one profile compatibility, realization-domain membership, amplitude, fixed edge extraction, edge realization, field declaration, equivalence policy, claim admissibility, manifest consistency, and immutable artifact identity.

Definition 3.99 (Higher eligibility manifest extension)**.** A proof object invoking a degree-n Ext clause records at least:

(i) the repaired evaluation and pre-threshold profile identifiers;

(ii) the detector source stage and zero-evidence route;

(iii) the representative identifier, construction artifact, and actual degree-n profile isomorphism;

(iv) n, W, τ, endpoint convention, and monitored scope;

(v) the realization domain, functor, version, and artifact identity;

(vi) the independently meaningful realized object;

(vii) the amplitude certificate or registered replacement;

(viii) the profile class and membership proof;

(ix) the fixed edge extractor, object/morphism scope, and version;

(x) zero compatibility;

(xi) the edge-realization certificate and its `objectwise` or `natural_family` classification;

(xii) when a converse is used, the complete natural zero-reflection package;

(xiii) representative-change and realization-change compatibility;

(xiv) the non-adaptive policy record;

(xv) the degreewise and aggregate categorical statuses;

(xvi) the theorem or direct-computation identities supporting pass or reject;

(xvii) exact/metric records independently supporting the topological zero statement;

(xviii) detector completeness, profile stage, class, and artifact records when a detector is consumed;

(xix) a `natural`, `synthetic`, `toy`, or `spec` classification for every shifted detector or encoded Ext object;

(xx) all Claim, CM, TB, proof, hash, and provenance links required for replay.

For $n = 1$, this extension contains the inherited v19 eligibility fields.

Definition 3.100 (Higher eligible-realization obligation bundle)**.** The non-scalar obligation bundle contains:

(i) representative availability and degree-n profile compatibility;

(ii) realization-domain membership and independent semantic meaning;

(iii) amplitude or admissible replacement;

(iv) profile-class declaration and membership;

(v) fixed edge extractor and zero compatibility;

(vi) edge-realization isomorphism;

(vii) naturality when claimed;

(viii) zero-reflection when a converse is consumed;

(ix) detector completeness at the exact profile stage when detector-based;

(x) representative- and realization-change compatibility;

(xi) non-adaptivity;

(xii) field coefficient declaration;

(xiii) claim-status admissibility;

(xiv) manifest completeness and consistency;

(xv) immutable artifact identity.

Remark 3.101 (Appendix F/G handoff)**.** Appendix F may integrate the eligible-realization obligations into status, dependency, and non-compensation logic. Appendix G may serialize their fields and immutable references. Neither appendix creates a missing mathematical proof by recording it.

Remark 3.102 (Failure routing)**.** An invoked clause with missing required data is `undefined`, not a mathematical counterexample and not `not_invoked` unless an explicit scope exclusion applies. An internally inconsistent manifest is routed to Type V rejection under the canonical audit architecture.

3.14 Higher eligible realization and Ext-edge package

Theorem 3.103 (Appendix C eligible-realization package)**.** *Within the inherited degree-one eligible regime:*

(i) the repaired degree-one input is $\mathrm{Eval}_{1,W,\tau}(F)$*;*

(ii) an admissible representative realizes that profile;

(iii) the realized object lies in $D^{[-1,0]}(k\text{-mod})$*;*

(iv) the fixed residual edge satisfies $\mathrm{Edge}_1(0) = 0$*;*

(v) the edge-realization certificate identifies

$$H^{-1}(A_{F,W,\tau}) \cong \mathrm{Edge}_1(\mathrm{Eval}_{1,W,\tau}(F));$$

(vi) the canonical two-term theorem gives

$$\mathrm{Ext}^1(A,k) \cong \mathrm{Hom}_k(H^{-1}(A),k);$$

(vii) consequently,

$$\mathrm{Eval}_{1,W,\tau}(F) = 0 \Longrightarrow \mathrm{Ext}^1(\mathcal{R}(C_{\tau,W}F),k) = 0;$$

(viii) the result is invariant under recorded realization-equivalent representative changes;

(ix) the reverse implication fails without zero-reflection;

(x) incomplete eligibility is undefined, not categorical vanishing.

Proof. Items (i)–(v) are Definitions 3.22, 3.23, 3.30, 3.32, and 3.33. Item (vi) is Theorem 3.15. Item (vii) is Theorem 3.53. Item (viii) is Proposition 3.91. Item (ix) is Proposition 3.76. Item (x) is Definition 3.83. □

Theorem 3.104 (Appendix C higher eligible Ext-edge package)**.** *In the v20 field-coefficient Core:*

(i) the inherited v19 degree-one eligible bridge remains valid at its original strength;

(ii) for every $n \geq 0$ *and bounded derived object,*

$$\mathrm{Ext}^n(A,k) \cong \mathrm{Hom}_k(H^{-n}(A),k)$$

canonically and naturally;

(iii) the standard amplitude $D^{[-n,0]}$ *is a conservative eligibility policy, not an algebraic necessity for item (ii);*

(iv) a standard n-eligible record supplies a degree-n compatible representative, independently meaningful realization, fixed edge extractor, and artifact-backed edge certificate;

(v) under zero compatibility,

$$\mathrm{Eval}_{n,W,\tau}(F) = 0 \Longrightarrow \mathrm{Ext}^n(A_{n;F,W,\tau},k) = 0;$$

(vi) under a complete natural zero-reflection package, the preceding implication upgrades to an equivalence of vanishing predicates;

(vii) a complete fixed-stencil detector supplies a natural zero-reflecting edge only on its certified class and exact profile stage;

(viii) birth and germ certificates supply objectwise zero-reflection at their registered strengths, but require additional functorial packaging for a natural bridge;

(ix) standard n-eligibility alone does not give the reverse implication, even for a fixed natural point-evaluation edge;

(x) a shifted detector object has a canonical encoded Ext duality but is synthetic without an independent natural realization theorem;

(xi) representative independence holds only relative to an actual realization equivalence and compatible edge data;

(xii) no clause in this package replaces metric, Type IV, transient-defect, finite-to-infinite, manifest, claim-status, or external Return-Theorem obligations.

Proof. Item (i) is Theorem 3.53. Item (ii) is Theorem 3.11. Item (iii) is Remark 3.18. Item (iv) is Definition 3.46. Item (v) is Theorem 3.51. Item (vi) is Theorem 3.60. Item (vii) is Theorem 3.65. Item (viii) is Remark 3.68. Item (ix) is Proposition 3.75. Item (x) is Proposition 3.80 and Theorem 3.81. Item (xi) is Proposition 3.94. Item (xii) is the non-compensation discipline recorded throughout the eligibility and manifest contracts. □

3.15 Explicit exclusions

The following are not asserted.

(i) No unconditional object-level equivalence

$$\mathrm{PH}_n \cong \mathrm{Ext}^n$$

is asserted in any degree.

(ii) No reverse implication from Ext vanishing to repaired-profile vanishing is asserted without a separately proved zero-reflection package.

(iii) No higher edge extractor is assumed faithful, conservative, or zero-reflecting merely because it is functorial or zero-compatible.

(iv) No objectwise edge certificate is a natural transformation without a natural-family theorem.

(v) No realization, edge extractor, stencil, sample point, or profile class may be chosen adaptively after inspecting the desired verdict.

(vi) No invoked Ext clause is typed without its representative, realization, amplitude policy, profile class, edge certificate, claim, and manifest data.

(vii) No missing eligibility, detector completeness, naturality, or artifact evidence is interpreted as zero.

(viii) No finite detector supplies a reverse bridge outside its exact profile stage, endpoint convention, coverage, birth/germ, or family scope.

(ix) No soundness-only detector is treated as complete.

(x) No synthetic shifted detector object is promoted to the principal natural higher bridge merely because its Ext group is dual to the detector output.

(xi) No general-ring, sheaf, stack, motivic, dg, A_∞, or spectral-sequence extension inherits the field theorem without separate hypotheses.

(xii) No standard amplitude certificate is claimed to be mathematically necessary for the canonical field-coefficient Ext isomorphism.

(xiii) No global collapse functor, unconditional crop–collapse commutation, exactness, or Serre localization is used.

(xiv) No threshold-gap, repaired-zero transport, detector-rank stability, or crop theorem is reproved by the Ext edge.

(xv) No categorical discharge cancels a failed detector, metric, representative, Type IV, transient-defect, manifest, or claim-status clause.

(xvi) No detector is a metric-ledger component merely because its output is finite-dimensional.

(xvii) No point-evaluation edge is assumed to detect transient bars.

(xviii) No representative-profile equivalence automatically implies realization equivalence.

(xix) No arithmetic Ext group, Selmer group, PDE obstruction, motivic Ext, or external categorical invariant is automatically identified with the Core group $\mathrm{Ext}^n(A_{n;F,W,\tau}, k)$.

(xx) No external mathematical conclusion follows without a separately proved Interface and Return Theorem.

(xxi) No Search artifact, AI confidence value, ranking, or advisory invariant discharges a categorical clause.

3.16 Provenance and status

The semisimplicity of finite-dimensional vector spaces and the natural hyper-Ext spectral sequence are standard derived-category inputs. The v19 contribution inherited here is the typed degree-one eligible bridge. The v20 conservative extension consists of:

- the canonical higher field-coefficient Ext edge;
- standard n-eligibility;
- the higher one-way repaired discharge;
- natural zero-reflecting vanishing equivalence;
- detector-induced zero-reflection at exact stage and class;
- the natural nonfaithful-edge counterexample;
- the synthetic-versus-natural promotion boundary;
- degreewise and aggregate categorical audit semantics.

Status labels for Appendix C. *Standard derived-category results:* Lemma 3.9, Lemma 3.10, Theorem 3.11, Corollaries 3.12, 3.13, and 3.14.

Inherited degree-one Core results: Proposition 3.52, Theorem 3.53, Proposition 3.87, Proposition 3.72.

New higher Core results: Proposition 3.50, Theorem 3.51, Proposition 3.74, Proposition 3.75.

Conditional natural zero-reflection results: Definition 3.59, Theorem 3.60, Theorem 3.65, Corollary 3.66, Propositions 3.63 and 3.67.

Synthetic supporting algebra, not principal natural bridge: Definition 3.79, Proposition 3.80, Theorem 3.81.

Core definitions and operational records: Definitions 3.4, 3.5, 3.6, 3.8, 3.22, 3.23, 3.25, 3.39, 3.40, 3.41, 3.42, 3.43, 3.44, 3.45, 3.46, 3.58, 3.62, 3.84, 3.85, 3.86, 3.92, 3.99, 3.100, 3.71.

Conservative closure package: Theorem 3.104.

Explicit exclusions: Subsection 3.15.

4 Appendix D: Tower Diagnostics, Transient-Defect Certification, and Finite-to-Infinite Reduction

4.1 Scope and dependence on the repaired Core

This appendix supplies the detailed Foundation layer for the failure taxonomy, terminal-generic Type IV diagnostics, transient-defect certification, and exact finite-to-infinite tower reductions of Chapter 5. It consumes:

(i) the repaired windowed evaluation and threshold-gap discipline of Chapters 2–3 and Appendix A;

(ii) the representative and actual-morphism discipline of Appendix B;

(iii) the inherited degree-one and v20 higher eligible one-way bridges of Chapter 4 and Appendix C;

(iv) the stage-typed finite detector calculus of Chapter 3 and Appendix B, including the dedicated `kernel_defect` and `cokernel_defect` stages;

(v) the typed audit statuses and Minimal Manifest hierarchy of Chapters 1 and 7.

All tower diagnostics are formed after the repaired evaluation

$$\mathrm{Eval}_{i,W,\tau}(F) := T_\tau^{\mathrm{bar}}\big(\mathsf{W}_W \mathbf{P}_i(F)\big).$$

The diagnostic is not an invariant of a bare sequence of barcodes. It is an invariant of a declared comparison morphism between a typed tower-side source profile and a repaired apex profile.

The inherited v19 distinction is

$$(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) = (0,0)$$

under status `certified_zero`. This means absence of monitored *terminal-generic* kernel and cokernel defect relative to the declared artifact. The v20 extension keeps that status unchanged and adds a separate long-transient verdict on full or noncofinal defect profiles at an independently declared threshold $\tau_{\mathrm{def}} > 0$. Terminal zero and transient zero remain distinct; neither is inferred from missing evidence, and neither alone implies full comparison-map isomorphism.

Remark 4.1 (Appendix D does not enlarge the Core)**.** This appendix does not restore a global exact collapse functor, manufacture a tower comparison morphism from a barcode matching, identify an undeclared apex with a categorical colimit, or promote zero terminal diagnostics to full comparison-map isomorphism. It gives the window-relative definitions and proof obligations already required by the Chapter 5 contract.

Remark 4.2 (The bounded-window repair)**.** The former shorthand

$$\mathrm{gdim}(M) = \lim_{t \to +\infty} \dim_k M(t)$$

is valid only on a right-unbounded window. On a bounded window, extension by zero outside the window would force this limit to vanish and would make Type IV vacuous. The canonical inherited invariant is therefore the window-relative terminal-generic dimension tgdim_W.

Remark 4.3 (v20 terminal/transient separation)**.** A nonzero terminal-generic pair is Type IV. A certified nonzero detector on an interior defect profile is a distinct long-transient obstruction for claims requiring transient cleanliness; it is not renamed Type IV. Detector inapplicability, missing completeness, or wrong-stage evidence is Type III undefinedness, not a mathematical absence or presence verdict.

Remark 4.4 (Finite observation is not infinite proof)**.** Observed agreement through finitely many tower levels does not prove eventual stabilization, a colimit identity, apex agreement, or terminal cleanliness. A finite-to-infinite conclusion requires a theorem or certificate whose quantified tail, finite bound, comparison map, apex semantics, and failure mode are explicit.

4.2 Directed towers and repaired profile systems

Definition 4.5 (Directed tower)**.** A *directed tower* in the declared filtered regime is a sequence

$$F_0 \longrightarrow F_1 \longrightarrow F_2 \longrightarrow \cdots$$

of objects of $\mathsf{FiltCh}(k)$ and filtration-preserving morphisms. We write $F_\bullet = \{F_n\}_{n \in \mathbb{N}}$.

Definition 4.6 (Declared apex object)**.** A filtered object F_∞ is a *declared apex object* if it is supplied as the object against which the repaired tower-side source profile is compared. Its provenance may be an actual filtered colimit, a verified stabilization, a completion, or an implementation artifact. The provenance status is part of the tower record and is never inferred from the symbol F_∞.

Definition 4.7 (Apex identity and authority record)**.** An apex identity record states whether the declared target is a genuine realized limiting object, a completion, a formally declared implementation apex, a finite-stage proxy, or an advisory/specification placeholder. It also records the construction, the claim scope for which the target is admissible, and any finite-to-infinite theorem needed to promote a finite proxy. An advisory or specification placeholder cannot support an invoked Core tower artifact.

Remark 4.8 (Apex is declared, not automatic)**.** The notation F_∞ does not assert existence of an objectwise filtered colimit inside the finite-type constructible regime. If the repaired apex evaluation, provenance, or comparison typing is absent, the apex is not a Core-readable Type IV input.

Definition 4.9 (Monitored degrees and parameter-indexed scope)**.** For one fixed window–threshold pair (W, τ), let

$$I_{\mathrm{mon}} \subset \mathbb{Z}$$

be the finite set of monitored homological degrees. Degreewise numerical totals are formed only over this fixed pair.

A run may additionally declare a finite family of pairs

$$\mathfrak{S} = \{(W_a, \tau_a, I_{\mathrm{mon},a})\}_{a \in A},$$

but the corresponding diagnostic pairs remain indexed by a. Diagnostics from different windows or thresholds are not silently added into one scalar. An additional cross-parameter aggregation requires a separately defined codomain, normalization, aggregation law, and theorem proving that the aggregation has the claimed meaning.

Remark 4.10 (Constructibility caveat)**.** A raw tower or limit may leave the constructible regime. Type IV is defined only when all repaired profiles, transition morphisms, source and apex profiles, comparison morphisms, and defect profiles belong to a declared abelian constructible profile category on the relevant window.

Definition 4.11 (Windowed collapsed finite-stage profile)**.** For every stage F_n, monitored degree i, window W, and threshold $\tau > 0$, define

$$M_{n;i,W,\tau} := \mathcal{P}_{i,W,\tau}(F_n) := \mathrm{Eval}_{i,W,\tau}(F_n).$$

Definition 4.12 (Repaired windowed collapsed tower-profile system)**.** A *repaired tower-profile system* at (i, W, τ) consists of:

(i) the repaired stage profiles $M_{n;i,W,\tau}$;

(ii) actual morphisms in the declared profile category

$$\rho_{n,m} : M_{n;i,W,\tau} \longrightarrow M_{m;i,W,\tau} \qquad (n \le m);$$

(iii) the identity and composition laws

$$\rho_{n,n} = \mathrm{id}, \qquad \rho_{m,\ell}\rho_{n,m} = \rho_{n,\ell};$$

(iv) provenance and artifact identifiers for the morphisms.

A sequence of barcode multisets, or a family of pairwise bottleneck matchings, is not by itself a repaired tower-profile system.

Definition 4.13 (Collapsed source profile)**.** Given a repaired tower-profile system, a source profile

$$\mathrm{ColimProfile}_{i,W,\tau}(F_\bullet)$$

is supplied with exactly one provenance status:

`genuine_colimit`, `verified_stabilization`, `implementation_artifact`.

In the first case it is a genuine colimit in the declared profile category. In the second it is a certified stabilized profile and is identified with a categorical colimit only when a stabilization-to-colimit or finite-to-infinite theorem is recorded. In the third its relation to the tower is itself part of the artifact and is not promoted to a categorical colimit or verified stabilization.

Every consuming claim declares the source semantics it requires. The source is admissible for that claim only when its provenance has sufficient semantic authority. In particular, an `implementation_artifact` may support an implementation-scoped comparison, but it cannot discharge a claim about a genuine categorical colimit merely because it is named ColimProfile.

Definition 4.14 (Source-semantic authority record)**.** A source-semantic authority record contains:

(i) the source provenance status;

(ii) the semantic strength required by the claim;

(iii) the theorem, construction, or scope rule relating that provenance to the required semantics;

(iv) exactly one verdict

$$\texttt{sufficient}, \qquad \texttt{insufficient}, \qquad \texttt{undefined};$$

(v) immutable provenance and artifact references.

A Type IV or transient-defect verdict is claim-readable only under `sufficient` authority.

Definition 4.15 (Collapsed apex profile)**.** For a declared apex F_∞, define

$$\mathrm{ApexProfile}_{i,W,\tau}(F_\infty) := \mathrm{Eval}_{i,W,\tau}(F_\infty),$$

provided the repaired evaluation record for F_∞ is declared.

Remark 4.16 (No automatic colimit–collapse interchange)**.** No identity

$$\operatorname*{colim}_n \mathrm{Eval}_{i,W,\tau}(F_n) \cong \mathrm{Eval}_{i,W,\tau}(F_\infty)$$

or

$$\operatorname*{colim}_n T_\tau^{\mathrm{bar}}(M_n) \cong T_\tau^{\mathrm{bar}}\left(\operatorname*{colim}_n M_n\right)$$

is available in the general Core. Any such comparison is precisely what the artifact must construct or justify.

4.3 Tower comparison artifacts

Definition 4.17 (Tower comparison artifact)**.** A *tower comparison artifact* for (i, W, τ) consists of:

(i) a repaired tower-profile system;

(ii) a source profile with one of the provenance statuses of Definition 4.13;

(iii) a repaired apex profile;

(iv) an actual morphism in the declared constructible profile category,

$$\phi_{i,W,\tau} : \mathrm{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \mathrm{ApexProfile}_{i,W,\tau}(F_\infty);$$

(v) a construction or proof that $\phi_{i,W,\tau}$ is well-typed;

(vi) the claim-required source semantics and a `sufficient` source-semantic authority record;

(vii) the apex identity and semantic-authority record;

(viii) every repaired-evaluation, comparison-mode, threshold-gap, crop-budget, representative, and metric-ledger record used to construct its source, target, or comparison; each metric comparison transported through hard deletion carries its two-sided threshold-gap record, whereas an exact comparison carries its exact proof;

(ix) the terminal mode `bounded_terminal` or `right_unbounded`;

(x) an explicitly declared abelian constructible profile category on W, closed under the kernels and cokernels used here;

(xi) terminal-tameness certificates and kernel/cokernel computation artifacts, or a categorical proof establishing them;

(xii) when transient detection is invoked, immutable full defect-profile identifiers sufficient to register new Appendix B detector records at stages `kernel_defect` and `cokernel_defect`; no detector record is relabeled across stages;

(xiii) immutable artifact and claim identifiers.

Remark 4.18 (Comparison artifact, not automatic map)**.** The map $\phi_{i,W,\tau}$ is not produced by the existence of a filtered tower, a barcode decomposition, or a bottleneck matching. It must be explicitly constructed, or supplied canonically by a separately proved exact-admissible regime. Its source provenance must also have sufficient authority for the claim's declared source semantics.

Remark 4.19 (A bottleneck matching is not a persistence morphism)**.** A bottleneck matching controls metric discrepancy between barcodes. It is not, by itself, a natural transformation of persistence modules and therefore does not define the kernel or cokernel needed by Type IV.

Definition 4.20 (Exact-admissible tower artifact)**.** A tower comparison artifact is *exact-admissible* if its transition morphisms, source construction, comparison map, kernels, and cokernels are supplied by a declared regime in which the retained-long-bar assignment is functorial and preserves every exact sequence used by the diagnostic.

Remark 4.21 (Order of operations)**.** The normative order is

$$F_n \longmapsto \mathbf{P}_i(F_n) \longmapsto \mathsf{W}_W\mathbf{P}_i(F_n) \longmapsto T_\tau^{\mathrm{bar}}\big(\mathsf{W}_W\mathbf{P}_i(F_n)\big),$$

followed by construction of the repaired profile system, source profile, apex profile, and actual comparison morphism. Windowing and threshold deletion are not silently interchanged.

Remark 4.22 (Why repaired collapse precedes diagnostics)**.** The terminal defect is diagnosed on the repaired profile selected by the Core. This does not mean that threshold deletion is exact or globally functorial; the artifact must separately supply the morphisms that make the repaired tower well-typed.

4.4 Terminal-tame profiles and terminal-generic dimension

Definition 4.23 (Terminal-tame profile)**.** Let $W = [u, v)$, where $v \in \mathbb{R} \cup \{+\infty\}$, and let M be a constructible persistence profile on W. We call M *terminal-tame* if:

(i) its fibers near the terminal end are finite-dimensional;

(ii) if $v < +\infty$, there exists $s < v$ such that every structure map $M(t) \to M(t')$ with $s \le t \le t' < v$ is an isomorphism;

(iii) if $v = +\infty$, there exists $s \in \mathbb{R}$ such that every structure map $M(t) \to M(t')$ with $s \le t \le t'$ is an isomorphism.

Definition 4.24 (Generic and terminal-generic dimension)**.** For a terminal-tame profile M on $W = [u, v)$, define

$$\mathrm{tgdim}_W(M) := \begin{cases} \lim_{t \uparrow v} \dim_k M(t), & v < +\infty, \\ \lim_{t \to +\infty} \dim_k M(t), & v = +\infty. \end{cases}$$

When $v = +\infty$, the same quantity may be written $\mathrm{gdim}(M)$. On a bounded window, gdim after zero extension is not the canonical diagnostic.

Proposition 4.25 (Well-definedness of terminal-generic dimension)**.** *For every terminal-tame profile M on W, the quantity* $\mathrm{tgdim}_W(M)$ *exists and is a finite nonnegative integer.*

Proof. Terminal tameness supplies a terminal subinterval on which all structure maps are isomorphisms. The finite-dimensional fibers on that subinterval therefore have one constant dimension. This value is exactly the displayed limit. □

Definition 4.26 (Cofinal bar)**.** A bar I in a barcode profile on $W = [u, v)$ is *cofinal in W* if

$$\sup I = v \quad \text{when } v < +\infty,$$

or if it is unbounded above when $v = +\infty$. A cofinal bar on a bounded window is a boundary-reaching bar relative to that window; it is not a global infinite bar.

Proposition 4.27 (Terminal dimension counts cofinal bars)**.** *Let M be terminal-tame and interval-decomposable on W. Then* $\mathrm{tgdim}_W(M)$ *equals the multiplicity of bars cofinal in W. In particular, on a right-unbounded window it equals the multiplicity of infinite bars.*

Proof. Choose a terminal tail on which every structure map is an isomorphism. An interval summand contributes one dimension to every fiber on that tail exactly when it is cofinal in W. Every noncofinal interval dies before the tail. Summing the surviving summands proves the claim. □

Remark 4.28 (Endpoint conventions)**.** Endpoint bookkeeping at finite endpoints does not alter terminal-generic dimension. What matters is whether the bar reaches the declared terminal end cofinally.

Proposition 4.29 (Terminal tameness passes to defects)**.** *Let $\phi : M \to N$ be a morphism of constructible profiles on W. If M and N are terminal-tame, then* $\ker \phi$ *and* $\operatorname{coker} \phi$ *are terminal-tame.*

Proof. Choose a common terminal tail on which the structure maps of M and N are isomorphisms. Naturality of ϕ gives commuting squares on that tail. The induced maps on pointwise kernels and cokernels are therefore isomorphisms, and finite-dimensionality is inherited pointwise. □

4.5 Defect profiles and canonical diagnostics

Definition 4.30 (Kernel and cokernel defect profiles)**.** For a valid tower comparison artifact, define

$$\mathsf{Defect}^{\mathrm{ker}}_{i,W,\tau} := \ker(\phi_{i,W,\tau}), \qquad \mathsf{Defect}^{\mathrm{coker}}_{i,W,\tau} := \operatorname{coker}(\phi_{i,W,\tau}).$$

These are objects of the declared constructible profile category on W.

Remark 4.31 (Kernel and cokernel require a profile regime)**.** The defect profiles are not inferred from a tower of filtered objects or a barcode matching. They require the actual comparison morphism and an abelian profile regime in which pointwise kernels and cokernels are defined.

Remark 4.32 (Interpretation of kernel and cokernel defects)**.** The kernel records repaired tower-side classes killed by the comparison. The cokernel records repaired apex classes not supplied by the tower-side source. Both are artifact-relative, after-window, after-threshold defects.

Definition 4.33 (Full defect profile)**.** The full defect profile is

$$\mathsf{Defect}_{i,W,\tau} := \left(\mathsf{Defect}^{\mathrm{ker}}_{i,W,\tau}, \mathsf{Defect}^{\mathrm{coker}}_{i,W,\tau}\right).$$

Definition 4.34 (Degreewise tower diagnostics)**.** Assume that both defect profiles are terminal-tame. Define

$$\mu_{i,W,\tau} := \mathrm{tgdim}_W\left(\mathsf{Defect}^{\mathrm{ker}}_{i,W,\tau}\right), \qquad \nu_{i,W,\tau} := \mathrm{tgdim}_W\left(\mathsf{Defect}^{\mathrm{coker}}_{i,W,\tau}\right).$$

Thus μ measures terminal-generic kernel defect and ν measures terminal-generic cokernel defect.

Definition 4.35 (Total collapse diagnostics)**.** For fixed (W,τ) and finite I_{mon}, define

$$\mu_{\mathrm{Collapse}}(W,\tau) := \sum_{i\in I_{\mathrm{mon}}} \mu_{i,W,\tau}, \qquad \nu_{\mathrm{Collapse}}(W,\tau) := \sum_{i\in I_{\mathrm{mon}}} \nu_{i,W,\tau}.$$

When the fixed pair (W,τ) is clear, the arguments may be suppressed. A run containing several windows or thresholds stores the corresponding pairs $(\mu_{\mathrm{Collapse}}(W_a,\tau_a), \nu_{\mathrm{Collapse}}(W_a,\tau_a))$ separately. No cross-pair scalar total is defined by default, and no infinite aggregate is admitted without a separate aggregation theorem.

The symbols $u_{i,W,\tau}$ and u_{Collapse} are deprecated. The canonical cokernel notation is $\nu_{i,W,\tau}$ and ν_{Collapse}.

Remark 4.36 (Meaning of μ)**.** On a bounded window, $\mu_{i,W,\tau}$ counts kernel defect bars cofinal to the right endpoint. On a right-unbounded window it counts infinite kernel bars.

Remark 4.37 (Meaning of ν)**.** On a bounded window, $\nu_{i,W,\tau}$ counts cokernel defect bars cofinal to the right endpoint. On a right-unbounded window it counts infinite cokernel bars.

Proposition 4.38 (Diagnostics count terminal-cofinal defects)**.** *For every valid terminal-tame artifact, $\mu_{i,W,\tau}$ and $\nu_{i,W,\tau}$ equal the multiplicities of cofinal interval summands in the kernel and cokernel defect profiles, respectively. On a right-unbounded window these are precisely infinite-bar defects.*

Proof. Apply Proposition 4.27 to the two defect profiles. □

Definition 4.39 (Type IV diagnostic record)**.** A Type IV diagnostic record contains the fixed pair (W,τ), the finite monitored degree set, the declared tower scope, and exactly one terminal status

`not_invoked`, `undefined`, `certified_zero`, `obstructed`.

Its remaining fields are status-dependent.

(i) For `not_invoked`, the record contains an explicit no-tower scope rationale. No numerical μ, ν, full-interior, or transient value is asserted.

(ii) For `undefined`, the record contains the partial or invalid artifact identifiers that exist and a missing/invalid-obligation marker for every absent, insufficient-authority, or ill-typed input. No Core numerical diagnostic, full-interior absence, or transient absence is asserted.

(iii) For `certified_zero` or `obstructed`, the record contains a valid artifact identifier, source provenance, claim-required source semantics, a `sufficient` semantic-authority verdict, apex identity, ambient abelian category, terminal mode, terminal-tameness certificates, full kernel/cokernel defect-profile artifacts, and all degreewise and fixed-(W, τ) total μ, ν values.

(iv) In a valid invoked record, the full-interior-defect field is recorded separately with exactly one value

$$\texttt{not\_checked}, \quad \texttt{certified\_absent}, \quad \texttt{certified\_present}.$$

The value `certified_absent` requires proof that both entire noncofinal defect profiles vanish; `certified_present` requires an actual nonzero noncofinal witness.

(v) When transient detection is invoked, the record additionally contains $\tau_{\mathrm{def}} > 0$, detector component, exact detector stage, information class, certificate status, value status, and one aggregate long-transient verdict

$$\texttt{not\_invoked}, \quad \texttt{undefined}, \quad \texttt{certified\_absent}, \quad \texttt{certified\_present}.$$

Long-transient `certified_absent` excludes only noncofinal bars of length strictly greater than τ_{def}; it is not the full-interior verdict.

The terminal status is `certified_zero` exactly when all invoked terminal inputs are valid for the claim's source semantics and $(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) = (0, 0)$. It is `obstructed` exactly when those inputs are valid and $(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) \neq (0, 0)$. The transient field does not alter the terminal Type IV status. Missing, wrong-stage, or insufficient-authority input yields `undefined`, never absence or numerical zero.

4.6 Typed Type IV status

Definition 4.40 (Monitored terminal-generic Type IV obstruction)**.** Relative to a valid tower comparison artifact, a run has a *Type IV obstruction* if

$$(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) \neq (0, 0).$$

On a bounded window this is a defect cofinal to the declared right endpoint; on a right-unbounded window it is an infinite-bar defect.

Definition 4.41 (Type IV-free in the monitored terminal-generic sense)**.** Relative to a valid terminal-tame artifact, a run is *Type IV-free in the monitored terminal-generic sense* if

$$(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) = (0, 0).$$

Theorem 4.42 (Characterization of certified Type IV-free status)**.** *A tower diagnostic record has status `certified_zero` if and only if:*

(i) the tower comparison artifact is valid;

(ii) its kernel and cokernel defect profiles are terminal-tame;

(iii) $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$.

It has status `obstructed` *when the first two clauses hold and* $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) \neq (0, 0)$. *Missing or ill-typed input yields* `undefined`*, never zero.*

Proof. These are exactly the validity, terminal-tameness, and numerical clauses in Definition 4.39. The first two clauses are necessary to ensure that the displayed numbers are defined Core diagnostics rather than missing values. □

Remark 4.43 (No full-isomorphism conclusion)**.** The equality $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$ says that no defect bar reaches the terminal end of the declared window. It does not imply

$$\ker \phi_{i,W,\tau} = 0 \quad \text{or} \quad \operatorname{coker} \phi_{i,W,\tau} = 0.$$

Interior defect bars may remain.

Remark 4.44 (Undefined is not zero)**.** Absence of an actual morphism, invalid source provenance, failure of terminal tameness, or unavailable kernel/cokernel artifacts yields `undefined`. Reporting that state as $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$ is a Type V manifest failure.

Remark 4.45 (No scalar repair of Type IV typing)**.** No metric budget or threshold clearance can manufacture a missing profile system, comparison morphism, terminal-tameness certificate, or defect artifact. Those are non-scalar proof obligations.

4.7 Interior defects and the full-isomorphism upgrade

Definition 4.46 (Interior or noncofinal defect)**.** An *interior defect* is a noncofinal interval summand of either $\ker \phi_{i,W,\tau}$ or $\operatorname{coker} \phi_{i,W,\tau}$. On a right-unbounded window these are exactly the finite-bar defects. On a bounded window, a globally finite bar reaching the right endpoint is cofinal and is therefore counted by tgdim_W, not classified as an interior defect.

Definition 4.47 (No-interior-defect condition)**.** A comparison artifact satisfies the *no-interior-defect condition* if its kernel and cokernel have no noncofinal interval summands. The legacy phrase "no-finite-defect" is retained only as a label alias; the window-relative notion is no-interior-defect.

Proposition 4.48 (Zero diagnostics plus no interior defects imply isomorphism)**.** *Assume the defect profiles are terminal-tame and interval-decomposable. If*

$$(\mu_{i,W,\tau}, \nu_{i,W,\tau}) = (0, 0)$$

and the no-interior-defect condition holds, then $\phi_{i,W,\tau}$ *is an isomorphism in the declared abelian constructible profile category.*

Proof. Zero terminal diagnostics exclude every cofinal summand of the kernel and cokernel. The no-interior-defect condition excludes every noncofinal summand. Thus both defect profiles have empty interval decompositions and are zero. A morphism with zero kernel and cokernel is an isomorphism in the declared abelian category. □

Example 4.49 (Zero terminal diagnostics with a nonisomorphism)**.** Let $W = [0, 10)$ and consider

$$0 \longrightarrow k_{[2,3)}.$$

Its cokernel is $k_{[2,3)}$, so the map is not an isomorphism. The bar is not cofinal in W, hence

$$\operatorname{tgdim}_W\big(k_{[2,3)}\big) = 0.$$

Thus zero terminal diagnostics do not exclude an interior defect.

Remark 4.50 (Why the interior condition is separated)**.** Type IV intentionally monitors terminal-generic failure. Full comparison isomorphism is stronger and requires a separately recorded interior-defect verdict. This prevents a zero terminal count from being overread as complete agreement.

4.8 Cofinal/interior decomposition and transient-defect certification

Definition 4.51 (Cofinal and interior barcode profiles)**.** Let M be an interval-decomposable constructible profile on $W = [u, v)$. Define $\mathrm{Cof}_W(M)$ to be the barcode subprofile consisting of all bars cofinal in W, and define $\mathrm{Int}_W(M)$ to consist of all noncofinal bars, with multiplicity. At barcode-profile isomorphism strength,

$$M \cong \mathrm{Cof}_W(M) \oplus \mathrm{Int}_W(M).$$

The decomposition is canonical on barcode isomorphism classes. It is not asserted to be functorial on arbitrary persistence morphisms.

Proposition 4.52 (Terminal/interior separation)**.** *If M is terminal-tame and interval-decomposable, then*

$$\mathrm{tgdim}_W(M) = \#\mathcal{B}\big(\mathrm{Cof}_W(M)\big)$$

with multiplicity, and

$$\mathrm{tgdim}_W(M) = 0 \iff \mathrm{Cof}_W(M) = 0 \iff M \cong \mathrm{Int}_W(M).$$

Proof. The first assertion is Proposition 4.27. A direct sum of cofinal intervals has terminal-generic dimension zero exactly when it is empty. The barcode decomposition gives the final equivalence. □

Definition 4.53 (Kernel/cokernel component notation)**.** Write

$$K_{i,W,\tau} := \ker \phi_{i,W,\tau}, \qquad Q_{i,W,\tau} := \operatorname{coker} \phi_{i,W,\tau},$$

and define the interior components

$$K^{\circ}_{i,W,\tau} := \mathrm{Int}_W(K_{i,W,\tau}), \qquad Q^{\circ}_{i,W,\tau} := \mathrm{Int}_W(Q_{i,W,\tau}).$$

These are barcode-profile constructions attached to the actual comparison artifact. Their immutable identifiers are distinct from those of the full defect profiles.

Definition 4.54 (Defect threshold and detector component)**.** A transient-defect run declares a threshold

$$\tau_{\mathrm{def}} > 0$$

independently of the collapse threshold τ, and declares exactly one component for each detector application:

$$\texttt{full\_profile} \quad \text{or} \quad \texttt{interior\_noncofinal}.$$

The full component reads $K_{i,W,\tau}$ or $Q_{i,W,\tau}$; the interior component reads $K^{\circ}_{i,W,\tau}$ or $Q^{\circ}_{i,W,\tau}$. Changing τ_{def}, component, endpoint convention, information class, or detector stage creates a new detector claim and new artifact.

Definition 4.55 (Kernel and cokernel transient detectors)**.** Let $\mathcal{D}$ be an Appendix B finite detector with all applicability, soundness, and completeness data registered anew at source stage `kernel_defect` or `cokernel_defect`. For the declared component set

$$K^{\det} \in \{K_{i,W,\tau}, K^{\circ}_{i,W,\tau}\}, \qquad Q^{\det} \in \{Q_{i,W,\tau}, Q^{\circ}_{i,W,\tau}\},$$

and write

$$\mathcal{D}^{\ker}_{\tau_{\mathrm{def}}} := \mathcal{D}_{\tau_{\mathrm{def}}}(K^{\det}), \qquad \mathcal{D}^{\mathrm{coker}}_{\tau_{\mathrm{def}}} := \mathcal{D}_{\tau_{\mathrm{def}}}(Q^{\det}).$$

Stencil, birth, germ, and certified family-mesh detectors may be used only at their exact Appendix B strength. A topological-profile detector artifact is not relabeled as a defect detector.

Theorem 4.56 (Complete transient-defect characterization)**.** *Assume a valid tower comparison artifact, full constructible defect profiles, and complete detector certificates at the declared defect stages and components. Then*

$$\mathcal{D}^{\ker}_{\tau_{\mathrm{def}}} = 0 \quad \Longleftrightarrow \quad T^{\mathrm{bar}}_{\tau_{\mathrm{def}}}(K^{\det}) = 0,$$

and

$$\mathcal{D}^{\mathrm{coker}}_{\tau_{\mathrm{def}}} = 0 \quad \Longleftrightarrow \quad T^{\mathrm{bar}}_{\tau_{\mathrm{def}}}(Q^{\det}) = 0.$$

A certified nonzero detector value is equivalent to the existence of a bar of length strictly greater than τ_{def} in the corresponding declared component.

Proof. Apply the Appendix B detector soundness/completeness theorem at threshold τ_{def} to the supplied constructible defect profile. The source stage, component, endpoint convention, coverage or birth/germ exhaustiveness, family membership, excess provenance, and immutable artifact remain explicit hypotheses. This is a fresh defect-stage certificate, not transport of a detector record from another stage. □

Remark 4.57 (Defect-stage registration)**.** A detector registered at `windowed_prethreshold` or `repaired_postthreshold` cannot be reused by changing its stage label. Kernel and cokernel profiles, information classes, coverage data, and certificate hashes are reconstructed and registered anew.

Corollary 4.58 (Full and interior detectors coincide under terminal zero)**.** *Assume the kernel and cokernel defect profiles are terminal-tame and interval-decomposable. If*

$$(\mu_{i,W,\tau}, \nu_{i,W,\tau}) = (0,0),$$

then

$$K_{i,W,\tau} \cong K^{\circ}_{i,W,\tau}, \qquad Q_{i,W,\tau} \cong Q^{\circ}_{i,W,\tau}$$

at barcode-profile strength. Consequently complete full-profile and corresponding interior detectors have identical zero/nonzero verdicts, after the detector invariance under exact profile isomorphism is recorded.

Proof. Apply Proposition 4.52 to the kernel and cokernel profiles and then Appendix B, Proposition 2.43. □

Definition 4.59 (Aggregate long-transient verdict)**.** For fixed $(W, \tau, \tau_{\mathrm{def}}, I_{\mathrm{mon}})$, the aggregate long-transient verdict is exactly one of

`not_invoked`, `undefined`, `certified_absent`, `certified_present`.

It is `certified_absent` exactly when every required kernel and cokernel detector certificate is complete and every detector value is zero. It is `certified_present` when all artifacts needed to interpret the witness are well-typed and at least one required detector value is nonzero. Missing, incomplete, wrong-component, or wrong-stage evidence yields `undefined`, never absence.

Theorem 4.60 (Joint terminal/transient defect classification)**.** *Assume a valid tower comparison artifact, terminal-tame interval-decomposable defect profiles, and complete interior defect detectors at τ_{def}. Exactly the following mathematical cases occur degreewise and after finite aggregation over the fixed monitored degrees:*

(i) if $(\mu, \nu) \neq (0, 0)$, a terminal/cofinal Type IV defect is present; the interior detector independently records whether a long noncofinal defect is also present;

(ii) if $(\mu, \nu) = (0, 0)$ and the long-transient verdict is `certified_present`, there is no terminal defect but there is a noncofinal kernel or cokernel bar of length greater than τ_{def};

(iii) if $(\mu, \nu) = (0, 0)$ and the long-transient verdict is `certified_absent`, there is no terminal defect and no noncofinal defect bar longer than τ_{def}. Shorter nonzero defects may remain.

No undefined detector certificate is assigned a mathematical zero verdict.

Proof. Terminal/cofinal presence is characterized by Proposition 4.52; long noncofinal presence or absence is characterized by Theorem 4.56. The conclusions are independent conjunctions. □

Definition 4.61 (Short-defect exclusion certificate)**.** A short-defect exclusion certificate at τ_{def} proves that every nonzero bar in the declared interior kernel and cokernel profiles has length strictly greater than τ_{def}. It is a structural theorem or artifact; it is not inferred from detector zero, terminal-generic zero, metric slack, or finite sampling.

Theorem 4.62 (Conditional full-isomorphism upgrade)**.** *Let $\phi_{i,W,\tau}$ be a valid morphism in the declared abelian profile category, and assume its kernel and cokernel are terminal-tame and interval-decomposable. Suppose:*

(i) $(\mu_{i,W,\tau}, \nu_{i,W,\tau}) = (0, 0)$;

(ii) complete interior kernel and cokernel detectors at τ_{def} are certified zero;

(iii) a short-defect exclusion certificate holds.

Then

$$\ker \phi_{i,W,\tau} = 0, \qquad \operatorname{coker} \phi_{i,W,\tau} = 0,$$

and $\phi_{i,W,\tau}$ is an isomorphism.

Proof. Terminal zero excludes all cofinal kernel and cokernel bars. Complete interior detector zero excludes every interior bar longer than τ_{def}. The short-defect exclusion certificate says that any remaining nonzero interior bar would have length strictly greater than that threshold, a contradiction. Hence both full defect profiles vanish, and a morphism with zero kernel and cokernel is an isomorphism in an abelian category. □

Example 4.63 (Zero terminal and transient diagnostics with a short residual defect)**.** Let $W = [0, 10)$, choose $\tau_{\mathrm{def}} > 0$, and consider

$$0 \longrightarrow k_{[2,\, 2+\tau_{\mathrm{def}}/2)}.$$

The terminal pair is zero, and every complete defect detector at threshold τ_{def} is zero because the sole defect bar is deleted by that threshold. The morphism is not an isomorphism. Thus the short-defect exclusion in Theorem 4.62 is essential.

Remark 4.64 (No automatic metric transport of defect profiles)**.** A bottleneck or interleaving bound between source profiles and a separate bound between apex profiles do not by themselves induce a bound between their kernels or cokernels. Metric transport of a defect-detector verdict requires actual comparison data between the defect profiles and an independently proved kernel/cokernel stability theorem. Only then may Appendix A's threshold-gap and zero-separation machinery be applied at stage `kernel_defect` or `cokernel_defect`.

4.9 Pure kernel, pure cokernel, and mixed modes

Definition 4.65 (Pure kernel Type IV)**.** A valid artifact has *pure kernel Type IV* if

$$\mu_{\mathrm{Collapse}} > 0, \qquad \nu_{\mathrm{Collapse}} = 0.$$

Definition 4.66 (Pure cokernel Type IV)**.** A valid artifact has *pure cokernel Type IV* if

$$\mu_{\mathrm{Collapse}} = 0, \qquad \nu_{\mathrm{Collapse}} > 0.$$

Definition 4.67 (Mixed Type IV)**.** A valid artifact has *mixed Type IV* if

$$\mu_{\mathrm{Collapse}} > 0, \qquad \nu_{\mathrm{Collapse}} > 0.$$

Remark 4.68 (Interpretation of the three modes)**.** Let K_W denote one cofinal interval bar: $k_{[u,v)}$ for bounded $W = [u, v)$, and $k_{[u,+\infty)}$ for a right-unbounded window. Then the zero maps

$$K_W \to 0, \qquad 0 \to K_W, \qquad K_W \oplus 0 \to 0 \oplus K_W$$

have diagnostics $(1, 0)$, $(0, 1)$, and $(1, 1)$, respectively. These are artifact-level kernel loss, apex-only birth, and mixed failure. No external geometric or arithmetic interpretation follows without an interface theorem.

4.10 Finite-stage admissibility does not determine Type IV

Definition 4.69 (Finite-level collapse admissibility)**.** A finite stage F_n is *finite-level collapse-admissible* on (W, τ) if its repaired topological clause passes and, when the Ext clause is invoked, a valid eligible realization record discharges that clause. This stagewise notion contains no undeclared tower-comparison verdict.

Proposition 4.70 (Finite admissibility does not imply monitored tower admissibility)**.** *There exists a valid artifact-level witness in which every finite stage is finite-level collapse-admissible while the declared source-to-apex comparison has pure cokernel Type IV.*

Proof. Fix $W = [u, v)$ and let every repaired finite-stage profile be zero, with zero transition morphisms. Take the source profile to be the genuine colimit zero. Declare an apex profile K_W consisting of one cofinal bar and take the actual comparison morphism

$$0 \longrightarrow K_W.$$

Every finite stage has zero repaired degree-one profile and may be represented by the zero filtered complex. The comparison cokernel is K_W, so

$$(\mu_{i,W,\tau}, \nu_{i,W,\tau}) = (0, 1).$$

The finite profiles and the cofinal apex barcode admit filtered realizations by the finite-barcode realization theorem of Appendix B, Theorem 2.65. This is a declared apex-mismatch witness; it does not claim that an arbitrary externally chosen apex is the categorical colimit of the tower. □

Remark 4.71 (Why Type IV is invisible to finite stages)**.** Finite-stage evaluations do not determine the kernel or cokernel of an undeclared source-to-apex morphism. Type IV becomes meaningful only after the comparison artifact has been supplied.

4.11 Exact finite-to-infinite reduction and eventual constancy

Definition 4.72 (Finite-to-infinite reduction certificate)**.** A finite-to-infinite reduction certificate for one fixed (i, W, τ) is a theorem-backed record proving that a declared finite subsystem or finite computation determines a stated infinite tower, colimit, apex, full defect, terminal diagnostic, or transient-defect conclusion. It records:

(i) the exact infinite statement being reduced;

(ii) a finite bound N and the finite artifacts consumed;

(iii) quantified tail hypotheses and their proof;

(iv) the source and apex semantics;

(v) the induced comparison morphism;

(vi) whether the conclusion is exact or metric;

(vii) the theorem, failure mode, and immutable references.

A finite experiment, repeated equality, or numerical plateau is not such a certificate.

Definition 4.73 (Exact eventual-constancy reduction)**.** An exact eventual-constancy reduction consists of a repaired tower-profile system $(M_n, \rho_{n,m})$ in the declared abelian constructible profile category, an integer N, and:

(i) every transition $\rho_{n,n+1}$ is an isomorphism for $n \geq N$;

(ii) a genuine colimit $L = \operatorname{colim}_n M_n$ exists in the declared profile category;

(iii) the canonical map $\iota_N : M_N \to L$ is part of the artifact;

(iv) an actual apex agreement isomorphism

$$a_N : M_N \xrightarrow{\sim} \mathrm{ApexProfile}_{i,W,\tau}(F_\infty)$$

is supplied.

The comparison morphism is constructed in the theorem below after the canonical map ι_N is proved invertible.

Theorem 4.74 (Eventually constant exact finite-to-infinite reduction)**.** *Under the hypotheses of Definition 4.73, the canonical map*

$$\iota_N : M_N \longrightarrow \operatorname*{colim}_n M_n$$

is an isomorphism. The maps

$$b_n := a_N \circ c_n : M_n \longrightarrow \mathrm{ApexProfile}_{i,W,\tau}(F_\infty),$$

where

$$c_n = \begin{cases} \rho_{n,N}, & n \leq N, \\ \rho_{N,n}^{-1}, & n \geq N, \end{cases}$$

form a compatible apex cocone. Let

$$\phi_{i,W,\tau} : \operatorname*{colim}_n M_n \longrightarrow \mathrm{ApexProfile}_{i,W,\tau}(F_\infty)$$

be the unique morphism induced by that cocone. Then

$$\phi_{i,W,\tau} = a_N \circ \iota_N^{-1}$$

and is an isomorphism. Its full kernel and cokernel profiles vanish. Hence, whenever the remaining Type IV record fields are valid, the terminal status is `certified_zero`*, the full-interior-defect status is* `certified_absent`*, and every complete well-typed long-transient detector has verdict* `certified_absent`*.*

Proof. For $n \le N$, define

$$c_n := \rho_{n,N} : M_n \longrightarrow M_N,$$

and for $n \ge N$, define

$$c_n := \rho_{N,n}^{-1} : M_n \longrightarrow M_N.$$

The composition law and the tail-isomorphism hypothesis give

$$c_m \circ \rho_{n,m} = c_n \qquad (n \le m),$$

so $(M_N, \{c_n\})$ is a cocone on the full diagram. For every cocone $(X, \{x_n : M_n \to X\})$, cocone compatibility gives

$$x_n = x_N \circ c_n$$

for all n, and the factor $x_N : M_N \to X$ is unique because its component at N is fixed. Thus M_N is a colimit of the full diagram. By uniqueness of colimits, $\iota_N : M_N \to L$ is an isomorphism.

The family $b_n = a_N \circ c_n$ is a compatible cocone, so it induces a unique comparison map $\phi_{i,W,\tau} : L \to$ ApexProfile. At stage N,

$$\phi_{i,W,\tau} \circ \iota_N = a_N,$$

whence

$$\phi_{i,W,\tau} = a_N \circ \iota_N^{-1}.$$

It is therefore an isomorphism. Its kernel and cokernel are zero profiles, which are terminal-tame and have no cofinal, interior, or long-transient bars. All stated diagnostic consequences follow at their declared status strength. □

Remark 4.75 (Foundation proof and Chapter 8 ownership)**.** The theorem above supplies the detailed tower-defect consequences of the internal eventual-constancy reduction. The normative main-body finite-to-infinite locator and its role in the final Projection/Return architecture remain in Chapter 8. Appendix D does not create a second, stronger reduction principle or an external-domain theorem.

Remark 4.76 (Observed stabilization is not eventual constancy)**.** Isomorphism or numerical agreement through the inspected levels $0, \dots, N$ does not establish the quantified tail hypothesis $\rho_{n,n+1}$ is an isomorphism for every $n \ge N$. The latter must come from a theorem, induction, control argument, recurrence, monotonicity principle with termination, or another registered finite-to-infinite reduction.

Remark 4.77 (Finite reduction does not change source semantics silently)**.** A proved eventual-constancy reduction may justify identifying a `verified_stabilization` source with the genuine colimit for the registered claim. Without that proof, the provenance statuses remain distinct. No finite proxy acquires categorical authority merely by matching the observed levels.

4.12 Artifact-coherent stable bands

Definition 4.78 (Degreewise artifact-coherent stable band)**.** Fix i, a window W, a tower-profile construction rule, source provenance, and a terminal mode. An interval $J \subset (0, +\infty)$ is a *degreewise stable band* if for every $\tau \in J$:

(i) the same declared construction rule supplies a valid artifact $\phi_{i,W,\tau}$;

(ii) source provenance and terminal mode do not change silently;

(iii) the defect profiles are terminal-tame;

(iv) every repaired comparison used by the artifact has its required two-sided threshold-gap, crop-budget, ledger, and scalarization records;

(v) the Type IV diagnostic status is `certified_zero`.

Definition 4.79 (Global artifact-coherent stable band)**.** A threshold interval J is a *global stable band* for a finite monitored degree set if Definition 4.78 holds coherently for every monitored degree and the total record is `certified_zero` for every $\tau \in J$.

Proposition 4.80 (Degreewise stable bands exclude degreewise Type IV)**.** *If J is a degreewise stable band for i, then no terminal-generic Type IV obstruction occurs in degree i for any $\tau \in J$, relative to the declared artifact family.*

Proof. Every threshold in the band has a valid record with status `certified_zero`. Theorem 4.42 applies pointwise. □

Proposition 4.81 (Global stable bands exclude monitored Type IV)**.** *If J is a global stable band, then every $\tau \in J$ is Type IV-free in the monitored terminal-generic sense relative to its declared artifact.*

Proof. Apply Proposition 4.80 in every monitored degree and sum over the finite monitored scope. □

Remark 4.82 (Uniform reserve is a separate claim)**.** If one metric budget is claimed uniformly over J, the manifest must provide a common ledger $\mathfrak{d}_{\mathrm{met},J}$ and prove

$$\mathrm{val}_{\mathrm{B}}(\mathfrak{d}_{\mathrm{met},J}) < \inf_{\tau \in J} \mathrm{gap}_\tau(\mathcal{B}_\tau).$$

Pointwise positive reserves do not imply that this infimum is positive.

Remark 4.83 (Stable band does not mean full-isomorphism band)**.** A stable band controls terminal-generic defects. Interior defects may remain at some or all thresholds.

Definition 4.84 (Full-isomorphism band)**.** A threshold interval J is a *full-isomorphism band* if every artifact $\phi_{i,W,\tau}$, for every monitored degree and every $\tau \in J$, is an isomorphism in the declared constructible profile category.

Proposition 4.85 (Full-isomorphism band implies stable band)**.** *Every artifact-coherent full-isomorphism band is a global stable band.*

Proof. An isomorphism has zero kernel and cokernel, hence zero terminal diagnostics and a `certified_zero` record, provided the remaining artifact fields are present. □

Remark 4.86 (Converse requires interior-defect data)**.** A global stable band becomes a full-isomorphism band only after the no-interior-defect condition is verified throughout the band.

Definition 4.87 (Artifact-coherent joint defect-clean band)**.** An interval $J \subset (0, +\infty)$ is a joint defect-clean band at fixed $\tau_{\mathrm{def}} > 0$ if it is an artifact-coherent Type IV stable band and, for every $\tau \in J$:

(i) the same defect threshold, detector stages, endpoint convention, component policy, information class, and completeness rule are used;

(ii) complete kernel and cokernel interior detectors are `certified_zero`;

(iii) detector artifacts, source semantics, and semantic authority remain auditable across the band.

Proposition 4.88 (Uniform terminal and long-transient cleanliness)**.** *On an artifact-coherent joint defect-clean band, every threshold is free of monitored terminal defects and monitored noncofinal defects longer than τ_{def}. Full isomorphism does not follow without a uniform short-defect exclusion theorem or direct full-defect vanishing.*

Proof. Apply Proposition 4.81 and Theorem 4.60 pointwise. □

4.13 Integration with admissibility and B-Gate^{+}

Definition 4.89 (Tower-clean structural collapse admissibility)**.** A declared run is *tower-clean structurally collapse-admissible* if:

(i) $\mathrm{Eval}_{1,W,\tau}(F) = 0$;

(ii) if the Ext clause is invoked, an eligible realization record exists and the clause is discharged;

(iii) the Type IV status is either `not_invoked` under an explicit no-tower scope decision or `certified_zero`;

(iv) no Type V claim-status or manifest defect occurs in the structural record.

Proposition 4.90 (Eligible reduction of tower-clean admissibility)**.** *In the eligible regime of Chapter 4 and Appendix C, the categorical clause is discharged by repaired degree-one vanishing. Hence the independent structural conditions reduce to*

$$\mathrm{Eval}_{1,W,\tau}(F) = 0$$

and a correctly typed Type IV status equal to `not_invoked` or `certified_zero`, together with absence of Type V failure.

Proof. The one-way bridge, Theorem 3.53, discharges the Ext clause from the repaired topological clause in the eligible regime. Type IV remains independent because it concerns a source-to-apex comparison morphism. A no-tower claim records `not_invoked`; an invoked tower claim requires a valid `certified_zero` record. □

Definition 4.91 (v20 monitored tower-clean structural collapse admissibility)**.** Fix a finite monitored degree set I_{mon}. A declared run is *tower-clean structurally collapse-admissible* if:

(i) $\mathrm{Eval}_{n,W,\tau}(F) = 0$ for every invoked $n \in I_{\mathrm{mon}}$;

(ii) every invoked degree-n Ext clause has a complete n-eligibility record and is discharged;

(iii) the Type IV status is either `not_invoked` under an explicit no-tower scope decision or `certified_zero`;

(iv) if absence of long transient defects is required, the full defect profiles, τ_{def}, and complete kernel/cokernel detector records exist and the aggregate long-transient verdict is `certified_absent`;

(v) if full comparison-map isomorphism is required, either direct full defect-profile vanishing or the hypotheses of Theorem 4.62 are certified;

(vi) no Type V claim-status or manifest defect occurs in the structural record.

Proposition 4.92 (v20 eligible reduction of monitored tower-clean admissibility)**.** *In the degreewise eligible regime of Chapter 4 and Appendix C, every invoked categorical clause is discharged by repaired degree-n vanishing. The remaining independent structural conditions are therefore: repaired zero in every invoked degree; a correctly typed Type IV status equal to `not_invoked` or `certified_zero`; every required long-transient verdict equal to `certified_absent`; every requested full-isomorphism upgrade fully certified; and absence of Type V failure.*

Proof. Apply Theorem 3.51 degreewise, using the inherited Theorem 3.53 when $n = 1$. Type IV and transient-defect clauses remain independent because they concern the full kernel/cokernel profiles of a source-to-apex comparison morphism. A no-tower claim records `not_invoked`; an invoked tower claim requires a valid `certified_zero` terminal record and every separately required transient/full-isomorphism record. □

Remark 4.93 (Relation to B-Gate^{+})**.** A final audited B-Gate^{+} pass additionally requires every applicable two-sided threshold-gap, crop-budget, scalarized metric-ledger, representative, claim-status, and manifest clause. Type IV zero cannot compensate for failure of those independent obligations.

4.14 Diagnostic equivalence and invariance

Definition 4.94 (Diagnostically equivalent tower artifacts)**.** Two valid tower artifacts are *diagnostically equivalent* at (i, W, τ) if their kernel and cokernel defect profiles are isomorphic in the declared profile category by isomorphisms compatible with the terminal mode and terminal-tameness records.

Proposition 4.95 (Invariance of diagnostics)**.** *Diagnostically equivalent artifacts have equal degreewise diagnostics. If the equivalence holds over the same finite monitored scope, their total μ_{Collapse} and ν_{Collapse} values agree.*

Proof. Terminal-generic dimension is invariant under profile isomorphism. Apply this to the kernel and cokernel defect profiles and sum over the common finite scope. □

Remark 4.96 (Representative independence)**.** The result is artifact-relative, not a claim that arbitrary strict filtered representatives are canonically identical. Representative changes must preserve the actual comparison artifact and its defect profiles in the declared sense.

4.15 Manifest extension and obligation handoff

Definition 4.97 (Type IV manifest entry)**.** Every run records a Type IV scope status. When tower semantics are invoked, the manifest additionally records:

(i) the monitored scope and terminal mode;

(ii) tower-profile-system and transition-morphism artifacts;

(iii) source provenance, claim-required source semantics, semantic-authority verdict, apex identity, and apex evaluation records;

(iv) the actual comparison morphism, ambient abelian category, and typing proof;

(v) terminal-tameness and full/cofinal/interior defect-profile artifacts;

(vi) degreewise and fixed-(W, τ) total μ, ν values;

(vii) the separately typed full-interior-defect status;

(viii) whenever transient detection is invoked: τ_{def}, component, exact detector stage, information class, endpoint convention, stencil or birth/germ data, coverage/mesh/excess evidence, certificate status, value status, and aggregate long-transient verdict;

(ix) every finite-to-infinite reduction record, tail theorem, finite bound, and apex-agreement artifact when finite evidence is promoted to an infinite or terminal conclusion;

(x) every threshold-gap, crop-budget, ledger, representative, eligibility, claim, and immutable provenance reference consumed by the artifact.

An incomplete invoked entry yields `undefined`; an inconsistent entry or undefined-as-zero declaration is a Type V rejection.

Definition 4.98 (Type IV obligation bundle)**.** Appendix D forwards the following non-scalar obligations to the Appendix F status calculus and Appendix G manifest schema:

```
tower_system,                   actual_comparison_morphism,
source_provenance,              apex_evaluation,
terminal_tameness,              defect_profiles,
diagnostic_scope,               full_interior_defect_status,
defect_threshold,               defect_detector_completeness,
transient_verdict,              short_defect_exclusion,
finite_to_infinite_reduction,   semantic_authority,
artifact_identity,              claim_status.
```

No scalar budget purchases any of these obligations.

4.16 Terminal-generic Type IV package

Theorem 4.99 (Appendix D terminal-generic Type IV package)**.** *Within the repaired constructible tower regime, the following package is available.*

(i) Type IV is defined only relative to a repaired tower-profile system, typed source provenance, a repaired apex profile, and an actual comparison morphism.

(ii) On a bounded window, tgdim_W *counts defect bars cofinal to the right endpoint; on a right-unbounded window it counts infinite bars.*

(iii) Terminal tameness makes tgdim_W *a well-defined finite integer and passes from terminal-tame source and target profiles to kernel and cokernel defects.*

(iv) The canonical diagnostics are

$$\mu_{i,W,\tau} = \mathrm{tgdim}_W(\ker \phi_{i,W,\tau}), \qquad \nu_{i,W,\tau} = \mathrm{tgdim}_W(\mathrm{coker}\, \phi_{i,W,\tau}).$$

(v) Total diagnostics are finite-scope sums; infinite aggregation requires a separate theorem.

(vi) The four statuses are `not_invoked`, `undefined`, `certified_zero`, and `obstructed`.

(vii) An invoked record is `certified_zero` exactly when the artifact is valid, the defects are terminal-tame, and $(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) = (0, 0)$.

(viii) Zero terminal diagnostics are weaker than comparison-map isomorphism; the latter additionally requires absence of interior defects.

(ix) Finite-stage admissibility does not determine Type IV because the diagnostic belongs to the declared source-to-apex comparison.

(x) A stable band is an artifact-coherent family of passing records, not a pointwise collection of unrelated numerical zeros.

(xi) In the eligible regime, repaired degree-one vanishing discharges the Ext clause but does not discharge Type IV or any metric/manifest clause.

(xii) Missing morphisms, terminal-tameness, provenance, or defect artifacts yield `undefined`, never numerical zero.

Proof. Items (i)–(v) are Definitions 4.12, 4.17, 4.24, 4.34, and 4.35, together with Propositions 4.25–4.38. Items (vi)–(vii) are Definition 4.39 and Theorem 4.42. Item (viii) is Proposition 4.48 and Example 4.49. Item (ix) is Proposition 4.70. Item (x) is the stable-band contract. Item (xi) is Proposition 4.90. Item (xii) is the typed failure routing of Remarks 4.44 and 4.45. □

Theorem 4.100 (Appendix D v20 tower-defect package)**.** *Within the repaired constructible tower regime, the following conservative package is available.*

(i) Type IV is defined only relative to a repaired tower-profile system, typed source provenance, sufficient source-semantic authority, a repaired apex profile, an ambient abelian profile category, and an actual comparison morphism.

(ii) On a bounded window, tgdim_W *counts defect bars cofinal to the right endpoint; on a right-unbounded window it counts infinite bars.*

(iii) Terminal tameness makes tgdim_W *a well-defined finite integer and passes from source and target to kernel and cokernel defects.*

(iv) The canonical terminal diagnostics are

$$\mu_{i,W,\tau} = \mathrm{tgdim}_W(\ker \phi_{i,W,\tau}), \qquad \nu_{i,W,\tau} = \mathrm{tgdim}_W(\operatorname{coker} \phi_{i,W,\tau}).$$

(v) Total diagnostics sum only over finite monitored degrees at one fixed (W, τ); different windows or thresholds remain indexed separately.

(vi) The terminal status is one of `not_invoked`, `undefined`, `certified_zero`, `obstructed`; missing artifacts or insufficient semantic authority never produce numerical zero.

(vii) Full defect profiles decompose, at barcode-isomorphism strength, into cofinal and interior components.

(viii) *Appendix B detectors may be registered anew at* `kernel_defect` *and* `cokernel_defect`*; complete certificates characterize absence or presence of bars longer than the independently declared* τ_{def}.

(ix) *Terminal/cofinal and long-transient verdicts form the joint classification of Theorem 4.60; shorter nonzero defects may remain after both terminal zero and long-transient absence.*

(x) *Full comparison-map isomorphism follows from terminal zero, complete interior detector zero, and a short-defect exclusion certificate, or directly from full kernel/cokernel vanishing.*

(xi) *Source/apex object-level metric closeness does not automatically transport to kernel/cokernel defect profiles.*

(xii) *Finite-stage admissibility does not determine an undeclared tower diagnostic or apex comparison.*

(xiii) *The exact eventual-constancy theorem supplies one registered finite-to-infinite reduction: quantified tail isomorphisms, a genuine colimit, and apex agreement imply full comparison isomorphism and all defect verdicts absent.*

(xiv) *An artifact-coherent stable band is a family of coherently typed artifacts, and a joint defect-clean band additionally carries uniform detector policy and long-transient absence.*

(xv) *Higher Ext discharge remains independent of Type IV, transient, metric, manifest, and finite-to-infinite obligations.*

Proof. Items (i)–(vi) are the inherited tower-artifact and terminal-generic calculus, strengthened by Definitions 4.14 and 4.35. Items (vii)–(ix) are Definition 4.51, Theorem 4.56, and Theorem 4.60. Item (x) is Theorem 4.62 together with Proposition 4.48. Item (xi) is Remark 4.64. Item (xii) is Proposition 4.70. Item (xiii) is Theorem 4.74. Item (xiv) is the stable-band and joint-band calculus. Item (xv) is Definition 4.91, Proposition 4.92, and the non-compensation discipline. □

4.17 Explicit exclusions

The following are not asserted in Appendix D.

(i) No barcode matching is treated as the tower comparison morphism.

(ii) No comparison $\operatorname{colim}_n \operatorname{Eval}(F_n) \to \operatorname{Eval}(F_\infty)$ is automatic.

(iii) No colimit–collapse interchange is asserted in the general Core.

(iv) No global exactness, Serre localization, or strict global filtered collapse functor is used.

(v) No bounded-window diagnostic is computed from the dimension after zero extension to $+\infty$.

(vi) No globally finite but window-cofinal bar is excluded from tgdim_W.

(vii) No missing or ill-typed artifact is interpreted as $(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) = (0, 0)$.

(viii) No $(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) = (0, 0)$ implies full comparison-map isomorphism without a no-interior-defect theorem.

(ix) No pointwise family of unrelated zero diagnostics defines a stable band.

(x) No finite-stage pass determines an undeclared tower diagnostic.

(xi) No metric budget repairs a missing morphism, provenance record, terminal-tameness certificate, or manifest field.

(xii) No terminal Type IV zero is interpreted as long-transient absence.

(xiii) No long-transient `certified_absent` verdict excludes bars of length at most τ_{def}.

(xiv) No soundness-only, wrong-stage, or incomplete detector is used to certify defect absence.

(xv) No detector artifact is relabeled from a topological profile to a kernel or cokernel defect profile.

(xvi) No metric bound on source and apex profiles is promoted to a bound on their kernel or cokernel profiles without a morphism-level stability theorem.

(xvii) No finite observed plateau proves eventual constancy, a genuine colimit, apex agreement, or a terminal conclusion.

(xviii) No implementation source or finite proxy acquires stronger semantic authority without a registered theorem.

(xix) No diagnostics from distinct windows or thresholds are silently summed into one scalar total.

(xx) No local PH–Ext reverse bridge is introduced.

(xxi) No Search, AI, spectral, or advisory proxy replaces the actual artifact.

(xxii) No geometric, arithmetic, PDE, cryptographic, or categorical interpretation follows without a separately proved interface theorem.

4.18 Provenance and status

Appendix A fixes the repaired evaluation, actual-morphism requirement, and canonical μ/ν notation contract. Appendix B supplies representative tower and comparison-artifact obligations. Appendix C supplies only the eligible one-way categorical discharge. The present appendix gives the full window-relative terminal-generic, transient-defect, joint-classification, and exact finite-to-infinite calculus required by Chapter 5.

Status labels for Appendix D. *Foundation definitions:* Definitions 4.5, 4.6, 4.9, 4.11, 4.12, 4.13, 4.15, 4.17, 4.20, 4.23, 4.24, 4.26, 4.30, 4.33, 4.34, 4.35, 4.39, 4.40, 4.41, 4.46, 4.47, 4.65, 4.66, 4.67, 4.69, 4.78, 4.79, 4.84, 4.89, 4.91, 4.94, 4.97, 4.98.

Foundation propositions and theorems: Propositions 4.25, 4.27, 4.29, 4.38, 4.48, 4.70, 4.80, 4.81, 4.85, 4.90, 4.92, 4.95; Theorems 4.42, 4.99, and 4.100.

Operational cautions: Remarks 4.1, 4.2, 4.8, 4.10, 4.16, 4.18, 4.19, 4.21, 4.31, 4.43, 4.44, 4.45, 4.50, 4.71, 4.82, 4.83, 4.86, 4.93, 4.96, 4.75.

v20 transient-defect and reduction results: Definition 4.14, Definition 4.7, Definition 4.51, Proposition 4.52, Definitions 4.53, 4.54, 4.55, Theorem 4.56, Corollary 4.58, Definition 4.59, Theorem 4.60, Definition 4.61, Theorem 4.62, Definition 4.72, Definition 4.73, Theorem 4.74, Definition 4.87, and Proposition 4.88.

Explicit exclusions: Subsection 4.17.

5 Appendix E: Window, Overlap, Detector Atlases, κ-Restart, and Record-Level Gluing

5.1 Scope and dependence on the repaired Core

This appendix is the Foundation expansion of Chapter 6 in the AK-HPDST v20 conservative extension. It inherits the complete v19 window, overlap, κ-Restart, normalized summability, and record-atlas calculus at its original strength. It additionally consumes:

(a) the finite detector and profile-stage discipline of Appendix B;

(b) the higher eligible realization boundary of Appendix C;

(c) the terminal/transient defect separation of Appendix D.

Its purpose is to specify exactly when local certificate records may be compared, continued, assembled, and promoted to a global repaired-zero predicate without silently asserting an unsupported global-object theorem.

The central separations are:

(i) a right-open partition cover is used for sequential continuation and Restart;

(ii) an overlap-admitting cover is used for pairwise and higher compatibility;

(iii) the Overlap Gate compares post-threshold profiles, while threshold safety is certified on pre-threshold profiles;

(iv) pairwise overlap compatibility is not higher Čech coherence;

(v) Restart is licensed only by an explicit κ_k-admissible continuation artifact;

(vi) ordinary unweighted summability is not sufficient under contracting κ_k;

(vii) finite and countable gluing conclusions concern certificate records and atlases, not automatically the underlying mathematical objects;

(viii) local detector zero becomes global repaired zero only through a declared global profile interface and a retained-bar coverage theorem;

(ix) source-profile overlap data do not transport kernel or cokernel defect profiles without a separately proved defect comparison;

(x) exact simplex cocycles and metric-reference simplex records are distinct coherence modes and cannot be silently interchanged.

Every windowed persistence profile is evaluated in the fixed order

$$\mathbf{P}_i(F) \longmapsto \mathsf{W}_W\mathbf{P}_i(F) \longmapsto T^{\mathrm{bar}}_\tau(\mathsf{W}_W\mathbf{P}_i(F)) = \mathrm{Eval}_{i,W,\tau}(F).$$

Remark 5.1 (Appendix E does not enlarge the Core)**.** This appendix does not create a universal Restart constant, an automatic continuation map, an unrestricted local-to-global theorem, a global persistence profile from pairwise data, or an object-level descent theorem. It develops a typed record architecture whose non-scalar obligations cannot be repaired by enlarging a numerical budget. Search-side maps, AI-generated proposals, and advisory quantities remain non-verdict unless they re-enter through independently verified Core records.

5.2 Right-open windows, partitions, and overlap covers

Definition 5.2 (Right-open window)**.** A *right-open Core window* is an interval

$$W = [u, v), \qquad -\infty < u < v \le +\infty.$$

It is bounded when $v < +\infty$ and right-unbounded when $v = +\infty$.

Definition 5.3 (Declared window range)**.** A *declared window range* is an interval $\Omega \subseteq \mathbb{R}$ together with the cover type, index set, endpoint convention, and audit scope on which local certification is performed.

Remark 5.4 (Why right-open windows)**.** If

$$W_k = [u_k, u_{k+1}), \qquad W_{k+1} = [u_{k+1}, u_{k+2}),$$

then $W_k \cap W_{k+1} = \varnothing$. Hence a partition does not double-count boundary parameters. This convention does not itself provide a continuation law across the boundary.

Definition 5.5 (Window restriction)**.** For $M \in \mathsf{Pers}_k^{\mathrm{cons}}$, the window restriction $\mathsf{W}_W M$ is the restriction of M to W, extended by zero outside W only when a globally indexed profile is required. At barcode level, each interval is clipped to W and empty intersections are discarded.

Definition 5.6 (Core window-evaluation order)**.** For a filtered object F, monitored degree i, window W, and threshold $\tau > 0$, define

$$B_{i,W}(F) := \mathsf{W}_W \mathbf{P}_i(F), \qquad \mathrm{Eval}_{i,W,\tau}(F) := T_\tau^{\mathrm{bar}}\big(B_{i,W}(F)\big).$$

The pre-threshold profile $B_{i,W}(F)$ is the object on which clearance is measured.

Remark 5.7 (No unconditional crop–collapse commutation)**.** In general,

$$\mathsf{W}_W T_\tau^{\mathrm{bar}} \ncong T_\tau^{\mathrm{bar}} \mathsf{W}_W.$$

A crop-after-collapse substitution is permitted only under the exact all-retained-bars criterion of Appendix A, namely Definition 1.60 and Proposition 1.61, including globally retained infinite bars clipped by bounded windows.

Definition 5.8 (MECE windowing / right-open partition cover)**.** A family

$$\mathcal{W}_{\mathrm{MECE}} = \{W_k = [u_k, u_{k+1})\}_{k \in K}$$

is a *right-open partition cover* of Ω if the windows are pairwise disjoint, their union is Ω, and the index order is part of the data whenever sequential continuation is invoked. The term *MECE windowing* is retained as an operational alias.

Definition 5.9 (MECE certificate sequence)**.** A *MECE certificate sequence* is an ordered family $\{C_k\}_{k \in K}$ of complete local certificate records indexed by a right-open partition cover. Ordering alone does not certify continuation.

Remark 5.10 (MECE windowing is not overlap gluing)**.** A disjoint partition has no nonempty overlap on which an Overlap Gate can be evaluated. Conversely, an overlap cover does not by itself supply an ordered continuation path. The two cover types have different proof obligations.

Definition 5.11 (Overlap-admitting cover)**.** A family $\mathcal{U} = \{U_\alpha\}_{\alpha\in A}$ of right-open windows is an *overlap-admitting cover* of Ω if

$$\Omega = \bigcup_{\alpha\in A} U_\alpha$$

and every nonempty finite intersection used by the certificate logic is explicitly indexed and recorded.

Definition 5.12 (Overlap stratum)**.** For a finite index set $\sigma = \{\alpha_0, \dots, \alpha_p\}$, define

$$U_\sigma := U_{\alpha_0} \cap \cdots \cap U_{\alpha_p}.$$

A stratum participates only when it is nonempty and included in the declared nerve record.

Definition 5.13 (Declared overlap structure)**.** A *declared overlap structure* records the cover, every nonempty nerve simplex used by the run, its face relations, common thresholds, monitored degrees, metric ledgers and scalarizations, representative and realization provenance, and typed Type IV scope information.

Remark 5.14 (Why overlap data must be declared)**.** Without an indexed nerve and immutable overlap artifacts, an auditor cannot determine which local records, thresholds, face restrictions, budgets, or Type IV artifacts were compared. Mere set-theoretic overlap is not a certificate.

5.3 Scalarized reserves and non-scalar obligations

Definition 5.15 (Bottleneck scalarization for Appendix E)**.** Let $(Q, \oplus, \preceq, 0_Q)$ be the declared commutative ledger quantale. A bottleneck scalarization is a monotone subadditive map

$$\mathrm{val}_{\mathrm{B}} : Q \longrightarrow [0, +\infty], \qquad \mathrm{val}_{\mathrm{B}}(0_Q) = 0, \qquad \mathrm{val}_{\mathrm{B}}(q \oplus q') \le \mathrm{val}_{\mathrm{B}}(q) + \mathrm{val}_{\mathrm{B}}(q').$$

Only after this scalarization may a ledger value be compared with a real threshold clearance.

Definition 5.16 (Gate-scoped threshold reserve)**.** Let $\mathfrak{B} = \{B_j\}_{j\in J}$ be the finite pre-threshold profile family consumed by one local, overlap, simplex, or Restart gate, and let ε be its certified scalarized bottleneck budget. Set

$$\mathrm{gap}_\tau(\mathfrak{B}) := \min_{j\in J} \mathrm{clear}_\tau(B_j), \qquad \mathrm{res}_\tau(\mathfrak{B}; \varepsilon) := \mathrm{gap}_\tau(\mathfrak{B}) - \varepsilon.$$

This reserve belongs only to the declared gate and profile family.

Definition 5.17 (Non-scalar continuation and coherence obligations)**.** Representative existence, actual comparison or continuation morphisms, threshold reconciliation, κ-admissibility, higher-face coherence, terminal tameness, Type IV typing, claim status, manifest completeness, and artifact readability are non-scalar obligations. No increased budget, larger gap, or summability estimate can replace them.

Remark 5.18 (Obligation handoff to Appendices F and G)**.** The statuses and dependency closure of the non-scalar obligations in Definition 5.17 are forwarded to the typed status calculus of Appendix F. Their immutable identifiers, evidence references, scope fields, and failure routes are forwarded to the Manifest schema of Appendix G. This handoff records obligations; it does not convert missing evidence into a pass.

5.4 Certificate records

Definition 5.19 (Window-restricted certificate record)**.** A *window-restricted certificate record* on $(F; W, \tau)$ records at least:

(i) immutable object, window, threshold, and monitored-degree identifiers;

(ii) the pre-threshold profiles $B_{i,W}(F)$ and post-threshold profiles $\mathrm{Eval}_{i,W,\tau}(F)$;

(iii) clearance, gap, metric-ledger provenance, scalarization, and positive-reserve verdicts;

(iv) the local B-Gate$^+$, representative, eligibility, and typed Type IV statuses;

(v) every invoked representative, realization, tower, comparison, crop, or continuation artifact;

(vi) a local status `pass`, `reject`, or `undefined`;

(vii) artifact references sufficient for independent reconstruction.

Definition 5.20 (Globally coherent certificate record)**.** A family of local records forms a *globally coherent certificate record system* only if every required local, pairwise, higher-simplex, Restart, budget, Type IV, and manifest clause is present and typed. The phrase does not denote a single global B-Gate$^+$ pass unless an independent global object and global gate record are supplied.

Remark 5.21 (Certificate record versus proof object)**.** A certificate record is the audit-readable representation of declared proof data. Schema completeness does not by itself prove the mathematical clauses stored in the record, and record gluing does not construct a new mathematical object.

5.5 Direct common-window comparison and the sharp Overlap Gate

Definition 5.22 (Overlap budget)**.** For a nonempty recorded simplex σ, let $\mathfrak{d}_\sigma \in Q$ be the gate-scoped metric ledger and define its scalarized budget

$$\varepsilon_\sigma := \mathrm{val}_{\mathrm{B}}(\mathfrak{d}_\sigma) \in [0, +\infty].$$

For a pair $\sigma = \{\alpha, \beta\}$, this is the overlap budget formerly denoted $\Sigma\delta_{\alpha\beta}$.

Definition 5.23 (Preferred repaired overlap profile)**.** For $U_{\alpha\beta} := U_\alpha \cap U_\beta \neq \varnothing$, define the direct common-window profiles

$$B^{\alpha}_{i,\alpha\beta} := \mathsf{W}_{U_{\alpha\beta}}\mathbf{P}_i(F_\alpha), \qquad B^{\beta}_{i,\alpha\beta} := \mathsf{W}_{U_{\alpha\beta}}\mathbf{P}_i(F_\beta).$$

After declaring a common overlap threshold $\tau_{\alpha\beta}$, set

$$O_{\alpha\beta,i}(F_\alpha) := T^{\mathrm{bar}}_{\tau_{\alpha\beta}}(B^{\alpha}_{i,\alpha\beta}), \quad O_{\beta\alpha,i}(F_\beta) := T^{\mathrm{bar}}_{\tau_{\alpha\beta}}(B^{\beta}_{i,\alpha\beta}).$$

Definition 5.24 (Overlap comparison data)**.** The overlap data consist of the two direct pre-threshold profiles, the common threshold and threshold- reconciliation record, a pre-threshold bottleneck certificate, two-sided clearance certificates, post-threshold comparison data, representative provenance when invoked, typed Type IV status, and all artifact references.

Definition 5.25 (Threshold reconciliation record)**.** If local records use thresholds τ_α and τ_β, an Overlap Gate is well typed only after a common threshold $\tau_{\alpha\beta}$ is declared and both local objects are directly re-evaluated on $U_{\alpha\beta}$ at that threshold. The data justifying this choice form the threshold-reconciliation record.

Remark 5.26 (Preferred overlap convention)**.** The default comparison is performed directly on $U_{\alpha\beta}$. Cropping profiles already thresholded on U_α or U_β is forbidden unless the exact criterion of Definition 1.60 and Proposition 1.61 is certified.

Definition 5.27 (Two-sided overlap metric certificate)**.** For each monitored degree i, a two-sided overlap metric certificate at budget $\varepsilon_{\alpha\beta,i}$ consists of

$$d_B\left(B^{\alpha}_{i,\alpha\beta}, B^{\beta}_{i,\alpha\beta}\right) \le \varepsilon_{\alpha\beta,i},$$

$$\mathrm{clear}_{\tau_{\alpha\beta}}\left(B^{\alpha}_{i,\alpha\beta}\right) > \varepsilon_{\alpha\beta,i}, \qquad \mathrm{clear}_{\tau_{\alpha\beta}}\left(B^{\beta}_{i,\alpha\beta}\right) > \varepsilon_{\alpha\beta,i},$$

together with the ledger provenance and direct common-window records.

Proposition 5.28 (Sharp overlap transfer)**.** *Under a two-sided overlap metric certificate,*

$$d_B\big(O_{\alpha\beta,i}(F_\alpha), O_{\beta\alpha,i}(F_\beta)\big) \le \varepsilon_{\alpha\beta,i}.$$

In the declared one-parameter constructible regime, the same bound may be read using interleaving distance.

Proof. The two clearance inequalities exclude finite bars from the closed band $[\tau_{\alpha\beta} - \varepsilon_{\alpha\beta,i}, \tau_{\alpha\beta} + \varepsilon_{\alpha\beta,i}]$ on both sides. Apply Appendix A, Theorem 1.33, to the certified pre-threshold bottleneck bound, then use the bottleneck–interleaving isometry in the declared regime. □

Definition 5.29 (Overlap Gate)**.** The pairwise Overlap Gate on $U_{\alpha\beta}$ passes exactly when, for every monitored degree:

(i) a two-sided overlap metric certificate exists;

(ii) the post-threshold discrepancy bound of Proposition 5.28 holds;

(iii) representative and eligible-realization provenance used on the overlap is compatible;

(iv) if Type IV is in scope, its status is `certified_zero`;

(v) `undefined` is never interpreted as zero and `obstructed` is a failure;

(vi) all required crop, threshold-reconciliation, comparison, and artifact records are present.

Remark 5.30 (Overlap Gate is a compatibility test)**.** The discrepancy verdict is post-threshold, but the safety certificate is pre-threshold. Small post-threshold distance alone does not prove stable threshold classification. Passing a pairwise gate also does not create a local pass or higher coherence.

Proposition 5.31 (Overlap Gate preserves pairwise audit readability)**.** *If the pairwise Overlap Gate passes and all of its records are immutable and referenced, then the pairwise compatibility claim is independently reconstructible. No higher-simplex conclusion follows from this proposition.*

Proof. The gate records both pre-threshold inputs, the scalarized metric bound, the two clearance inequalities, the post-threshold verdict, typed statuses, and all required provenance. These data reconstruct precisely the pairwise claim and nothing stronger. □

5.6 Finite Čech depth and simplex coherence

Definition 5.32 (Čech nerve)**.** The Čech nerve $N(\mathcal{U})$ has a p-simplex for every nonempty intersection of $p + 1$ distinct cover windows.

Definition 5.33 (Finite Čech depth)**.** The cover has *Čech depth at most* K if every intersection of $K + 2$ distinct windows is empty. Equivalently, $\dim N(\mathcal{U}) \leq K$. Thus nonempty records occur on at most $(K + 1)$-fold intersections.

Definition 5.34 (Simplex compatibility record)**.** For a nonempty simplex σ, a simplex compatibility record contains direct pre-threshold profiles on U_σ, a common threshold, all required face comparison data, sharp two-sided gap certificates, face-compatibility statements, typed Type IV statuses, scalarized budgets, and artifact references.

Proposition 5.35 (Finite overlap reduction)**.** *If the cover has Čech depth at most K, no compatibility record is required above simplicial degree K.*

Proof. There are no nonempty intersections indexed by higher-dimensional simplices. This bounds the order of coherence conditions, but not the number of simplices when the cover is infinite. □

Remark 5.36 (Why finite depth is a Core condition)**.** Finite depth bounds overlap order only. It does not imply local finiteness, summability, face compatibility, or finiteness of the record set. Those are separate obligations.

Definition 5.37 (Compatible local certificate family)**.** A family $\{C_\alpha\}$ is *Čech-compatible* if every local record passes, every nonempty nerve simplex carries a complete simplex compatibility record, all face restrictions agree under the declared equivalence policy, every scalarized simplex reserve is positive, and no required status is undefined.

Definition 5.38 (Čech-coherent certificate atlas)**.** A *Čech-coherent certificate atlas* is a Čech-compatible family indexed by a declared nerve, with immutable local and simplex records and mutually compatible face restrictions.

Theorem 5.39 (Finite-cover record-level local-to-global principle)**.** *Let $\mathcal{U}$ be a finite overlap-admitting cover of finite Čech depth. Assume:*

(i) every window carries a passing local certificate record;

(ii) every nonempty nerve simplex carries a simplex compatibility record;

(iii) every Type IV scope has status `not_invoked` or `certified_zero`, never `undefined`;

(iv) every scalarized simplex budget has positive sharp threshold reserve;

(v) all face restrictions and artifact references are audit-readable.

Then the family is a Čech-coherent certificate atlas on the covered range.

Proof. The local records provide the vertices. The simplex records supply every nonempty overlap through the maximal nerve degree, and the face conditions establish coherence under restriction. Finiteness of the cover and finite depth make the required registry finite. Positive reserves and typed Type IV statuses prevent unstable or undefined comparisons from being reported as coherent. □

Remark 5.40 (No unrestricted object-level gluing)**.** Theorem 5.39 glues records into an atlas. It does not glue filtered complexes, persistence modules, derived objects, sheaves, or other mathematical objects.

5.7 κ-admissible Restart

Definition 5.41 (Sequential continuation step)**.** For adjacent partition windows W_k and W_{k+1}, a continuation step is a declared artifact linking the source and target local records. Adjacency itself supplies no inequality and no morphism.

Definition 5.42 (Per-window budget)**.** Let $\mathfrak{B}_k$ be the finite pre-threshold profile family used by the local gate. Choose a finite working gap

$$0 < g_k \le \mathrm{gap}_{\tau_k}(\mathfrak{B}_k),$$

using a finite audit cap when the exact clearance is $+\infty$. Let $\varepsilon_k < +\infty$ be the gate-scoped scalarized local budget. These quantities must have immutable provenance.

Definition 5.43 (Restart margin)**.** The finite local reserve is

$$r_k := g_k - \varepsilon_k \in \mathbb{R}.$$

The local record is reserve-positive exactly when $r_k > 0$. A finite working gap avoids undefined extended-real arithmetic in the Restart recursion.

Definition 5.44 (Continuation artifact)**.** A continuation artifact from W_k to W_{k+1} records source and target identifiers and thresholds, all endpoint-state, initialization, representative, and comparison data used by the transition, a finite continuation loss $\eta_k \ge 0$ not already counted in the target local budget, a degradation factor $\kappa_k \in (0, 1]$, and independently checkable evidence for the reserve inequality below. Missing non-scalar obligations may not be encoded numerically inside η_k.

Definition 5.45 (κ_k-admissible continuation)**.** The artifact is *κ_k-admissible* if it certifies

$$r_{k+1} \ge \kappa_k r_k - \eta_k.$$

Neither κ_k nor the inequality is inferred from adjacency, threshold deletion, or a generic "admissible continuation class."

Definition 5.46 (Restart transition status)**.** Each transition has exactly one status:

`not_invoked`, `undefined`, `certified`, `failed`.

The status is `certified` only when the artifact is complete and the declared κ_k-inequality is verified. A certified transition is necessary but not sufficient for a complete Restart pass: the target local record must also pass and have positive reserve.

Lemma 5.47 (Finite-horizon Restart bound)**.** *Assume the transitions* $0 \to 1, \dots, n-1 \to n$ *are respectively* $\kappa_0, \dots, \kappa_{n-1}$*-admissible. Then*

$$r_n \ge \left(\prod_{j=0}^{n-1} \kappa_j\right) r_0 - \sum_{m=0}^{n-1} \left(\prod_{j=m+1}^{n-1} \kappa_j\right) \eta_m,$$

where an empty product is 1.

Proof. For $n = 1$ this is the certified continuation inequality. If the formula holds at n, substitute it into $r_{n+1} \ge \kappa_n r_n - \eta_n$, distribute κ_n, and collect the product-weighted loss terms. □

Remark 5.48 (Restart is conditional)**.** There is no universal $\kappa > 0$, no default $\kappa = 1$, and no unconditional Restart lemma. Window restriction, threshold changes, and unrelated target profiles can destroy inherited reserve. A missing continuation artifact gives status `undefined`, not harmless continuation.

Definition 5.49 (Normalized Restart loss)**.** Set

$$K_0 := 1, \qquad K_n := \prod_{j=0}^{n-1} \kappa_j \quad (n \geq 1),$$

and

$$L_n^{\mathrm{norm}} := \sum_{m=0}^{n-1} \frac{\eta_m}{K_{m+1}}.$$

Then Lemma 5.47 implies

$$\frac{r_n}{K_n} \geq r_0 - L_n^{\mathrm{norm}}.$$

5.8 Countable Restart and normalized summability

Definition 5.50 (Summability of window budgets)**.** A certified countable Restart run is *κ-normalized summable* if

$$\sum_{k=0}^{\infty} \frac{\eta_k}{K_{k+1}} < \infty.$$

It is *reserve-safe* if, in addition,

$$\sum_{k=0}^{\infty} \frac{\eta_k}{K_{k+1}} < r_0.$$

Ordinary summability $\sum_k \eta_k < \infty$ is not an equivalent condition when K_k contracts.

Definition 5.51 (Summable restart sequence)**.** A MECE certificate sequence is a *reserve-safe Restart sequence* if the initial reserve is positive, every invoked transition is `certified`, the normalized loss is reserve-safe, every target has a complete passing local record including all invoked representative, eligibility, and Type IV clauses, and no transition or target is undefined.

Theorem 5.52 (Conditional countable Restart–Summability principle)**.** *For a reserve-safe Restart sequence, every finite-stage reserve r_n is positive and the ordered family forms an auditable sequential certificate record on the partition cover. More precisely, if*

$$L_\infty^{\mathrm{norm}} := \sum_{k=0}^{\infty} \frac{\eta_k}{K_{k+1}}, \qquad c := r_0 - L_\infty^{\mathrm{norm}} > 0,$$

then

$$r_n \geq K_n c > 0$$

for every finite n. A uniform lower bound follows only under the additional condition $\inf_n K_n > 0$.

Proof. For each finite n, Definition 5.49 gives

$$\frac{r_n}{K_n} \geq r_0 - L_n^{\mathrm{norm}} \geq r_0 - L_\infty^{\mathrm{norm}} = c.$$

Since $K_n > 0$, the pointwise bound follows. The continuation artifacts and complete target records make the record chain reconstructible. Taking infima yields a positive uniform reserve only when $\inf_n K_n > 0$. □

Remark 5.53 (Why ordinary summability is not enough)**.** Even when $\sum_k \eta_k < \infty$, the normalized series can diverge if $K_{k+1} \to 0$. Conversely, pointwise positivity $r_n > 0$ at every finite stage does not imply a uniform positive reserve. The theorem certifies record inheritance, not convergence to a global object.

5.9 Countable overlap atlases

Definition 5.54 (Definable window)**.** A window is *definable* if it is definable in the declared ambient structure. Definability may support reconstructibility but does not replace local finiteness, simplex records, or coherence.

Definition 5.55 (Countable definable cover)**.** A *countable definable overlap cover* is a countable overlap-admitting cover whose windows are definable, whose Čech depth is finite, and whose complete recorded nerve is included in the audit scope. To support a countable atlas theorem it must additionally be audit-locally finite.

Definition 5.56 (Audit-locally finite cover)**.** A countable overlap cover is *audit-locally finite* if every bounded subinterval meets only finitely many windows and only finitely many recorded nerve simplices.

Definition 5.57 (Summable simplex budget)**.** Let $\varepsilon_\sigma \geq 0$ be the scalarized metric budget of each recorded nonempty simplex. Define the unordered nonnegative sum

$$\sum_{\sigma \in N_{\mathrm{rec}}} \varepsilon_\sigma := \sup_{S \subseteq N_{\mathrm{rec}},\, S \text{ finite}} \sum_{\sigma \in S} \varepsilon_\sigma .$$

The simplex budget is summable when this quantity is finite. This controls total declared exposure but does not create missing face maps or replace the simplexwise inequalities $\varepsilon_\sigma < \mathrm{gap}_{\tau_\sigma}(\mathfrak{B}_\sigma)$.

Theorem 5.58 (Countable definable-cover gluing)**.** *Let $\{U_n\}_{n \geq 1}$ be a countable, definable, audit-locally finite overlap cover of finite Čech depth. Assume:*

(i) every local window has a complete passing certificate record;

(ii) every nonempty recorded nerve simplex has a complete simplex compatibility record;

(iii) every simplex has positive sharp threshold reserve;

(iv) the scalarized simplex budgets are summable;

(v) all invoked Type IV scopes are defined and correctly typed;

(vi) all face restrictions and artifact references are audit-readable.

Then the family forms an audit-locally finite Čech-coherent certificate atlas on the covered range and has finite declared scalarized simplex exposure.

Proof. Audit-local finiteness makes every bounded audit query finite. Finite Čech depth bounds overlap order. The simplex records and face conditions establish coherence, positive reserves license each thresholded comparison, and the unordered sum records finite total scalarized exposure. Hence the atlas is locally reconstructible and globally indexed as a coherent record system. □

Remark 5.59 (Definability is not decorative, but is not sufficient)**.** Definability constrains the description language. It does not imply audit-local finiteness, summability, positive reserve, or higher coherence. All of those remain explicit theorem hypotheses.

5.10 Budget aggregation across windows and simplices

Definition 5.60 (Global window budget)**.** For a declared sequence of scalarized local or continuation exposures $\varepsilon_k \geq 0$, define the global window exposure by the unordered or ordered nonnegative sum specified by the run. This quantity is not a quantale element unless the manifest separately records the corresponding $\oplus$-aggregation.

Definition 5.61 (Global overlap budget)**.** For a finite or countable recorded nerve, the global overlap exposure is the scalarized simplex sum of Definition 5.57. It includes every declared simplex, not merely pairwise overlaps.

Definition 5.62 (Total gluing budget)**.** The *total gluing exposure* is a manifest-declared aggregation of window, continuation, crop, and simplex metric components after preserving their provenance. When represented by a quantale element it is aggregated using $\oplus$; when compared with real clearance it is first scalarized by $\mathrm{val}_{\mathrm{B}}$.

Proposition 5.63 (Budget auditability under summability)**.** *Finite scalarized window exposure and a summable simplex budget make the declared total metric exposure finite and audit-readable. They do not establish any local or global gate pass without the corresponding profile-scoped strict reserve inequalities and non-scalar obligations.*

Proof. The nonnegative sums are finite by hypothesis, while the provenance-preserving aggregation and scalarization make the value reconstructible. Gate validity remains separate because a finite total may still exceed a local clearance or coexist with a missing artifact. □

Remark 5.64 (Relation to gap_τ)**.** There is no single untyped global inequality $\Sigma\delta_{\mathrm{glue}} < \mathrm{gap}_\tau$ unless the run declares a global profile family, threshold, aggregation law, and scalarization. The canonical requirements are the simplexwise and gatewise strict reserve inequalities actually consumed by the proof. A separately claimed uniform reserve requires an independent positive infimum certificate.

5.11 Compatibility with typed Type IV diagnostics

Definition 5.65 (Overlap Type IV cleanliness)**.** A local, overlap, simplex, or continuation-target record is *Type IV-clean* when Type IV is in scope and its status is `certified_zero` relative to the declared comparison artifact and terminal-generic data. When Type IV is explicitly outside scope, `not_invoked` is permitted. Status `undefined` is never clean.

Proposition 5.66 (Overlap cleanliness is required for gluing)**.** *No simplex may participate in a successful certificate atlas with Type IV status `obstructed` or `undefined`. If Type IV is in scope it must be `certified_zero`; otherwise the scope policy must justify `not_invoked`.*

Proof. This is the typed Type IV participation rule of Appendix D, Theorem 4.42, applied to every vertex, simplex, and continuation target. Compatibility cannot repair a rejected or undefined local clause. □

Remark 5.67 (Type IV cleanliness is terminal-generic)**.** On a bounded window, terminal-generic diagnostics count bars cofinal to the right endpoint, even though such bars may be globally finite. Therefore Type IV cleanliness is not accurately described as merely "absence of infinite-bar defect." It also does not exclude interior noncofinal defects or prove full isomorphism.

5.12 Object-level gluing as a non-Core strengthening

Definition 5.68 (Object-level gluing claim)**.** An *object-level gluing claim* asserts that local filtered complexes, persistence modules, derived objects, sheaves, or categorical objects descend to a global object in a declared category.

Proposition 5.69 (Certificate gluing does not imply object gluing)**.** *A Čech-coherent certificate atlas does not, by itself, imply an object-level gluing claim.*

Proof. The atlas stores verdicts, profiles, budgets, statuses, face records, and artifact references. Object-level descent additionally requires morphisms or equivalences satisfying the descent axioms of a declared category, stack, or higher stack. Those hypotheses are not consequences of record coherence. □

Remark 5.70 (Where object-level gluing belongs)**.** Object-level descent belongs to a separately imported technical extension and must be reproved under the Core guard rails. Even then, descent does not replace the audit manifest.

5.13 Window, overlap, Restart, and atlas package

Theorem 5.71 (Appendix E conditional closure package)**.** *Within the repaired Core, the following package is available.*

(i) Overlap profiles are evaluated directly on the common window before threshold deletion.

(ii) Metric ledger data are compared with clearance only after bottleneck scalarization.

(iii) A sharp two-sided overlap certificate transfers a pre-threshold bottleneck bound to the post-threshold evaluations.

(iv) Pairwise Overlap Gates do not imply higher Čech coherence.

(v) A finite-depth finite cover yields a coherent certificate atlas only when every simplex record and face condition is supplied.

(vi) Restart is conditional on an explicit κ_k-admissible continuation artifact.

(vii) Finite-horizon reserves satisfy the product-weighted inequality of Lemma 5.47.

(viii) Countable Restart requires normalized reserve-safety; ordinary summability alone is insufficient.

(ix) Pointwise positive reserves do not imply a uniform positive lower bound without control of $\inf_n K_n$.

(x) A countable atlas additionally requires audit-local finiteness, finite Čech depth, simplex coherence, positive simplex reserves, and summable scalarized exposure.

(xi) Undefined artifacts or statuses are never promoted to successful overlap, Restart, or gluing verdicts.

(xii) Record-level gluing does not construct a global mathematical object or a single global B-Gate$^+$ pass.

Proof. Items (i)–(iii) are Definitions 5.23–5.29 and Proposition 5.28. Items (iv)–(v) are Definitions 5.34–5.38 and Theorem 5.39. Items (vi)–(ix) are Definitions 5.44–5.51, Lemma 5.47, and Theorem 5.52. Item (x) is Theorem 5.58. Item (xi) follows from the typed status rules, and item (xii) is Proposition 5.69. □

5.14 Explicit exclusions

The following are not asserted in Appendix E.

(i) No unconditional Restart lemma or automatically existing $\kappa > 0$ is asserted.

(ii) No missing continuation artifact is interpreted as $\kappa = 1$, zero loss, or a certified transition.

(iii) No ordinary unweighted summability is sufficient under arbitrary contracting κ_k.

(iv) No pointwise reserve statement is promoted to a uniform positive lower bound without an additional product lower bound.

(v) No pairwise Overlap Gate implies triple or higher coherence.

(vi) No differently thresholded outputs are compared without threshold reconciliation and direct common-threshold re-evaluation.

(vii) No crop-after-collapse profile replaces direct common-window evaluation without a commutation certificate.

(viii) No unscalarized quantale element is compared directly with real threshold clearance.

(ix) No budget, gap, or summability estimate repairs a missing representative, morphism, coherence record, Type IV certificate, claim status, or manifest field.

(x) No finite Čech depth alone implies a finite record set or audit-local finiteness.

(xi) No definability alone implies coherence, summability, or local finiteness.

(xii) No finite prefix is treated as a complete countable audit certificate.

(xiii) No summable atlas is automatically a single global B-Gate$^+$ pass.

(xiv) No certificate atlas is automatically a global filtered object, persistence module, sheaf, or derived object.

(xv) No undefined local, overlap, Restart, representative, or Type IV status is treated as zero or pass.

(xvi) No Search-side map, score, or AI proposal supplies a missing overlap, continuation, or gluing artifact.

(xvii) No global exactness of positive-threshold deletion or global strict collapse functor is assumed.

(xviii) No local detector zero family implies global repaired zero without a global profile interface and retained-bar coverage.

(xix) No pairwise local compatibility constructs the global profile used by a detector-atlas theorem.

(xx) No metric-reference simplex record creates transition morphisms, cocycles, or object descent.

(xxi) No κ-Restart reserve inequality transports a repaired-zero predicate without a separately certified zero route.

(xxii) No terminal Type IV zero implies long-transient absence.

(xxiii) No source-profile overlap certificate transports kernel or cokernel defect profiles.

(xxiv) No boundary-crossing retained bar is ruled out by disjoint local detector zero unless boundary-bridge or another global coverage theorem is supplied.

(xxv) No post hoc reading of a complete global barcode is advertised as nontrivial finite proof compression.

5.15 Provenance and status

The sharp overlap-transfer theorem is inherited from Appendix A, Theorem 1.33. Finite detectors and stage discipline are inherited from Appendix B. Representative and eligible-realization clauses are inherited from Appendices B–C. Typed terminal Type IV and long-transient participation are inherited from Appendix D, especially Theorem 4.42 and Definition 4.59. The non-scalar obligation states are forwarded to Appendix F and their Manifest serialization to Appendix G.

The inherited v19 contribution is the record architecture joining sharp overlap comparison, simplex coherence, conditional κ-Restart, product-weighted reserve bounds, normalized countable summability, and audit-locally finite atlases. The v20 conservative extension adds exact/metric simplex modes, detector-aware global retained-bar coverage, detector-safe Restart, explicit normalized counterexamples, transient-clean atlas participation, and the strict boundary between predicate globalization and object descent.

Status labels for Appendix E. *Foundation definitions:* Definitions 5.2, 5.3, 5.5, 5.6, 5.8, 5.9, 5.11, 5.12, 5.13, 5.19, 5.20, 5.22, 5.23, 5.24, 5.29, 5.32, 5.33, 5.37, 5.41, 5.42, 5.43, 5.50, 5.51, 5.54, 5.55, 5.60, 5.61, 5.62, 5.65, 5.68.

Foundation lemmas/propositions/theorems: Propositions 5.31, 5.35, Theorem 5.39, Lemma 5.47, Theorems 5.52, 5.58, Propositions 5.63, 5.66, 5.69, and Theorem 5.71.

Operational cautions: Remarks 5.1, 5.4, 5.7, 5.10, 5.14, 5.21, 5.26, 5.40, 5.48, 5.53, 5.59, 5.67, 5.70, and 5.18.

v20 detector-atlas and coherence extension: Definitions 5.73, 5.74, 5.75, 5.77, 5.78, 5.79, 5.80, 5.83, 5.84, 5.85, 5.93, 5.94, 5.97, 5.100, 5.101; Propositions 5.81, 5.102; Theorems 5.86, 5.98, 5.104; Corollary 5.95; Examples 5.88, 5.91, 5.92.

Explicit exclusions: Subsection 5.14.

5.16 v20 conservative extension overview

Remark 5.72 (Immutable inherited locator discipline)**.** All preceding numbered definitions, propositions, lemmas, and theorems retain their inherited v19 statements and order. The v20 material below is a conservative extension with new labels. It does not silently change the source strength of an inherited Claim Register entry.

5.17 v20 exact and metric-reference coherence modes

Definition 5.73 (v20 overlap comparison mode)**.** Every overlap inference declares exactly one runtime comparison mode:

`exact`, `metric_gap_certified`, `not_compared`.

The mode `exact` consumes an actual equality or persistence-profile isomorphism on the directly evaluated common window. The mode `metric_gap_certified` consumes a pre-threshold bottleneck bound and the two-sided threshold-gap theorem at the same budget. The mode `not_compared` is permitted only when the governing claim consumes no overlap inference. An exact comparison does not acquire a positive robustness radius merely because it is exact.

Definition 5.74 (Exact overlap certificate)**.** An exact overlap certificate on $(i, U_{\alpha\beta}, \tau_{\alpha\beta})$ contains:

(i) the direct common-window pre-threshold profiles;

(ii) the common threshold and reconciliation record;

(iii) the two repaired profiles;

(iv) an actual equality or profile isomorphism

$$g_{\alpha\beta,i} : O_{\alpha\beta,i}(F_\alpha) \xrightarrow{\sim} O_{\beta\alpha,i}(F_\beta);$$

(v) immutable source, target, proof, and artifact identities.

No threshold-gap reserve is inferred from this record unless a separate metric certificate is supplied.

Definition 5.75 (v20 pairwise Overlap Gate record)**.** A v20 pairwise Overlap Gate record has exactly one status

$$\texttt{not\_invoked}, \qquad \texttt{undefined}, \qquad \texttt{pass}, \qquad \texttt{reject}.$$

Its status is determined as follows.

(i) `not_invoked`: the governing claim consumes no overlap inference, the comparison mode is `not_compared`, and an explicit scope rationale is recorded;

(ii) `undefined`: overlap inference is invoked but the comparison mode, threshold reconciliation, proof, provenance, Type IV, transient, crop, or immutable artifact data are missing or ill-typed;

(iii) `pass`: the comparison mode is exactly one of `exact` or `metric_gap_certified`, its complete certificate is valid in every monitored degree, all consumed representative and realization provenance is compatible, terminal Type IV is `not_invoked` or `certified_zero`, every required long-transient verdict is `certified_absent`, and no required clause is false or undefined;

(iv) `reject`: the record is well-typed and at least one required exact equality, metric bound, threshold-gap inequality, Type IV, transient, or other mathematical participation condition is evidenced false.

The mode `not_compared` never supports `pass`.

Remark 5.76 (Source overlap does not transport defect profiles)**.** An exact or metric comparison between local repaired source profiles does not by itself compare

$$\ker\phi \qquad \text{or} \qquad \operatorname{coker}\phi$$

for tower artifacts attached to those windows. Terminal or transient defect transport requires actual defect-profile comparison data and a separately proved kernel/cokernel stability theorem.

Definition 5.77 (Simplex coherence mode)**.** Every nonempty nerve simplex used by a coherence claim declares exactly one mode:

$$\texttt{exact\_cocycle}, \qquad \texttt{metric\_reference}, \qquad \texttt{not\_compared}.$$

The first mode consumes actual isomorphisms and cocycle identities. The second consumes a fixed reference profile and certified metric comparisons to that reference. It supplies bounded record-level discrepancy but no transition morphisms or categorical cocycle.

Definition 5.78 (Exact simplex cocycle record)**.** Let $\sigma = \{\alpha_0, \dots, \alpha_p\}$ index a nonempty simplex. An exact simplex cocycle record contains actual repaired-profile isomorphisms

$$g^\sigma_{\alpha\beta} : \mathrm{Eval}^\alpha_{i,\sigma} \xrightarrow{\sim} \mathrm{Eval}^\beta_{i,\sigma}$$

with

$$g^\sigma_{\alpha\alpha} = \mathrm{id}, \qquad g^\sigma_{\beta\alpha} = (g^\sigma_{\alpha\beta})^{-1}, \qquad g^\sigma_{\beta\gamma} \circ g^\sigma_{\alpha\beta} = g^\sigma_{\alpha\gamma}.$$

Restriction to every face agrees with the previously registered face cocycle.

Definition 5.79 (Metric-reference simplex record)**.** For a nonempty simplex σ, a metric-reference record contains:

(i) one pre-threshold reference profile $R_{i,\sigma}$ on U_σ;

(ii) one common threshold τ_σ;

(iii) for every $\alpha \in \sigma$, an artifact-supported bound

$$d_B(B^\alpha_{i,\sigma}, R_{i,\sigma}) \le \varepsilon_{\alpha,\sigma,i};$$

(iv) the two-sided threshold-gap certificate for $(B^\alpha_{i,\sigma}, R_{i,\sigma})$ at that same budget;

(v) reference identity, scalarization, threshold reconciliation, and face-restriction provenance.

No map between two local repaired profiles and no cocycle is inferred.

Definition 5.80 (Metric-reference face compatibility)**.** Let $\rho \subseteq \sigma$ be nonempty recorded simplices in `metric_reference` mode. Their records are face-compatible only when:

(i) the threshold on ρ is the restriction of the declared threshold policy on σ;

(ii) there is an actual equality or persistence-profile isomorphism

$$\mathsf{W}_{U_\rho} R_{i,\sigma} \xrightarrow{\sim} R_{i,\rho};$$

(iii) every local-to-reference comparison on σ, after restriction to U_ρ, agrees with the registered comparison on ρ at the declared provenance strength;

(iv) scalarization, gap, reference identity, and artifact hashes commute with the face map.

Without this record, simplexwise metric bounds do not form a coherent atlas. A weaker metric comparison between the two references would require a new three-way threshold-gap and exposure theorem and is not inferred here.

Proposition 5.81 (Pairwise bound through a metric simplex reference)**.** *Under a metric-reference simplex record,*

$$d_B\Big(\mathrm{Eval}^\alpha_{i,\sigma}, \mathrm{Eval}^\beta_{i,\sigma}\Big) \le \varepsilon_{\alpha,\sigma,i} + \varepsilon_{\beta,\sigma,i}$$

for every $\alpha, \beta \in \sigma$.

Proof. Apply the Appendix A threshold-gap-safe theorem to each local-to-reference comparison and use the triangle inequality through $T^{\mathrm{bar}}_\tau(R_{i,\sigma})$. □

Remark 5.82 (Metric coherence is not an approximate cocycle)**.** A system of distance bounds has no composable transition morphisms. Metric-reference coherence concerns reference identity, threshold, face records, and certified bounds. It is not an exact cocycle modulo an unspecified error.

5.18 Detector-aware local-to-global coverage

Definition 5.83 (Global profile and local restriction interface)**.** Fix a covered range X, degree i, and one threshold $\tau > 0$. A global profile interface contains a constructible pre-threshold profile

$$B_{i,X}$$

and, for every atlas window W_α, the exact restriction

$$R_i^\alpha := \mathsf{W}_{W_\alpha} B_{i,X}.$$

The observed local pre-threshold profile B_i^α is connected to R_i^α by exactly one branch:

(i) `exact`: an actual isomorphism $R_i^\alpha \cong B_i^\alpha$;

(ii) `metric_zero_transport`: an artifact-supported bound

$$d_B(R_i^\alpha, B_i^\alpha) \le \varepsilon_\alpha,$$

the Appendix A two-sided threshold-gap hypotheses for both profiles, and

$$2\varepsilon_\alpha \le \tau.$$

The operational policy may require the strict reserve $2\varepsilon_\alpha < \tau$.

The second branch consumes Theorem 1.43. Pairwise compatibility among local profiles does not construct $B_{i,X}$.

Definition 5.84 (Certified local detector atlas)**.** Relative to a global profile interface, a certified local detector atlas assigns to every W_α:

(i) a finite stencil $T_\alpha \subseteq (W_\alpha)_\tau$, or the canonical witness family of a certified birth or right-germ detector;

(ii) an Appendix B detector at source stage `windowed_prethreshold`;

(iii) endpoint, critical-set, family, applicability, completeness, and immutable artifact records;

(iv) one interface branch from Definition 5.83;

(v) one mathematical detector value and an independent detector certificate status.

A mathematical zero is consumed globally only when the local certificate is complete at its exact declared strength.

Definition 5.85 (Global retained-bar coverage)**.** A certified local detector atlas has global retained-bar coverage when every bar I of $B_{i,X}$ retained by T_τ^{bar}, including every genuinely infinite retained bar, has a certified local witness: there exist α and $t \in T_\alpha$ such that

$$t \in I \cap W_\alpha, \qquad t + \tau \in I \cap W_\alpha.$$

For a birth or right-germ detector, this clause is read using its canonical interval witness. The coverage status is one of

`undefined`, `certified`, `failed`.

It is `certified` only when the artifact proves the universal quantifier over all globally retained bars. It is `failed` when the coverage claim is well-typed and a retained bar without a certified local witness is exhibited. Missing applicability, global-barcode, witness, or provenance data yield `undefined`, never vacuous coverage.

Theorem 5.86 (Detector-atlas local-to-global zero theorem)**.** *Let a global profile interface, a certified local detector atlas, and a* `certified` *global retained-bar coverage artifact be supplied in the declared constructible interval-decomposable regime. Assume every metric interface branch satisfies its complete repaired-zero transport hypotheses and every local detector is applicable and complete on its declared profile class. Then*

$$T_\tau^{\mathrm{bar}}(B_{i,X}) = 0 \quad \Longleftrightarrow \quad \mathcal{D}_\alpha^{\mathrm{loc}}(B_i^\alpha) = 0 \quad \text{for every } \alpha.$$

Proof. Suppose first that the global repaired profile is zero. Restriction cannot increase the length of a finite interval, and a globally zero repaired profile contains no infinite interval. Therefore

$$T_\tau^{\mathrm{bar}}(R_i^\alpha) = 0$$

for every α. The exact or metric-safe interface branch gives

$$T_\tau^{\mathrm{bar}}(B_i^\alpha) = 0.$$

Local detector completeness then gives detector zero.

Conversely, suppose every local detector is zero and assume, for contradiction, that a globally retained interval I exists. Coverage supplies α and a witness t with $t, t+\tau \in I \cap W_\alpha$. Hence

$$T_\tau^{\mathrm{bar}}(R_i^\alpha) \neq 0.$$

The exact or metric-safe interface branch preserves and reflects repaired zero, so

$$T_\tau^{\mathrm{bar}}(B_i^\alpha) \neq 0.$$

Completeness of the local detector forces a nonzero detector value, contrary to hypothesis. □

Remark 5.87 (Coverage is the global proof obligation)**.** Coverage may be verified by reading the complete global barcode, but that post hoc route supplies no proof-compression advantage. A nontrivial finite-certificate theorem derives coverage independently from geometry, an endpoint lattice, bounded birth data, a family theorem, boundary-bridge windows, or another registered reduction.

Example 5.88 (Local zero and exact overlap coherence do not imply global zero)**.** Fix $\tau > 0$, set

$$X = [0, 2\tau), \qquad B_{i,X} = \{[0, 2\tau)\},$$

and cover X by

$$W_1 = [0, \tau), \qquad W_2 = \left[\frac{\tau}{2}, \frac{3\tau}{2}\right), \qquad W_3 = [\tau, 2\tau).$$

Every restricted interval has length exactly τ, so every local repaired profile is zero. Every nonempty overlap repaired profile is also zero and admits an exact cocycle. Nevertheless the global interval has length $2\tau > \tau$ and survives. The missing hypothesis is global retained-bar coverage: no listed window contains a full τ-span strictly inside the retained global bar under the hard deletion convention.

Definition 5.89 (Object-level descent interface)**.** An object-level descent claim additionally declares:

(i) a category or higher category with restriction functors;

(ii) actual local objects;

(iii) actual transition equivalences;

(iv) an exact cocycle or higher coherence datum;

(v) an effective-descent theorem for the declared cover;

(vi) an artifact constructing the descended object.

Metric-reference records, retained-bar coverage, and a certificate atlas do not substitute for these data.

Remark 5.90 (Predicate globalization is not object descent)**.** Theorem 5.86 proves a predicate about a separately declared global profile. It neither constructs that profile from local data nor descends filtered, persistence, or derived objects.

Example 5.91 (Ordinary summability can coexist with reserve failure)**.** Let

$$r_0 = 1, \qquad \kappa_k = \frac{1}{2}, \qquad \eta_k = 2^{-k-2}.$$

Then

$$\sum_{k\geq 0} \eta_k = \frac{1}{2} < r_0,$$

but

$$K_{k+1} = 2^{-k-1}, \qquad \frac{\eta_k}{K_{k+1}} = \frac{1}{2}.$$

Thus the normalized loss diverges. If equality holds in every Restart recurrence, then

$$\frac{r_n}{K_n} = 1 - \frac{n}{2},$$

so the reserve becomes nonpositive at a finite stage.

Example 5.92 (Pointwise positive reserve need not be uniform)**.** Let

$$\eta_k = 0, \qquad \kappa_k = \frac{1}{2},$$

and take equality in every Restart step. Then

$$r_n = 2^{-n} r_0 > 0$$

for every finite n, but

$$\inf_n r_n = 0.$$

5.19 Detector-aware Restart and boundary-bridge coverage

Definition 5.93 (Detector-aware Restart handoff)**.** A Restart transition that consumes a repaired-zero predicate records, in addition to reserve transport, exactly one target-zero route:

(i) a fresh complete detector certificate on the target window whose mathematical value is zero;

(ii) an exact profile isomorphism from a source profile already certified zero to the target profile;

(iii) an Appendix A metric-safe zero-transport record from a source profile already certified zero, including the two-sided threshold gap and post-threshold zero separation.

The zero-route status is exactly one of

$$\texttt{not\_invoked}, \qquad \texttt{undefined}, \qquad \texttt{certified}, \qquad \texttt{failed}.$$

It is `certified` only when the source-zero premise, the complete route artifact, and the target-zero conclusion are all verified. It is `failed` when the route is well-typed but a required mathematical premise or conclusion is evidenced false. Missing or ill-typed route data yield `undefined`; an explicit claim that does not consume zero transport uses `not_invoked`. The record identifies source and target stages, threshold, endpoint convention, comparison mode, proof identity, and immutable artifact. A κ_k-admissible inequality alone is not a zero-transport theorem.

Definition 5.94 (Boundary-bridge coverage)**.** For an ordered partition $\{W_k\}$, let $\{V_k\}$ be declared bridge windows meeting both sides of the boundary between W_k and W_{k+1}. The augmented family

$$\mathcal{W}_{\mathrm{aug}} := \{W_k\}_k \cup \{V_k\}_k$$

has boundary-bridge coverage at τ when every globally retained interval admits a certified τ-span witness in one partition or bridge window. Every bridge window has its own direct evaluation and complete detector certificate.

Corollary 5.95 (Detector-safe partition globalization)**.** *Assume a global profile interface for an augmented partition/bridge family, boundary-bridge coverage, and complete zero detector certificates on every partition and bridge window. Then*

$$T_\tau^{\mathrm{bar}}(B_{i,X}) = 0.$$

Proof. The augmented family is a certified local detector atlas with global retained-bar coverage. Apply Theorem 5.86. □

Remark 5.96 (Reserve transport does not detect crossing bars)**.** A positive Restart reserve controls the threshold classification of the profiles named by the continuation artifact. It does not prove that a retained interval crossing a partition boundary is detected inside either disjoint cell. Fresh target detection controls the target predicate; boundary-bridge coverage controls the global predicate.

Definition 5.97 (Metric-reference simplex exposure)**.** Let N_{met} be the set of recorded simplices in `metric_reference` mode. For $\sigma \in N_{\mathrm{met}}$, let $e_\sigma \geq 0$ be a fixed nonnegative quantity dominating every metric comparison charged to that simplex. Define the unordered sum

$$\sum_{\sigma \in N_{\mathrm{met}}} e_\sigma := \sup_{\substack{S \subseteq N_{\mathrm{met}} \\ S \text{ finite}}} \sum_{\sigma \in S} e_\sigma.$$

Exact-cocycle simplices are supported by exact proof artifacts and are not assigned fictitious metric exposure.

Theorem 5.98 (v20 countable record-atlas principle)**.** *Let $\{W_n\}_{n\geq 1}$ be a countable audit-locally finite overlap cover of finite Čech depth. Assume:*

(i) every local window has a complete certificate record;

(ii) every nonempty nerve simplex has one complete record in `exact_cocycle` or `metric_reference` mode;

(iii) all exact-cocycle identities and exact face restrictions are proved;

(iv) all metric-reference comparisons satisfy their individual two-sided threshold-gap conditions and every metric-reference face satisfies Definition 5.80;

(v) metric-reference simplex exposure is summable;

(vi) terminal Type IV is correctly `not_invoked` or `certified_zero`;

(vii) if long-transient cleanliness is required, Appendix D records the aggregate verdict `certified_absent`;

(viii) all records and artifact references are audit-readable.

Then the family is an audit-locally finite Čech-coherent certificate atlas with finite declared metric-reference exposure.

If a global repaired-zero conclusion is additionally invoked, the atlas must also carry:

(a) a global profile interface;

(b) complete local detector certificates;

(c) exact or metric-safe local zero transport;

(d) global retained-bar coverage;

(e) zero local detector values.

Under these additional clauses, the global repaired profile is zero.

Proof. Audit-local finiteness makes every bounded audit query finite, while finite Čech depth bounds overlap order. Exact-cocycle records provide actual coherent morphisms at record level. Metric-reference records provide reference-consistent bounds without claiming a cocycle. The face conditions and local records therefore form a coherent atlas, and the unordered exposure sum is finite by hypothesis. When the detector-globalization clauses are invoked, Theorem 5.86 gives the global repaired-zero predicate. □

Remark 5.99 (Summability is not coherence)**.** Exposure summability does not create missing face maps, cocycles, reference identities, local completeness, or retained-bar coverage. Conversely, a coherent atlas may fail a separately declared exposure policy.

Definition 5.100 (v20 detector-atlas manifest extension)**.** A proof object invoking local detector globalization additionally records:

(i) the global profile and covered-range identity;

(ii) every exact local restriction profile;

(iii) every observed local profile;

(iv) the exact or metric-zero-transport interface branch;

(v) every local detector identifier, stage, witness set, endpoint convention, family, completeness source, and value/status pair;

(vi) the global retained-bar coverage theorem and artifact;

(vii) the exact-cocycle or metric-reference mode of every consumed simplex;

(viii) terminal and long-transient participation fields;

(ix) every Restart zero route and boundary-bridge artifact when invoked;

(x) claim, proof, manifest, hash, and provenance references.

Missing coverage or an incomplete local detector yields `undefined`, not global zero.

Definition 5.101 (Terminal/transient participation policy)**.** Each local, overlap, simplex, Restart-target, or atlas record declares separately:

(i) terminal Type IV scope and status;

(ii) whether long-transient absence is required;

(iii) when required, the fixed $(W, \tau, \tau_{\mathrm{def}}, I_{\mathrm{mon}})$ and the Appendix D aggregate transient verdict.

Terminal `certified_zero` does not imply transient `certified_absent`, and neither implies full comparison-map isomorphism.

Proposition 5.102 (Transient-clean participation rule)**.** *When a gluing claim requires absence of long transient kernel or cokernel defects, no participating record may have transient status `certified_present` or `undefined`. Every required record must have `certified_absent` at the same declared detector component and threshold policy.*

Proof. This is the aggregate verdict discipline of Appendix D, Definition 4.59. Record compatibility cannot repair a present or undefined defect clause. □

Remark 5.103 (Source-profile coherence is not defect coherence)**.** Even a complete exact source-profile cocycle does not prove that tower comparison kernels or cokernels glue, agree, or remain absent. Such a conclusion requires actual compatible tower artifacts and defect-profile comparison theorems.

Theorem 5.104 (Appendix E v20 detector-atlas and gluing package)**.** *Within the v20 conservative extension:*

(i) the complete inherited v19 overlap, Restart, summability, and record-atlas package remains valid at its original strength;

(ii) pairwise and simplex coherence distinguish exact-cocycle from metric-reference modes;

(iii) metric-reference data do not create morphisms or cocycles;

(iv) local detector zero becomes global repaired zero only through a declared global profile interface, complete local detectors, safe local zero transport, and global retained-bar coverage;

(v) local zero and exact overlap coherence alone do not imply global zero;

(vi) a κ-Restart reserve inequality does not transport a zero predicate without a separate detector or zero-transport route;

(vii) boundary-bridge coverage repairs the blind spot of disjoint partition cells only when each bridge has its own complete detector;

(viii) ordinary summability may coexist with finite-stage reserve failure;

(ix) pointwise positive reserves need not admit a uniform positive lower bound;

(x) terminal Type IV cleanliness and long-transient cleanliness are separate atlas clauses;

(xi) source-profile coherence does not transport kernel/cokernel defect profiles;

(xii) countable atlas coherence, metric exposure, detector globalization, and object-level descent are four distinct conclusions.

Proof. Item (i) is Theorem 5.71. Items (ii)–(iii) are Definitions 5.77–5.79 and Remark 5.82. Items (iv)–(v) are Theorem 5.86 and Example 5.88. Items (vi)–(vii) are Definition 5.93 and Corollary 5.95. Items (viii)–(ix) are Examples 5.91 and 5.92. Items (x)–(xi) are Definition 5.101 and Remark 5.103. Item (xii) follows from Theorem 5.98, Theorem 5.86, and Definition 5.89. □

6 Appendix F: Typed Obligations, Status Algebra, Dependency Closure, and Certificate Readiness

6.1 Scope, dependency position, and design boundary

This appendix supplies the Foundation-level typed obligation calculus supporting Chapter 7. The complete v19 calculus below is inherited without weakening or silent reinterpretation. The v20 extension added after the inherited package integrates detector, transient-defect, global-zero, finite-to-infinite, object-descent, semantic-replay, and release-acceptance obligations. It consumes:

(i) the representative lifecycle, representative obligation bundle, and manifest handoff of Appendix B, especially Definition 2.109;

(ii) the eligible-realization, Ext-clause, and obligation bundle of Appendix C, especially Definition 3.98;

(iii) the Type IV diagnostic record, four typed Type IV statuses, and obligation bundle of Appendix D, especially Definition 4.98;

(iv) the scalar/non-scalar separation, overlap and simplex obligations, and κ-admissible Restart obligations of Appendix E, especially Definition 5.17.

Its output is a typed status-and-dependency semantics suitable for serialization by Appendix G. The central rule is:

> A Core verdict is a dependency-closed conjunction of independently evidenced clauses. No numerical reserve, passing sibling clause, or complete-looking manifest may create missing evidence.

Remark 6.1 (Appendix F does not enlarge the mathematical Core)**.** This appendix proves no new persistence, derived-category, tower, overlap, or Restart theorem. It governs the admissibility and aggregation of evidence already defined elsewhere. A status record, registry entry, or manifest field never replaces the theorem, proof, or verified artifact whose existence it records.

Remark 6.2 (Normative Foundation semantics)**.** Most statements below are definitions, typing contracts, or elementary consequences of those contracts. They are normative for v20 audit objects while preserving the inherited v19 audit semantics, but they are not external-domain mathematical claims.

6.2 Immutable obligation atoms and registries

Definition 6.3 (Typed obligation atom)**.** A *typed obligation atom* is a tuple

$$\begin{aligned}\mathfrak{o} := (&\text{ObligationID}, \text{owner}, \text{scope}, \text{kind},\\ &\text{evidence policy}, \text{dependencies}, \text{failure routes},\\ &\text{manifest fields}).\end{aligned}$$

Its fields identify:

(i) the immutable clause identity;

(ii) the chapter or appendix owning its mathematical meaning;

(iii) the object, degree, window, threshold, tower, overlap, transition, or atlas scope;

(iv) the class of evidence needed to evaluate it;

(v) the prerequisite obligations;

(vi) every broad Type I–V route that may apply on failure;

(vii) the fields required for manifest serialization and replay.

Definition 6.4 (Obligation kinds)**.** Each obligation atom has one primary kind:

(i) *mathematical verdict*: for example repaired profile vanishing or eligible Ext vanishing;

(ii) *existence/construction*: for example a representative, comparison morphism, or continuation artifact;

(iii) *compatibility/coherence*: for example profile compatibility, edge realization, face compatibility, or composition coherence;

(iv) *metric-bearing safety*: a clause whose evidence is a declared upper bound consumed by a metric gate;

(v) *operational transition/computation*: for example terminal tameness, scalarization, or a Restart inequality;

(vi) *governance/manifest*: claim status, repair history, artifact identity, scope justification, schema consistency, or replayability.

The primary kind does not prevent one failed atom from receiving several failure routes.

Definition 6.5 (Declared obligation registry)**.** For an immutable audit run $\mathfrak{R}$, a *declared obligation registry* is a finite or countable family

$$\mathsf{Obl}(\mathfrak{R}) = \{\mathfrak{o}_a\}_{a\in A}$$

with immutable identifiers, together with:

(i) the required subset $\mathsf{Req}(\mathfrak{R}) \subseteq \mathsf{Obl}(\mathfrak{R})$;

(ii) an explicit scope-exclusion record for every atom outside $\mathsf{Req}(\mathfrak{R})$;

(iii) the dependency edges and their provenance;

(iv) evidence identifiers and evidence-policy versions;

(v) the registered claim and repair-record versions governing every theorem-level use.

A silent change in any of these data defines a new run or revision.

Definition 6.6 (Evidence satisfaction predicate)**.** Each obligation atom $\mathfrak{o}$ has a declared predicate

$$\mathrm{Sat}_{\mathfrak{o}}(E_{\mathfrak{o}}) \in \{\mathrm{true}, \mathrm{false}\}$$

which is evaluable only when the evidence object $E_{\mathfrak{o}}$ is present, typed, version-compatible, and attached to the same immutable run scope. If those typing conditions fail, the atom is not assigned a mathematical truth value by this predicate.

Remark 6.7 (Registry completeness is not proof)**.** A registry lists what must be checked. It does not prove the listed atoms. A complete registry with empty or invalid evidence remains incomplete as a proof object.

6.3 Scalarizable and non-scalarizable obligations

Definition 6.8 (Scalarizable obligation)**.** An obligation is *scalarizable* only when its evidence policy is a declared metric upper bound that may enter the typed ledger and whose scalarization is proved to dominate the actual bottleneck discrepancy consumed by a specific gate.

Definition 6.9 (Non-scalarizable obligation)**.** The following are non-scalarizable:

(i) representative existence, profile compatibility, monitored-scope correctness, and equivalence strength;

(ii) realization compatibility, amplitude, residual edge extraction, and edge-realization certificates;

(iii) actual profile morphisms, source provenance, terminal tameness, defect profiles, and Type IV typing;

(iv) threshold reconciliation, higher-face coherence, continuation artifacts, and κ-admissibility;

(v) local-pass, target-pass, atlas-completeness, and countable-registry coverage obligations;

(vi) claim status, repair records, artifact identity, manifest completeness, consistency, and replayability.

Proposition 6.10 (No scalar manufacture of non-scalar evidence)**.** *If a required non-scalarizable obligation is absent, ill-typed, inconsistent, or attached to another run, then no ledger value, scalarized tolerance, threshold clearance, or positive reserve makes that obligation pass.*

Proof. A scalar metric record bounds only the declared metric discrepancy named by its evidence policy. It does not construct, type, or verify any of the non-scalar objects in Definition 6.9. □

Remark 6.11 (Non-compensation begins at the typing boundary)**.** A large reserve may absorb a larger certified metric discrepancy. It may not absorb missing semantics. This distinction is prior to any aggregate verdict rule.

6.4 Canonical atomic statuses

Definition 6.12 (Atomic audit status)**.** Every obligation atom in the declared registry has exactly one canonical status

$$\mathsf{AuditStatus} = \{\texttt{pass}, \texttt{reject}, \texttt{undefined}, \texttt{not\_invoked}\}.$$

Their meanings are:

(i) `pass`: the atom is required, well-typed, fully evidenced, and its satisfaction predicate is true;

(ii) `reject`: the atom is required, well-typed, fully evidenced, and its satisfaction predicate is false or the declared obstruction is nonzero;

(iii) `undefined`: the atom is required but incomplete, ill-typed, inconsistent, blocked by an undischarged dependency, or not computable from the declared evidence;

(iv) `not_invoked`: the immutable scope positively declares that the atom is outside the claim.

Remark 6.13 (Missing is not not-invoked)**.** A missing atom is not automatically outside scope. Without a valid scope-exclusion record it is `undefined`.

Definition 6.14 (Justified scope exclusion)**.** A scope exclusion is *justified* when it identifies the excluded atom, the governing policy, the reason it is irrelevant to the declared claim, and the immutable run fields on which that exclusion depends. An exclusion inferred only after observing an inconvenient result is invalid.

Definition 6.15 (Effective aggregation normalization)**.** For aggregation only, define

$$\mathrm{Norm}(s, j) := \begin{cases} \texttt{pass}, & s = \texttt{not\_invoked} \text{ and } j \text{ is a valid scope exclusion},\\ \texttt{undefined}, & s = \texttt{not\_invoked} \text{ and no valid exclusion exists},\\ s, & s \in \{\texttt{pass}, \texttt{reject}, \texttt{undefined}\}. \end{cases}$$

The original `not_invoked` status remains stored; normalization does not erase the exclusion record.

6.5 Specialized status embeddings from Appendices B–E

Definition 6.16 (Representative lifecycle embedding)**.** The representative lifecycle statuses of Definition 2.70 map as follows when a representative is required:

$$\texttt{verified} \mapsto \texttt{pass}, \qquad \texttt{failed} \mapsto \texttt{reject},$$

$$\texttt{constructed}, \texttt{missing} \mapsto \texttt{undefined}.$$

The status `not_required` maps to `not_invoked` only under a valid scope exclusion.

Definition 6.17 (Ext-clause embedding)**.** The typed Ext-clause statuses of Definition 3.83 map identically to the canonical statuses with the same names.

Definition 6.18 (Type IV embedding)**.** The Type IV statuses of Definition 4.39 map by

$$\texttt{certified\_zero} \mapsto \texttt{pass}, \qquad \texttt{obstructed} \mapsto \texttt{reject}, \qquad \texttt{undefined} \mapsto \texttt{undefined}.$$

Type IV `not_invoked` remains canonical `not_invoked` only under an explicit no-tower scope decision.

Definition 6.19 (Restart transition embedding)**.** The Restart transition statuses of Definition 5.46 embed at the level of the *continuation-law subclause*:

$$\texttt{certified} \mapsto \texttt{pass}, \qquad \texttt{failed} \mapsto \texttt{reject}, \qquad \texttt{undefined} \mapsto \texttt{undefined}.$$

This map does not assign the complete Restart bundle a passing status.

Definition 6.20 (Complete Restart bundle)**.** A complete Restart step contains the following required atoms:

(i) a certified continuation-law subclause;

(ii) a defined positive target reserve;

(iii) a passing target-local certificate record;

(iv) every required endpoint, representative, threshold-reconciliation, scalarization, and comparison artifact;

(v) no required undefined non-reserve obligation.

Its canonical status is the aggregate status of this bundle.

Remark 6.21 (Certified transition is necessary but not sufficient)**.** A certified transition proves only

$$r_{k+1} \geq \kappa_k r_k - \eta_k.$$

The complete Restart step may still reject because the target reserve is nonpositive or the target-local record rejects, and it may remain undefined because another required artifact is missing.

Definition 6.22 (Manifest status embedding)**.** Manifest statuses embed by

$$\texttt{complete} \mapsto \texttt{pass}, \qquad \texttt{incomplete} \mapsto \texttt{undefined}, \qquad \texttt{inconsistent} \mapsto \texttt{reject}.$$

A complete manifest supports, but does not itself prove, the mathematical atoms it records.

6.6 Claim-use preflight

Definition 6.23 (Registered claim-use status)**.** Each invoked claim has exactly one registered status

$$\mathsf{ClaimStatus} = \{\texttt{proved}, \texttt{conditional}, \texttt{operational}, \texttt{spec}, \texttt{deprecated}, \texttt{rejected}, \texttt{unknown}\}.$$

Their meanings are the canonical Chapter 7 meanings: theorem-level permission, conditional permission, procedural use only, target/specification, provenance-only legacy status, explicit mathematical rejection, or absence of authoritative registration.

Definition 6.24 (Claim-use atom status)**.** For one intended use of a claim:

(i) `proved` with matching hypotheses maps to `pass`;

(ii) `conditional` maps to `pass` only when every registered condition is discharged, otherwise to `undefined`;

(iii) `operational` maps to `pass` only for procedural use and to `reject` if used as theorem evidence;

(iv) `spec`, `deprecated`, or `rejected` used as theorem evidence maps to `reject`;

(v) `unknown` maps to `undefined`;

(vi) a registered status attached to the wrong version, theorem location, or repair state maps to `undefined` or `reject` according to whether the mismatch is unresolved or evidenced inconsistent.

Definition 6.25 (Audit preflight bundle)**.** The preflight bundle contains:

(i) complete run identity and immutable scope;

(ii) admissible claim-use atoms;

(iii) every relevant v18-to-v19 repair record;

(iv) required artifacts with matching types, versions, and hashes;

(v) absence of inconsistent artifact resolution;

(vi) the manifest status atom.

Its aggregate status is denoted $\mathsf{Preflight}(\mathfrak{R})$.

Proposition 6.26 (Preflight failure blocks Core readiness)**.** *If the preflight bundle is not* $\mathtt{pass}$*, the complete Foundation readiness bundle cannot be* $\mathtt{pass}$*.*

Proof. The preflight atoms are required dependencies of every theorem-level and manifest-level final verdict. Apply the dependency-closure and aggregate rules below. □

6.7 Well-founded dependency registries

Definition 6.27 (Immediate dependency relation)**.** For obligation atoms $\mathfrak{p}, \mathfrak{o}$, write

$$\mathfrak{p} \prec \mathfrak{o}$$

when $\mathfrak{p}$ is an immediate prerequisite of $\mathfrak{o}$. The transitive dependency closure is denoted

$$\mathrm{Cl}(\mathfrak{o}) = \{\mathfrak{p} : \mathfrak{p} \prec^* \mathfrak{o}\}.$$

Only dependencies active under the immutable run scope enter the active closure.

Definition 6.28 (Well-founded dependency registry)**.** A registry is *well-founded* when either:

(i) it is finite and its dependency graph is acyclic; or

(ii) it is countable and carries a rank function

$$\mathrm{rk} : \mathsf{Obl}(\mathfrak{R}) \longrightarrow \mathbb{N}$$

such that $\mathfrak{p} \prec \mathfrak{o}$ implies $\mathrm{rk}(\mathfrak{p}) < \mathrm{rk}(\mathfrak{o})$.

Self-dependence and cyclic mutual certification are forbidden.

Definition 6.29 (Dependency-complete pass)**.** A required atom $\mathfrak{o}$ has a *dependency-complete pass* when:

(i) its own evidence is present, typed, run-consistent, and satisfies its evidence predicate;

(ii) every active required atom in $\mathrm{Cl}(\mathfrak{o})$ has status $\mathtt{pass}$;

(iii) every inactive atom in the declared dependency template has a valid scope exclusion;

(iv) all claim, artifact, and repair-record dependencies agree with the immutable run identity.

Proposition 6.30 (Pass is downward closed)**.** *If* $\mathfrak{o}$ *has a dependency-complete pass, then every active required atom in* $\mathrm{Cl}(\mathfrak{o})$ *has status* $\mathtt{pass}$*.*

Proof. This is clause (ii) of Definition 6.29. □

Definition 6.31 (Blocked theorem-application atom)**.** Suppose an atom records the application of a theorem or construction whose active prerequisite is not `pass`. The application atom is *blocked* and has canonical status `undefined`, even when the failed prerequisite has status `reject`. The rejected or undefined prerequisite remains separately recorded.

Remark 6.32 (Why blocked application is not automatically reject)**.** Failure of a theorem hypothesis blocks that theorem application; it does not necessarily prove the intended conclusion false. The surrounding conjunction or bundle still rejects whenever the rejected prerequisite is itself a required atom.

Proposition 6.33 (Dependency blocker propagation)**.** *If an active prerequisite is `undefined` or `reject`, then a dependent theorem-application atom cannot pass. It is `undefined` as an application atom, while the aggregate bundle preserves any rejected prerequisite.*

Proof. The dependency-complete pass criterion fails. Definition 6.31 types the application atom as `undefined`. Aggregate rejection, when applicable, follows from the rejected prerequisite itself. □

6.8 The status aggregation algebra

Definition 6.34 (Effective precedence order)**.** After applying the normalization of Definition 6.15, use the total order

$$\texttt{pass} < \texttt{undefined} < \texttt{reject}.$$

For effective statuses s, t, define

$$s \otimes t := \max\{s, t\}$$

in this order.

Proposition 6.35 (Status-algebra laws)**.** *The operation $\otimes$ is commutative, associative, and idempotent, with neutral element `pass`.*

Proof. These are the corresponding properties of the maximum operation on a total order. □

Definition 6.36 (Aggregate audit verdict)**.** Write

$$A_{\mathrm{req}}(\mathfrak{R}) := \{a \in A \mid \mathfrak{o}_a \in \mathsf{Req}(\mathfrak{R})\}, \qquad A_{\mathrm{exc}}(\mathfrak{R}) := A \setminus A_{\mathrm{req}}(\mathfrak{R}).$$

Let j_a be the scope-exclusion record, when any, for atom $\mathfrak{o}_a$. The aggregate verdict is

$$\mathrm{Agg}(\mathfrak{R}) := \left(\bigotimes_{a \in A_{\mathrm{req}}(\mathfrak{R})} \mathrm{Norm}\big(s(\mathfrak{o}_a), j_a\big)\right) \otimes \left(\bigotimes_{a \in A_{\mathrm{exc}}(\mathfrak{R})} \mathrm{Norm}\big(s(\mathfrak{o}_a), j_a\big)\right),$$

provided the finite or countable registry satisfies the completeness conditions of this appendix. The two factors make the required/scope-excluded partition explicit; they do not permit an excluded atom to bypass the justified-scope condition of Definition 6.14. Equivalently:

(i) `reject` if at least one atom in $\mathsf{Req}(\mathfrak{R})$ rejects;

(ii) `undefined` if none rejects but at least one atom in $\mathsf{Req}(\mathfrak{R})$ is undefined, or one purported `not_invoked` atom outside $\mathsf{Req}(\mathfrak{R})$ lacks valid scope justification;

(iii) `pass` if every atom in $\mathsf{Req}(\mathfrak{R})$ passes and every atom outside $\mathsf{Req}(\mathfrak{R})$ has justified `not_invoked` status.

Theorem 6.37 (Non-compensation)**.** *No number of passing atoms, zero diagnostics, positive reserves, or favorable scalar values cancels a rejected or required undefined atom in the aggregate verdict.*

Proof. The aggregate is the maximum in the effective precedence order. Passing atoms are neutral, while `undefined` and `reject` remain respectively above `pass`. Proposition 6.10 separately prevents numerical evidence from changing the type of a non-scalar atom. □

Proposition 6.38 (Partition invariance of hierarchical aggregation)**.** *Partition a complete registry into disjoint complete subregistries. Aggregate each subregistry and then aggregate the resulting statuses. The headline verdict equals the direct aggregate of all atoms, provided all rejected, undefined, and justified* `not_invoked` *identifiers remain attached to the intermediate records.*

Proof. Associativity and commutativity of $\otimes$ give equality of headline statuses. Retention of atom identifiers preserves the complete diagnostic detail. □

Remark 6.39 (Headline compression does not erase detail)**.** If one atom rejects and another is undefined, the headline is `reject`, but both atomic facts, dependency blockers, and failure routes remain in the record.

6.9 Failure routing and nonexclusive diagnostics

Definition 6.40 (Failure-routing set)**.** Every non-pass run records all applicable routes

$$\mathsf{Routes}(\mathfrak{R}) \subseteq \{\mathrm{I}, \mathrm{II}, \mathrm{III}, \mathrm{IV}, \mathrm{V}\}.$$

Their meanings are:

(i) Type I: nonzero repaired degree-one topological profile;

(ii) Type II: nonzero categorical obstruction in a complete eligible realization regime;

(iii) Type III: failed or undefined metric, crop, representative, realization, overlap, continuation, Restart, terminal-tameness, or other operational condition;

(iv) Type IV: nonzero terminal-generic tower defect relative to a valid comparison artifact;

(v) Type V: claim-status, scope, manifest, repair-record, artifact-identity, version, replayability, or proposer/verifier governance failure.

The routes are nonexclusive and are not a severity ranking.

Definition 6.41 (Route assignment discipline)**.** Route assignment obeys the following distinctions:

(i) a nonzero valid Type IV diagnostic is Type IV, not merely missing evidence;

(ii) an absent Type IV artifact makes the Type IV atom `undefined` and is routed operationally as Type III and, when omitted or falsely serialized, also as Type V;

(iii) missing eligibility is not Type II; it is Type III and possibly Type V;

(iv) an inconsistent manifest is Type V and canonical `reject`;

(v) an incomplete manifest is canonical `undefined`, unless another evidenced atom already rejects;

(vi) reporting `undefined` as zero or pass creates an additional Type V route.

Proposition 6.42 (Atomic status and failure route are different types)**.** *Two atoms may share a failure route while having different canonical statuses. In particular, a Type III operational clause may be evidenced false and* `reject`*, or incomplete and* `undefined`*.*

Proof. A route identifies the domain of failure. The atomic status identifies whether the clause was evidenced false or could not be validly evaluated. □

6.10 Bundle readiness without a fifth verdict

Definition 6.43 (Bundle readiness)**.** For a complete declared obligation bundle $\mathcal{B}$, its *certificate-readiness status* is its aggregate canonical status:

$$\mathsf{Ready}(\mathcal{B}) := \mathrm{Agg}(\mathcal{B}).$$

The words *ready*, *blocked*, and *rejected* may describe the cases `pass`, `undefined`, and `reject`, but they do not form a competing fifth status system.

Definition 6.44 (Representative readiness bundle)**.** The representative readiness bundle is the obligation family of Definition 2.109 with the lifecycle embedding of Definition 6.16 and every active dependency required for the intended use.

Definition 6.45 (Eligible Ext readiness bundle)**.** When the Ext clause is invoked, the Ext readiness bundle is the obligation family of Definition 3.98, including the repaired topological input, representative, realization, amplitude, residual-edge, edge-realization, field, theorem-use, and manifest atoms.

Definition 6.46 (Type IV readiness bundle)**.** When tower semantics are invoked, the Type IV readiness bundle is the obligation family of Definition 4.98, including actual morphism, source provenance, terminal mode, terminal tameness, defect profiles, interior-defect typing, and the canonical Type IV status.

Definition 6.47 (Overlap and Restart readiness bundle)**.** When overlap, atlas, or sequential inheritance is invoked, the corresponding readiness bundle contains the non-scalar obligations of Definition 5.17, all metric reserve atoms, all required simplex records, and every complete Restart bundle of Definition 6.20.

Definition 6.48 (Foundation readiness bundle)**.** The complete Foundation readiness bundle contains, as applicable:

(i) the audit identity, immutable scope, claim-use, repair, and manifest-preflight atoms;

(ii) the gate-scoped metric ledger, scalarization, actual upper-bound theorem, gap, and reserve atoms;

(iii) the representative readiness bundle;

(iv) the eligible Ext readiness bundle;

(v) the Type IV readiness bundle;

(vi) all local, overlap, simplex, atlas, continuation, and complete Restart bundles;

(vii) manifest completeness, consistency, replayability, aggregate verdict, and failure-routing atoms.

Theorem 6.49 (Certificate-readiness criterion)**.** *A Foundation bundle has readiness status* `pass` *exactly when:*

(i) *its registry is complete and well-founded;*

(ii) *preflight passes;*

(iii) *every required atom has a dependency-complete pass;*

(iv) *every excluded atom has justified* `not_invoked` *status;*

(v) *the manifest fragment is complete, consistent, and run-compatible;*

(vi) *the aggregate verdict is* `pass`*.*

Proof. The forward direction follows from the meanings of `pass`, dependency-complete pass, and manifest completeness. The reverse direction follows from Definition 6.36 after all listed conditions are satisfied. □

Remark 6.50 (Readiness is scope-relative)**.** A passing readiness bundle certifies only the declared Core audit scope. It does not create a stronger claim, a global object, an object-level gluing theorem, or an external arithmetic, geometric, PDE, cryptographic, or computational conclusion.

6.11 Complete Restart and atlas composition rules

Proposition 6.51 (Complete Restart verdict)**.** *For an invoked Restart step:*

(i) *the complete bundle is* `pass` *exactly when the transition-law subclause passes, the target reserve is defined and positive, the target-local record passes, and all required non-reserve artifacts pass;*

(ii) *it is* `reject` *if the certified inequality is false, the defined target reserve is nonpositive, or the target-local record rejects;*

(iii) *it is* `undefined` *if no required item rejects but a required continuation, reserve, target, representative, comparison, or manifest atom is undefined;*

(iv) *it is* `not_invoked` *only under an explicit scope exclusion of sequential inheritance.*

Proof. Apply the aggregate verdict to the complete Restart bundle of Definition 6.20. □

Proposition 6.52 (Atlas composition rule)**.** *A finite or countable certificate atlas has readiness status* `pass` *only when:*

(i) *every required local vertex record passes;*

(ii) *every required pairwise and higher-simplex record passes;*

(iii) *the declared Čech-depth, audit-local-finiteness, threshold reconciliation, and simplex-budget atoms pass;*

(iv) *every invoked Type IV and Restart bundle has passing status;*

(v) *the countable registry, when any, has a complete coverage certificate;*

(vi) *the manifest is complete and consistent.*

A rejected required atom rejects the atlas. If none rejects but one required atom is undefined, the atlas is undefined.

Proof. Apply the aggregate and dependency-closure rules to the atlas obligation registry of Appendix E. □

Remark 6.53 (Atlas readiness is not global-object readiness)**.** A passing atlas bundle certifies record-level coherence. It does not construct a single global filtered object, persistence module, derived object, or global B-Gate^{+} record.

6.12 Countable registries and hidden-tail prevention

Definition 6.54 (Audit-readable countable registry)**.** A countable obligation registry is *audit-readable* when it has:

(i) a declared countable index set and immutable enumeration or generator;

(ii) a coverage certificate proving that every obligation in scope occurs in the registry;

(iii) a canonical status, evidence record, and scope decision for every index;

(iv) a replayable dependency generator and well-founded rank;

(v) a universal status certificate sufficient to verify that all required atoms pass or to identify all non-pass atoms;

(vi) no hidden, unindexed, or implementation-dependent tail.

Proposition 6.55 (Finite-prefix checks do not certify countable readiness)**.** *A finite prefix of a countable registry cannot support a pass verdict for the full registry unless a separate coverage proof establishes that the omitted tail contains no required atom or only atoms with valid scope exclusions.*

Proof. Without such a proof, an unchecked tail may contain a rejected, undefined, or unjustifiably excluded atom. The full aggregate verdict is therefore not determined by the finite prefix. □

Theorem 6.56 (Countable aggregate soundness)**.** *For an audit-readable countable registry, the aggregate verdict of Definition 6.36 is sound under the universal status certificate. In particular, a pass verdict requires universal dependency-complete pass over all required indices.*

Proof. The coverage certificate identifies the full registry, the rank proves dependency well-foundedness, and the universal status certificate supplies the required universal quantification. The aggregate cases then follow from Definition 6.36. □

6.13 Repair calculus and immutable verdict history

Definition 6.57 (Repair record)**.** A repair record contains:

(i) the predecessor RunID or revision identifier;

(ii) every repaired rejected or undefined atom;

(iii) the changed evidence, dependency, scope, theorem, claim-status, or manifest fields;

(iv) the old and new artifact hashes and schema versions;

(v) the reason the repair is admissible;

(vi) the resulting new atomic statuses, failure routes, and aggregate verdict.

Proposition 6.58 (No retroactive verdict mutation)**.** *Resolving an undefined atom or correcting a rejected atom creates a new run or revision record. It does not rewrite the historical verdict of the immutable earlier run.*

Proof. The obligation registry, evidence identities, and repair state belong to the immutable audit scope. Changing them changes the audited object or its revision identity. □

Remark 6.59 (Repairs need not be monotone in headline status)**.** New evidence may resolve `undefined` to `pass` or `reject`; correction of an invalid rejection may also change the result. Immutability concerns historical records, not monotonicity of future revisions.

6.14 Appendix G serialization handoff

Definition 6.60 (Status-record serialization contract)**.** For every obligation atom, the Appendix G schema must be able to serialize at least:

(i) ManifestID, RunID, obligation ID, owner, and immutable scope;

(ii) requiredness, scope-exclusion status, and scope-exclusion evidence;

(iii) obligation kind and evidence-policy version;

(iv) immediate dependencies, closure references, and rank data;

(v) specialized source status and canonical atomic status;

(vi) evidence, theorem, ClaimID, artifact, version, and hash references;

(vii) scalar ledger, scalarization, actual upper-bound, gap, and reserve fields when applicable;

(viii) every Type I–V failure route and every blocker identifier;

(ix) predecessor and repair-record identifiers;

(x) aggregate bundle verdict, registry-coverage status, consistency, and replayability fields.

Proposition 6.61 (Schema conformance preserves semantics)**.** *A chapter-local or appendix-local record is admissible for the canonical manifest only if its serialized fields preserve the obligation identity, status meaning, dependency relation, failure routes, and immutable artifact references fixed by this appendix.*

Proof. Changing one of those meanings would serialize a different audit atom or run under the same identifier, contradicting immutable registry semantics. □

Remark 6.62 (Semantics here, schema there)**.** Appendix F fixes status meanings, dependency closure, aggregation, readiness, and failure routing. Appendix G fixes canonical field names, serialization constraints, conformance, and replayability. Schema completeness alone is not mathematical proof, and a mathematically true unrecorded clause does not produce an auditable pass until its required evidence is serialized consistently.

6.15 Illustrative audit patterns

Example 6.63 (Constructed but unverified representative)**.** A representative candidate exists, but its profile isomorphisms are not verified. Its lifecycle status is `constructed`; its canonical status is `undefined`. A small bottleneck budget and a large threshold gap do not change that status.

Example 6.64 (Undefined Type IV is not zero)**.** An actual comparison morphism or terminal-tameness certificate is absent. The Type IV atom is `undefined`, and no numerical μ, ν pair is asserted. Recording

$$(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) = (0, 0)$$

would add a Type V manifest failure.

Example 6.65 (Certified transition with rejected target)**.** A continuation artifact verifies the κ-inequality, so the transition-law subclause passes. The target-local record has a nonzero repaired topological profile and rejects. The complete Restart bundle therefore rejects.

Example 6.66 (Rejected prerequisite and blocked theorem application)**.** Suppose a theorem application requires an eligible realization, but the amplitude certificate is well-typed and false. The amplitude atom rejects. The theorem-application atom is blocked and undefined; the full bundle rejects because it still contains the rejected amplitude atom.

Example 6.67 (Pairwise overlap without higher coherence)**.** Every pairwise Overlap Gate passes, but a required triple-overlap simplex record is missing. The pairwise bundle passes, while the atlas bundle is `undefined`. No numerical overlap reserve manufactures the missing higher-face record.

Example 6.68 (Complete-looking but inconsistent manifest)**.** All expected fields are populated, but two artifact hashes are declared for the same immutable identifier. The manifest is inconsistent and maps to `reject`; the run has a Type V route even if every numerical computation can be replayed separately.

6.16 Appendix F Foundation package

Theorem 6.69 (Appendix F typed-obligation and readiness package)**.** *Within the declared v20 audit scope, preserving the inherited v19 governance kernel:*

(i) every obligation has an immutable typed identity and evidence policy;

(ii) scalarizable and non-scalarizable obligations are separated;

(iii) every atom has exactly one canonical audit status;

(iv) specialized statuses of Appendices B–E embed without changing their mathematical meaning;

(v) theorem-level use is controlled by claim-use preflight;

(vi) dependency registries are well-founded and a pass is downward closed over active dependencies;

(vii) blocked theorem applications remain distinct from rejected prerequisites;

(viii) status aggregation is commutative, associative, idempotent, and non-compensatory;

(ix) hierarchical aggregation preserves the headline verdict and all diagnostic detail;

(x) failure routes preserve nonexclusive Type I–V classifications without collapsing `undefined` *into* `reject`*;*

(xi) readiness is the aggregate canonical status of a complete dependency-closed bundle;

(xii) complete Restart and atlas verdicts are conjunctions of their required subclauses;

(xiii) countable readiness requires a coverage certificate and universal dependency-complete status;

(xiv) repairs create new immutable verdict records;

(xv) Appendix G serializes but does not alter these semantics.

Proof. Items (i)–(ii) follow from Definitions 6.3, 6.8, and 6.9. Item (iii) is Definition 6.12. Item (iv) is Definitions 6.16–6.19. Item (v) follows from Definitions 6.23–6.25. Items (vi)–(vii) follow from Definitions 6.28, 6.29, and Proposition 6.33. Items (viii)–(ix) follow from Propositions 6.35 and 6.38, together with Theorem 6.37. Item (x) follows from Definition 6.40 and Proposition 6.42. Item (xi) is Theorem 6.49. Item (xii) follows from Propositions 6.51 and 6.52. Item (xiii) is Theorem 6.56. Item (xiv) is Proposition 6.58. Item (xv) follows from Definition 6.60 and Proposition 6.61. □

6.17 v20 canonical token sorts and cross-sort safety

Definition 6.70 (Canonical token-sort registry)**.** Every serialized token consumed by a v20 audit object has exactly one declared semantic sort. The registry includes at least:

(i) atomic audit status:

$$\texttt{pass}, \quad \texttt{reject}, \quad \texttt{undefined}, \quad \texttt{not\_invoked};$$

(ii) runtime comparison mode:

$$\texttt{exact}, \quad \texttt{metric\_gap\_certified}, \quad \texttt{not\_compared};$$

(iii) detector profile stage:

$$\texttt{windowed\_prethreshold}, \quad \texttt{repaired\_postthreshold}, \\ \texttt{kernel\_defect}, \quad \texttt{cokernel\_defect};$$

(iv) detector certificate status:

$$\texttt{detector\_certified}, \quad \texttt{detector\_failed}, \quad \texttt{detector\_pending};$$

(v) detector mathematical value:

$$\texttt{zero}, \quad \texttt{nonzero}, \quad \texttt{undefined};$$

(vi) terminal Type IV status, full/interior inspection status, aggregate transient status, source semantics, simplex-coherence mode, Restart-transition status, Restart zero-route status, manifest status, claim evidence status, and invocation role.

Equal character strings occurring in different sorts are not identified. Every serialized occurrence records its field name and semantic sort.

Proposition 6.71 (No cross-sort token substitution)**.** *A token of one semantic sort cannot discharge a field of another sort. In particular, a detector value, profile stage, comparison mode, source semantics, simplex mode, manifest status, or Restart status is not an atomic audit verdict. An unresolved cross-sort assignment makes the affected clause* `undefined`*; an evidenced inconsistent or concealed substitution rejects as a Type V governance failure.*

Proof. The fields have different source types and governing predicates. A value of one type supplies neither an inhabitant nor evidence for another. An inconsistent declaration additionally contradicts immutable manifest semantics. □

Definition 6.72 (v20 typed evidence record)**.** A typed evidence record contains:

$$\begin{aligned}&(\text{EvidenceID}, \text{ClauseID}, \text{evidence type}, \text{source}, \text{target},\\ &\ \text{theorem/policy}, \text{hypotheses}, \text{token sorts}, \text{transformations},\\ &\ \text{artifacts}, \text{hashes}, \text{versions}, \text{atomic status}).\end{aligned}$$

A bare numerical value, detector value without certificate status, bare isomorphism symbol, comparison mode, or citation without claim-use preflight is not a complete evidence record.

6.18 v20 specialized status embeddings

Definition 6.73 (Two-axis detector status embedding)**.** The Appendix B detector certificate status and mathematical value remain independent axes. When a detector obligation is required, its certificate axis embeds by

$$\begin{aligned}\texttt{detector\_certified} &\mapsto \texttt{pass},\\ \texttt{detector\_failed} &\mapsto \texttt{reject},\\ \texttt{detector\_pending} &\mapsto \texttt{undefined}.\end{aligned}$$

The mathematical value `zero`, `nonzero`, or `undefined` does not itself map to an atomic verdict.

For a required zero clause:

(i) certified detector with value `zero` passes the detector-based zero atom;

(ii) certified detector with value `nonzero` rejects that zero atom and supplies a Type I witness at a topological stage or the applicable defect witness at a defect stage;

(iii) failed or pending certification cannot be repaired by a favorable value and yields respectively `reject` or `undefined`;

(iv) detector semantics outside immutable scope map to `not_invoked` only through a justified exclusion.

Remark 6.74 (Detector value is consumable only after certification)**.** A sampled value of zero with incomplete applicability, endpoint, family, coverage, stage, or completeness evidence is not a zero proof. A nonzero value from an inapplicable or ill-typed detector is likewise not a typed obstruction witness.

Definition 6.75 (Transient-defect status embedding)**.** For a required Appendix D aggregate long-transient clause:

$$\texttt{certified\_absent} \mapsto \texttt{pass}, \qquad \texttt{certified\_present} \mapsto \texttt{reject}, \qquad \texttt{undefined} \mapsto \texttt{undefined}.$$

The status `not_invoked` remains canonical `not_invoked` only under a justified scope decision. The detector component `full_profile` or `interior_noncofinal`, the defect threshold, window, collapse threshold, and monitored degrees are part of the clause identity.

Definition 6.76 (v20 Overlap Gate status embedding)**.** The Appendix E pairwise Overlap Gate status embeds identically:

$$\texttt{pass} \mapsto \texttt{pass}, \quad \texttt{reject} \mapsto \texttt{reject}, \quad \texttt{undefined} \mapsto \texttt{undefined}, \quad \texttt{not\_invoked} \mapsto \texttt{not\_invoked}.$$

The runtime comparison mode is stored separately and never substitutes for this atomic status.

Definition 6.77 (Restart zero-route embedding)**.** When a Restart step consumes a repaired-zero predicate, the target-zero route is a required independent atom with exactly one specialized status:

$$\texttt{not\_invoked}, \qquad \texttt{undefined}, \qquad \texttt{certified}, \qquad \texttt{failed}.$$

Its canonical embedding is

$$\texttt{certified} \mapsto \texttt{pass}, \qquad \texttt{failed} \mapsto \texttt{reject}, \qquad \texttt{undefined} \mapsto \texttt{undefined}.$$

The status `not_invoked` is permitted only when the Restart claim does not consume target zero. A certified κ_k-inequality does not certify this atom.

6.19 Coverage atoms and the five-layer obligation architecture

Definition 6.78 (Coverage-obligation taxonomy)**.** The following are distinct non-scalar obligations:

(i) *registry coverage*: every obligation in a finite or countable audit scope occurs in the registry;

(ii) *retained-bar coverage*: every globally retained bar has a detector witness in the declared local atlas;

(iii) *boundary-bridge coverage*: every globally retained interval crossing a partition boundary has a witness in a declared bridge or another certified coverage window;

(iv) *artifact closure*: every referenced artifact and transitive interpretation dependency is immutable and resolvable;

(v) *release-document coverage*: every mandatory document role and release clause occurs in the advertised release registry.

A certificate for one coverage type does not discharge another.

Definition 6.79 (Five-layer obligation architecture)**.** Every v20 readiness graph is organized into the following typed layers:

(L1) *mathematical layer*: repaired profiles, Ext clauses, comparison morphisms, defect profiles, exact equalities, reduction and descent theorems;

(L2) *detector layer*: detector stage, applicability, completeness, witness family, endpoint convention, value, and certification;

(L3) *atlas/transition layer*: overlap, simplex, retained-bar coverage, continuation, Restart, boundary bridge, and local-to-global interfaces;

(L4) *readiness layer*: dependency closure, atomic statuses, non-compensatory aggregation, failure routing, and bundle verdicts;

(L5) *manifest/release layer*: claim use, token sorts, artifact closure, schema conformance, replayability, repair history, and release acceptance.

Edges may skip layers only when a registered theorem or policy explicitly authorizes the dependency. No higher layer manufactures missing lower-layer evidence.

Proposition 6.80 (Layered non-compensation)**.** *A passing detector, atlas, readiness, manifest, or release-layer atom cannot repair an absent, undefined, or rejected prerequisite in a lower layer. Likewise, a true mathematical clause does not produce an auditable pass until its required upper-layer identity, dependency, and manifest obligations are satisfied.*

Proof. The first assertion follows from dependency-complete pass and Theorem 6.37. The second follows because audit pass is a conjunction of mathematical and governance obligations rather than mathematical truth alone. □

6.20 v20 readiness bundles

Definition 6.81 (Detector readiness bundle)**.** A detector readiness bundle contains:

(i) exact source profile identity and authorized detector stage;

(ii) detector mode, information class, endpoint convention, witness set, family, and applicability record;

(iii) soundness and, when a zero conclusion is consumed, completeness;

(iv) detector certificate status and independent mathematical value;

(v) every metric-safe transport dependency when the detector conclusion crosses a comparison or hard deletion;

(vi) artifact identities, theorem locators, claim-use preflight, and manifest fields.

Its readiness status is the aggregate status of these atoms.

Definition 6.82 (Detector-atlas readiness bundle)**.** A detector-atlas readiness bundle contains:

(i) every local detector readiness bundle;

(ii) exact local restrictions from a declared global profile;

(iii) the exact or metric-safe local interface branch;

(iv) pairwise and higher-simplex records in their declared exact-cocycle or metric-reference modes;

(v) retained-bar coverage and, when needed, boundary-bridge coverage;

(vi) terminal and long-transient participation atoms;

(vii) complete registry coverage, artifact closure, and manifest fields.

Pairwise local compatibility without the global profile and coverage atoms does not make this bundle pass.

Definition 6.83 (Global repaired-zero readiness bundle)**.** A global repaired-zero claim has a passing readiness bundle exactly when:

(i) the global pre-threshold profile and covered range are identified;

(ii) every local restriction and observed local profile is typed;

(iii) every local zero detector is certified and complete;

(iv) every exact or metric-safe zero-transport branch passes;

(v) retained-bar coverage passes;

(vi) every consumed overlap/simplex clause passes;

(vii) no required terminal, transient, artifact, claim-use, or manifest atom rejects or remains undefined;

(viii) the detector-atlas local-to-global theorem is admissibly invoked.

A family of local zero values without these dependencies is `undefined`, not a global zero proof.

Definition 6.84 (v20 complete Restart bundle)**.** When a Restart step consumes only reserve inheritance, the inherited complete Restart bundle applies. When it additionally consumes target repaired zero, the v20 complete bundle also requires:

(i) a passing target-zero route atom;

(ii) a passing target detector or zero-transport subbundle;

(iii) boundary-bridge coverage when the claimed predicate is global rather than target-local;

(iv) stage, threshold, endpoint, source, target, and artifact consistency.

The reserve and zero-route subbundles are independently mandatory.

Definition 6.85 (Finite-to-infinite reduction readiness bundle)**.** A finite-to-infinite reduction claim contains:

(i) the actual directed system and transition morphisms;

(ii) the finite witness or stabilization index;

(iii) the genuine colimit or other theorem-authorized source semantics;

(iv) a compatible apex cocone and induced comparison morphism;

(v) apex agreement and every reduction-theorem hypothesis;

(vi) the resulting terminal, transient, and full-isomorphism consequences actually claimed;

(vii) claim-use, artifact-closure, replay, and manifest atoms.

An observed finite plateau, repeated finite barcode, or implementation apex is not a reduction certificate.

Definition 6.86 (Object-descent readiness bundle)**.** An object-level descent claim contains:

(i) the declared category or higher category and restriction functors;

(ii) actual local objects and transition equivalences;

(iii) exact cocycle or higher coherence;

(iv) an effective-descent theorem with discharged hypotheses;

(v) the constructed descended object and comparison artifacts;

(vi) independent certificate-atlas and manifest clauses.

Metric-reference coherence, record-level atlas readiness, or global-zero readiness does not substitute for this bundle.

Definition 6.87 (v20 Foundation readiness bundle)**.** The v20 Foundation readiness bundle is the inherited Foundation bundle together with every applicable:

(i) token-sort and typed-evidence atom;

(ii) detector and detector-atlas readiness bundle;

(iii) transient-defect bundle;

(iv) global repaired-zero bundle;

(v) v20 complete Restart bundle;

(vi) finite-to-infinite reduction bundle;

(vii) object-descent bundle;

(viii) artifact-closure, semantic-replay, schema-migration, and release-acceptance atom.

Theorem 6.88 (v20 certificate-readiness criterion)**.** *A v20 Foundation bundle has status* `pass` *if and only if:*

(i) its immutable registry and token-sort assignments are complete and well-founded;

(ii) claim-use preflight passes;

(iii) every required specialized status is validly embedded;

(iv) every required atom has a dependency-complete pass;

(v) every excluded atom has justified `not_invoked` *status;*

(vi) every applicable detector, atlas, transient, Restart, reduction, and descent bundle passes;

(vii) artifact closure and semantic replay pass;

(viii) the canonical manifest is complete, consistent, and schema-conformant;

(ix) the aggregate verdict is `pass`*.*

Proof. The forward direction follows from the meaning of a v20 pass and dependency-complete evidence. For the reverse direction, all required atoms normalize to `pass`, every excluded atom has a valid exclusion, and the non-compensatory aggregate is therefore `pass`. □

6.21 Artifact closure, schema migration, and semantic replay

Definition 6.89 (Artifact closure)**.** A manifest has artifact closure when:

(i) every referenced artifact has one immutable identifier, type, provenance, version, and content hash;

(ii) every transitive dependency needed to interpret a consumed artifact is referenced or belongs to the registered theory release;

(iii) no identifier is dangling, ambiguous, mutable under the same hash claim, or resolved differently by two fragments;

(iv) no undeclared artifact replacement occurs after run identity is fixed.

Artifact closure supports replayability but is not mathematical proof.

Definition 6.90 (Semantic schema-migration record)**.** A migration records source and target schema identifiers, a total map on every consumed field, token-sort preservation, status and scope preservation, explicit defaults only for newly out-of-scope fields, semantic-difference evidence, artifact hashes, and proof that verdict reconstruction at the old supported strength is preserved. A change in theorem strength, evidence, scope, or claim use creates a new run rather than a schema-only migration.

Proposition 6.91 (No lossy migration of theorem evidence)**.** *A migration that drops a required hypothesis, dependency, token sort, lower-level status, missing-artifact marker, scope boundary, or forbidden stronger reading cannot preserve audit equivalence. The migrated record is incomplete or inconsistent until repaired by a new evidence-backed run.*

Proof. The omitted field participates in clause meaning or verdict reconstruction. Deleting it changes or prevents reconstruction of the consumed claim. □

Definition 6.92 (Computational and semantic replayability)**.** A run is computationally replayable when artifact closure permits an independent verifier to reconstruct every deterministic profile, detector, ledger, diagnostic, dependency, and serialization computation. It is semantically replayable when the verifier can additionally reconstruct each clause's source type, governing claim, hypotheses, token sort, specialization map, dependency transition, and headline aggregation. Every consumed machine-produced verdict requires semantic replayability for a Core pass.

Proposition 6.93 (Replay invariance under fixed inputs)**.** *Two semantic replays using the same immutable artifacts, token registry, claim-register snapshot, schema semantics, and verifier version yield the same atomic and headline verdicts. A disagreement evidences an implementation, schema, artifact, or manifest inconsistency and blocks pass.*

Proof. Under fixed inputs, every evidence predicate, dependency transition, specialized embedding, normalization, and aggregation operation is deterministic. □

6.22 v20 failure routing and release acceptance

Definition 6.94 (v20 extended failure routing)**.** The inherited Type I–V routes remain nonexclusive, with the following v20 readings:

(i) Type I includes a certified complete topological detector with nonzero value for a required repaired-zero clause;

(ii) Type II remains a nonzero eligible categorical obstruction and does not include missing eligibility or a failed detector;

(iii) Type III includes failed or undefined detector stage, applicability, soundness, completeness, exact/metric comparison, simplex coherence, retained-bar coverage, transient clause, target-zero route, finite-to-infinite reduction, object descent, or Return theorem;

(iv) Type IV remains a valid semantically authorized tower comparison with nonzero terminal-generic diagnostic;

(v) Type V includes cross-sort substitution, silent detector-stage relabeling, unsupported zero/pass, lossy schema migration, artifact inconsistency, silent claim supersession, unauthorized scope exclusion, or proposer output treated as verifier evidence.

Mere missing mathematical evidence first yields an undefined Type III clause. Type V is added when the governance record is incomplete, inconsistent, concealed, or semantically false.

Definition 6.95 (Release-acceptance bundle)**.** A document set advertised as a complete v20 release has a release-acceptance bundle containing:

(i) mandatory document-role and version coverage;

(ii) canonical Claim Register and supersession closure;

(iii) all mandatory Foundation, interface, manifest, and reproducibility artifacts;

(iv) token-sort, schema, artifact-closure, and semantic-replay clauses;

(v) absence of unresolved mandatory rejected or undefined atoms;

(vi) the v20 Foundation readiness verdict.

The release passes exactly when every mandatory clause passes. A mathematically strong chapter does not compensate for an incomplete release registry or inconsistent manifest.

Definition 6.96 (Proposer–verifier re-entry protocol)**.** A human, Search-side, Platform-side, Companion-side, algorithmic, or AI proposal may influence a Core verdict only after:

(i) normalization into a declared Core type;

(ii) claim-status and invocation-role classification;

(iii) production of required mathematical and artifact evidence;

(iv) comparison-mode, ledger, detector-stage, and threshold-gap typing;

(v) semantic-difference or transformation-preservation evidence when the proposal was translated, summarized, formalized, or migrated;

(vi) manifest registration, token-sort validation, dependency closure, artifact closure, and immutable identity;

(vii) execution of every applicable verifier clause.

Proposal confidence, ranking, or apparent plausibility has no direct status effect.

Theorem 6.97 (Appendix F v20 governance-kernel package)**.** *Within the v20 conservative extension:*

(i) the complete inherited v19 obligation, status, dependency, aggregation, readiness, countable-registry, repair, and serialization package remains valid at its original strength;

(ii) token sorts prevent cross-type status substitution;

(iii) detector certificate status and detector mathematical value remain independent axes;

(iv) terminal Type IV, long-transient, Overlap, Restart transition, and Restart zero-route statuses embed without semantic collapse;

(v) registry, retained-bar, boundary-bridge, artifact, and release coverage are distinct obligations;

(vi) the five-layer architecture prevents upper-layer evidence from manufacturing lower-layer proof;

(vii) detector, detector-atlas, global-zero, complete Restart, finite-to-infinite, and object-descent readiness are distinct bundles;

(viii) a v20 pass is a dependency-closed, non-compensatory conjunction of all applicable bundles;

(ix) artifact closure, lossless schema migration, and semantic replay are mandatory for consumed machine-produced verdicts;

(x) v20 failure routing preserves Type I–V detail;

(xi) release acceptance is an independent conjunction and is not inferred from mathematical strength alone;

(xii) proposer output affects a verdict only through the typed re-entry protocol.

Proof. Item (i) is Theorem 6.69. Items (ii)–(iv) follow from Definitions 6.70–6.77. Items (v)–(vi) are Definitions 6.78 and 6.79. Items (vii)–(viii) follow from Definitions 6.81–6.87 and Theorem 6.88. Item (ix) follows from Definitions 6.89, 6.90, and 6.92. Items (x)–(xii) are Definitions 6.94, 6.95, and 6.96. □

6.23 Explicit exclusions

Appendix F does not assert:

(i) that an obligation registry proves the atoms it lists;

(ii) that a complete manifest proves the mathematical claims it records;

(iii) that numerical slack repairs missing non-scalar evidence;

(iv) that `undefined` means zero, false, harmless, or not invoked;

(v) that `not_invoked` is valid without an explicit immutable scope decision;

(vi) that one passing atom cancels a rejected or required undefined atom;

(vii) that a rejected hypothesis proves every dependent conclusion false;

(viii) that a certified Restart transition implies a complete Restart pass;

(ix) that pairwise overlap passes imply higher Čech coherence;

(x) that zero terminal-generic Type IV diagnostics imply full comparison-map isomorphism;

(xi) that a finite prefix certifies a countable registry;

(xii) that a repaired run retroactively changes an earlier immutable verdict;

(xiii) that schema conformance replaces proof;

(xiv) that Search, AI, ranking, confidence, or proposer output has verifier authority;

(xv) that certificate readiness creates a global object or an external-domain theorem;

(xvi) that a detector value is an atomic status or is consumable without a certified detector record;

(xvii) that local detector zero implies global repaired zero without a declared global profile and retained-bar coverage;

(xviii) that registry coverage, retained-bar coverage, boundary-bridge coverage, artifact closure, and release coverage are interchangeable;

(xix) that a passing κ-Restart reserve clause certifies target zero;

(xx) that terminal Type IV zero implies long-transient absence or full isomorphism;

(xxi) that exact or metric source-profile coherence transports kernel or cokernel defect profiles;

(xxii) that a complete record atlas proves object-level descent;

(xxiii) that a lossy schema migration preserves theorem evidence;

(xxiv) that computational replay alone validates theorem semantics;

(xxv) that a passing Foundation bundle automatically makes an advertised document set a complete release;

(xxvi) that proposer, Search, Platform, Companion, or AI output has direct verifier authority.

6.24 Provenance and status classification

This appendix integrates the representative and detector obligations of Appendix B, the eligible higher-Ext obligations of Appendix C, the terminal and transient Type IV obligations of Appendix D, and the exact/-metric overlap, detector-atlas, coverage, descent-boundary, and Restart obligations of Appendix E under the canonical audit semantics of Chapter 7. It supplies the semantic input for the Appendix G manifest schema. The inherited v19 package remains the lower baseline; the v20 governance kernel is the separately labeled conservative extension above.

Foundation definitions. Definitions 6.3–6.6, 6.8, 6.9, 6.12, 6.14, 6.15, 6.16–6.22, 6.23–6.25, 6.27–6.31, 6.34, 6.36, 6.40, 6.41, 6.43–6.48, 6.54, 6.57, 6.60.

Foundation propositions and theorems. Propositions 6.10, 6.26, 6.30, 6.33, 6.35, 6.38, 6.42, 6.51, 6.52, 6.55, 6.58, 6.61, Theorems 6.37, 6.49, 6.56, 6.69.

Operational cautions. Remarks 6.1, 6.2, 6.7, 6.11, 6.13, 6.21, 6.32, 6.39, 6.50, 6.53, 6.59, 6.62.

v20 governance-kernel extension. Definitions 6.70, 6.72, 6.73, 6.75, 6.76, 6.77, 6.78, 6.79, 6.81–6.87, 6.89, 6.90, 6.92, 6.94, 6.95, 6.96; Propositions 6.71, 6.80, 6.91, 6.93; Theorems 6.88, 6.97.

Explicit exclusions. Subsection 6.23.

7 Appendix G: Canonical Manifest Schema, Proof Objects, and Replayability

7.1 Scope, dependency position, and canonical boundary

This appendix is the canonical manifest layer of the v20.0.0 Foundation Appendices. The complete v19 schema below remains the immutable lower baseline. It consumes the mathematical and record-theoretic disciplines of Appendices A–E and the typed obligation and governance-kernel semantics of Appendix F. Its purpose is not to invent a fifth verdict system, but to serialize without semantic loss the obligation atoms, token sorts, dependency relations, evidence, statuses, failure routes, coverage certificates, artifact closure, schema migration, semantic replay, release history, and aggregate readiness fixed by Appendix F.

Chapter 1 supplies the foundational Minimal Manifest requirement. Chapter 7 supplies the canonical typed audit semantics and Core sovereignty contract. Appendix F fixes the status meanings and dependency calculus. The present appendix fixes the canonical logical field schema, conformance rules, identity, immutability, and replayability requirements.

Appendix CR-v20 supplies claim-status governance, Appendix CM-v20 supplies Core micro-theorem references, and Appendix TB-v20 supplies toy-bridge references. Whenever such material is invoked, the manifest records

$$\mathrm{ClaimRef}, \qquad \mathrm{CMRef}, \qquad \mathrm{TBRef},$$

together with the exact permitted use and every discharged hypothesis.

Remark 7.1 (Appendix G is logically machine-readable, not format-prescriptive)**.** This appendix fixes canonical logical fields, field types, conditional requiredness, identity constraints, and replayability conditions. It does not mandate one concrete syntax such as JSON, YAML, CSV, SQL, or a particular proof-store layout. A concrete encoding is admissible only if it preserves the logical schema bijectively enough for independent reconstruction.

Remark 7.2 (Appendix G does not enlarge the mathematical Core)**.** This appendix creates no new persistence, derived, tower, Restart, overlap, or external-domain theorem. It governs admissible records. A complete schema is not a proof, and serialization cannot strengthen the status of a claim, micro-theorem, toy bridge, or proposed external interface.

Remark 7.3 (Relation to Appendix CR-v20)**.** A registered claim is usable only through the claim-use preflight semantics of Appendix F. Registration is not proof. The manifest records the claim identifier, snapshot version, registered status, permitted use, discharged conditions, and non-claim boundary.

Remark 7.4 (Relation to Appendix CM-v20)**.** A Core micro-theorem reference records its identifier, statement version, hypotheses, hypothesis evidence, and exact invocation site. Citation does not discharge hypotheses and does not turn an internal Core lemma into an external-domain theorem.

Remark 7.5 (Relation to Appendix TB-v20)**.** A toy-bridge reference records the toy source category, realization route, repaired profiles, actual comparison artifact, diagnostic conclusion, and explicit toy-scope boundary. No toy theorem is promoted to a problem-specific bridge theorem without a separately proved transport theorem.

7.2 Semantic authority: Appendix F first, Appendix G second

Definition 7.6 (Semantic-preserving serialization)**.** A serialization of an Appendix F obligation registry is *semantic-preserving* if it preserves:

(i) every obligation identity and immutable scope;

(ii) requiredness and justified scope exclusion;

(iii) the dependency relation, closure references, and well-founded rank data;

(iv) specialized source status and canonical atomic status;

(v) evidence-policy version, artifact identity, and theorem/claim references;

(vi) scalar ledger data only for scalarizable obligations;

(vii) all Type I–V failure routes and blocker identifiers;

(viii) predecessor, revision, and repair-record links;

(ix) registry coverage, consistency, replayability, and aggregate verdict.

Proposition 7.7 (Appendix F semantics are authoritative)**.** *No Appendix G field, default, omission rule, encoding convention, or implementation shortcut may alter the status meanings, dependency closure, non-compensation rule, readiness criterion, or failure routing fixed by Appendix F, in particular Definition 6.60 and Proposition 6.61.*

Proof. An encoding that changes one of those meanings represents a different obligation registry or verdict while reusing the same identifiers. That violates immutable run semantics and schema conformance. □

Remark 7.8 (Schema completeness is not proof)**.** A schema may be complete and replayable while recording a rejected run. Conversely, a mathematically true but unserialized clause does not yield an auditable pass. Appendix F fixes the semantics; Appendix G fixes how those semantics are exposed for audit.

7.3 Audit identity, schema identity, and immutable scope

Definition 7.9 (Audit run identity)**.** An audit run identity is the tuple

$$\mathfrak{I} = (\mathrm{RunID}, \mathrm{ManifestID}, \mathrm{SchemaID}, \mathrm{Version}_{\mathrm{theory}}, \mathrm{Version}_{\mathrm{claim}}, \mathrm{Version}_{\mathrm{artifact}}, \mathrm{RevisionID}).$$

It identifies the exact theory release, canonical schema release, claim-register snapshot, artifact namespace, and revision under which all fields are interpreted.

Definition 7.10 (Immutable audit scope)**.** The immutable scope of a run records at minimum:

(i) the object or object family under audit;

(ii) monitored degrees, windows, thresholds, and terminal modes;

(iii) invoked clauses and justified non-invoked clauses;

(iv) collapse policy and evaluation order;

(v) claim, theorem, micro-theorem, and toy-bridge identifiers used;

(vi) immutable identifiers of all source, profile, representative, realization, comparison, diagnostic, continuation, and atlas artifacts.

Changing any item creates a new run or revision; it does not silently mutate the old verdict.

Definition 7.11 (Schema version boundary)**.** A schema version boundary is a declared change in field names, field types, requiredness rules, normalization, or conformance semantics. A record from another schema version is admissible only through an explicit, auditable migration map that preserves Appendix F semantics.

Proposition 7.12 (Identity collision is manifest inconsistency)**.** *If two records use the same* RunID *or* ManifestID *but assign different immutable scope, theory version, schema version, artifact identity, or revision semantics, then the manifest is inconsistent and routes to Type V rejection.*

Proof. One immutable identifier cannot denote two distinct runs under the canonical single-run semantics. The conflict is therefore an internal governance contradiction rather than mere missing data. □

7.4 Proof objects and certificate records

Definition 7.13 (Proof object)**.** A *proof object* is an abstract mathematical-and-audit entity intended to support a declared claim. It contains or references the mathematical evidence, typed obligation registry, dependencies, ledger records, diagnostics, artifacts, claim-use preflight, and verdict data required by the applicable Core clauses.

Definition 7.14 (Certified proof object)**.** A proof object is *certified* for a declared scope only if:

(i) its Appendix F foundation readiness bundle has aggregate status `pass`;

(ii) its manifest is complete, consistent, conformant, and replayable;

(iii) every required obligation in the dependency closure has canonical status `pass`;

(iv) every justified `not_invoked` clause has a valid scope-exclusion record;

(v) no hidden registry tail or unresolved blocker remains.

Manifest completeness alone does not certify a proof object.

Definition 7.15 (Certificate record)**.** A *certificate record* is the canonical manifest-level representation of one proof object or one scope-relative proof-object fragment. It serializes evidence and verdict semantics but is not identical to the underlying mathematical object.

Remark 7.16 (Proof object vs. certificate record)**.** The proof object is the evidenced entity; the certificate record is its audit-readable representation. A record may be complete and record a rejection. A record may also be incomplete even when an unrecorded mathematical argument happens to be true.

Definition 7.17 (Local certificate record)**.** A *local certificate record* is attached to one declared window or overlap stratum and contains the local obligation subregistry, repaired evaluation data, local ledger, typed diagnostics, local verdicts, dependencies, and artifact references.

Definition 7.18 (Global certificate record)**.** A *global certificate record* is a record assembled from local records through a declared record-level Čech atlas, Restart sequence, finite union, or other proved certificate-record composition rule. It retains all local identities, simplex records, transition records, budgets, routes, and dependencies used in the assembly.

Remark 7.19 (Global record is not object-level gluing)**.** A globally coherent certificate atlas does not construct a global filtered complex, persistence module, derived object, sheaf, or external-domain solution. Object-level descent requires separate mathematical hypotheses.

7.5 Canonical manifest object and field cardinalities

Definition 7.20 (Canonical manifest object)**.** A canonical manifest is the typed record

$$\begin{aligned}\mathsf{Manifest} = \big(&\mathsf{Identity}, \mathsf{Scope}, \mathsf{Objects}, \mathsf{Windows}, \mathsf{Policies}, \mathsf{Registry}, \mathsf{Evidence}, \mathsf{Artifacts},\\ &\mathsf{Ledger}, \mathsf{DomainRecords}, \mathsf{References}, \mathsf{Verdicts}, \mathsf{Routes}, \mathsf{Repairs}, \mathsf{Replay}\big).\end{aligned}$$

Each component is itself typed and versioned. Conditional components are absent only through a justified scope-exclusion record; silent omission is not permitted.

Definition 7.21 (Canonical cardinality markers)**.** For schema statements, the markers

$$[1], \qquad [0..1], \qquad [0..*], \qquad [1..*]$$

mean respectively exactly one, optional singleton, finite or countable family possibly empty, and nonempty finite or countable family. Countable families require a coverage certificate and replayable generator.

Definition 7.22 (Conditional requiredness)**.** A field is *conditionally required* when an invocation predicate is true. The record stores both the predicate and its evidence. If the predicate is false, the field is absent only together with a justified `not_invoked` scope-exclusion record. If the predicate is true and the field is missing, the relevant obligation is `undefined`.

Proposition 7.23 (No silent default theorem)**.** *No omitted representative, comparison morphism, eligibility certificate, continuation artifact, claim status, ledger upper-bound theorem, or coherence record may be reconstructed from a default value.*

Proof. Each listed item is a non-scalarizable proof obligation. A default would manufacture evidence and violate Appendix F non-compensation. □

7.6 Minimal Manifest Axiom

Axiom 7.1 (Minimal Manifest Axiom)**.** *A proof object may support an auditable Core verdict only if its manifest contains, at minimum:*

(i) audit, manifest, schema, theory, claim-register, artifact-root, and revision identities;

(ii) immutable scope, declared objects, windows, thresholds, policies, and monitored degrees;

(iii) the full required obligation registry, dependency relation, well-founded rank, and coverage certificate;

(iv) every specialized source status and canonical atomic status;

(v) evidence-policy versions and immutable evidence/artifact references;

(vi) scalar ledger, scalarization, actual metric upper-bound, gap, and reserve records where applicable;

(vii) representative, realization, eligibility, Type IV, overlap, Restart, and atlas records when invoked;

(viii) claim-use, micro-theorem, and toy-bridge preflight records when invoked;

(ix) every blocker and every applicable Type I–V failure route;

(x) aggregate bundle verdict, manifest status, readiness status, consistency, and replayability fields;

(xi) predecessor and repair-record identifiers for revised runs.

Remark 7.24 (Meaning of minimality)**.** Minimality means necessary for the declared Core scope, not small in byte size. Concrete systems may add runtime logs, timestamps, proof-assistant traces, signatures, or redundant indexes, but may not omit a conditionally required field.

Proposition 7.25 (Necessity of manifest data)**.** *If a required field of Axiom 7.1 is absent or not reconstructible, then the manifest is incomplete and the corresponding audit clause is* `undefined`*, unless another evidenced clause already forces aggregate rejection.*

Proof. The missing field prevents reconstruction of at least one required obligation or its evidence. Appendix F maps missing required evidence to `undefined`; the aggregate precedence rule may still report `reject` if a separate clause is evidenced false, while retaining the undefined detail. □

7.7 Obligation registry serialization and dependency closure

Definition 7.26 (Obligation atom record)**.** For every obligation atom, the manifest stores exactly one primary atom record containing:

(i) ObligationID, owner, immutable scope, and obligation kind;

(ii) requiredness and scope-exclusion state;

(iii) evidence-policy identifier and version;

(iv) immediate dependency identifiers and closure reference;

(v) well-founded rank or finite acyclic-graph certificate;

(vi) specialized source status and canonical atomic status;

(vii) evidence references, blockers, routes, predecessor, and repair links.

Definition 7.27 (Dependency record)**.** A dependency record is a directed edge

$$\mathrm{ObligationID}(\mathfrak{p}) \longrightarrow \mathrm{ObligationID}(\mathfrak{o})$$

meaning that $\mathfrak{p}$ is an immediate prerequisite of $\mathfrak{o}$. The manifest stores the edge source, target, dependency kind, evidence for the dependency, and rank check.

Definition 7.28 (Registry coverage certificate)**.** A registry coverage certificate proves that the serialized atom family is exactly the declared finite registry, or for a countable registry, that an immutable generator enumerates every required atom without hidden tail. It includes the generator version, index type, rank rule, and universal status certificate.

Definition 7.29 (Closure reference)**.** A closure reference identifies a reproducible computation or certificate for the transitive dependency closure of an atom or bundle. It does not replace the immediate dependency edges.

Proposition 7.30 (Finite prefix cannot certify a countable registry)**.** *A manifest containing only a finite prefix of a countable obligation registry cannot support a pass for the full registry without a valid coverage certificate and universal status certificate.*

Proof. An unchecked tail may contain a required undefined or rejected atom. Finite-prefix success therefore does not determine the countable aggregate. □

7.8 Canonical statuses, classifications, and aggregate verdicts

Definition 7.31 (Canonical status field)**.** Every obligation atom has exactly one canonical atomic status

$$\mathsf{Status}(\mathfrak{o}) \in \{\texttt{pass}, \texttt{reject}, \texttt{undefined}, \texttt{not\_invoked}\},$$

with the meaning fixed by Appendix F. The field also stores the specialized source status, normalization rule, evidence reference, timestamp or sequence index if used, and responsible verifier version.

Definition 7.32 (Claim/use classification field)**.** The claim/use classification is a separate field whose values may include

$$\texttt{proved}, \texttt{conditional}, \texttt{operational}, \texttt{spec}, \texttt{deprecated}, \texttt{rejected}, \texttt{unknown},$$

or declared layer tags such as Core, Interface, Bridge Candidate, Bridge Program, Toy Bridge, Search Artifact, Companion Template, and Non-Claim. These are not atomic audit statuses.

Definition 7.33 (Aggregate verdict record)**.** An aggregate verdict record stores:

(i) the required atom set and justified scope-excluded set;

(ii) the normalized status of every included atom;

(iii) the aggregation rule and Appendix F version;

(iv) the headline aggregate $\mathrm{Agg}(\mathfrak{R})$;

(v) all retained lower-level statuses, blockers, and routes.

The headline does not erase an undefined atom merely because another atom rejects.

Definition 7.34 (Manifest status field)**.** The manifest itself has one of the source statuses

$$\texttt{complete}, \quad \texttt{incomplete}, \quad \texttt{inconsistent}.$$

Under the Appendix F embedding, these map respectively to

$$\texttt{pass}, \quad \texttt{undefined}, \quad \texttt{reject}.$$

This status is one obligation atom inside the full readiness bundle, not a replacement for the bundle verdict.

Proposition 7.35 (Status/classification separation)**.** *A strong claim classification cannot compensate for an undefined or rejected audit status, and an audit pass cannot promote a claim beyond its registered classification.*

Proof. The fields have different types and govern different questions. Appendix F non-compensation and claim-use preflight prohibit cross-type substitution. □

7.9 Declared objects and windows

Definition 7.36 (Declared object data)**.** Declared object data include object identity, ambient type, source provenance, monitored degrees, coefficient field, constructibility or finiteness conditions, and immutable source artifacts. For an interface run, both the external object and its projected Core object are recorded together with the interface version.

Definition 7.37 (Declared window data)**.** Declared window data include window identifiers, endpoints, endpoint convention, finite or right-unbounded terminal mode, MECE or overlap-cover role, cover/partition identifier, and all nonempty recorded overlap strata.

Definition 7.38 (Window identifier)**.** A window identifier is immutable within one run and links all evaluations, representatives, diagnostics, ledgers, statuses, simplex records, continuation records, and artifacts for that window.

Remark 7.39 (Window declarations and Appendix E)**.** The MECE/overlap distinction, simplex coherence, Restart, and record-level gluing semantics are inherited from Appendix E. The historical label name of this remark is retained, but the governing main-body chapter is Chapter 6 in v19.

Proposition 7.40 (Window data are audit-critical)**.** *Without declared window and terminal-mode data, the repaired evaluation, threshold clearance, terminal-generic Type IV diagnostic, overlap comparison, and Restart scope are not determined.*

Proof. Each listed construction is window-relative. Omitting the window therefore leaves required mathematical inputs undefined. □

7.10 Threshold, collapse, and repaired evaluation records

Definition 7.41 (Threshold policy)**.** A threshold policy records the threshold value or indexed family, fixed-threshold or sweep mode, common-threshold reconciliation rules, stable-band claims, window dependence, and the threshold-policy version.

Definition 7.42 (Collapse policy)**.** A collapse policy records the permitted level of use, including

`barcode_level`, `representative_verified`, `tau_split`, `exact_admissible`,

together with every artifact and hypothesis required by the chosen level. The default is barcode-level hard deletion after window restriction.

Definition 7.43 (Fixed-threshold certificate)**.** A fixed-threshold record evaluates all declared clauses at one immutable $\tau > 0$, with gate-scoped profile families and clearance records.

Definition 7.44 (Threshold-sweep certificate)**.** A threshold-sweep record stores the threshold index set or band, artifact-coherent construction policy, pointwise records, bandwise quantifiers, and any claimed uniform reserve proof. Pointwise positivity is not serialized as a uniform positive infimum without separate evidence.

Remark 7.45 (Threshold policy and repaired collapse)**.** The threshold policy fixes barcode-level deletion semantics. It does not silently provide global exactness, Serre localization, or a strict filtered collapse functor.

Proposition 7.46 (Threshold and collapse data are audit-critical)**.** *Without threshold, policy version, evaluation order, and collapse status, the repaired profile and the set of licensed downstream operations are not reconstructible.*

Proof. Different thresholds and collapse statuses produce different profiles and different admissible theorem uses. □

Definition 7.47 (Evaluation order record)**.** An evaluation order record declares

$$\mathbf{P}_i(F) \longmapsto \mathsf{W}_W\mathbf{P}_i(F) \longmapsto T_\tau^{\mathrm{bar}}\big(\mathsf{W}_W\mathbf{P}_i(F)\big) = \mathrm{Eval}_{i,W,\tau}(F).$$

It stores the operator versions and every conditional commutation certificate if another route is used.

Definition 7.48 (Repaired evaluation record)**.** For each monitored triple (i, W, τ), a repaired evaluation record stores the pre-threshold profile, barcode or reconstructive profile artifact, thresholded profile, evaluation status, clearance data, and immutable artifact references.

Definition 7.49 (Overlap evaluation record)**.** An overlap evaluation record is computed directly on the common overlap stratum at the reconciled threshold. It includes both local-source identities, the common-window pre-threshold profiles, discrepancy certificate, clearance certificates, and post-threshold comparison.

Remark 7.50 (No hidden crop-collapse commutation)**.** A restriction of an already-collapsed larger-window profile cannot replace direct common-window evaluation without the window-adapted condition of Definition 1.60 and the conditional criterion of Proposition 1.61, or another explicitly proved exact-admissible route.

Proposition 7.51 (Evaluation records are audit-critical)**.** *A record without reconstructible repaired evaluations cannot determine the topological gate atom.*

Proof. The atom is the typed assertion $\mathrm{Eval}_{1,W,\tau}(F) = 0$ or its evidenced negation. Without the profile and artifacts, neither outcome is auditable. □

7.11 Representative, realization, and eligibility records

Definition 7.52 (Representative policy)**.** A representative policy records whether a filtered representative is invoked, its lifecycle status, canonical atomic status, monitored degrees, equivalence policy, representative theorem/artifact, and realization compatibility when required.

Definition 7.53 (Representative record)**.** A representative record serializes $C_{\tau,W}F$, the relation

$$\mathbf{P}_i(C_{\tau,W}F) \cong \mathrm{Eval}_{i,W,\tau}(F),$$

its proof artifact, construction provenance, equivalence class, status, blockers, and repair links.

Definition 7.54 (Eligibility record)**.** When the Ext^1-clause is invoked, the eligibility record stores the representative, realization functor or procedure, realized object, amplitude certificate, typed residual-edge extractor, edge-realization isomorphism, zero-compatibility, and the canonical clause status.

Remark 7.55 (No representative, no derived clause)**.** A missing representative or eligibility artifact makes the derived theorem-application atom `undefined`. It is not repaired by topological vanishing, numerical slack, or a claimed Ext value.

Proposition 7.56 (Representative and eligibility records are audit-critical when invoked)**.** *If a run invokes filtered or derived clauses, omission of the applicable representative or eligibility records prevents a dependency-complete pass.*

Proof. Those records are immediate prerequisites in the obligation registry. Appendix F dependency completeness therefore blocks the downstream application. □

7.12 Typed ledger and non-scalar obligations

Definition 7.57 (Ledger policy)**.** A ledger policy declares an ordered commutative monoid or, when fully justified, commutative quantale; each metric-bearing component; aggregation; bottleneck scalarization; actual upper-bound theorem; gate-scoped pre-threshold profile family; clearance; scalarized budget; and reserve.

Definition 7.58 (Budget component)**.** A budget component is a scalarizable metric discrepancy such as algorithmic, discretization, measurement, representative-choice, realization, comparison, crop, continuation, overlap, or Restart discrepancy. Each component stores semantic type, scope, provenance, value, and upper-bound evidence.

Definition 7.59 (Non-scalar obligation record)**.** A non-scalar obligation record covers representative existence, eligibility, actual morphisms, terminal tameness, claim status, threshold reconciliation, simplex coherence, continuation and κ-admissibility, target-local pass, manifest identity, and comparable proof obligations. Such a record has a typed status and evidence; it is not a numerical budget component.

Definition 7.60 (Total budget record)**.** A total budget record stores the typed metric ledger element, component decomposition, scalarization map, scalarized value, and theorem proving that the value dominates the actual gate discrepancy.

Definition 7.61 (Safety margin record)**.** A safety margin record stores the gate-scoped pre-threshold family, each clearance, the minimum gap, scalarized metric budget, reserve, and strict comparison result.

Remark 7.62 (Strict inequality)**.** The boundary equality between scalarized discrepancy and gap is not a pass. The manifest stores the exact comparison operator and does not round equality into positivity.

Proposition 7.63 (Ledger data are audit-critical)**.** *A numerical gate cannot pass from a ledger value alone; it also requires typed component provenance, scalarization, an actual metric upper-bound theorem, and a positive gate-scoped reserve.*

Proof. A number without a theorem connecting it to the consumed discrepancy is not evidence. The reserve must be computed from the relevant pre-threshold profiles, not from an unrelated global scalar. □

7.13 Type IV diagnostics and comparison artifacts

Definition 7.64 (Diagnostic policy)**.** A diagnostic policy records monitored degrees, windows, thresholds, terminal modes, source-profile provenance, actual profile-system morphisms, source-to-apex comparison morphisms, terminal-tameness certificates, kernel and cokernel defect profiles, and the canonical μ/ν diagnostics.

Definition 7.65 (Tower comparison artifact record)**.** A tower comparison artifact record serializes

$$\phi_{i,W,\tau} : \mathrm{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \mathrm{ApexProfile}_{i,W,\tau}(F_\infty),$$

including actual source and target objects, morphism identity, construction provenance, terminal-tameness, defect artifacts, and immutable hashes or equivalent integrity identifiers.

Definition 7.66 ($(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) = (0,0)$ record)**.** A $(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) = (0,0)$ record is permitted only when the Type IV specialized status is `certified_zero` and the canonical atomic status is `pass`. The numeric pair alone is insufficient without a valid comparison morphism and terminal-tameness evidence.

Definition 7.67 ($(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) \neq (0,0)$ record)**.** A $(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) \neq (0,0)$ record stores specialized status `obstructed`, atomic status `reject`, the nonzero degreewise diagnostics, defect artifacts, and Type IV failure route.

Definition 7.68 (Undefined Type IV record)**.** If Type IV is invoked but the comparison morphism, source provenance, terminal-tameness, or defect artifacts are missing, the record stores specialized status `undefined` and atomic status `undefined`; it must not serialize $(0,0)$ as a substitute.

Remark 7.69 (Terminal-generic Type IV meaning)**.** The diagnostic semantics follow Appendix D. On bounded windows, cofinal bars reaching the right endpoint are counted even though they are globally finite. Therefore "infinite-bar only" is not the canonical bounded-window meaning. Zero terminal diagnostics do not imply full comparison isomorphism without the no-interior-defect condition.

Proposition 7.70 (Diagnostic and comparison-artifact data are audit-critical)**.** *A Type IV atom is not well formed without the actual comparison artifact and the terminal-generic diagnostic prerequisites of Appendix D.*

Proof. Kernel, cokernel, and terminal-generic dimensions are defined relative to that typed morphism and regime. Without them, the clause is undefined rather than zero. □

7.14 Overlap, Čech, Restart, and atlas records

Definition 7.71 (Simplex compatibility record)**.** For every recorded nonempty Čech simplex, the manifest stores the simplex identifier, common overlap window, face identifiers, common-threshold reconciliation, pre-threshold profiles, discrepancy and sharp clearance certificates, face compatibility, typed Type IV status, non-scalar coherence obligations, and artifact links.

Definition 7.72 (Restart transition record)**.** A Restart transition record stores source and target windows, source reserve, κ_k, η_k, the certified recurrence inequality, continuation artifact, transition substatus, target reserve, target-local status, and complete Restart verdict.

Definition 7.73 (Countable Restart record)**.** A countable Restart record additionally stores K_n, normalized losses, the reserve-safe series certificate, pointwise reserve conclusions, and any separately proved uniform lower bound. Ordinary unnormalized summability is not recorded as sufficient evidence.

Definition 7.74 (Atlas record)**.** An atlas record stores the cover identity, finite Čech depth, audit-local-finiteness certificate when countable, every required simplex record, face maps, coherence status, ledger exposure, coverage certificate, and the scope-limited conclusion "certificate atlas".

Proposition 7.75 (Pairwise records do not imply higher coherence)**.** *A manifest containing all pairwise overlap passes but missing a required higher-simplex record cannot support an atlas pass.*

Proof. Higher coherence is a distinct non-scalar obligation in Appendix E. It cannot be inferred from pairwise data or numerical slack. □

7.15 Gate and bundle verdict records

Definition 7.76 (Gate verdict)**.** A gate verdict is a typed bundle record, not merely a Boolean. It stores the gate scope, required atom set, justified non-invoked set, atomic statuses, aggregate status, blockers, reserves, failure routes, and dependency closure used to compute the headline verdict.

Definition 7.77 (B-Gate$^+$ verdict record)**.** A B-Gate$^+$ verdict record stores distinct atoms for repaired topological vanishing, eligible Ext use when invoked, Type IV status, metric reserve, claim-use preflight, and manifest conformance. The aggregate is computed by Appendix F; no passing atom cancels an undefined or rejected atom.

Definition 7.78 (Overlap Gate verdict record)**.** An Overlap Gate verdict record stores the common-window evaluation and sharp two-sided transfer atoms, threshold reconciliation, metric reserve, Type IV status, actual artifacts, and simplex/face obligations.

Definition 7.79 (Foundation readiness record)**.** A foundation readiness record serializes the Appendix F readiness bundle and its aggregate status. Readiness is not a fifth atomic status; it is the aggregate status of a declared required bundle.

Remark 7.80 (Verdicts are not scores)**.** A confidence, ranking, probability, heuristic score, or AI assessment may appear only as proposer provenance. It is never a canonical atomic status or gate verdict.

Proposition 7.81 (Gate verdicts are audit-critical)**.** *A record that stores only a headline verdict without the atomic statuses, dependency closure, blockers, and failure routes is not replayable and cannot support a canonical Core pass.*

Proof. The headline is a compression of typed detail. Without the detail, the verifier cannot reconstruct the aggregation or distinguish rejection from undefinedness. □

7.16 Claim, micro-theorem, and toy-bridge references

Definition 7.82 (Claim-register reference)**.** A claim-register reference stores ClaimID, claim-register snapshot, stable statement pointer, registered status, layer, permitted use, discharged conditions, forbidden use, exact invocation site, and artifacts.

Definition 7.83 (Micro-theorem reference)**.** A micro-theorem reference stores CMID, statement version, hypotheses, evidence for each hypothesis, exact invocation site, conclusion consumed, and non-claim boundary.

Definition 7.84 (Toy-bridge reference)**.** A toy-bridge reference stores TBID, toy source category, source object/tower, realization route, repaired profiles, actual comparison artifact, diagnostics, ledger assumptions, exact invocation site, and toy-scope boundary.

Definition 7.85 (Reference consistency check)**.** Reference consistency is the claim-use preflight that verifies version identity, registered classification, permitted strength, discharged conditions, and invocation-site compatibility for every reference.

Proposition 7.86 (Reference data are audit-critical when invoked)**.** *An invoked reference without complete preflight data makes the corresponding theorem-application atom* `undefined`*; a forbidden or status-inconsistent use makes the claim-use atom* `reject`*.*

Proof. This is the canonical Appendix F claim-use embedding serialized at the invocation site. □

Remark 7.87 (Reference is not proof)**.** A reference identifies a source and permitted use. It neither proves the source nor discharges its hypotheses.

7.17 Artifacts, integrity, and evidence roles

Definition 7.88 (Artifact)**.** An artifact is an immutable or versioned evidential object supporting one or more manifest fields, including source data, barcodes, profiles, filtered representatives, realization records, comparison morphisms, defect profiles, ledgers, simplex records, formal proofs, computation traces, or governance snapshots.

Definition 7.89 (Artifact reference)**.** An artifact reference stores ArtifactID, URI or storage locator, media/type identifier, version, integrity algorithm, digest or equivalent immutable identity, producer, creation provenance, and supported field roles.

Definition 7.90 (Artifact reference policy)**.** The artifact reference policy declares which artifacts are required for verification, recomputation, explanatory context, or external transport; which fields they support; and how identity and integrity are checked.

Definition 7.91 (Evidence role)**.** Every evidence reference has a declared role such as theorem proof, hypothesis discharge, metric upper bound, artifact existence, identity attestation, deterministic computation, provenance, or explanatory context. Explanatory context cannot satisfy a proof role.

Remark 7.92 (Artifact references are not themselves theorems)**.** Integrity and provenance establish which artifact was used; they do not establish that the artifact proves the claimed mathematics.

Proposition 7.93 (Artifact references are audit-critical)**.** *A required evidence claim lacking an immutable artifact reference or an equivalent self-contained proof object cannot be independently reconstructed.*

Proof. The verifier cannot identify or inspect the alleged evidence. The relevant obligation remains undefined. □

7.18 Repair records and immutable verdict history

Definition 7.94 (Repair record)**.** A repair record stores predecessor run and manifest identifiers, changed fields, changed evidence, reason, responsible schema/theory versions, affected obligations, re-evaluated dependency closure, and new verdict.

Definition 7.95 (Revision relation)**.** The revision relation is a directed acyclic predecessor relation between immutable run records. It is not an in-place overwrite operation.

Proposition 7.96 (No retroactive verdict mutation)**.** *A repair may change the verdict of a new revision but cannot alter the stored status or evidence semantics of its predecessor.*

Proof. The predecessor is identified by immutable run, manifest, schema, and artifact identities. Changing its contents would destroy audit history and violate Appendix F repair semantics. □

7.19 Completeness, consistency, replayability, and audit readability

Definition 7.97 (Audit-readable certificate record)**.** A certificate record is *audit-readable* if an independent verifier can identify the run, reconstruct the required registry and dependency closure, inspect all required evidence, recompute deterministic records, reproduce atomic statuses and the aggregate verdict, and verify every failure route and repair link.

Definition 7.98 (Audit reconstruction)**.** Audit reconstruction is the replay procedure that, under the registered theory and schema versions, recomputes or checks all deterministic profile, ledger, diagnostic, status-normalization, aggregation, and conformance steps used by the run.

Definition 7.99 (Manifest completeness)**.** A manifest is complete if every conditionally required field and required atom is present, typed, versioned, and linked to reconstructible evidence, and every omitted conditional field has a valid scope-exclusion record.

Definition 7.100 (Manifest consistency)**.** A manifest is consistent if shared identifiers have one meaning, duplicate field assignments agree, dependency ranks are well founded, status normalizations agree with source records, artifact identities match, and no field contradicts immutable scope or version boundaries.

Definition 7.101 (Replayability)**.** A manifest is replayable if an independent verifier using only declared immutable artifacts, generators, and registered versions can reconstruct every deterministic verdict computation and identify every non-deterministic or theorem-dependent step with its evidence.

Proposition 7.102 (Minimal Manifest implies audit readability)**.** *If a manifest satisfies the Minimal Manifest Axiom and is complete, consistent, conformant, and replayable, then it is audit-readable. This conclusion does not imply that its aggregate verdict is pass.*

Proof. The hypotheses provide all typed fields, identities, dependencies, evidence, and replay instructions required for reconstruction. The reconstructed statuses may still contain rejection. □

Proposition 7.103 (Audit readability enables proposer-independent evaluation)**.** *An audit-readable record can be evaluated by a verifier that did not participate in proposing or producing the original candidate, provided the verifier implements the registered theory and schema semantics and has access to the referenced immutable artifacts.*

Proof. Under these hypotheses, the verdict depends on immutable Core inputs, typed evidence, registered semantics, and artifact access, not on proposer identity. □

Remark 7.104 (Relation to Core sovereignty)**.** Replayability supports but does not replace mathematical validity. Search-side confidence, proposal history, or AI ranking is not a verdict field.

7.20 Proposer–verifier separation

Definition 7.105 (Proposer-side data)**.** Proposer-side data include candidate objects, windows, thresholds, maps, rankings, terrain models, validity scores, bridge programs, generated proof sketches, and AI recommendations.

Definition 7.106 (Verifier-side data)**.** Verifier-side data are the typed Core objects, obligation atoms, evidence, artifacts, policies, ledgers, diagnostics, references, statuses, dependencies, and repair records required by the registered audit semantics.

Proposition 7.107 (Only verifier-side data determine Core verdicts)**.** *Proposer-side data may suggest or locate evidence, but cannot alter a Core verdict until normalized into the required Core types, registered, evidenced, and re-evaluated by the verifier.*

Proof. This is Core sovereignty: proposal authority and verdict authority are distinct. □

Remark 7.108 (Recording proposer data)**.** Optional proposer provenance must be tagged as non-verdict data and may not fill a missing verifier field.

7.21 Abstract schema and concrete execution encodings

Definition 7.109 (Abstract certificate schema)**.** The canonical logical schema is

$$\begin{aligned}C = \big(&\mathsf{Identity}, \mathsf{Scope}, \mathsf{Object}, \mathsf{Window}, \mathsf{Threshold}, \mathsf{Collapse}, \mathsf{Evaluation},\\ &\mathsf{Registry}, \mathsf{Dependency}, \mathsf{Evidence}, \mathsf{Representative}, \mathsf{Eligibility}, \mathsf{Diagnostic}, \mathsf{Comparison},\\ &\mathsf{OverlapRestartAtlas}, \mathsf{Ledger}, \mathsf{Reference}, \mathsf{Artifact}, \mathsf{AtomicStatus},\\ &\mathsf{AggregateVerdict}, \mathsf{FailureRoutes}, \mathsf{Repair}, \mathsf{Replay}\big).\end{aligned}$$

Every component is typed, versioned, and conditionally required according to the declared scope.

Remark 7.110 (Schema is abstract but canonical)**.** The tuple fixes logical content and field roles, not punctuation or storage syntax. A JSON object, relational schema, proof-assistant structure, or other encoding may refine it only through a declared conformance map.

Definition 7.111 (Execution schema)**.** An execution schema is a concrete encoding with a total decoding/conformance procedure into the canonical logical schema, plus validation rules for requiredness, types, identities, versions, and integrity references.

Remark 7.112 (Appendix G vs. execution companion)**.** Appendix G fixes what a conformant record means. The Execution/Reproducibility Companion may fix concrete field names, file layouts, APIs, signatures, or replay commands, but may not weaken the logical requirements.

Proposition 7.113 (Concrete execution schemas must refine the abstract schema)**.** *A concrete schema is Core-admissible only if decoding preserves identities, types, statuses, dependencies, evidence roles, routes, repairs, and aggregate semantics.*

Proof. Otherwise two non-equivalent audit records could decode to the same canonical record or one canonical field could be lost, violating semantic-preserving serialization. □

7.22 Schema conformance and validation profiles

Definition 7.114 (Schema conformance)**.** A record is schema-conformant if it passes structural type validation, conditional-requiredness validation, identifier uniqueness, version compatibility, dependency well-foundedness, status normalization, artifact integrity, and cross-field consistency checks.

Definition 7.115 (Validation profile)**.** A validation profile is a named set of required field families for a declared scope, such as `topological_only`, `eligible_ext`, `typeIV`, `overlap_atlas`, `restart`, or `full_foundation`. Profiles control conditional requiredness only; they do not change field meanings.

Proposition 7.116 (Canonical manifest conformance)**.** *A chapter-local or appendix-local fragment is admissible only if it embeds into one canonical manifest without changing shared field meanings, identifiers, statuses, version boundaries, or artifact identities.*

Proof. The canonical manifest represents one immutable run. Incompatible fragments produce manifest inconsistency. □

Definition 7.117 (Status field, legacy label retained)**.** The historical phrase "final status field", introduced in v19 and inherited unchanged in v20, is interpreted as the pair

$$(\mathsf{AtomicStatusRegistry}, \mathsf{AggregateVerdict})$$

together with a separate claim/use classification field. No mixed enumeration of pass/reject with operational, interface, toy, or spec layers is canonical.

Proposition 7.118 (Schema sufficiency for Core audit)**.** *A conformant, complete, consistent, and replayable record is sufficient to reconstruct a Core audit verdict, but not necessarily sufficient to obtain a pass.*

Proof. The record exposes all inputs and rules required for reconstruction. Reconstruction may yield reject or undefined. □

7.23 Incomplete, inconsistent, and failed records

Definition 7.119 (Incomplete certificate record)**.** A record is incomplete when at least one required field, atom, dependency, evidence reference, coverage certificate, or justified scope exclusion is missing or not reconstructible.

Definition 7.120 (Failed certificate record)**.** A record is failed when enough data are present to evidence at least one required atom as false, giving canonical status `reject`. A failed record may simultaneously contain other undefined atoms.

Definition 7.121 (Inconsistent certificate record)**.** A record is inconsistent when shared identifiers, immutable scope, versions, artifact identities, dependency ranks, source statuses, or duplicate field assignments contradict one another.

Remark 7.122 (Incomplete vs. failed)**.** Incomplete means audit undefinedness for the affected clause; failed means evidenced rejection. Inconsistency is a Type V rejection. The aggregate headline follows Appendix F precedence while retaining all details.

Proposition 7.123 (Failure is preferable to incompleteness)**.** *For diagnosis and repair, an audit-readable failed record is more informative than an incomplete record because it exposes a reconstructible counterevidence path. This does not make rejection normatively preferable to pass.*

Proof. A failed record identifies the atom, evidence, and route. An incomplete record lacks at least one required reconstructive component. □

7.24 Appendix G manifest package

Theorem 7.124 (Appendix G manifest package)**.** *Within the v20 Core audit regime, preserving the inherited v19 semantics:*

(i) Appendix F fixes verdict semantics and Appendix G serializes them without mutation;

(ii) immutable run, manifest, schema, theory, artifact, and revision identities delimit one audit meaning;

(iii) the Minimal Manifest includes the full required registry, dependency closure, evidence, artifacts, statuses, routes, repairs, and replay data;

(iv) canonical atomic statuses are exactly `pass`, `reject`, `undefined`, and justified `not_invoked`;

(v) claim/use classification is separate from atomic status;

(vi) scalar ledgers contain only metric-bearing discrepancies and cannot encode missing non-scalar evidence;

(vii) Type IV zero requires a valid actual comparison artifact and terminal-generic certificates;

(viii) overlap, higher coherence, Restart, target-local pass, and atlas records remain distinct obligations;

(ix) a complete manifest may record rejection, and manifest completeness never implies theorem truth;

(x) countable registries require coverage and universal-status certificates;

(xi) repairs create new immutable revisions rather than mutating historical verdicts;

(xii) conformance, completeness, consistency, and replayability yield audit readability but not automatic pass;

(xiii) proposer data cannot alter verdicts without Core re-entry;

(xiv) concrete execution schemas must refine the canonical logical schema.

Proof. The items collect the definitions and propositions of this appendix together with Appendix F's authoritative status, dependency, aggregation, readiness, and repair semantics. □

7.25 v20 token-sort and typed-evidence serialization

Definition 7.125 (Token-sort registry record)**.** The manifest stores one immutable token-sort registry containing:

(i) TokenSortID, registry version, owning theory release, and digest;

(ii) every token literal, semantic sort, governing field family, and admissible source types;

(iii) the canonical audit-status sort, runtime comparison-mode sort, detector-stage sort, detector-certificate sort, detector-value sort, terminal Type IV sort, transient-verdict sort, source-semantics sort, simplex-mode sort, Restart-transition sort, Restart-zero-route sort, manifest-status sort, claim-status sort, and invocation-role sort;

(iv) every permitted embedding into atomic audit status;

(v) every prohibited cross-sort conversion.

Every token occurrence in a consumed record stores both its literal and its TokenSortID-qualified semantic sort.

Definition 7.126 (Typed evidence record)**.** For every evidence-bearing atom, the manifest stores one typed evidence record:

$$\begin{aligned}&(\,\text{EvidenceID}, \text{ObligationID}, \text{evidence type}, \text{source type}, \text{target type},\\ &\ \text{governing theorem/policy}, \text{hypotheses}, \text{transformations}, \text{token sorts},\\ &\ \text{artifacts}, \text{versions}, \text{digests}, \text{specialized status}, \text{atomic status}\,).\end{aligned}$$

The record distinguishes theorem proof, hypothesis discharge, exact isomorphism, metric upper bound, detector certificate, coverage certificate, artifact identity, deterministic computation, provenance, and explanatory context.

Proposition 7.127 (Cross-sort inconsistency is detectable)**.** *A record assigning a token to a field of the wrong semantic sort is schema-inconsistent. If the mismatch is unresolved, the affected obligation is* `undefined`*; if the manifest asserts the mismatched assignment as a valid verdict or conceals the mismatch, the manifest is* `inconsistent` *and routes to Type V rejection.*

Proof. The token registry and field schema determine the unique admissible source sort. An incompatible assignment cannot decode to the canonical Appendix F semantics. □

7.26 v20 detector and transient records

Definition 7.128 (Detector policy record)**.** A detector policy record stores:

(i) detector identifier, mode, information class, profile stage, family, endpoint convention, and monitored scope;

(ii) exact source profile identity and source artifact;

(iii) witness set, stencil, birth, right-germ, or canonical family data;

(iv) soundness, applicability, completeness, and sharpness references;

(v) metric-safe transport dependencies when the detector conclusion is transferred across a comparison or hard deletion;

(vi) detector-certificate status and detector mathematical value as separate fields;

(vii) canonical atomic status, blockers, routes, and immutable artifacts.

Definition 7.129 (Detector certificate record)**.** A detector certificate record serializes

$$(\text{CertificateStatus}, \text{DetectorValue})$$

without identifying the two axes. A required zero atom may pass only when the certificate status is `detector_certified`, the value is `zero`, and every applicability and completeness dependency passes. A sampled zero with pending or failed certification is never serialized as a zero proof.

Definition 7.130 (Transient-defect record)**.** A transient-defect record stores:

(i) window, collapse threshold, defect threshold, monitored degrees, and detector component;

(ii) kernel and cokernel defect-profile identities;

(iii) complete detector policies and values for both profiles;

(iv) specialized aggregate status

$$\texttt{not\_invoked}, \quad \texttt{undefined}, \quad \texttt{certified\_absent}, \quad \texttt{certified\_present};$$

(v) canonical atomic status, witness bars when present, blockers, theorem locators, and artifacts.

Terminal Type IV status is stored separately.

Definition 7.131 (Terminal/transient participation record)**.** Every local, overlap, simplex, Restart-target, atlas, and global-predicate record stores:

(i) terminal Type IV invocation and specialized status;

(ii) whether long-transient absence is required;

(iii) the exact transient clause identity when required;

(iv) independent canonical statuses and artifact references.

Terminal zero is not serialized as transient absence or full isomorphism.

7.27 v20 exact/metric coherence and coverage records

Definition 7.132 (v20 overlap comparison record)**.** A v20 overlap record stores separately:

(i) atomic Overlap Gate status;

(ii) runtime comparison mode: `exact`, `metric_gap_certified`, or `not_compared`;

(iii) direct common-window source profiles and reconciled threshold;

(iv) in exact mode, the actual repaired-profile isomorphism and proof;

(v) in metric mode, the pre-threshold bottleneck bound, both gap certificates, scalarization, and post-threshold bound;

(vi) terminal/transient participation and all artifacts.

The comparison mode never substitutes for the atomic status.

Definition 7.133 (v20 simplex-coherence record)**.** A v20 simplex record declares exactly one mode:

$$\texttt{exact\_cocycle}, \qquad \texttt{metric\_reference}, \qquad \texttt{not\_compared}.$$

In exact-cocycle mode it stores all transition isomorphisms, identities, inverses, cocycle equations, and face restrictions. In metric-reference mode it stores the reference profile, local-to-reference bounds, two-sided gap certificates, reference face-compatibility isomorphisms, and pairwise bounds. Metric mode does not serialize transition morphisms or a categorical cocycle.

Definition 7.134 (Coverage certificate record)**.** Every coverage record contains CoverageID, coverage type, governing scope, universal quantifier, witness generator or proof, theorem/policy reference, artifacts, status, and dependencies. The coverage type is exactly one of:

(i) `registry_coverage`;

(ii) `retained_bar_coverage`;

(iii) `boundary_bridge_coverage`;

(iv) `artifact_closure`;

(v) `release_document_coverage`.

A record of one type cannot discharge another.

Definition 7.135 (Global profile interface record)**.** A global profile interface record stores:

(i) global pre-threshold profile, covered range, degree, threshold, and profile artifact;

(ii) every exact local restriction profile;

(iii) every observed local profile;

(iv) for each local profile, either an exact isomorphism branch or a metric-safe zero-transport branch;

(v) every two-sided gap and $2\varepsilon \leq \tau$ certificate used;

(vi) local detector and retained-bar coverage references.

Pairwise local compatibility does not construct this record.

Definition 7.136 (Detector-atlas record)**.** A detector-atlas record serializes:

(i) the global profile interface;

(ii) every local detector policy and certificate;

(iii) every consumed overlap and simplex record;

(iv) retained-bar and boundary-bridge coverage records;

(v) terminal/transient participation;

(vi) local and global zero atoms;

(vii) atlas readiness, global-zero readiness, blockers, routes, and artifacts.

Local zero values without complete detector and coverage records do not support a global-zero verdict.

Example 7.137 (Manifest distinction for local zero without global coverage)**.** A record may contain certified zero detector values for every listed local window and exact zero overlap profiles while retaining-bar coverage is `undefined` or `reject`. The local clauses may pass, but the global repaired-zero atom remains `undefined` or rejects according to the evidenced coverage clause. The manifest must retain both facts rather than compressing them to "global zero."

7.28 v20 Restart, reduction, and descent records

Definition 7.138 (Restart zero-route record)**.** When a Restart claim consumes target repaired zero, the transition record additionally stores:

(i) zero-route status:

$$\texttt{not\_invoked}, \quad \texttt{undefined}, \quad \texttt{certified}, \quad \texttt{failed};$$

(ii) route kind: fresh complete detector, exact zero transport, or metric-safe zero transport;

(iii) source and target zero atoms;

(iv) profile stages, thresholds, endpoint conventions, comparison mode, theorem/policy, and artifacts;

(v) boundary-bridge coverage when a global predicate is claimed.

The κ_k-reserve substatus and zero-route substatus remain independent.

Definition 7.139 (Finite-to-infinite reduction record)**.** A finite-to-infinite reduction record stores:

(i) the actual directed system, transition morphisms, and source semantics;

(ii) stabilization index or finite witness;

(iii) genuine colimit or theorem-authorized source object;

(iv) compatible apex cocone, induced comparison morphism, and apex agreement;

(v) every reduction-theorem hypothesis;

(vi) terminal, transient, or full-isomorphism consequences actually consumed;

(vii) claim-use, artifact closure, replay, and verdict data.

A finite observed plateau or repeated barcode is serialized only as proposer or empirical context unless a reduction theorem is proved.

Definition 7.140 (Object-descent record)**.** An object-descent record stores:

(i) category or higher category and restriction functors;

(ii) actual local objects and transition equivalences;

(iii) exact cocycle or higher coherence;

(iv) effective-descent theorem and discharged hypotheses;

(v) constructed descended object and comparison artifacts;

(vi) independent certificate-atlas, readiness, and manifest clauses.

A metric-reference atlas or global-zero predicate is not serialized as object descent.

7.29 Artifact closure and semantic replay

Definition 7.141 (Artifact-closure record)**.** An artifact-closure record contains:

(i) the artifact-root identifier and complete reachable artifact graph;

(ii) for each node, type, version, producer, provenance, locator, integrity algorithm, digest, and supported evidence roles;

(iii) every transitive interpretation dependency not supplied by the registered theory release;

(iv) uniqueness and non-dangling checks;

(v) ambiguity, collision, and silent-replacement checks;

(vi) closure status and replay instructions.

Artifact closure identifies the complete evidence graph; it does not prove the mathematics stored in that graph.

Definition 7.142 (Semantic replay record)**.** A semantic replay record contains ReplayID, verifier implementation and version, token-sort registry, claim-register snapshot, schema version, theory version, artifact closure, replay environment, and commands or proof-checking procedure sufficient to reconstruct:

(i) deterministic profiles, detectors, ledgers, diagnostics, and dependency graphs;

(ii) source types, governing claims, theorem hypotheses, specialization maps, and invocation roles;

(iii) specialized-status embeddings and canonical atomic statuses;

(iv) normalization, aggregation, blockers, routes, and bundle verdicts;

(v) every non-deterministic or theorem-dependent step with its evidence.

Computational replay is recorded as a subfield; semantic replay is the stronger Core requirement for consumed machine-produced verdicts.

Proposition 7.143 (Semantic replay is stronger than byte reproduction)**.** *Reproducing files or numerical outputs without reconstructing their source types, theorem hypotheses, token sorts, dependencies, and status embeddings does not establish semantic replayability.*

Proof. Identical bytes may be interpreted under different schemas, theorem versions, or claim statuses. Semantic identity therefore requires the registered interpretation data in addition to file reproduction. ◻

Definition 7.144 (Replay discrepancy record)**.** If two replays disagree, the manifest stores both replay identities, immutable inputs, first divergent atom or deterministic step, candidate causes, affected dependency closure, and revised verdict. Until resolved, the affected replayability atom is `undefined`; an evidenced contradiction under identical declared semantics makes the manifest `inconsistent`.

7.30 Lossless schema migration

Definition 7.145 (Schema migration record)**.** A migration record contains:

(i) MigrationID, source and target SchemaID, theory releases, and digests;

(ii) a total map for every consumed field and token sort;

(iii) requiredness, scope-exclusion, status, dependency, artifact, route, and repair-history preservation;

(iv) explicit defaults only for newly introduced fields proved outside the old claim scope;

(v) semantic-difference evidence;

(vi) source and target replay records;

(vii) a proof that the old supported verdict is reconstructible at exactly its old strength.

A theorem-strength, evidence, scope, or claim-use change is a new run or revision, not a schema-only migration.

Proposition 7.146 (Lossy migration is non-conformant)**.** *A migration dropping a required hypothesis, dependency, token sort, lower-level status, missing-artifact marker, scope boundary, blocker, route, or forbidden stronger reading is non-conformant. The target record is incomplete or inconsistent until repaired as a new evidence-backed revision.*

Proof. The dropped data participate in clause meaning or verdict reconstruction, so the decoding cannot preserve Appendix F semantics. □

7.31 Release manifest and acceptance

Definition 7.147 (Release manifest)**.** A release manifest is the typed record

$$\mathfrak{M}_{\mathrm{rel}} = (\mathsf{ReleaseID}, \mathsf{Documents}, \mathsf{Claims}, \mathsf{Schemas}, \mathsf{Artifacts}, \mathsf{Supersession}, \mathsf{Replay}, \mathsf{Acceptance}).$$

For every mandatory document role it stores DocumentID, title, role, version, digest, schema, dependencies, advertised scope, supersession status, and artifact closure.

Definition 7.148 (Mandatory release-document roles)**.** A complete v20 release registry declares the mandatory roles applicable to its advertised scope, including:

(i) Main;

(ii) Foundation Appendices;

(iii) Claim Register;

(iv) Core micro-theorems and toy bridges when advertised;

(v) problem-interface or search-platform appendices when included;

(vi) canonical manifest and execution/reproducibility artifacts;

(vii) release notes, supersession map, and integrity index.

A role may be excluded only by a release-level scope decision consistent with the advertised product.

Definition 7.149 (Release-acceptance record)**.** A release-acceptance record stores:

(i) release-document coverage;

(ii) canonical Claim Register and supersession closure;

(iii) schema conformance and migration closure;

(iv) artifact closure and semantic replay;

(v) every mandatory Foundation readiness verdict;

(vi) all unresolved rejected and undefined mandatory atoms;

(vii) headline acceptance status.

The release passes exactly when every mandatory clause passes. A strong mathematical theorem does not compensate for an incomplete release registry or inconsistent manifest.

Definition 7.150 (Supersession record)**.** A supersession record identifies predecessor and successor documents, claims, schemas, and artifacts; classifies each relation as replacement, refinement, deprecation, withdrawal, or coexistence; states preserved and changed semantics; and prevents an obsolete claim from silently re-entering an active release.

Proposition 7.151 (Document completeness and theorem strength are independent)**.** *A release with a proved theorem but missing a mandatory manifest, claim snapshot, replay artifact, or supersession record does not pass release acceptance. Conversely, a complete release manifest may faithfully record a mathematically rejected or incomplete program.*

Proof. Release acceptance and mathematical clause status are distinct required atoms combined by non-compensatory aggregation. □

7.32 v20 canonical schema and validation profiles

Definition 7.152 (v20 canonical manifest object)**.** The v20 canonical manifest extends Definition 7.20 to

$$\begin{aligned}\mathsf{Manifest}_{20} = \big(&\mathsf{Identity}, \mathsf{TokenSorts}, \mathsf{Scope}, \mathsf{Objects}, \mathsf{Windows}, \mathsf{Policies},\\ &\mathsf{Registry}, \mathsf{Dependencies}, \mathsf{TypedEvidence}, \mathsf{Artifacts}, \mathsf{ArtifactClosure},\\ &\mathsf{Ledger}, \mathsf{Evaluations}, \mathsf{Detectors}, \mathsf{Representatives}, \mathsf{Eligibility},\\ &\mathsf{TerminalTransient}, \mathsf{OverlapSimplex}, \mathsf{Coverage}, \mathsf{Restart},\\ &\mathsf{Reduction}, \mathsf{Descent}, \mathsf{References}, \mathsf{Verdicts}, \mathsf{Routes},\\ &\mathsf{Repairs}, \mathsf{Migration}, \mathsf{SemanticReplay}, \mathsf{Release}\big).\end{aligned}$$

Every field family remains conditionally required under immutable scope.

Definition 7.153 (v20 validation profiles)**.** In addition to inherited profiles, v20 provides:

`detector`, `detector_atlas`, `global_zero`, `transient_clean`,
`restart_zero`, `finite_to_infinite`, `object_descent`,
`semantic_replay`, `release_acceptance`, `full_v20_foundation`.

A profile selects conditionally required fields; it never changes their meaning or weakens dependencies.

Proposition 7.154 (v20 schema sufficiency for audit reconstruction)**.** *A v20 record that is token-sort valid, conformant, complete, consistent, artifact-closed, semantically replayable, and dependency-complete is sufficient to reconstruct every declared v20 Foundation verdict. It is not sufficient by itself to make those verdicts pass.*

Proof. The hypotheses expose the complete typed evidence and deterministic semantics needed for reconstruction. The reconstructed atoms may still reject or remain undefined. □

Theorem 7.155 (Appendix G v20 manifest and replay package)**.** *Within the v20 conservative extension:*

(i) the complete inherited v19 manifest package remains valid at its original strength;

(ii) token-sort and typed-evidence records prevent cross-type verdict substitution;

(iii) detector certificate status and detector mathematical value are serialized independently;

(iv) terminal Type IV and long-transient records remain distinct;

(v) exact-cocycle and metric-reference coherence remain distinct;

(vi) registry, retained-bar, boundary-bridge, artifact, and release coverage have separate record types;

(vii) detector-atlas and global-zero records expose every local-to-global dependency;

(viii) Restart reserve and target-zero routes are independently serialized;

(ix) finite-to-infinite reduction and object descent require actual mathematical artifacts;

(x) artifact closure and semantic replay expose the complete interpretation graph of machine-produced verdicts;

(xi) schema migration is admissible only when semantic and verdict reconstruction are lossless at the inherited strength;

(xii) release acceptance is a separate non-compensatory conjunction;

(xiii) the v20 canonical schema is sufficient for audit reconstruction but never creates theorem truth.

Proof. Item (i) is Theorem 7.124. Items (ii)–(iv) are Definitions 7.125–7.131. Items (v)–(vii) follow from Definitions 7.132–7.136. Items (viii)–(ix) are Definitions 7.138–7.140. Item (x) follows from Definitions 7.141 and 7.142. Item (xi) is Definition 7.145 and Proposition 7.146. Item (xii) follows from Definitions 7.147–7.149. Item (xiii) is Definition 7.152 and Proposition 7.154. □

7.33 Explicit exclusions

The following are not asserted in Appendix G.

- No concrete JSON, YAML, CSV, database, API, signature, or file layout is mandated.
- No schema field creates mathematical evidence.
- No complete manifest is automatically a passing proof.
- No passing mathematical atom repairs an incomplete or inconsistent manifest atom.

- No numerical budget contains claim-status, representative-existence, morphism-existence, eligibility, coherence, or manifest obligations.
- No omitted field receives a theorem-producing default.
- No `undefined` Type IV record is serialized as $(0, 0)$.
- No `not_invoked` value is accepted without immutable scope-exclusion evidence.
- No claim/use classification is conflated with atomic audit status.
- No headline verdict erases lower-level undefined, reject, blocker, or failure-route detail.
- No finite prefix certifies a countable registry.
- No repair mutates a historical run in place.
- No pairwise overlap family automatically supplies higher Čech coherence.
- No certified transition alone supplies a complete Restart pass.
- No certificate atlas implies object-level descent or a global mathematical object.
- No Search, Platform, Companion, or AI output updates a verdict without Core re-entry.
- No Claim Register entry is proof merely because it is registered.
- No micro-theorem reference discharges its own hypotheses.
- No toy bridge is promoted to an external bridge theorem by serialization.
- No global exactness of positive-threshold collapse or strict filtered lift is assumed.
- No tower comparison artifact is automatic.
- No token literal is interpreted without its semantic sort.
- No detector mathematical value substitutes for detector certification or atomic status.
- No terminal Type IV zero is serialized as long-transient absence or full isomorphism.
- No local detector-zero family is serialized as global zero without a global profile interface and retained-bar coverage.
- No registry coverage substitutes for retained-bar, boundary-bridge, artifact, or release coverage.
- No metric-reference simplex record creates transition morphisms or a categorical cocycle.
- No κ-Restart reserve pass certifies a target-zero route.
- No observed finite plateau is serialized as eventual constancy or finite-to-infinite reduction.
- No certificate atlas or global-zero predicate is serialized as object descent.
- No artifact hash alone proves the mathematical role of the artifact.
- No computational replay alone establishes semantic replay.

- No lossy schema migration preserves an audit verdict at the same claimed strength.
- No mathematically strong document compensates for missing mandatory release records.
- No proposer, Search, Platform, Companion, or AI field has verifier status without typed Core re-entry.

7.34 Provenance and status

This appendix consumes the complete inherited v19 record architecture and the v20 conservative extensions of Appendices A–F. In particular, it serializes detector stages and certificates, terminal/transient separation, exact/metric atlas modes, retained-bar and boundary-bridge coverage, Restart zero routes, finite-to-infinite reduction, object-descent boundaries, artifact closure, semantic replay, and release acceptance.

This appendix consumes:

(i) Chapter 1 for the foundational Minimal Manifest requirement;

(ii) Chapters 2–3 and Appendices A–B for evaluation, threshold, and representative records;

(iii) Chapter 4 and Appendix C for eligible Ext records;

(iv) Chapter 5 and Appendix D for terminal-generic Type IV records;

(v) Chapter 6 and Appendix E for overlap, Čech, Restart, and atlas records;

(vi) Chapter 7 and Appendix F for canonical typed audit semantics, dependency closure, aggregation, readiness, non-compensation, and repair history;

(vii) Appendix CR-v20, Appendix CM-v20, and Appendix TB-v20 for governed downstream references.

Status labels for Appendix G. *Normative schema definitions and axiom:* Definitions 7.6, 7.9, 7.10, 7.13, 7.14, 7.20, 7.26, 7.31, 7.34, 7.109, and Axiom 7.1.

Contract consequences: Propositions 7.7, 7.12, 7.23, 7.30, 7.35, 7.75, 7.96, 7.102, 7.116, 7.118, and Theorem 7.124.

v20 manifest and replay extension: Definitions 7.125, 7.126, 7.128–7.131, 7.132–7.136, 7.138–7.140, 7.141, 7.142, 7.144, 7.145, 7.147–7.150, 7.152, 7.153; Propositions 7.127, 7.143, 7.146, 7.151, 7.154; Theorem 7.155.

Operational cautions and exclusions: Remarks 7.1, 7.2, 7.8, 7.16, 7.19, 7.55, 7.69, 7.80, 7.87, 7.92, 7.104, and Subsection 7.33.